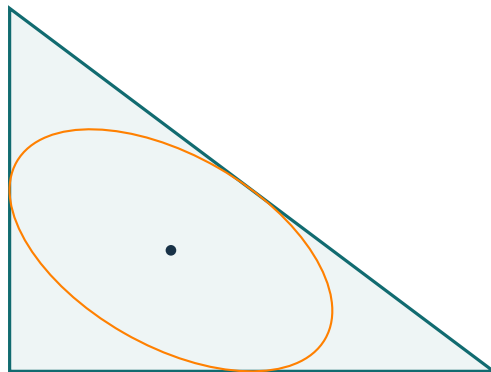


A MONOGRAPH OF GEOMETRIC EXPERIMENTS

Triangle Centers Inspired by Physics

From physical laws to central points and conics



Six physical studies with a common mathematical framework

Mass and charge · magnetostatics · elasticity
Quantum and random motion · diffusion and reaction · optics

Consolidated research edition
3 October 2026

Preface

This book collects the six dedicated physics chapters into one continuously numbered monograph. It assumes undergraduate vectors, calculus, linear algebra and elementary field physics. Boundary-value equations and numerical methods are derived as needed. The introduction connects their common definitions; the appendices collect proofs and computation conventions. The individual atlas entries preserve the detailed location, ETC and attractivity investigations rather than replacing them with a shorter lecture summary.

The physical inventory includes successful constructions, parameter families and experiments that fail to select a unique center. A negative result can be as informative as a new named point. Stable numerical evidence and interval-certified assertions have distinct labels. In particular, the later E0 certificate corrects older positive-candidate wording.

The evidence snapshot comprises the September 2026 chapter studies and their 30 September addendum. The conic-axis appendix incorporates the subsequent analytical work of 3 October. This consolidation performs an editorial, cross-reference and typesetting audit, not a new full-ETC or continuum certification campaign.

Notation and reading conventions

Symbol	Meaning
$T, \partial T$	Filled triangle and its perimeter
$A, B, C; a, b, c$	Vertices and opposite side lengths
$\mathcal{A}, P, L, D_T$	Area, perimeter, dimensionless-family length scale, diameter
I, G, W	Incenter X1, area centroid X2, wire centroid X10
w_i	Normalized barycentric observation coordinates
Σ, Q	Covariance matrix; ellipse shape matrix with squared-semiaxis eigenvalues
$J_i, \text{sym } J_i$	Vertex Jacobian and its symmetric part
Δ	Usually Laplacian in a PDE; explicitly identified area in some inherited formulas
$\mu, \eta, \tau, F, \beta$	Dimensionless family parameters; definitions belong to the model

Inherited chapters use different local symbols for area and angles. Their definitions are preserved to avoid changing equations silently. Parameter labels such as Abs0.0 and Mem3.0 are machine-readable atlas identifiers. Only frozen-snapshot ETC ranks are asserted. Earlier bundle paths refer to the project research workspace; the source companion's README gives the portable typesetting paths. Figure numbers inherited from the individual studies are local to their chapter; their captions and atlas identifiers preserve the research records.

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Chapter 1

Physical experiments as a language for triangle geometry

A triangle has no single physical center. A uniformly massive sheet balances at its area centroid; a uniform wire frame balances at its perimeter centroid. A stretched membrane reaches its greatest deflection elsewhere. A confined quantum particle has a most likely position and a mean position, and those observations need not agree. Each experiment is a question addressed to the same geometry.

This monograph develops six substantial studies of those questions. The route begins with mass, charge and moments, proceeds through steady currents and elasticity, and then reaches quantum and random motion, diffusion, reaction and optics. The physical setting is elementary; the resulting mathematical problems range from quadratic minimization to singular integral equations, conformal mapping, eigenvalue perturbation and nonlinear partial differential equations. An especially useful comparison is between two experiments that share an operator but observe different aspects of its solution.

1.1 What is a triangle center?

Let $T = \text{conv}\{A, B, C\}$ be a nondegenerate triangle. An intrinsic triangle-center rule $\mathcal{C}$ must respect similarities and vertex relabeling:

$$\mathcal{C}(sRT + t) = sR\mathcal{C}(T) + t, \quad s > 0, \quad R^T R = \text{Id}.$$

Here R may be a reflection. Producing a point from three coordinates is insufficient: a dependence on vertex numbering, a laboratory direction or an arbitrary eigenbasis can violate this condition. A pair of unordered foci or a whole maximizing set is a useful central object, but not automatically a single center.

Central lines, circles, conics and tensors obey the analogous transformation rule. The Euler line and Steiner ellipses transform with the triangle. A principal-axis pair is meaningful when the associated symmetric matrix has distinct eigenvalues; at an isotropic matrix the direction is undefined. An arbitrarily horizontal axis at an equilateral triangle does not come from its geometry.

The classical reference points are $I = X_1$, $G = X_2$ and $W = X_{10}$. The common numerical teaching example is

$$A = (0, 0), \quad B = (4, 0), \quad C = (0, 3), \quad I = (1, 1), \quad G = (4/3, 1), \quad W = (3/2, 1).$$

An acute 4–5–6 triangle and an obtuse 6–9–13 triangle appear in comparison panels. The latter also serves a different purpose in some retained counterexamples; it is not interchangeable with the teaching triangle.

1.2 Barycentric coordinates and the ETC

Put $a = |BC|$, $b = |CA|$, $c = |AB|$. A point p has normalized barycentric coordinates

$$p = w_A A + w_B B + w_C C, \quad w_A + w_B + w_C = 1.$$

Each $w_i(p)$ is affine. For an interior point its coordinates are positive and equal the opposite subtriangle areas divided by the full area. Negative weights describe exterior points.

A usual barycentric center function is written

$$\mathcal{C}(T) = f(a, b, c) : f(b, c, a) : f(c, a, b).$$

The colon notation means that the weights are normalized by their sum. Appropriate symmetry and homogeneity are needed; the cyclic expression alone does not guarantee a well-defined, reflection-equivariant center on every shape.

The *Encyclopedia of Triangle Centers* (ETC) gives numbered rules, relations and source information [1]. This project compares against a frozen X1–X72807 snapshot. A number is a catalogue identifier, not a physical energy or an importance ranking. An implicit integral or PDE definition also determines barycentric functions. Absence of a convenient algebraic formula is neither proof of transcendence nor a general ETC exclusion rule. This book uses *implicit*, *spectral* or *PDE-defined* where those are the established facts.

1.3 Four kinds of physical selector

Moment. A nonnegative density gives a normalized first moment. Uniform mass, equilibrium charge, quantum probability and detected optical intensity are examples.

Maximum or minimum. An instrument selects the hottest, deepest, darkest or least-field point. Uniqueness of the field does not imply uniqueness of this point.

Equilibrium. A force balance, stationary potential, variational equation or harmonic-probability condition selects a location.

Controlled limit. A thickness, core radius, time, wavelength-scaled distance or spectral cutoff approaches a specified endpoint. The limiting convention is part of the definition.

The ground-state probability mean and its maximum are different selectors. A singular magnetic weight with divergent numerator and denominator is not a first moment until a regularization has been specified.

Proposition 1.1 (A general location certificate). *The barycenter of a positive probability measure supported on the closed triangle lies in the triangle. It is strictly interior if the measure is not supported on any one side.*

Proof. Affineness gives

$$w_i \left(\int_T p \, d\mu(p) \right) = \int_T w_i(p) \, d\mu(p).$$

The integrand is nonnegative and the integral is positive unless the measure is supported on the side where $w_i = 0$. $\square$

This certificate supplies no numerical distance to the boundary and no vertex-attractivity theorem. Stronger certificates include inner parallel triangles, a harmonic-measure hexagon and eigenvalue-dependent distance bounds.

1.4 Uniformity, dimension and dimensionless parameters

Uniformity must name a measure: area, arclength, volume, surface area or probability. Three equal vertex masses share the lamina centroid but not its covariance. Equal-total-mass rods differ from equal-linear-density rods. A conductor is equipotential; its charge density generally is not uniform.

A triangle embedded in $\mathbb{R}^n$ remains a two-dimensional plate or a one-dimensional skeleton. Changing n changes the Green kernel and capacity threshold; it does not turn the triangle into an n -simplex. In two dimensions the logarithmic kernel has a conformal description. In higher dimensions an exactly thin set may have zero Newtonian capacity. The adopted uniform-thickness limit specifies a meaningful conductor family.

Use $\mathcal{A}$ for area, $P = a + b + c$ for perimeter, and distinguish

$$L^2 = \frac{a^2 + b^2 + c^2}{3}, \quad D_T = \max(a, b, c).$$

Physical families use L to nondimensionalize parameters, while ETC errors use D_T . Examples are $\kappa t/L^2$ for heat, a removal coefficient multiplied by L^2 for reaction, and height divided by L for illumination. Holding a dimensional time or height fixed while resizing generally destroys scale covariance. The stated dimensionless parameter stays fixed during a shape-response test.

1.5 The mathematical sequence behind an experiment

Specify the source, constitutive law, boundary condition, admissible configurations and observable. Derive the field or energy. Establish existence and uniqueness as far as possible. Compute the field, extract the point and check transformation behavior.

For $Au = f$, means involve integrals of u or u^2 , maxima involve stationary points, and compliance involves the diagonal of A^{-1} . These operations differ even with the same A . Torsion, a uniformly loaded tension membrane, fully developed Newtonian duct flow and mean Brownian exit time reduce, up to positive factors, to

$$-\Delta u = 1, \quad u|_{\partial T} = 0.$$

Their maximum is F1; their positive-density mean is Abs(0). A clamped plate instead has a biharmonic operator.

Conformal mapping does not turn every PDE into a constant-coefficient disk problem. A Poisson source and an eigenvalue equation acquire $|F'|^2$ after pullback. The chapters retain this weight in their equivalent formulations.

1.6 Fingerprints and comparison across triangle shapes

No finite fingerprint characterizes an arbitrary function on all shapes. Several complementary probes nevertheless reveal symmetry, singular limits and local sensitivity:

$$T_{\text{iso}}(h) = \{(-1, 0), (1, 0), (0, h)\}, \quad T_{\text{right}}(h) = \{(0, 0), (1, 0), (0, h)\}.$$

Their six diagrams contain the isosceles axial coordinate and derivative, and the two right-triangle coordinates and derivatives. Some archived right-family figures rotate or exchange coordinates; their captions state the convention.

The recent explorer uses the angular abscissa

$$\theta = \frac{180}{\pi} \arctan h, \quad \frac{dc}{d\theta} = \frac{dc}{dh} \frac{\pi}{180} \sec^2 \left(\frac{\pi\theta}{180} \right).$$

The default range is 0.5° – 89.5° . Changing the abscissa is not automatically changing derivative units. Original chapter figures retain their labeled height axes; the explorer supplies angular plots and a unit switch. Saved PITC data are not extrapolated.

The seventh diagram probes

$$T_{\text{thin}}(x, \varepsilon) = \{(-1, 0), (1, 0), (x, \varepsilon)\}, \quad 0 \leq x \leq 1.$$

Also examine the transverse coordinate divided by ε . Finite height is a probe, not a proof of the limit; joint apex-collision limits can differ.

For $C = (x, \sqrt{3} + y)$, smooth equivariant centers have

$$c_x = a_1x + a_2xy + O(r^3), \quad c_y = \frac{\sqrt{3}}{3} + b_1y + b_2x^2 + b_3y^2 + O(r^3), \quad a_1 + b_1 = \frac{2}{3},$$

where $r^2 = x^2 + y^2$. Energy-ordered eigenstates at a doublet need not possess this expansion. Numerical Taylor coefficients and fitted symbolic surrogates remain approximations.

The angle simplex is an equilateral shape diagram with distances to its sides proportional to the angles. Its ordered fundamental region $\vartheta_1 \leq \vartheta_2 \leq \vartheta_3$ occupies one sixth. The project uses 100 equal-area-cell representatives and 40 independent holdouts. Uniformity here differs from uniformity in side lengths or apex coordinates.

The location score for X_n is

$$E_{\text{loc}}(n) = \left(\frac{1}{N} \sum_{k=1}^N \frac{|\mathcal{C}(T_k) - X_n(T_k)|^2}{D_T(T_k)^2} \right)^{1/2}.$$

Derivative, thin-family and vertex-response diagnostics are separate. A location winner is not automatically the best full-behavior match. The restricted importer evaluates 70,853 entries finitely on the complete grid; 1,954 of 72,807 cached entries are excluded. A shortlist-only rerun is not a new whole-ETC search.

1.7 Attractivity is a geometric response property

Move just one vertex by h , holding the other two fixed. The requested property is

$$h \cdot [\mathcal{C}(T + h) - \mathcal{C}(T)] \geq 0.$$

$T + h$ means one specified vertex moved, not a translation. Both triangles and the intervening straight motion remain admissible. For vertex Jacobian J_i , the local condition is

$$\text{sym } J_i = \frac{J_i + J_i^T}{2} \succeq 0.$$

Only this part contributes to $h^T J_i h$. Integrating along an admissible motion gives the finite inequality. Positive derivatives along one special family do not test every direction.

Mechanical stability concerns second variation with the triangle fixed. Vertex attractivity concerns movement of the selected point when the triangle changes. Neither implies the other. A strictly convex objective can select a nonattractive center.

This edition recognizes four studied classical points with global proofs: X1, X2, X10 and X360. Exact aliases inherit them. E0, E1, E2 and M1 have certified failures. Other entries have refined negative evidence, positive screens or unresolved conclusions. M6 remains a positive conjecture. Abs fails for all sufficiently large removal parameters, and Solid fails for all sufficiently small positive heights; the arguments give no explicit certified thresholds.

1.8 How to read a numerical claim

Analytical identity or theorem. A derivation or cited theorem with applicable hypotheses establishes the claim.

Validated finite assertion. Outward-rounded intervals and a selection argument enclose the claimed coordinates or finite-motion sign.

Refined numerical evidence. Changes in mesh, degree, quadrature, spectral cutoff or precision preserve a result; this is not a continuum enclosure.

Positive screen. No failure appeared on the declared finite set; no universal sign follows.

Conditional selector. A maximum, eigenbranch, thickness convention or limit still needs a uniqueness or selection theorem.

Stored digits identify computations, not guaranteed continuum accuracy. Contours often use a coarser display mesh than the refinement study. Gaps preserve undefined or unevaluated values. Four certified failures do not validate every numerical contour.

1.9 Reading paths and recurring structures

Mass and charge supply moments and far fields. Magnetostatics introduces singular sources, cutoff means and Steiner foci. Elasticity separates well-posed fields from ill-posed sheet models. Quantum and random motion explain eigenspaces, time averages and harmonic probabilities. Transport and heat compare shared operators and distinct observations. Optics exposes interference, detector response and underspecified dark points.

For intuition follow scenarios, derivations and demonstrations. For reproduction follow weak formulations, quadrature, refinement and atlas sections. The appendices collect proofs, the latest attractivity audit and a software guide. A center index links all archived selectors to their treatment. Extensions and failed proposals are distinguished from the 66 numerical entries.

1.10 Classical centers selected by physical experiments

This chapter separates direct homogeneous experiments, limiting experiments, and deliberately calibrated constructions. The last category is useful for teaching but cannot support a claim that a classical center arose naturally from an independent physical law. Replacing a prescribed barycentric formula by three masses of those same weights is a realization, not an explanation of that formula.

1.10.1 The centroid $X(2)$: one point, many experiments

Let $G = (V_1 + V_2 + V_3)/3$. The following scenarios have different observables or physical mechanisms. Several share the same uniform-measure calculation, and are identified as such rather than counted as independent mathematical theorems.

1. **Balance of a uniform lamina.** A constant areal density gives $\int_T p dA/A_T = G$. An affine change of coordinates reduces the integral to the reference triangle, whose three barycentric means are $1/3$.
2. **Three equal vertex masses.** Static gravitational torque vanishes about their arithmetic mean G . This is a discrete measure, unlike the lamina.
3. **Three equal-mass straight rods along the sides.** Each rod's mass is the same, although their linear densities differ. Their midpoints average to G . Uniform linear density on all three rods instead gives $X(10)$.
4. **Three equal zero-rest-length Hookean springs.** A junction at p has energy $k \sum_i |p - V_i|^2/2$. Its force balance is $3kp = k \sum_i V_i$, hence $p = G$. A spring with nonzero natural length is a different model.
5. **Distributed quadratic attraction.** A junction coupled uniformly to the lamina has energy $\int_T |p - q|^2 dq/2$. Differentiation gives $A_T(p - G)$; the unique equilibrium is G .
6. **Uniform pressure on a horizontal triangular hatch.** The resultant normal force acts through G , since the first moment of a constant load is the area first moment. A tilted submerged plate has varying pressure and generally a different center of pressure.
7. **Buoyancy of a straight triangular prism.** A homogeneous, fully submerged right prism has displaced-volume centroid above G . Its projection is G ; a partially submerged or tilted prism requires a different calculation.
8. **Uniform axial DC conduction.** In a homogeneous straight prism with insulating side-walls and equipotential end faces, the axial electric field and current density are uniform. Their flux centroid in the triangular section is G . Contact and skin effects are excluded.
9. **A reflecting ideal gas at equilibrium.** With no external potential, its equilibrium spatial marginal is uniform in the container. The mean position is G , while the instantaneous spread is not zero.
10. **Reflecting Brownian motion.** Neumann diffusion conserves mass and has constant equilibrium density on a connected triangle. Its stationary expectation is G .
11. **Specular billiard equilibrium.** The invariant interior position-direction measure has uniform spatial marginal. Its spatial mean is G . Individual time averages require the ergodic qualifications in Chapter 5.
12. **Linear imaging of a uniform luminous aperture.** Convolution with a normalized point-spread function of zero first moment preserves the image centroid. Thus any symmetric, integrable blur with a finite first moment yields G . Nonlinear detection leads to the optical family in Chapter 7.
13. **Far-field aperture phase slope.** If $\hat{\chi}_T(k) = \int_T e^{-ik \cdot q} dq$, then $\nabla_k \log \hat{\chi}_T(0) = -iG$. The phase derivative identifies an aperture point, unlike the central Fraunhofer bright direction itself.

14. **Leading far-field current-loop translation.** A planar uniform loop is equivalent, in its solid-angle potential, to its spanning area. The first area moment in the far-distance expansion is at G ; this is an asymptotic multipole convention, not the near-field minimum M1.
15. **High-temperature quantum equilibrium.** The heat-kernel-diagonal spatial mean tends to G as inverse temperature tends to zero. Complete eigenspaces must be included.
16. **High-energy spectral averaging.** Averaging all quantum expectations below an energy cutoff tends to G by the local Weyl law; this is not an assertion about every individual high state.
17. **Strong bulk absorption with uniform production.** The concentration is nearly constant away from a boundary layer when $\mu \rightarrow \infty$. Its normalized mean tends to G ; see the absorption family.
18. **Strong logistic growth with a lethal boundary.** The positive population approaches its uniform carrying capacity in the interior as the dimensionless growth parameter grows. Its mean tends to G .
19. **A uniformly thickened plate in four dimensions.** The leading divergent electrostatic energy penalizes $\int_T \rho^2$ and selects uniform projected charge. Under the stated uniform-thickness convention the limiting center is G ; the geometric limiting argument is discussed with E4.
20. **Large-height solid-angle illumination.** Expanding $h(|p-q|^2 + h^2)^{-3/2}$ at large h leaves a position-dependent term proportional to $-\int_T |p-q|^2 dq$. Its limiting maximizer is G .
21. **Weak darkness preference.** When $\beta \rightarrow 0$, the darkness weights $e^{-\beta J}$ become uniform. The optical moment tends to G .
22. **Very narrow or very broad Gaussian nonlinear fluorescence.** Both endpoint limits of the quadratic-detection family are G , although intermediate widths need not give G .

Each identity involving a uniform measure follows from the same affine moment integral. The physical distinctions matter because nonuniform forcing, different boundary conditions or different detection laws break the equivalence. For $X(2)$, $D_{V_i}G = I_2/3$ proves global vertex attractivity directly.

1.10.2 The incenter $X(1)$: clearance, arrival and survival

Let $I = (aV_1 + bV_2 + cV_3)/P$, with inradius $r = 2A_T/P$. A disk contained in the triangle has radius at most the smallest distance from its center to a side. Since $\sum_i a_i d_i(p) = 2A_T$, that smallest distance is at most r , with equality precisely when all three distances equal r . Thus the largest inscribed disk has center I .

This gives several clean experiments. An expanding circular balloon constrained to remain circular first reaches maximal size at I . A wave propagating inward from all sides at equal speed reaches its last point at I : its arrival time is $d(p, \partial T)/v$. A uniform inward erosion of the triangle collapses at I . The short-time maximizer of Dirichlet heat content tends to I , because escape probability is exponentially sensitive to the nearest boundary distance. The limiting infinite-exponent torsion profile is the distance function, whose unique triangular maximum is I . These last two statements concern limits; they do not identify a finite-time heat maximum or a finite- p membrane maximum with I .

The incenter's global attractivity is part of the classical attractive-segment result in the project's Paper IV [21]. The simple fact that a largest disk is stable under an optimization algorithm is not an alternative proof.

1.10.3 The Spieker center $X(10)$: perimeter rather than area

For uniform linear density on the three sides,

$$S = \frac{a(V_2 + V_3) + b(V_3 + V_1) + c(V_1 + V_2)}{2P} = \frac{(P - a)V_1 + (P - b)V_2 + (P - c)V_3}{2P} = \frac{3G - I}{2}.$$

This is $X(10)$. A balanced wire frame, a uniform perimeter heat source's first moment, and an incoherent uniformly luminous wire's image centroid all select this point. The last two require linear observation and uniform emission per unit length. A reflecting diffusion constrained to the closed wire also has uniform arclength equilibrium when its diffusion coefficient and vertex transmission law are uniform.

Uniform-radius thin conducting wires in ambient dimensions three and four have the same limiting projected charge center; finite-radius corrections need not agree. A regularized magnetic-field area moment with a uniform near-edge cutoff has leading perimeter weights and the same limit, whereas the unregularized divergent integral is not a center. The Spieker point is also the incenter of the medial triangle, giving a largest-disk demonstration after physically constructing that smaller triangular chamber. The project's Paper IV establishes attractivity of $X(10)$ for triangles.

1.10.4 The symmedian point $X(6)$: three normal spring forces

Place a frictionless guide on each side line and connect a movable junction to its orthogonal projection by an identical zero-rest-length spring. Let $d_i(p)$ be signed distance to line i . The energy is

$$E(p) = \frac{k}{2} \sum_i d_i(p)^2.$$

For interior distances the affine constraint is $\sum_i a_i d_i = 2A_T$. Minimizing a sum of squares subject to this constraint gives $d_i = 2A_T a_i / \sum_j a_j^2$. These positive distances correspond to an interior point, and $w_i = a_i d_i / (2A_T) = a_i^2 / \sum_j a_j^2$. Thus the unique equilibrium is the symmedian point $X(6)$. The Hessian is the sum of three rank-one normal projections and is positive definite. This is a direct mechanical derivation of a nontrivial classical center. Its vertex attractivity remains a separate property and is not certified here.

1.10.5 The circumcenter $X(3)$, orthocenter $X(4)$, and nine-point center $X(5)$

Three simultaneous identical point emitters have equal direct travel time to the circumcenter, since it is the unique point equidistant from the three vertices. A rigid circular track passing through three pins has its rotation center there. These equal-range constraints determine $X(3)$ even when it lies outside the triangle.

The center of the smallest disk containing the three vertices agrees with $X(3)$ only for acute triangles (and the usual right-triangle boundary case). For an obtuse triangle it is the midpoint of the longest side. A minimax illumination or range experiment must retain this piecewise distinction.

There is an explicitly constructed spring experiment for $X(4)$. First construct each altitude line. Attach normal zero-rest-length springs to those three lines. Their quadratic energy is the

sum of squared distances to the altitudes, which vanishes only at their common intersection, the orthocenter. This is physically realizable but uses preconstructed geometric rails; it is less intrinsic than using the original sides for X(6). Similarly, three synchronized emitters at the side midpoints have equal travel times at the circumcenter of the medial triangle, the nine-point center X(5). These are labelled *derived-geometry realizations*, not unexplained new equilibrium laws.

1.10.6 The first Fermat center X(13): tension, networks and transport

For a junction connected to three vertex pulleys with equal hanging loads, the potential is proportional to $E(p) = \sum_i |p - V_i|$. At an interior equilibrium the three unit tension vectors sum to zero. Their pairwise angles are therefore 120° . If every triangle angle is less than 120° , this selects the first Fermat point X(13). The same junction minimizes the total length of a three-terminal Steiner network and the mean linear travel cost to equally weighted vertex destinations.

If a vertex angle is at least 120° , the convex geometric-median problem is minimized at that vertex. Its global physical selector is consequently not the unqualified analytic X(13) construction over every triangle. A soap-film triple junction has the same local force law for equal interfacial tensions, but boundary contact angles and the possibility of other film topologies must be controlled. These qualifications are essential in an experimental demonstration.

1.10.7 Contact-derived and calibrated realizations

The incircle and nine-point circle are tangent at the Feuerbach point X(11), away from cases where the usual construction loses uniqueness. Their contact point can be observed by fitting the two geometric circles, but it is a derived circle-contact realization, not a center selected by an unconstrained homogeneous sheet.

More generally, positive barycentric weights w_i can be realized by vertex masses w_i or spring constants w_i , since $\sum_i w_i |p - V_i|^2$ is minimized at their weighted mean. Examples include the Gergonne weights $1/(s - a)$ for X(7) and Nagel weights $s - a$ for X(8). Tangent lengths provide a way to calibrate those masses, but the weighting is part of the apparatus. Signed barycentrics cannot be implemented by positive masses or passive positive-stiffness springs in this way.

1.10.8 Scope of the first-100 catalogue

The direct and limiting constructions above concentrate on X(1), X(2), X(3), X(6), X(10) and X(13); the derived or calibrated examples add X(4), X(5), X(7), X(8) and X(11). This is not a claim that the other first-100 entries have no physical interpretation. An explicit inventory in the bundle records all 100 identifiers, with “no independent natural realization established in this survey” for the others. It prevents an unexamined entry from being mistaken for a negative theorem. The catalogue is broad in mechanisms, but does not manufacture a natural explanation for every number by restating its barycentric formula as prescribed masses.

For demonstrations, the most informative first comparison is a plate, a uniform wire frame, and the three side-normal springs cut to the same scalene outline. Their balance/equilibrium points are respectively X(2), X(10), and X(6). Add an inscribed expanding disk for X(1). All four are easy to measure and need no numerical PDE solution.

Chapter 2

Mass, charge and the geometry of a triangle

From balance and inertia to equilibrium charge and least capacity. This chapter develops one set of tools that will recur throughout the book: specify a physical measure, take its first and second moments, identify its governing field, and distinguish exact identities from numerical evidence. The reference triangle is always $A = (0, 0)$, $B = (4, 0)$, $C = (0, 3)$, unless another triangle is explicitly drawn. Its side lengths opposite A, B, C are $a = 5$, $b = 3$, $c = 4$.

The main actors are the area centroid $G = X2$, the perimeter centroid $W = X10$, the inner conformal center $E1$, the planar equilibrium-charge centroid $E2$, and the three-dimensional plate-charge centroid $E3$. The previously adopted uniform-thickness convention gives $E4 = X2$ and $S3 = S4 = X10$ in the limiting models explained below; $S2 = E2$ exactly. A separate notation L_n is introduced for the least-capacity center of an infinite triangular prism in ambient dimension n , with $L_2 = E1$. This avoids confusing dimension with the established E-series labels.

We will encounter three different questions. Where is the material balanced? How does its mass or charge look from far away? Where is a point best insulated from a grounded boundary? They are connected, but their answers need not coincide. Even a conductor whose potential is constant has a nontrivial charge distribution and a well-defined charge centroid.

Reading route. Sections 2.1-4 require vectors, elementary integration and eigenvectors of a symmetric two-by-two matrix. Section 2.5 introduces far-field expansions. Sections 2.6-9 develop electrostatic energy and conformal maps. Section 2.10 extends least capacity to higher dimensions. Section 2.11 gives the atlas, numerical comparisons and attractivity evidence. The appendices supply proofs, practical algorithms and reproducibility instructions.

Evidence convention. An identity follows from the displayed mathematics. A computed coordinate is the output of a specified discretization, accompanied by refinement information. A positive-weight location certificate is a proof of interiority, not a small error bar on a decimal coordinate. The higher-dimensional uniform-thickness conclusions are leading-order thin-conductor results under the stated geometric convention; the softened-kernel experiments in the earlier atlas are not physical finite-thickness simulations.

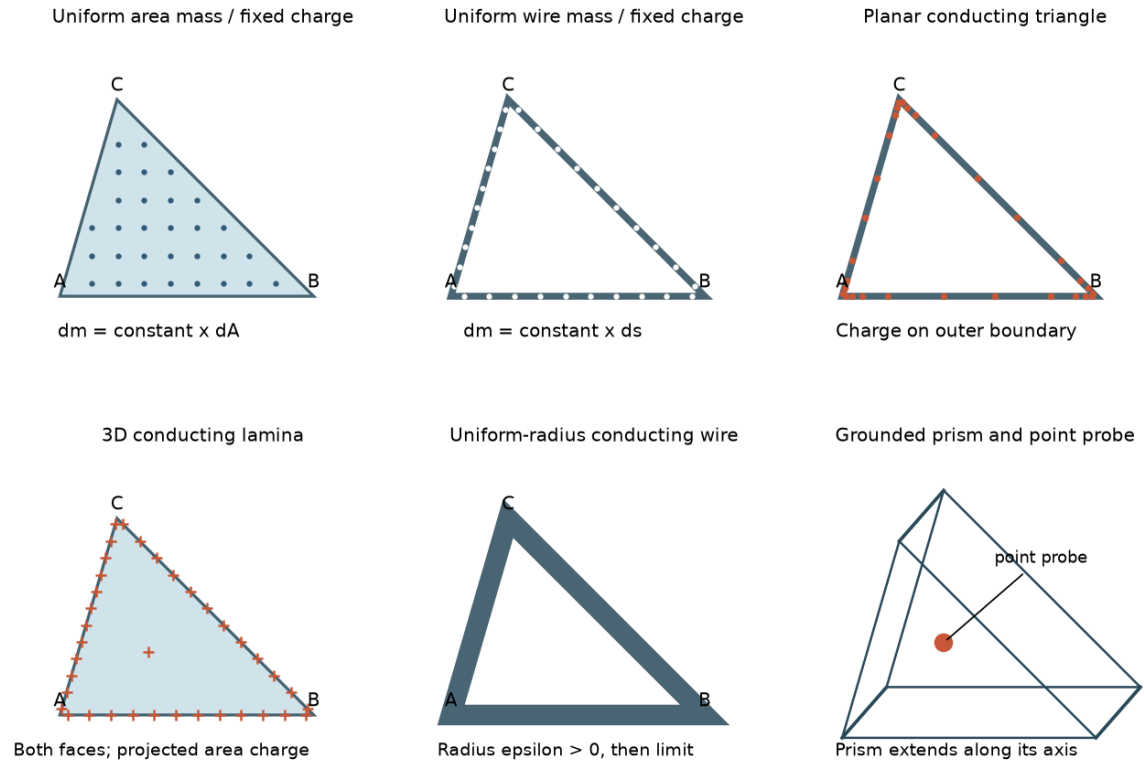


Figure 0. The measure and ambient field distinguish the physical models. Dots and plus signs are schematic, not quantitative charge samples. The finite prism sketch represents a segment of an infinite grounded prism. The conductor drawings omit the distant reference electrode or infinity condition.

2.1 What distribution is being centered?

2.1.1 A triangle, its boundary, and its ambient space

Let T be the closed filled triangle, and let $\Gamma = \partial T$ be its three sides. The area is $\mathcal{A}$, the perimeter is $P = a + b + c$, and $D = \max(a, b, c)$ is the diameter. A triangle embedded in $\mathbb{R}^n$ still has two-dimensional area and a one-dimensional boundary. Increasing the ambient dimension changes the field law; it does not turn the triangle into an n -dimensional simplex.

A lamina with constant surface mass density has mass element $dm = \sigma dA$. A wire frame with constant linear mass density has $dm = \lambda ds$. Equal masses placed at the three vertices form a third distribution. They happen to share the lamina's centroid, but their second moments are different. Here and below uniformity always specifies its reference measure: area, length, volume or surface area.

Write the total mass as $M = \int dm$, and introduce the probability measure $d\mu = dm/M$. The center and covariance matrix are

$$c_\mu = \int q d\mu(q), \quad C_\mu = \int (q - c_\mu)(q - c_\mu)^T d\mu(q).$$

The first moment gives a position. The second gives spread in each direction. The transpose makes an outer product: its ij entry is the product of the i th and j th coordinates. These definitions work unchanged for a prescribed nonnegative static charge distribution after replacing mass by total charge. Multiplying all masses or charges by the same positive constant does not change either normalized quantity.

Signed charge requires extra care. If the net charge is nonzero, one can form its signed first moment divided by net charge, but that point need not lie in the triangle. If the net charge is zero, this quotient does not exist: a dipole moment replaces the monopole as the leading distant signature. All conductor-centroid constructions here use an isolated conductor with a fixed positive total charge, no applied external field, and a homogeneous surrounding medium.

2.1.2 Why a first moment is a balance point

In a spatially uniform gravitational field g , the torque about a point p is the integral of $(q - p) \times g dm$. Linearity makes it equal to $M(c_\mu - p) \times g$. Thus the distributed load produces the same force and torque as one mass at c_μ . This statement is exact in a uniform field, irrespective of the object's shape. The far-field result in Section 2.5 is a different, approximate statement about the field produced by the object itself.

Another exact interpretation follows by expanding a square:

$$\int |q - p|^2 d\mu(q) = \text{tr } C_\mu + |p - c_\mu|^2.$$

The centroid uniquely minimizes mean squared distance. Equivalently, it minimizes the polar moment about a perpendicular axis through p . A uniform triangular sheet can therefore reveal X2 by balancing, by suspension from two points, or by finding the pivot that minimizes its perpendicular rotational inertia. The corresponding wire demonstrations reveal X10.

2.1.3 When does a physical construction define a triangle center?

A unique construction depending only on the triangle's Euclidean geometry should commute with translation, rotation, reflection, scaling and relabeling of vertices. The moment constructions

meet these requirements. For a nondegenerate triangle, every resulting point has barycentric coordinates: $p = uA + vB + wC$ with $u + v + w = 1$. Positive coordinates mean strict interiority.

A complicated center need not have an elementary barycentric formula. Its integral, variational problem or boundary-value problem can still characterize it exactly. Conversely, the use of a special function does not establish mathematical transcendence, and the ETC itself contains nonalgebraic centers. An ETC location match in a finite survey is evidence of similarity, not a proof of identity.

2.2 Two uniform distributions, two classical centers

2.2.1 The filled triangle gives X2

Parametrize a point by $q = A + u(B - A) + v(C - A)$, where $u, v \geq 0$ and $u + v \leq 1$. The Jacobian is constant. On this reference triangle the normalized integrals are $\mathbb{E}u = \mathbb{E}v = 1/3$. Hence

$$G = \frac{A + B + C}{3}, \quad G = (1 : 1 : 1)_{\text{bary}}.$$

This is the ordinary intersection of the medians, X2. In the reference triangle it is $(4/3, 1)$. Equal point masses at the three vertices also have this mean. So do three equal point masses at the side midpoints. These statements concern first moments only; spreading or concentrating mass changes inertia.

The derivation immediately covers a uniform immobile surface charge. An insulating triangular film with charges immobilized at uniform areal density has charge centroid X2 in every ambient dimension. The kernel of electrostatic interaction is irrelevant while the distribution is prescribed.

2.2.2 The skeleton gives X10

Each straight side has its own center at its midpoint. Under a common linear density, its mass is proportional to its length. Replacing each side by its midpoint mass gives

$$W = \frac{a(B + C) + b(C + A) + c(A + B)}{2P}.$$

Therefore $W = (b + c : c + a : a + b)_{\text{bary}}$, the Spieker center X10 [1]. Since the medial triangle has side lengths $a/2, b/2, c/2$, its incenter is exactly this weighted average of its vertices. The Spieker circle is the medial triangle's incircle. This geometric construction gives a simple way to locate the balance point of a triangular wire without integration.

For our triangle, $W = (3/2, 1)$. The difference from X2 is $(1/6, 0)$. The longer sloping side contributes more wire mass than either shorter side. Giving each side the same total mass would instead yield X2; it would require three different linear densities.

Let $I = (aA + bB + cC)/P$ denote the incenter X1. Rearranging the formula for W yields a useful affine identity:

$$W = \frac{3G - I}{2}.$$

Thus I, G and W are collinear, with G between I and W unless they coincide. In the 3-4-5 triangle $I = (1, 1)$, so the relation is visible without any sloping coordinate axes. This identity will also supply exact local shape fingerprints.

2.2.3 Elementary location certificates

For G , all three barycentric coordinates are $1/3$. For W , the normalized coordinate opposite side a is $(P - a)/(2P)$. The strict triangle inequality gives $0 < a < P/2$, so every coordinate of W lies strictly between $1/4$ and $1/2$. This is a useful certificate stronger than mere interiority: W lies inside the medial triangle. The distance of W from the side opposite A is its A -barycentric coordinate times the altitude from A .

For any positive probability distribution on the triangle, barycentric coordinates commute with averaging because they are affine functions. Therefore a charge centroid belongs to the triangle. It lies strictly inside when the charge is not supported entirely on a single side. This observation later certifies $E2$ and $E3$ without requiring an explicit formula for either distribution.

2.3 Inertia and the Steiner inellipse

2.3.1 Covariance is not the mechanical inertia tensor

Put the origin at the mass centroid and let e be a unit vector along an axis in the triangle's plane. The squared distance from q to that axis is $|q|^2 - (e \cdot q)^2$. Thus its moment of inertia is

$$I(e) = M \left(\text{tr } C - e^T C e \right), \quad J = M \left((\text{tr } C) \mathbf{1} - C \right).$$

Here $\mathbf{1}$ is the identity matrix. Covariance and inertia have the same eigenvectors, but their eigenvalues are interchanged in two dimensions. If $Ce_1 = \lambda_1 e_1$ and $Ce_2 = \lambda_2 e_2$, with $\lambda_1 \geq \lambda_2 > 0$, then $I(e_1) = M\lambda_2$ and $I(e_2) = M\lambda_1$. The direction of greatest spatial spread is the axis about which rotation has the smaller moment. Confusing these two matrices reverses the axis-ratio statement.

The perpendicular moment is $M \text{tr } C$. The parallel-axis theorem follows from the square expansion in Section 2.1. Translating a pivot by d adds $M|d|^2$ to the perpendicular moment. None of these results depends on whether the distribution consists of a sheet, a wire or discrete particles.

2.3.2 Integrating the area covariance

Write $v_i = V_i - G$ for the three centered vertex vectors. Uniform barycentric coordinates on a triangle satisfy $\mathbb{E}u_i^2 = 1/6$ and $\mathbb{E}u_i u_j = 1/12$ for $i \neq j$. For example, $2 \int_0^1 \int_0^{1-u} u^2 dv du = 1/6$. Since $v_1 + v_2 + v_3 = 0$, expanding the outer product of $\sum u_i v_i$ gives

$$C_T = \frac{1}{12} \sum_{i=1}^3 v_i v_i^T.$$

An equal three-vertex distribution has covariance four times as large. It has the same principal directions but different actual moments. A lamina's inertia therefore cannot be recovered merely by assigning equal vertex masses unless this factor is included.

For the reference triangle the matrix entries are $C_{xx} = 8/9$, $C_{xy} = C_{yx} = -1/3$, $C_{yy} = 1/2$. They have units of length squared. A matrix diagram is less informative than its action: $e^T C_T e$ gives the mean squared projection onto any direction e .

2.3.3 The ellipse determined by these moments

For a positive definite matrix Q , the ellipse centered at G with shape matrix Q is the set of points satisfying $(p - G)^T Q^{-1} (p - G) \leq 1$. Its squared semiaxes are the eigenvalues of Q . The triangle's Steiner inellipse has the particularly simple shape matrix

$$Q_{\text{St}} = 2C_T = \frac{1}{6} \sum_{i=1}^3 v_i v_i^T.$$

Equivalently, its boundary is $(p - G)^T C_T^{-1} (p - G) = 2$. It touches every side at its midpoint. One way to prove this is to start with an equilateral triangle: its covariance is a scalar multiple of the identity, and its incircle satisfies this equation. An invertible affine map carries that triangle to any other triangle, sends midpoints to midpoints, preserves tangency, and sends covariance to $LC_T L^T$. Thus the same equation describes the midpoint-tangent ellipse in every triangle.

The affine argument is particularly powerful because normalized uniform area remains uniform under an affine map: every small area is multiplied by the same determinant. We will see why uniform wire length does not share this property.

If $a_{\text{St}} \geq b_{\text{St}}$ are the semiaxes, their relation to the in-plane moments is

$$\frac{a_{\text{St}}}{b_{\text{St}}} = \sqrt{\frac{\lambda_1}{\lambda_2}} = \sqrt{\frac{I(e_2)}{I(e_1)}}.$$

It is the squared axis ratio that equals the ratio of the reversed principal moments. The axis ratio itself is the square root. This corrects the common temptation to equate two quantities merely because they share principal directions.

The Steiner inellipse is also the unique maximum-area ellipse contained in the triangle. Appendix A.1 gives a short determinant proof, avoiding an appeal to the general theory of John ellipsoids. Its area is $\pi\mathcal{A}/(3\sqrt{3})$. The concentric ellipse with twice its semiaxes passes through the three vertices and is the Steiner circumellipse. It is the minimum-area enclosing ellipse.

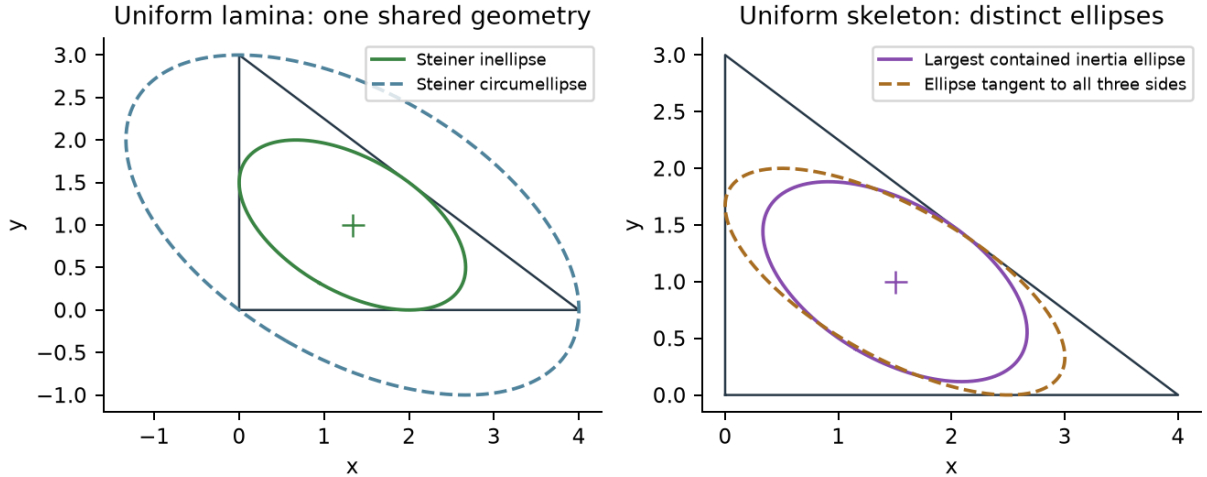


Figure 1. Left: the area inertia, Steiner inellipse and Steiner circumellipse have the same axes. Right: the skeleton's largest contained inertia ellipse and the X10-centered tangent ellipse are different. The inertia ellipse touches only one side at this scale. The two panels have independent axis limits.

2.3.4 Why this ellipse will reappear in magnetostatics

The moments summarize quadratic geometry. Surprisingly, the same ellipse controls a field generated by three equal parallel currents through the vertices. If the centered complex vertex coordinates are z_i and $P(z) = \prod_i (z - z_i)$, the transverse field vanishes where $\sum_i 1/(z - z_i) = P'(z)/P(z) = 0$. These are the two zeros of P' , and they are the foci of the Steiner inellipse. Appendix A.1 derives the equality directly from the covariance matrix.

This statement concerns a pair of points, not an arbitrarily labeled single triangle center. The unordered focal pair is natural under symmetries. At the equilateral shape the foci coincide at G, and there is no distinguished major-axis direction. The magnetostatics chapter develops the physical field and its other centers; the present calculation supplies its geometric foundation.

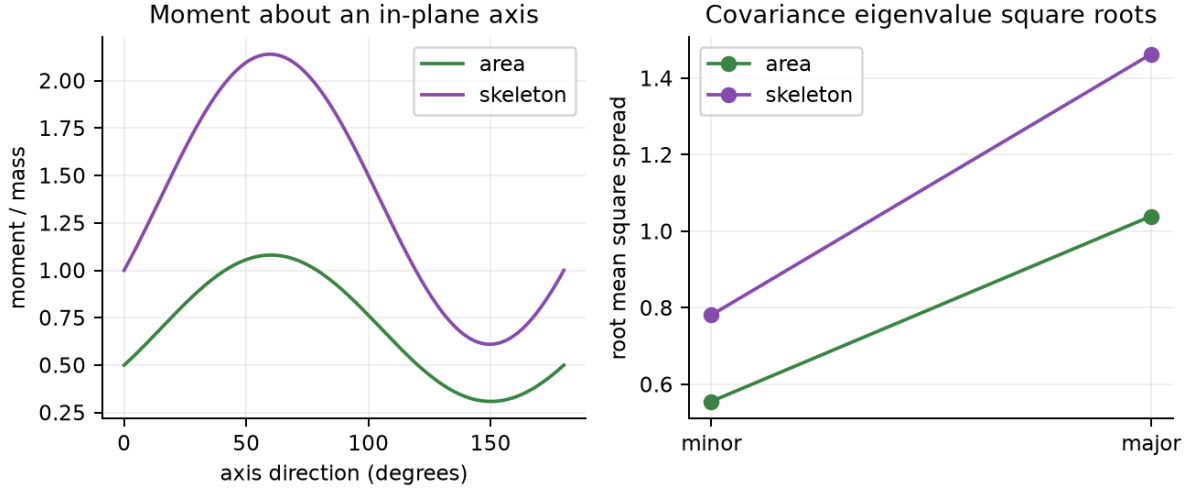


Figure 2. Directional moments for unit total mass on the reference triangle. A long spatial axis corresponds to a smaller moment about that axis. The area and skeleton distributions have different principal moments even though their centroids happen to share the same height.

2.4 What survives, and what fails, for a skeleton

2.4.1 Its covariance can also be integrated exactly

For one side joining U to V , let $m = (U + V)/2$, $d = V - U$, and length $\ell = |d|$. A uniform point on the side is $m + td$, where t is uniform on $[-1/2, 1/2]$. Its mean is m and $\mathbb{E}t^2 = 1/12$. Averaging the three sides by their lengths gives

$$C_{\Gamma} = \frac{1}{P} \sum_{\text{sides}} \ell \left((m - W)(m - W)^T + \frac{dd^T}{12} \right).$$

This is an exact finite formula for every triangle. It is often simpler and more reliable than a trigonometric formula for the principal-axis angle. Diagonalizing it gives both principal directions and moments. It also defines a family of inertia ellipses $(p - W)^T C_{\Gamma}^{-1} (p - W) = k$, with arbitrary positive scale k .

The same reversed square-root relation between ellipse axes and inertia holds for these ellipses. What fails is the additional assertion that one member must be an inellipse tangent to all three sides, or a circumellipse through all three vertices. There is no universal skeleton analogue combining those geometric and inertial properties.

2.4.2 An exact counterexample, rather than a visual impression

For the 3-4-5 skeleton, $W = (3/2, 1)$ and the covariance entries are $C_{xx} = 7/4$, $C_{xy} = -2/3$, $C_{yy} = 1$. Its eigenvalues are $(33 \pm \sqrt{337})/24$, so its inertia-ellipse semiaxis ratio is approximately 1.87282.

An ellipse of shape matrix kC_{Γ} centered at W is tangent to a line with unit normal n and distance d from W exactly when $k n^T C_{\Gamma} n = d^2$. Tangency to $x = 0$ requires $k = 9/7$. Tangency to $y = 0$ requires $k = 1$. These conditions already contradict each other. The third side cannot repair the contradiction. Consequently no scaling of the skeleton inertia ellipse is tangent to all three sides of this triangle.

There nevertheless is an inellipse centered at W . Solving the three tangency conditions for its shape matrix gives $Q_{xx} = 9/4$, $Q_{xy} = -1$, $Q_{yy} = 1$. Its eigenvectors are not the skeleton's principal directions: the upper-right entry of $C_\Gamma Q - QC_\Gamma$ is $1/12$, which is nonzero. Its semiaxis ratio is approximately 2.50820. The unique W -centered circumellipse similarly differs; its inverse shape matrix has entries $4/21$, $2/21$, $2/7$, and its axis ratio is the golden ratio.

Thus the conclusion is precise: inertia ellipses exist for the skeleton, and X_{10} -centered tangent and circumscribed ellipses exist in this example, but they do not coincide. It would be incorrect to state that the skeleton has no ellipse at all. Equilateral symmetry makes all these ellipses circular; isosceles symmetry aligns their axes but need not equalize their axis ratios.

2.4.3 Why the lamina argument breaks

Under an affine map L , a length element in direction t is multiplied by $|Lt|$. Different sides generally have different directions and different stretch factors. Therefore the image of a uniformly weighted equilateral wire does not have uniform linear density on the new triangle. Covariance is still affine-covariant for the pushed-forward measure, but that measure is not the uniformly weighted skeleton whose inertia we are studying.

This is a useful general lesson for physical triangle centers. Similarity transformations preserve a uniform isotropic physical model. General affine transformations often preserve geometric incidence while changing the physical density, conductivity or material law. An affine picture alone does not justify an affine physical equivalence.

2.5 What a distant observer sees

2.5.1 Deriving the dimension-dependent field law

A point mass produces a radial gravitational field. Gauss's law says that its flux through any surrounding sphere is proportional to the enclosed mass. The area of a sphere in $\mathbb{R}^n$ grows as r^{n-1} , so the field magnitude decays as r^{1-n} . Integrating radially gives a logarithmic potential in two dimensions and an inverse-power potential in dimensions three and above.

It is convenient to remove positive physical constants and use the kernels

$$K_2(r) = -\log r, \quad K_n(r) = r^{2-n} \quad (n \geq 3).$$

The reference length hidden inside the logarithm only changes an additive constant. Let $U(x) = M \int K_n(|x - q|) d\mu(q)$. For positive electrostatic charge the electric field points along $-\nabla U$ after restoring its positive coupling constant. Gravitational acceleration points along $+\nabla U$ with its gravitational coupling: attraction reverses the sign. Neither sign choice changes the first-moment calculation.

2.5.2 Monopole, dipole, and the distinguished origin

First choose an arbitrary origin and write the observation point as $x = Re$, $|e| = 1$. Suppose the source lies in a bounded region much smaller than R . Set $m = \int q d\mu(q)$ and $Q = \int qq^T d\mu(q)$. Taylor expansion of $|Re - q|^{2-n}$ gives, for $n \geq 3$,

$$\frac{U(Re)}{M} = R^{2-n} \left[1 + \frac{(n-2)e \cdot m}{R} + \frac{n-2}{2R^2} (ne^T Qe - \text{tr } Q) + O(R^{-3}) \right].$$

For a fixed source size, the remainder notation refers to $R \rightarrow \infty$; the hidden coefficient has the necessary powers of length. The first term is the monopole. The next is the dipole, which records the first moment. A single point mass at m has exactly these same first two terms. Moreover, translating the origin to m makes the dipole vanish in every direction. No other translation does so.

The gravitational far-field center is therefore the mass centroid: X_2 for the uniform lamina and X_{10} for the uniform skeleton, in every ambient dimension. For a conducting distribution it is instead that distribution's charge centroid. This is an operational interpretation of E2 and E3: they are the origins from which the exterior charge field has no dipole term.

The approximation is not exact to all orders. With the origin at the centroid, Q becomes C . Its quadrupole generally remains. A point mass has no central covariance and cannot reproduce it. The expression $ne^T Ce - \text{tr } C$ explains why the covariance principal directions, and hence the lamina's Steiner axes, also organize the angular far-field correction. When the observation direction leaves the triangle's plane, embed C as an n -by- n matrix with zero transverse rows and columns.

2.5.3 The two-dimensional expansion

Expanding the logarithm instead yields

$$\frac{U(Re)}{M} = -\log R + \frac{e \cdot m}{R} + \frac{2e^T Qe - \text{tr } Q}{2R^2} + O(R^{-3}).$$

The dipole again disappears precisely at the centroid. Unlike the higher-dimensional potential, the logarithmic potential does not approach zero at infinity. Its gradient is well defined and decays as $1/R$. Relative errors of the gravitational field are therefore more physically useful than relative errors of a logarithmic potential, whose additive constant is arbitrary.

For a centered bounded source, the monopole field has a relative error of order D^2/R^2 . Using a displaced point mass generally leaves an error of order D/R . These orders hold uniformly in direction when R stays sufficiently larger than the support radius, though special directions or symmetries may cancel a leading coefficient. Including the central quadrupole normally improves the potential error by another inverse power of distance.

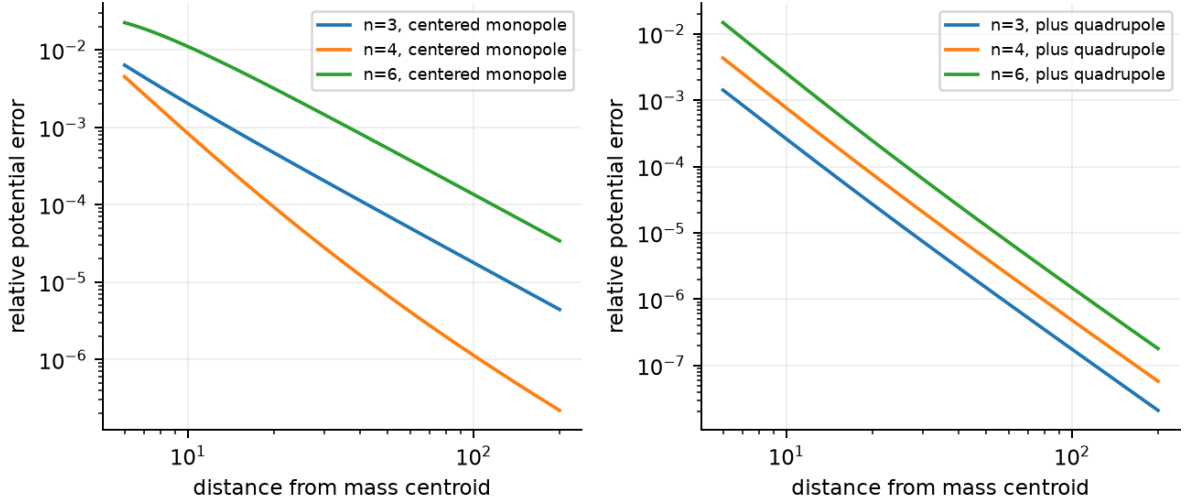


Figure 3. Unit area mass in the 3-4-5 triangle, observed along the three-dimensional direction (0.36,-0.48,0.8), embedded in each stated ambient dimension. Left: monopole placed at G. Right: the same expansion with its exact covariance correction. The comparison uses a 90-by-90 triangular subdivision for the source integral. The small high-distance numerical floor reflects that quadrature, not a new physical multipole. These are empirical convergence plots, not interval bounds.

2.5.4 Immobile charge versus mobile charge

For an insulating film carrying prescribed uniform charge, all the moment and multipole results above apply with no redistribution. For an isolated metallic conductor, charge moves until there is no tangential electric field along the conductor. Its potential becomes constant, and the required density is normally nonuniform. It is therefore essential to solve the equilibrium problem before taking the first moment. Reusing the uniform mass centroid without that step changes the physical problem.

2.6 Electrostatic equilibrium as a minimization problem

2.6.1 How constant potential emerges

Build a positive charge distribution gradually. Bringing a small amount of charge to a place with potential U costs work proportional to U . Summing all pairs, and dividing by two to avoid double counting, gives energy proportional to $\mathcal{E}(\mu) = \iint K_n(|p - q|) d\mu(p)d\mu(q)$. We fix total charge by requiring $\mu(K) = 1$, where K is the conductor.

Suppose a small amount of charge can be moved from p to q . The first-order energy change is twice that amount times $U(q) - U(p)$. At equilibrium, two occupied accessible regions must have equal potential; otherwise moving charge from the higher-potential region to the lower one reduces the energy. More precisely, the equilibrium potential equals an energy constant on its support and is at least that constant elsewhere on the admissible set, apart from exceptional sets of zero capacity.

For the conductors used here, the standard potential-theory equilibrium measure is unique when a finite-energy measure exists [2, 3]. The centroid is then unambiguous. Boundary singularities in the density do not imply divergent total charge: an integrable edge divergence is entirely compatible with a finite probability measure and a finite first moment.

2.6.2 Capacity and normalization

For our unnormalized Newtonian kernel, define $V_n(K) = \min \mathcal{E}(\mu)$ and $\text{cap}_n(K) = 1/V_n(K)$ when $n \geq 3$. In two dimensions use $\text{cap}_2(K) = \exp[-V_2(K)]$. Physical capacitance includes additional dimension-dependent constants. None affects the equilibrium probability measure or its centroid.

Capacity and centroid are different summaries. Capacity measures how much charge the body can hold at a given potential, with the appropriate convention at infinity. The centroid records the first moment of the minimizing distribution. A center called a least-capacity point will involve a family of transformed conductors indexed by a point; translating a fixed isolated conductor in homogeneous space cannot change its capacity.

2.6.3 Why ambient dimension matters for a thin set

Near a regular point of a k -dimensional support, a smooth trial density contributes a local radial energy integral proportional to $\int_0^\delta r^{k-1-s} dr$, where $s = n - 2$. It converges when $s < k$, and diverges at or above the threshold. The corresponding capacity criterion for the compact rectifiable sets here gives zero Newtonian capacity at the critical and supercritical dimensions.

Thus a flat plate has a genuine finite-energy Newtonian equilibrium in three dimensions, but not in four or more. A one-dimensional wire is already critical in three dimensions. A metallic object with a positive thickness always has a physical exterior problem; its exactly zero-thickness idealization may lose that problem. The remedy is to specify a limiting geometry and follow its normalized charge measures as thickness decreases.

This distinction is not optional numerical regularization. Adding an arbitrary softening to a singular kernel defines a different finite-size model. It may illustrate the limiting mechanism, but it does not automatically reproduce the charge distribution of a particular conductor with a particular cross section.

2.7 The planar conductor: $E2 = S2$

2.7.1 The charge lives on the boundary

In genuine two-dimensional electrostatics the kernel is logarithmic. A filled conducting triangle is equipotential inside. Its equilibrium charge is carried by the outer boundary, with nonuniform arclength density. If the conductor consists only of that same closed triangular boundary, the bounded interior region still has constant potential by the uniqueness of the harmonic Dirichlet problem. The exterior field and its equilibrium measure are unchanged. Consequently the planar plate and skeleton have the same center, $E2 = S2$.

This equivalence is special to the logarithmic planar problem. A metal film placed in ordinary three-dimensional space has a $1/r$ interaction. Charges can occupy both faces, and its projected equilibrium charge is distributed across the area. Calling both objects a two-dimensional triangle does not make their electrostatic kernels equal.

2.7.2 The exterior conformal formulation

Identify the plane with the complex numbers. Let f map the outside of the unit disk conformally onto the outside of the triangle, with its leading coefficient real and positive. Its expansion at infinity is

$$f(w) = \kappa w + a_0 + \frac{a_{-1}}{w} + \frac{a_{-2}}{w^2} + \cdots$$

The coefficient κ is the logarithmic capacity. If $g = f^{-1}$, the equilibrium potential outside is $U(z) = -\log \kappa - \log |g(z)|$. The unit circle has uniform harmonic measure $dt/(2\pi)$ as viewed from infinity. Conformal invariance transports this measure to the triangular boundary. Therefore

$$d\mu_2 = \frac{dt}{2\pi}, \quad E2 = \frac{1}{2\pi} \int_0^{2\pi} f(e^{it}) dt = a_0.$$

Every nonconstant Fourier mode averages to zero. This proves that the outer conformal center, the planar electrostatic centroid and the constant Laurent coefficient are exactly the same point. It also connects the coefficient a_0 to the dipole-free far-field origin of Section 2.5. See [2, 4] for the potential-theory and conformal-mapping background.

Where the boundary is smooth, $ds = |f'(e^{it})|dt$, so the charge per unit length is $1/(2\pi|f'|)$. At a triangle vertex of interior angle θ , the exterior opening is $2\pi - \theta$. The local conformal map has power $(2\pi - \theta)/\pi$, so density along a side behaves like distance to the vertex raised to $\pi/(2\pi - \theta) - 1$. This exponent is between $-1/2$ and 0 for a nondegenerate convex angle, hence the singularity is integrable. The charge is enhanced near the corners without becoming an atomic point charge.

2.7.3 A numerical route that does not require an exterior map

Partition each side into panels, more densely near the endpoints. Let q_j be the total charge on panel j , assumed constant per unit length within that panel. At the midpoint p_i of panel i , compute the exact panel average K_{ij} of $-\log |p_i - q|$. Solve $\sum_j K_{ij}q_j = V$ for all i , together with $\sum_j q_j = 1$. The unknown potential V is included in the linear system. This augmented system avoids a special scale at which solving only $Kq = 1$ would be singular.

For a segment parameterized by longitudinal coordinate s and perpendicular distance h , an antiderivative of $\log \sqrt{s^2 + h^2}$ is $s \log \sqrt{s^2 + h^2} - s + h \arctan(s/h)$, interpreted by its continuous

limiting values. This integrates the self panel as well as separated panels. Multiplying the solved panel charges by their midpoints gives the first moment exactly for the piecewise-constant density.

Refine the panel count, verify total charge, inspect positivity and compare centroids. A tiny residual at the chosen collocation points is not a discretization-error bound: the linear system was designed to make that residual tiny. Independent off-grid potential checks and mesh changes test different sources of error. The companion keeps the full panel charges, lengths, geometry and convergence history.

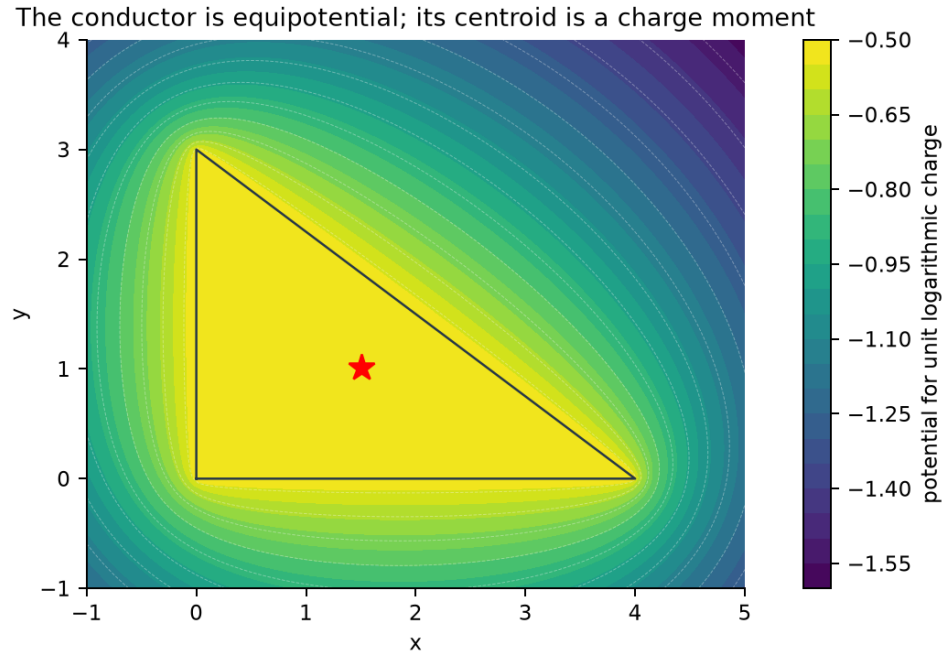


Figure 4. Equipotential contours for the planar conductor. The filled triangle is essentially a plateau of potential. The star is the first moment of boundary charge, not a unique extremum of the interior potential. A potential-minimum definition would select the entire conductor and would fail to define a point.

2.8 A conducting plate in three and more dimensions

2.8.1 The three-dimensional lamina: E3

A very thin isolated metal plate has two faces. Reflection across its plane makes their charges equal; adding the two faces gives a projected area measure on T . In the zero-thickness limit this measure solves

$$\int_T \frac{\rho(q)}{|p-q|} dA(q) = V \quad (p \in T), \quad \int_T \rho dA = 1.$$

The equality has its potential-theoretic interpretation at exceptional boundary points; an interior collocation solver evaluates it at regular points. The center is $E3 = \int_T q\rho(q) dA(q)$. This density is nonuniform. Along an open straight edge it has the familiar inverse-square-root edge singularity; corners require their own local analysis and should not be assigned the same one-dimensional exponent without justification.

There is no contradiction between a constant potential and a nonconstant charge density. The density adjusts precisely to compensate for the unequal availability of neighboring charge near edges and vertices. Nor is E3 obtained by taking the planar logarithmic density and spreading it through the plate: that would solve the wrong integral equation.

2.8.2 Integrating the self interaction correctly

Subdivide the plate into small triangles and represent charge by a constant density on each panel. Centroid collocation gives a dense linear system. The kernel $1/|p - q|$ has a finite area integral even when p belongs to the panel, but replacing the self interaction by a point value would incorrectly make it infinite.

In two integration dimensions, $\nabla_q \cdot [(q - p)/|q - p|] = 1/|q - p|$. Applying the divergence theorem converts the panel integral into three line integrals. If the panel edges are oriented counterclockwise and h_e is the signed distance of p to the inward side of edge e , then

$$\int_K \frac{dA(q)}{|q - p|} = \sum_{e \in \partial K} h_e \int_e \frac{ds(q)}{|q - p|}.$$

The sign convention follows because $(q - p)$ dotted with the outward normal equals h_e . The edge integral is a logarithm expressible using distances to its endpoints. The product has a continuous zero limit when p lies on the edge's supporting line. This formula handles self panels analytically and eliminates an otherwise arbitrary self-energy constant.

The algorithm in the bundle is collocation, not a certified variational Galerkin method. It gives a reproducible and converging first moment, but its final mesh change is an empirical diagnostic. Corner grading, symmetric Galerkin assembly or validated integral bounds would be appropriate refinements when rigorous coordinate enclosures are needed.

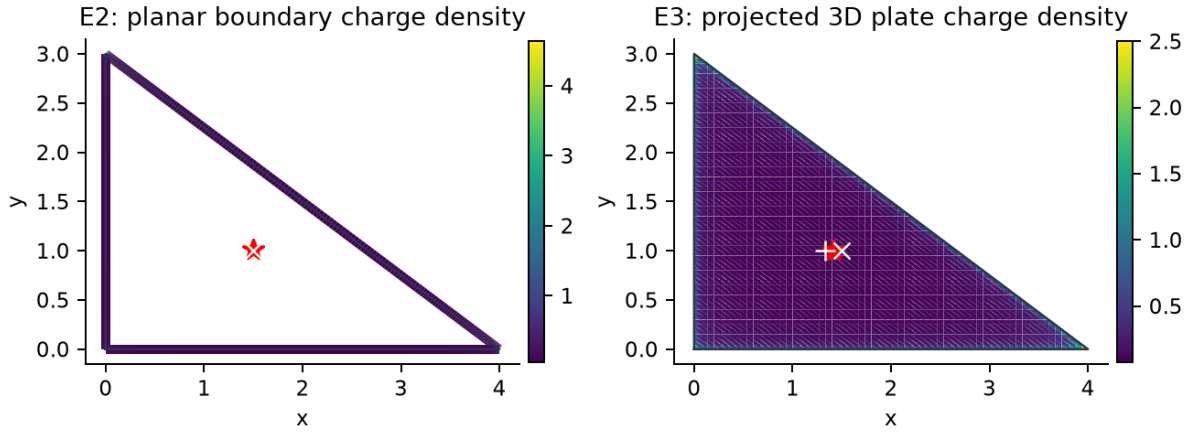


Figure 5. The two conductor distributions on the same triangle. Left: planar logarithmic charge per unit boundary length. Right: three-dimensional plate charge per unit projected area. The color scales differ because the units differ. The red stars mark their charge centroids; white comparison marks show X2 and X10. The right panel displays the $n=60$ subdivision used for the finest reference run.

2.8.3 Uniform-thickness limits above the capacity threshold

The convention throughout this project is a uniform geometric thickening. For a plate in $\mathbb{R}^n$, use its Euclidean epsilon-neighborhood, or an equivalent constant transverse cross section away from edges. For a skeleton, use tubes of a common radius, with compatible rounded junctions. Epsilon is then decreased relative to the triangle diameter. Varying the radius from side to side changes the model and can change the limit.

Let k be the intrinsic dimension, two for a plate and one for a skeleton. Near a regular point, integration over tangential separations gives a local leading energy proportional to $\int \rho^2$. At the critical exponent $s = k$ its coefficient grows as $\log(1/\epsilon)$. For $s > k$, scaling all local transverse and tangential distances by epsilon gives growth proportional to ϵ^{k-s} . The optimal transverse charge profile contributes a positive coefficient depending on the cross section; uniform cross section makes that coefficient independent of position along the underlying set.

For a probability density on a set of measure $|K|$, Cauchy-Schwarz gives $\int_K \rho^2 \geq 1/|K|$, with equality only at $\rho = 1/|K|$. The leading local energy thus favors uniform projected charge. A full limiting proof requires controlling nonlocal interactions and edge or junction layers; the scaling argument identifies the limit and its hypotheses, rather than a convergence rate or a finite-thickness error certificate. Thin-wire uniformity and its slow logarithmic approach in three dimensions are developed in [5]. Corners can be treated as shrinking junction regions in the stated uniform tubular model.

The resulting thin-limit center table is

Ambient dimension	Filled plate	Triangular skeleton
2	E2, logarithmic boundary charge	S2 = E2, exactly
3	E3, nonuniform area charge	S3 tends to X10
4	E4 tends to X2	S4 tends to X10
n greater than 4	Uniform-thickness limit X2	Uniform-radius projected limit X10

The entries labeled as limits are not finite-energy measures on the exactly zero-thickness Newtonian sets. They describe limits of normalized charges on positive-thickness conductors. Their capacitances tend to zero while their normalized first moments remain meaningful. At critical dimension the logarithmic divergence makes convergence particularly slow. The earlier softened-kernel data are bundled as illustrations of the local-energy mechanism and are explicitly excluded from claims of physical finite-thickness validation.

2.8.4 What conformal mapping can still do for E3

A planar conformal map $f : \mathbb{D} \rightarrow T$ may be used as a change of variables in the three-dimensional integral equation. If $d\nu(\zeta) = \rho(f(\zeta))|f'(\zeta)|^2 dA(\zeta)$, then the equation becomes $\int_{\mathbb{D}} |f(z) - f(\zeta)|^{-1} d\nu(\zeta) = V$, and $E3 = \int f d\nu$. This is an exact pulled-back formulation.

It is not the uniform-disk conductor problem: the transformed kernel retains f and is not simply $|z - \zeta|^{-1}$. Ordinary holomorphic conformal invariance belongs to the two-dimensional Laplace equation, not to the three-dimensional Coulomb kernel restricted to a plane. For higher-dimensional fields, Green functions, Kelvin inversion and integral equations provide the appropriate equivalents. Stating this limitation prevents a very attractive but incorrect shortcut.

2.9 The inner conformal center: E1

2.9.1 A grounded boundary and a small test electrode

Surround a tiny circular electrode centered at p by the grounded triangular boundary. In a genuinely two-dimensional model, its leading small-radius capacitance depends on the logarithm of a length associated with the whole domain. This length is the conformal radius $r_T(p)$, not just the nearest-side distance. For a small electrode radius ϵ , capacitance is asymptotically proportional to $1/\log[r_T(p)/\epsilon]$. Maximizing $r_T(p)$ therefore minimizes the small electrode's capacitance to the boundary at the first position-dependent order.

This is the least-capacity point E1. It is also called the inner conformal center or harmonic center in this planar setting. It does not redistribute the isolated conductor charge from Sections 2.7-8. The unknown is now a point inside a grounded cavity, rather than a probability measure on a conductor.

2.9.2 Green function and conformal radius

Let $\phi_p : T^\circ \rightarrow \mathbb{D}$ be a conformal map with $\phi_p(p) = 0$. The positive Dirichlet Green function, with logarithmic singularity normalized to one, is $G_T(z, p) = -\log|\phi_p(z)|$. It vanishes at the boundary. Near the pole, $\phi_p(z) = \phi'_p(p)(z - p) + O(|z - p|^2)$, so

$$G_T(z, p) = -\log|z - p| + \log r_T(p) + o(1), \quad r_T(p) = \frac{1}{|\phi'_p(p)|}.$$

In the convention $G = K - H$, the diagonal regular part is $R_T(p) = H(p, p) = -\log r_T(p)$. Minimizing R is equivalent to maximizing the radius. The word Robin function here names the regular part of a Dirichlet Green function; it does not mean that a Robin boundary condition has been imposed.

2.9.3 The literal least-capacity construction

Invert the exterior of the triangular cavity about p by $w = 1/(z - p)$, including infinity as the point zero. This produces a compact conductor K_p . If $F : \mathbb{D} \rightarrow T$ satisfies $F(0) = p$ and $F'(0) = r_T(p) > 0$, then $1/[F(1/w) - p]$ maps the disk exterior onto the exterior of K_p and has leading coefficient $1/r_T(p)$. Therefore

$$\text{cap}_2(K_p) = \frac{1}{r_T(p)}.$$

E1 literally minimizes the capacity of this family of inverted exterior conductors. One must invert the complement of the cavity, not take the capacity of the original triangle after a translation. The latter is constant and cannot select a center. A dimensional inversion radius can be restored; its fixed scale multiplies every capacity by the same factor and does not change the minimizer.

2.9.4 A computable formula for any triangle

Put the vertices at complex positions A, B, C in counterclockwise order, and let α, β, γ be their interior angles divided by π . They are positive and sum to one. The Schwarz-Christoffel map from the upper half-plane to the triangle, taking 0, 1 and infinity to A, B and C , is

$$F(z) = A + (B - A) \frac{B_z(\alpha, \beta)}{B(\alpha, \beta)}, \quad F'(z) = \frac{B - A}{B(\alpha, \beta)} z^{\alpha-1} (1 - z)^{\beta-1}.$$

The incomplete beta function means the integral of $t^{\alpha-1}(1-t)^{\beta-1}$ from 0 to z , with branches continued through the upper half-plane. It is a compact notation for a definite integration problem, not a new physical assumption. In the upper half-plane the conformal radius is $2 \operatorname{Im} z$, because a fractional linear map sends that point to the center of a unit disk. Local lengths under F are multiplied by $|F'|$, giving

$$r_T(F(u + iv)) = 2v|F'(u + iv)|.$$

Take its logarithm. Apart from a constant it is $\log v + (\alpha - 1) \log |u + iv| + (\beta - 1) \log |1 - u - iv|$. Differentiating with respect to u and v gives two rational equations. Their interior solution is

$$z_* = \frac{\alpha}{\alpha + \beta} + i \frac{\sqrt{\alpha\beta/\gamma}}{\alpha + \beta}, \quad E1 = F(z_*).$$

The radius tends to zero at the triangle boundary, and the stationary solution is unique. Thus this is the global maximum. Evaluation requires angles, a square root and one incomplete beta integral. High precision is useful for very thin triangles, where angle subtraction can lose digits. The literature on the least-capacity point provides special-triangle formulas and context [6]; the half-plane derivation here supplies a single general computational route.

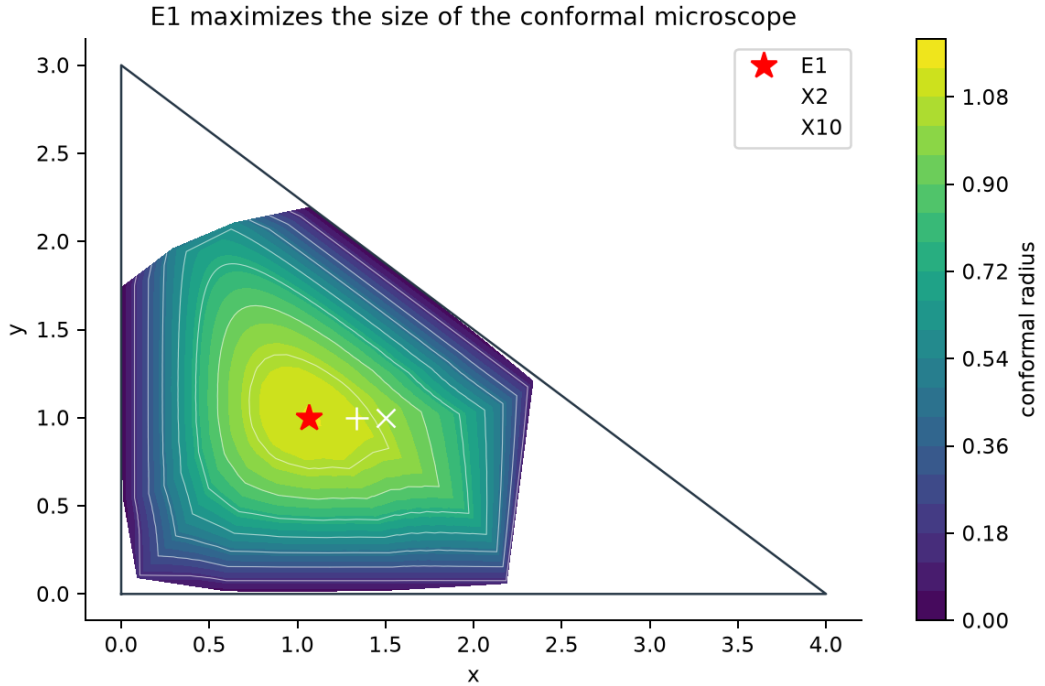


Figure 6. Conformal-radius contours in the 3-4-5 triangle. E1 is the unique maximum, distinct from both mass centroids. The contours are generated by sampling in the upper half-plane and mapping the samples into the triangle. The isolated maximum is exact in the defining problem; the plotted interpolation is illustrative.

2.9.5 Insulation does not mean stable particle trapping

The Green function's regular part describes the response of induced boundary charge to a point source. For an electric charge in a grounded cavity, the induced self-energy has the sign $-R_T(p)$, up to a positive factor and a position-independent divergent self term. At the Robin minimum its induced energy is a maximum. The least-capacity point is therefore not a stable electrostatic trap for a freely moving test charge. The mathematical minimum refers to capacity or boundary influence, not to the test charge's mechanical energy. It also says nothing by itself about vertex-motion attractivity.

2.10 Least capacity in higher-dimensional space

2.10.1 A cavity must be specified

A flat triangle has no interior as a subset of $\mathbb{R}^n$ when $n > 2$. Hence there is no intrinsic higher-dimensional Dirichlet Green function with that triangle as its cavity. The conductor problems of Section 2.8 remain meaningful through capacity or thickness limits, but they do not automatically define a higher-dimensional inner conformal center.

We adopt a parameter-free extension: $\Omega_n = T^\circ \times \mathbb{R}^{n-2}$, an infinite prism with grounded lateral boundary. Positions differing only in the axial coordinates are physically equivalent, so the selected point is its cross-sectional location p . For $n=2$ this reduces exactly to the triangular domain. A bounded, uniformly thickened cavity is a different extension, discussed below; it introduces an aspect ratio and cap geometry.

2.10.2 The Robin function and Kelvin inversion

For $n \geq 3$, normalize the Green function as $G_\Omega(x, p) = |x - p|^{2-n} - H_\Omega(x, p)$, with zero Dirichlet boundary values. Define $R_\Omega(p) = H_\Omega(p, p)$ and harmonic radius $r_\Omega(p) = R_\Omega(p)^{-1/(n-2)}$. The regular part is harmonic in x near p , and the diagonal value measures the boundary correction after removing the point-source singularity.

Let $\mathcal{I}_p(x) = (x - p)/|x - p|^2$ and let $K_p = \mathcal{I}_p(\Omega^c \cup \{\infty\})$. The Kelvin transform of a harmonic function is $|y|^{2-n}u(p + y/|y|^2)$. The multiplicative weight is essential: ordinary composition alone does not preserve harmonicity in dimension greater than two.

The equilibrium potential of K_p , normalized to equal one on it, is $|y|^{2-n}H_\Omega(p + y/|y|^2, p)$. On its boundary this equals one. As $|y| \rightarrow \infty$ it equals $R_\Omega(p)|y|^{2-n}$ to leading order. Comparing with the capacity coefficient of the far field gives

$$\text{cap}_n(K_p) = R_\Omega(p).$$

Thus the least-capacity center minimizes the Robin function, or maximizes harmonic radius, in every dimension under the stated kernel normalization. For a ball of radius a at its center, $R = a^{2-n}$ and the inverted exterior is a ball of radius $1/a$, providing an immediate normalization check. Restoring a fixed inversion radius only changes a common scale factor.

2.10.3 The prism family and a two-dimensional computational reduction

Define $L_n(T)$ as the cross-sectional minimizer of $R_{\Omega_n}(p, 0)$. Translation and rotation in the axial variables make their origin irrelevant. Similarity of the triangle rescales R but preserves its minimizer, and vertex relabeling changes nothing. Therefore each unique L_n is a triangle

center. These labels describe a proposed family within this chapter and do not claim priority of discovery.

A useful exact representation uses the Dirichlet heat kernel $k_T(t; p, p)$ for the generator Δ in the triangular cross section. The free axial heat kernel contributes a factor $(4\pi t)^{-(n-2)/2}$. With $c_n = 1/[(n-2)|S^{n-1}|]$, the coefficient in the fundamental solution of $-\Delta$, the unnormalized Robin function is

$$R_{\Omega_n}(p) = \frac{1}{c_n} \int_0^\infty (4\pi t)^{-(n-2)/2} \left[(4\pi t)^{-1} - k_T(t; p, p) \right] dt.$$

For an interior point the bracket is exponentially small at short time because the boundary is a positive distance away. At long time, the killed heat kernel decays while the free term is integrable for $n > 2$. This formula connects least capacity to the later heat chapter: it measures a weighted loss of return probability caused by the boundary. It is not the same statistic as the maximum of temperature from uniform initial data.

For computation we instead Fourier transform in the axial variables. Each frequency magnitude k produces a two-dimensional screened equation $(-\Delta + k^2)u = 0$. Its free fundamental solution is $K_0(k|p - q|)/(2\pi)$, where K_0 is the modified Bessel function. Write $h_k(p, p)$ for the boundary correction when the kernel is normalized as K_0 rather than $K_0/(2\pi)$. Then, up to a positive factor depending only on n ,

$$R_{\Omega_n}(p) \propto \int_0^\infty k^{n-3} h_k(p, p) dk.$$

This is a reduction to a family of planar boundary problems, not a holomorphic conformal equivalence. To compute h_k , solve a single-layer boundary integral equation whose boundary data are $K_0(k|q - p|)$, then evaluate the resulting harmonic correction at p . The companion uses graded line panels, analytic subtraction of the logarithmic self singularity, Gaussian quadrature for the remaining smooth panel integral, and Gaussian integration in $\log k$.

2.10.4 Existence, uniqueness, and numerical scope

The Robin function diverges when p approaches a side. For bounded convex cavities its convexity follows from harmonic-radius concavity [7]. Exhausting the infinite prism by increasing bounded convex prisms yields a convex Robin function on the cross section: the Green functions increase and their regular parts decrease to the prism limit. The limit remains convex. Interior regularity makes its restriction to a straight line real analytic. If two distinct points minimized it, convexity would make it constant on their segment; analyticity would extend that constancy along the entire interior chord, contradicting boundary divergence. Hence the cross-sectional minimizer exists and is unique.

This argument establishes the selector. The September 30 extension now supplies L3 and L4 with 100-shape and 40-holdout ETC comparisons, four-family diagnostics and refined negative finite-motion witnesses. Section 2.11 distinguishes these numerical witnesses from the interval certificates for E1 and E2. The bounded-cavity thickness family remains a separate, incompletely surveyed family.

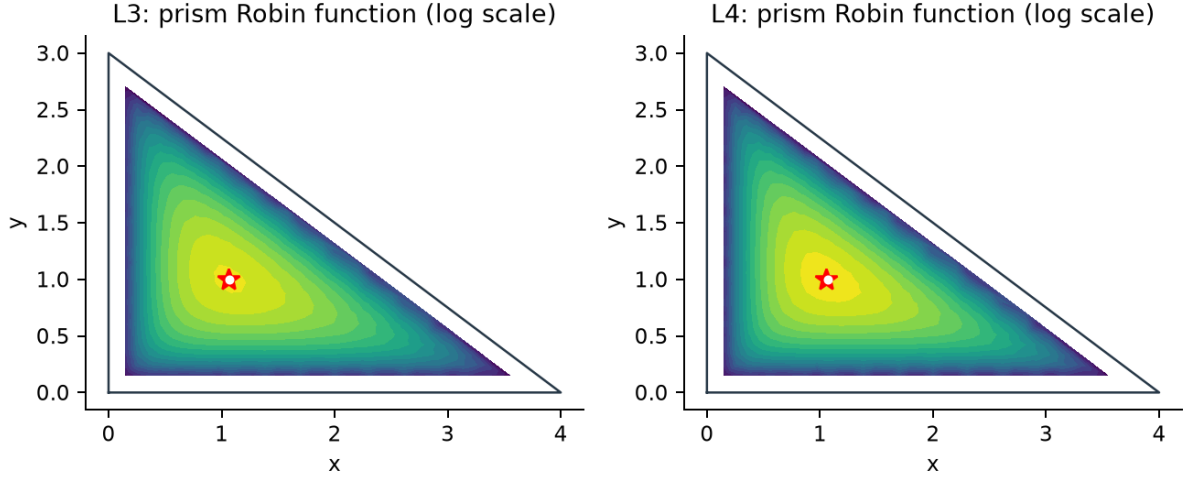


Figure 7. The cross-sectional Robin functions for three- and four-dimensional infinite prisms, shown on a logarithmic color scale. Red stars mark computed minima and white dots mark E1. The physical source is a point in the full ambient space; replacing it by a source uniform along the prism would give the two-dimensional logarithmic problem and E1 in every dimension.

2.10.5 A uniformly thickened cavity is a different family

For example, take $\Omega_{n,\eta} = T^\circ \times B_{\eta D}^{n-2}$, with all boundary parts grounded, and fix the dimensionless thickness $\eta > 0$. Its unique harmonic center lies on the axial symmetry set, giving a triangle center $L_{n,\eta}$. The same Green-function, Kelvin-capacity and small-electrode characterizations apply. Finite-element or boundary-element solvers must now include the end caps or transverse curved boundary.

Letting $\eta \rightarrow \infty$ recovers the infinite-prism cross-sectional problem. Letting $\eta \rightarrow 0$ is a separate singular cavity limit; it is not the uniform-charge thickening argument of Section 2.8. This chapter does not assign its limit to X1 or X2 without a dedicated asymptotic argument. Similarly, an isotropic rounded neighborhood of the plate differs at its side walls from the product cavity just defined. A dimension label alone does not fix these choices.

The physically important distinction is between mobility of source charge on a conductor, and the response of a grounded surrounding cavity to a point source. Their energy functionals, admissible objects and selected centers are different even when their drawings contain the same triangle.

2.11 Atlas: coordinates, fingerprints, and response to moving vertices

2.11.1 Coordinates and numerical refinement

Selector	x	y	Meaning of digits
X1	1	1	Exact comparison
X2	4/3	1	Exact area mean
X10	3/2	1	Exact length mean
E1	1.066232660	1.000366122	Special-function evaluation
E2	1.5024675	1.0157717	Last mesh change 1.4e-05
E3	1.4171419	1.0027539	Last mesh change 0.00016
L3	1.061201	1.000592	Last mesh change 0.0001
L4	1.056238	1.000829	Last mesh change 8.2e-05

All changes in this table are absolute coordinate distances in the reference units. The table retains extra digits as a reproducibility record, not as certified accuracy. In particular E3 continues to move under refinement, and the L3/L4 values have a measurable panel error. The exact thin-limit aliases $E4 = X2$ and $S3 = S4 = X10$ are not separate finite-thickness runs.

Calculation	Subdivision	Computed x	Computed y
E2: panels per side	24	1.502114245	1.015764362
E2: panels per side	48	1.502389553	1.015771941
E2: panels per side	96	1.502453101	1.015772008
E2: panels per side	192	1.502467522	1.015771658
E3: n squared panels	12	1.414435223	1.002942637
E3: n squared panels	20	1.415828085	1.002848310
E3: n squared panels	28	1.416396035	1.002806524
E3: n squared panels	36	1.416707027	1.002783827
E3: n squared panels	48	1.416978415	1.002764767
E3: n squared panels	60	1.417141904	1.002753921

The E2 sequence uses cosine endpoint grading. The E3 sequence uses uniform triangular panels, so corner and edge singularities limit its convergence rate. Do not interpret the last displayed value as an extrapolated continuum answer. The 3D plate capacity and the logarithmic capacity are recorded with their different normalizations in reference.json.

Prism panels per side	Frequency nodes	L3 coordinates	L4 coordinates
12	48	1.0580659, 1.0008109	1.0532800, 1.0013466
20	72	1.0604393, 1.0004142	1.0556587, 1.0006876
32	96	1.0609053, 1.0005178	1.0559995, 1.0007658
48	112	1.0611045, 1.0005670	1.0561584, 1.0008070
72	144	1.0612011, 1.0005919	1.0562378, 1.0008288

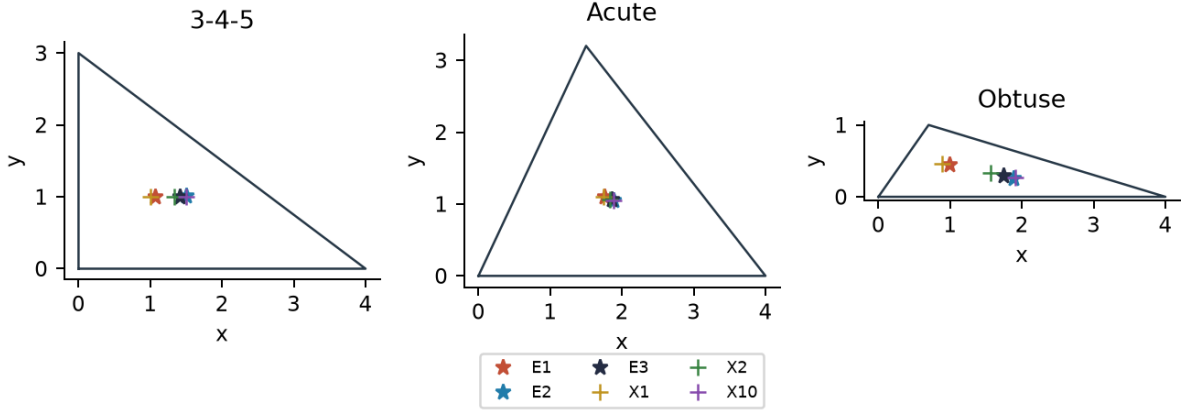


Figure 8. Centers of uniform area and perimeter mass compared with E1, E2 and E3 in right, acute and obtuse triangles. The markers are intentionally tied to the same physical definitions on all three shapes. Coincidence or close proximity on a single triangle is not an identification of center functions.

2.11.2 Four shape fingerprints

The isosceles family is $(-1, 0), (1, 0), (0, h)$. Its symmetry reduces each center to one height function. The right family used by the stored atlas is $(0, 1), (0, 0), (h, 0)$, so both position coordinates vary with h . This is a rotated and relabeled version of a right triangle whose variable leg is called its height. The convention is stated to make comparisons reproducible.

For the isosceles family, $G_y = h/3$. If $\ell = \sqrt{1 + h^2}$, then $W_y = h\ell/[2(1 + \ell)]$. Differentiating this expression gives its fingerprint derivative without finite differences. For the right family, with $\ell = \sqrt{1 + h^2}$ and $P = 1 + h + \ell$, $W_x = h(h + \ell)/(2P)$ and $W_y = (1 + \ell)/(2P)$. The area centroid is $(h/3, 1/3)$.

The nearly collinear family is $(-1, 0), (1, 0), (x, \epsilon)$ with $0 \leq x \leq 1$. Uniform area gives $G = (x/3, \epsilon/3)$. For fixed $x < 1$, uniform wire tends horizontally to zero and has $W_y/\epsilon \rightarrow 1/4$. At zero height the side segments double-cover the base interval, explaining the uniform longitudinal wire limit. For E1 the earlier asymptotic analysis gives

$$E1_x = x - \frac{\epsilon}{\pi} \operatorname{atanh} x + O(\epsilon^2), \quad E1_y = \frac{\epsilon}{2} + O(\epsilon^2).$$

This expansion is for fixed $x < 1$; its horizontal correction is not uniform at the endpoint. The endpoint $x = 1$ must be analyzed separately. E2 tends to the logarithmic equilibrium centroid of the limiting segment, which is its midpoint. That segment has an arcsine charge density, not uniform linear charge, despite sharing the same first moment. For E3 the finite-thin data are exploratory and should not be promoted to a proof of a simultaneous shape-and-capacity limit.

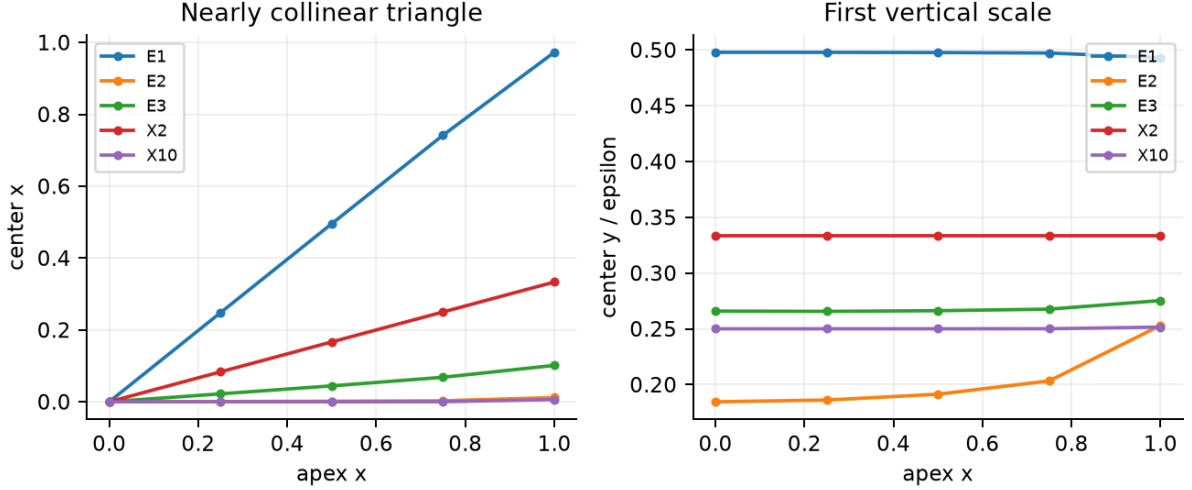


Figure 9. Nearly collinear fingerprints at $\epsilon = 0.025$, including the vertical coordinate divided by epsilon. This magnification reveals distinctions lost in the limiting horizontal picture. The curves are finite-epsilon computations from the earlier atlas, with their original diagnostics retained in the companion.

Near an equilateral triangle write $C = (x, \sqrt{3} + y)$ while $A = (-1, 0)$ and $B = (1, 0)$ remain fixed. Reflection symmetry gives an expansion $c_x = \alpha x + \beta xy + O(r^3)$ and $c_y = 1/\sqrt{3} + \gamma y + \delta x^2 + \eta y^2 + O(r^3)$, with $r = \sqrt{x^2 + y^2}$. Similarity and permutation covariance imply $\alpha + \gamma = 2/3$. Thus the two linear coefficients are not independent.

For X2, $\alpha = \gamma = 1/3$ and all higher coefficients vanish. For X10, $\alpha = 1/4$, $\gamma = 5/12$, $\beta = \sqrt{3}/16$, $\delta = \sqrt{3}/96$, and $\eta = 5\sqrt{3}/288$. The charge-center coefficients in the following atlas extract are numerical fits with stored step and resolution comparisons; they do not carry the same exact status as these two formulas.

2.11.3 Similarity to the ETC and attractivity

The inherited survey uses 100 shapes spread uniformly in area over the ordered angle simplex, plus 40 held-out shapes. Distances between centers are divided by D before averaging. Of the first 72,807 ETC entries in the local database, 70,853 evaluated finitely on the entire comparison grid; excluded expressions and nonfinite cases are recorded. The tables below rank locations. Fingerprint derivatives and response margins are additional diagnostics, not a hidden alteration of that ranking.

For a center c , let J_A be its derivative with respect to moving vertex A while B and C stay fixed. Local vertex attractivity means $h \cdot J_A h \geq 0$ for every vector h , equivalently positive semidefiniteness of $(J_A + J_A^T)/2$. There are analogous tests for B and C . A finite displacement test uses $h \cdot [c(A + h, B, C) - c(A, B, C)]$. This is the user's geometric meaning of attractivity, distinct from physical stability and from iterative convergence.

X2 is strictly attractive: its vertex derivative is the identity divided by three. X10 is also strictly attractive, as proved in the supplied Attractive Centers IV, Section 2.4.1; the derivative proof is reproduced below. E1 and E2 have interval-certified negative finite motions. E3, L3 and L4 have refined numerical negative witnesses. The adopted thin-conductor aliases E4, S3 and S4 inherit the exact classical attractivity theorems.

The September 30 survey visits all 72,807 entries of the supplied ETC snapshot. The restricted evaluator retains 70,853 candidates finite on all 100 training shapes; the other 1,954 entries are excluded, not proved dissimilar. Each top-ten list is checked on 40 held-out shapes and compared separately by position fingerprints, derivative fingerprints and three-vertex response

margins. Location RMS divided by triangle diameter is the ranking criterion. Thus this is a whole-snapshot computational survey, not an exhaustive symbolic classification or a proof of ETC nonmembership. E2 uses the independent exterior conformal evaluation. E3 retains the earlier resolved uniform-panel location grid; the new coarse strip-mesh grid is exploratory and excluded because it has poorer general-shape convergence. Its separate, finely resolved thin-triangle negative-motion experiment has positive charges at every level.

2.11.4 E1: updated individual atlas entry

Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.02086913).

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X53588	0.00253639	0.00246438	0.00530917
X66575	0.00293025	0.00297047	0.00670528
X66560	0.00403642	0.00399903	0.00582504
X29624	0.0042111	0.00389788	0.00830898
X8965	0.00435424	0.00433406	0.00686318

For X53588, position-fingerprint RMS / D is 0.00145761, derivative-fingerprint RMS is 0.00888637, and response-margin RMS difference is 0.0088342. The fingerprint comparison has 51 finite samples out of 51. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

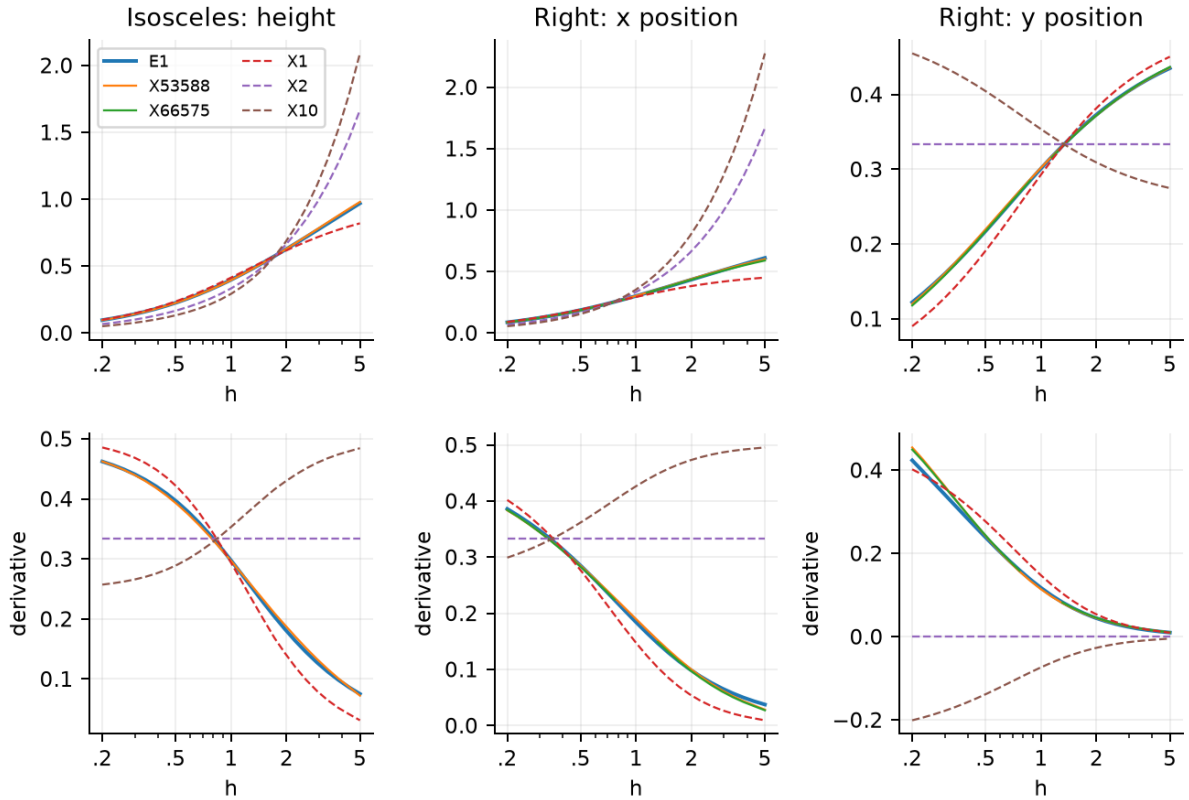


Figure. E1 and the two leading ETC candidates, with X1, X2 and X10. The diagrams retain the inherited numerical family samples; the refreshed full-snapshot rankings above use the declared grid and independent holdouts. Finite differences and mesh changes are numerical diagnostics.

2.11.5 E2: updated individual atlas entry

Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.04665096), independently evaluated with the exterior Schwarz-Christoffel map.

Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X59474	0.00134526	0.00130542	0.0024001
X5235	0.00222067	0.00213914	0.00350119
X49731	0.00228472	0.0021109	0.00373791
X31494	0.00255483	0.00248814	0.00454243
X25960	0.00286186	0.00266665	0.00566459

For X59474, position-fingerprint RMS / D is 0.00152878, derivative-fingerprint RMS is 0.00348358, and response-margin RMS difference is 0.00444283. The fingerprint comparison has 51 finite samples out of 51. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

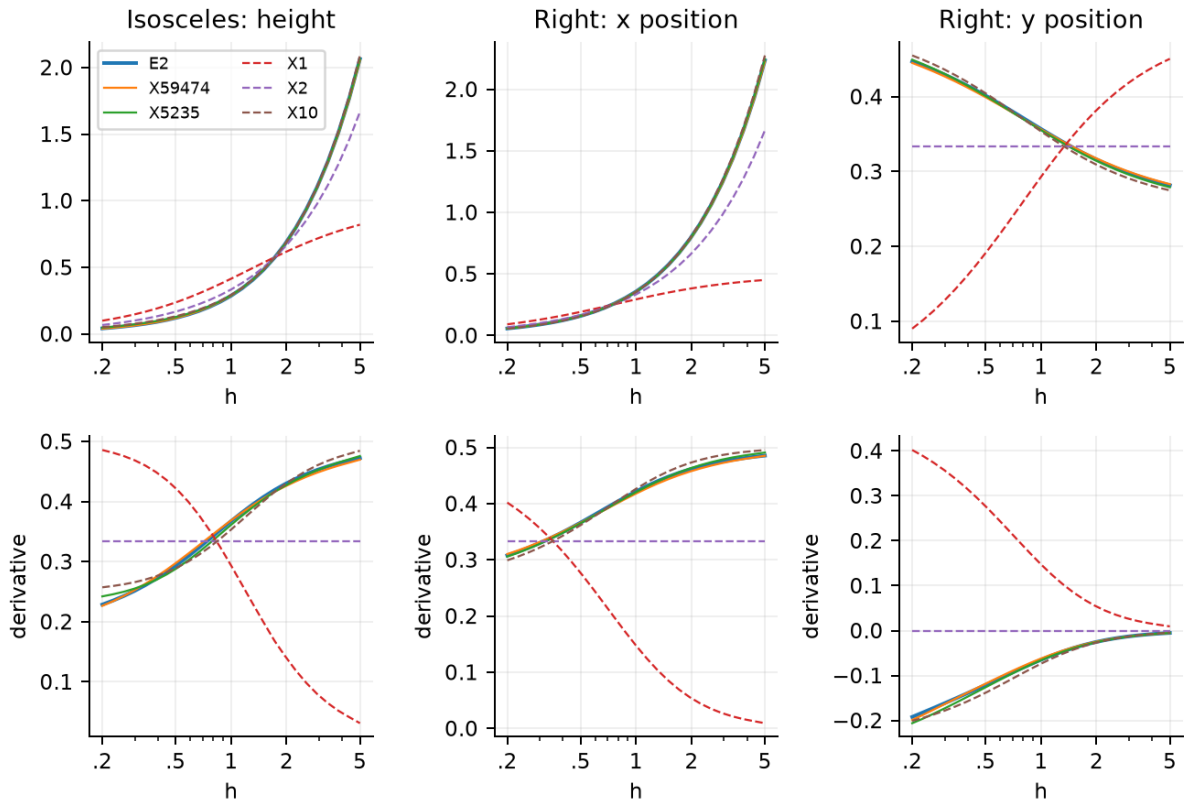


Figure. E2 and the two leading ETC candidates, with X1, X2 and X10. The diagrams retain the inherited numerical family samples; the refreshed full-snapshot rankings above use the declared grid and independent holdouts. Finite differences and mesh changes are numerical diagnostics.

2.11.6 E3: updated individual atlas entry

Refined numerical evidence of nonattractivity: negative finite motions persist on positive-charge anisotropic meshes up to 3,180 panels. The finest halved-step ratio is -0.55181093. A continuum interval certificate remains open.

Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X27760	0.00268211	0.00435905	0.0193914
X31351	0.00379642	0.0040384	0.0107557
X46896	0.00521082	0.0079198	0.0334464
X30915	0.00522478	0.00795365	0.0336552
X9780	0.00527946	0.00794309	0.0333402

For X27760, position-fingerprint RMS / D is 0.00237028, derivative-fingerprint RMS is 0.00583816, and response-margin RMS difference is 0.0245666. The fingerprint comparison has 51 finite samples out of 51. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

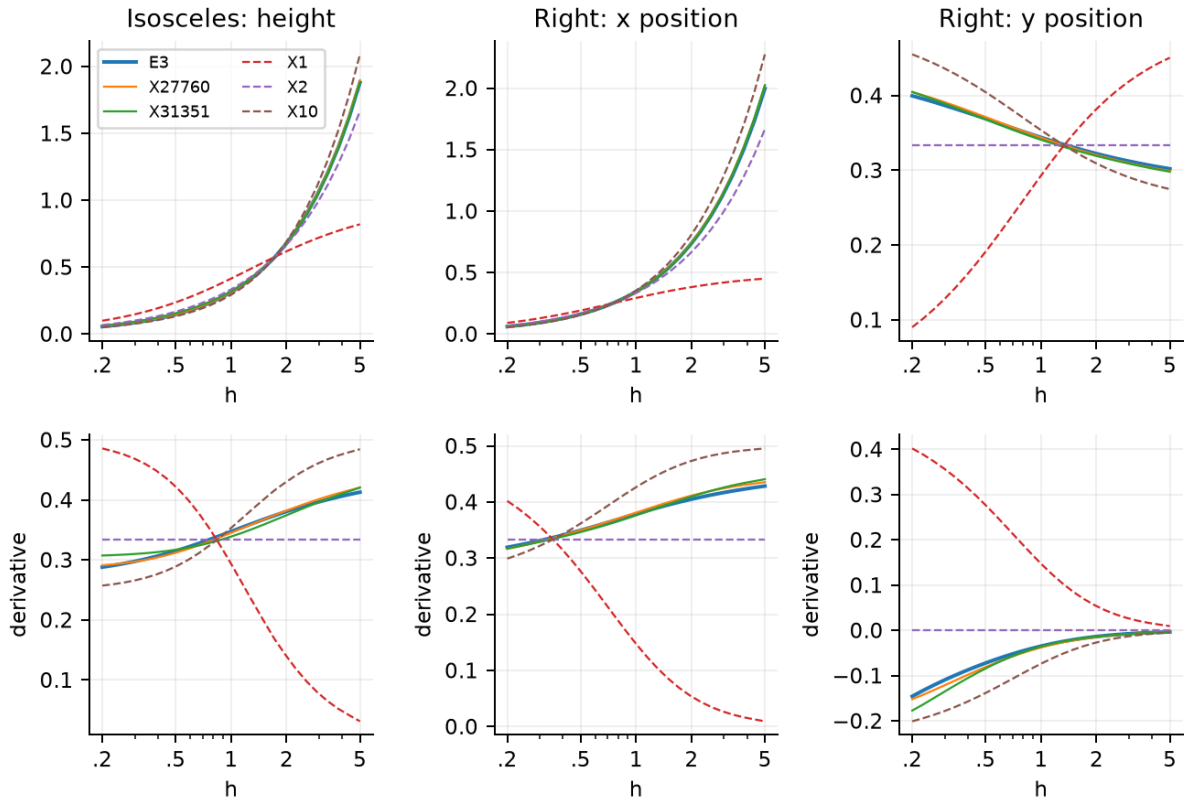


Figure. E3 and the two leading ETC candidates, with X1, X2 and X10. The diagrams retain the inherited numerical family samples; the refreshed full-snapshot rankings above use the declared grid and independent holdouts. Finite differences and mesh changes are numerical diagnostics.

2.11.7 L3 and L4: complete finite-sample atlas extensions

For $n = 3$ or 4 , ground the boundary of the infinite prism T times $\mathbb{R}^{n-2}$. Subtract the free-space singularity from its Dirichlet Green function at a point of the cross-section. Minimize the resulting Robin function, equivalently maximize harmonic radius. Translation invariance in the axial directions leaves a canonical point in T . L2 is E1. These selectors differ from the fixed-charge conductor centroids E2 and E3.

Fourier transformation in the axial coordinates reduces the problem to modified Helmholtz boundary problems in T . For each frequency k the planar kernel is proportional to $K_0(k \text{ times distance})$. Subtract its logarithmic self singularity analytically, solve the graded boundary system, and integrate in frequency with weight k^{n-3} . The new solver grades panels near both endpoints and source projections, scales frequency by the inradius, and optimizes inside the analytically certified region. Parameter m describes the grading density; merged panels are not simply m panels per side. Log-frequency quadrature covers -14 to $\log(30)$ in k times inradius. Cutoff and mesh errors remain numerical, not interval bounds.

The sample comprises 100 original shape points, 40 independent holdouts, 17 isosceles heights, 17 right-triangle heights, 25 thin configurations and two equilateral step sizes at two resolutions. All raw values, derivative steps and optimization diagnostics are retained.

L3: Three-dimensional prism Robin center

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01582971 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

The maximum training-grid change between grading parameters 10 and 18 was $0.000207269 D$; 75/100 changes were below $0.0001 D$. These changes are resolution diagnostics, not error bounds. At equilateral, the computed horizontal and vertical apex derivatives at grading 20 and step 0.005 are 0.4666520 and 0.2000183 . Symmetry forces zero mixed terms and trace $2/3$ in the smooth exact limit.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X66575	0.0021808	0.00208763	0.00460587
X53588	0.00262174	0.00243532	0.00413046
X16015	0.00376829	0.00389519	0.00819394
X29624	0.00380045	0.0032745	0.00812781
X59464	0.00384197	0.00388004	0.00733217

For X66575, position-fingerprint RMS / D is 0.00162041 , derivative-fingerprint RMS is 0.0137814 , and response-margin RMS difference is 0.00859114 . The fingerprint comparison has 17 finite samples out of 34. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

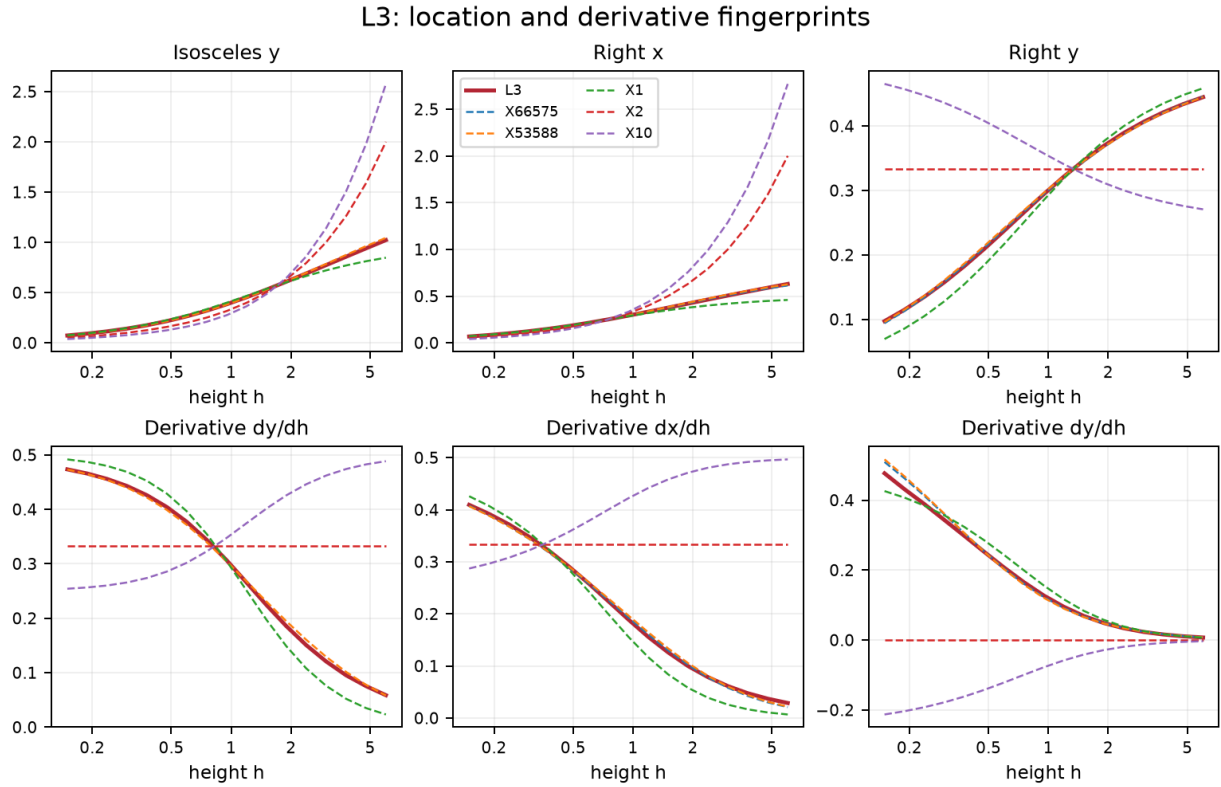


Figure. L3 position and derivative fingerprints, with its two leading ETC candidates and X1, X2, X10. The leading candidate X66575 is undefined in the supplied expression on the isosceles family, leaving only 17 of 34 fingerprint comparisons; its absent curve is intentional. No removable extension or identity is assumed.

L4: Four-dimensional prism Robin center

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01102191 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

The maximum training-grid change between grading parameters 10 and 18 was 0.000304191 D; 84/100 changes were below 0.0001 D. These changes are resolution diagnostics, not error bounds. At equilateral, the computed horizontal and vertical apex derivatives at grading 20 and step 0.005 are 0.4693248 and 0.1973406. Symmetry forces zero mixed terms and trace $2/3$ in the smooth exact limit.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X66575	0.00233082	0.00212511	0.00388145
X16015	0.0031703	0.00319691	0.0060394
X59464	0.00333147	0.00331106	0.00587152
X5703	0.00342174	0.00336797	0.00740775
X53588	0.00344042	0.00322926	0.00627888

For X66575, position-fingerprint RMS / D is 0.00261203, derivative-fingerprint RMS is 0.014949, and response-margin RMS difference is 0.00828281. The fingerprint comparison has 17 finite samples out of 34. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

L4: location and derivative fingerprints

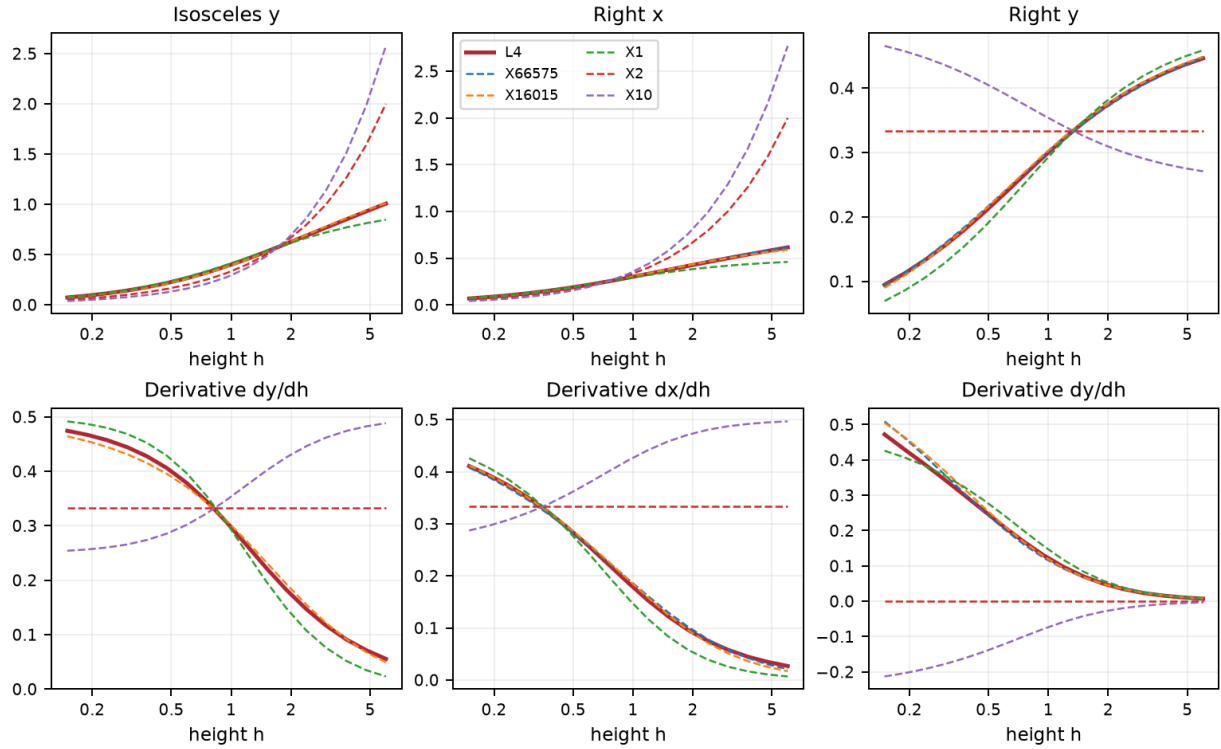


Figure. L4 position and derivative fingerprints, with its two leading ETC candidates and X1, X2, X10. The leading candidate X66575 is undefined in the supplied expression on the isosceles family, leaving only 17 of 34 fingerprint comparisons; its absent curve is intentional. No removable extension or identity is assumed.

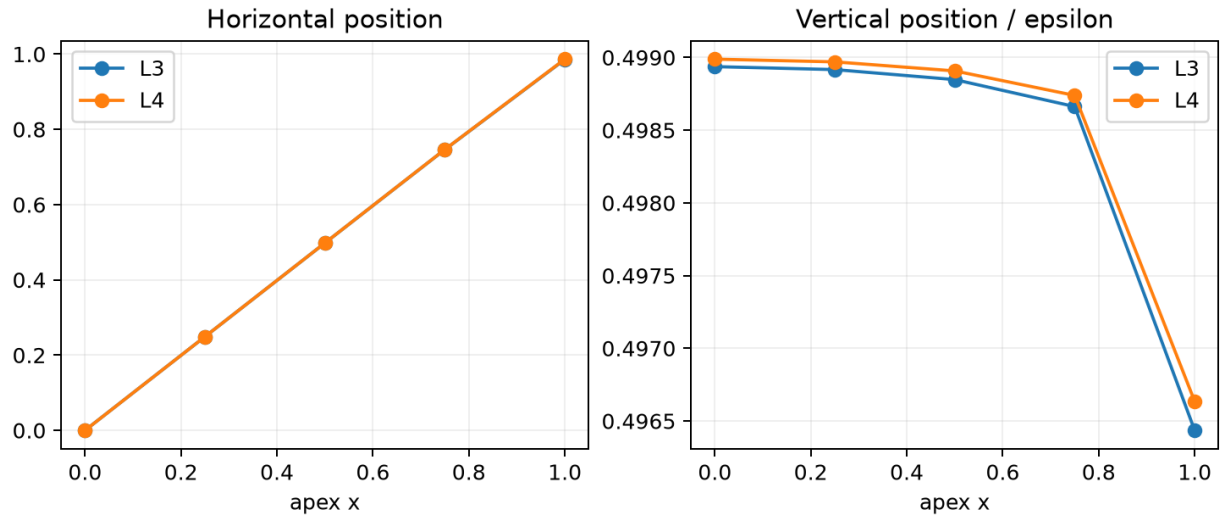


Figure. Finite-height collinear fingerprints at $\epsilon = 0.0125$ for L3 and L4, including the endpoint $x=1$. These are numerical thin configurations, not an asymptotic theorem.

The same refined witness is also negative for L6 (about -0.00254498). Its positive L10 value does not classify L10. No finite survey settles all dimensions n or every bounded-cavity thickness ϵ . Those parameter families retain separate open entries; they are not counted as fully surveyed centers.

Exact attractivity of X10 and its thin-conductor aliases

The uniform-wire centroid $W = X10$ is strictly vertex-attractive. This theorem was already in the supplied Attractive Centers IV, Section 2.4.1, and was missed in the earlier chapter. Here is the derivative calculation, so the conclusion does not depend on a positive numerical screen. Hold B and C fixed. Put $a = |B-C|$, $b = |C-A|$, $c = |A-B|$, $P = a+b+c$, $p = (B-A)/c$, $q = (C-A)/b$, and $u = p \cdot q$. Differentiate the length-weighted edge-midpoint mean. The symmetric part of its derivative with respect to A is

$$S_A(W) = \frac{b+c}{2P}I - \frac{bc}{2P^2}(pp^T + qq^T) + \frac{c(a+c) + b(a+b)}{4P^2}(pq^T + qp^T).$$

The directions $p+q$ and $p-q$ diagonalize this matrix. Their eigenvalues are

$$\mu_+ = \frac{2P(b+c) + (1+u)((b-c)^2 + a(b+c))}{4P^2}, \quad \mu_- = \frac{(b+c)(1+u)}{4P}.$$

Both are positive for a nondegenerate triangle. Integrating the positive quadratic form along any straight vertex displacement staying in the same nondegenerate half-plane gives a strictly positive finite-motion dot product. This proves the assertion globally on that domain. In particular the adopted uniform-thickness limits S3 and S4 inherit strict attractivity from X10. E4 inherits it from X2, whose derivative is $I/3$. A singular zero-thickness conductor with zero capacity is not itself being assigned an equilibrium distribution.

A location certificate for least-capacity centers

Let r be the triangle's inradius. Normalize the harmonic radius so a ball of radius r has harmonic radius r at its center. Domain monotonicity gives harmonic radius at least r at the incenter: the ball of radius r is contained in the infinite prism. On the other hand, the prism is contained in each side half-space. Reflection in the plane boundary gives harmonic radius exactly twice the distance d to that plane. A point maximizing harmonic radius must therefore satisfy

$$r \leq r_\Omega(L_n) \leq 2d_i(L_n), \quad d_i(L_n) \geq r/2.$$

The identical argument with conformal radius proves the planar E1 assertion. Thus all these centers lie in the closed inner parallel triangle obtained by shifting every side inward by $r/2$: the homothetic copy $I + (T-I)/2$. The proof applies to every infinite prism dimension, independently of a numerical mesh. For the bounded product cavity T times a ball of radius η D , replace r on the lower side by $\min(r, \eta D)$; the infinite-prism bound cannot be imported unchanged into a thin bounded cavity.

What the two new interval certificates prove

For E1, the incomplete-beta conformal map is evaluated by a hypergeometric series with outward-rounded complex intervals. In the retained disk $|z| < 1$, successive coefficient ratios are below one. The remaining tail after 201 terms is bounded by the last term times $|z|/(1-|z|)$. A 65-digit calculation encloses the finite-motion ratio in $[-0.020869126, -0.020869125]$.

For E2, the charge centroid equals the constant Laurent coefficient of the exterior map. An independent three-prevertex Schwarz-Christoffel construction, interval Taylor integration and explicit endpoint bounds enclose its finite-motion ratio in $[-0.046650959, -0.046650958]$. Both intervals lie strictly below zero. Decimal inputs, before/after coordinate enclosures, tail bounds

and executable verifiers are supplied. These certificates disprove global vertex attractivity; they do not assert that every triangle or every direction has negative response. The full exterior-map derivation is in the audit companion’s certificate proof note.

2.12 Physical interpretations that can be demonstrated in a classroom

A cardboard triangle of uniform thickness can be balanced on a pin or suspended from two different vertices. The vertical lines through the suspension points intersect at X2. A triangular frame made from uniform wire reveals X10 by the same procedure. Making the three rods have equal masses instead of equal linear density moves the balance point to X2; this is a direct demonstration of why the word uniform needs a reference measure.

A turntable with a light, accurately positioned mounting pivot can compare perpendicular moments around nearby points. The minimum occurs at the centroid appropriate to the mass distribution. Oscillations about axes in the plane, with gravity and suspension stiffness accounted for, can estimate principal moments. Their square-root ratio predicts the lamina’s Steiner axis ratio. On a scalene wire frame it predicts an inertia ellipse that fails the three-side tangency test of Section 2.4.

A two-dimensional electrostatic analogue can use a conducting sheet or resistive paper outside a triangular electrode, with a distant enclosing electrode approximating infinity. Current flux into the triangular boundary estimates harmonic measure, after accounting for the finite outer boundary. Equal intervals of boundary arclength do not collect equal flux. This demonstrates the measure behind E2; it does not measure the three-dimensional free-space plate distribution E3.

The least-capacity point can be visualized computationally by placing a small circular electrode at different interior positions and measuring its capacitance to the grounded triangle. The small electrode’s radius must be held fixed, and the asymptotic interpretation requires it to be much smaller than its distance to the boundary. A finite-size probe defines its own size-dependent selector. The conformal-radius contours reveal the limiting selector E1.

Finally, place three equal long parallel currents through the vertices and map the transverse magnetic field. Its two interior zeros identify the Steiner foci, reconnecting a mass-inertia ellipse to a different physical discipline. This demonstration belongs to the magnetostatics chapter in detail; it is included here to show why retaining the full second-moment matrix, rather than only its centroid, pays off later.

Chapter 3

Magnetostatics and triangle centers

This chapter asks how a triangle can select a point when the source is a steady electric current. Four experiments are considered: current circulating around a triangular wire, current in three long wires through the vertices, current spread over the lateral surface of a triangular prism, and current spread throughout that prism. Each has a uniform source in a specified sense. None has a uniform magnetic field.

We shall derive the fields, locate their extrema when possible, and ask where a sheet weighted by field magnitude or field energy would balance. Some answers recover familiar centers. Others give integral-defined centers, and some experiments produce two distinguished points rather than one. The distinction between a well-defined centroid and a divergent ratio is essential to the geometry.

The only results borrowed from the preceding mass-and-charge chapters are the centroid of a uniform triangular lamina, the centroid of its perimeter, and elementary second-moment tensors. We recall them before using them. Vector calculus is introduced where it is needed. The complex-number treatment and more advanced ETC relations are developed in [Appendix A.2](#), so the physical discussion can be read first.

Reading route. Sections [3.1-6](#) develop the four physical models. Sections [3.7-9](#) compare their centers and explain the computations. Section [3.10](#) contains the individual atlas entries M1-M7. [Appendix A.2](#) connects the Steiner inellipse to inertia and higher ETC geometry; [Appendix E.2](#) collects reproducibility details and exercises. A numerical result is explicitly distinguished from a theorem throughout.

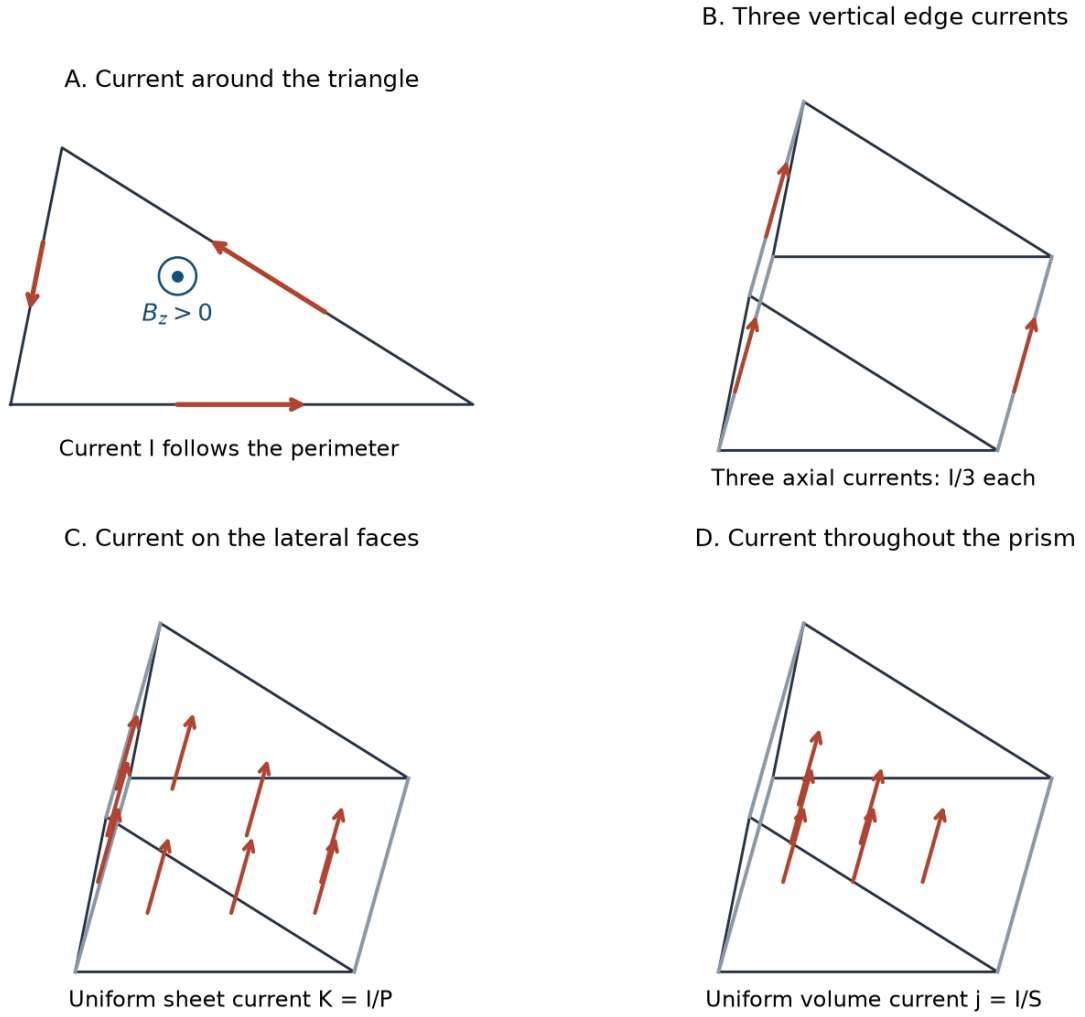


Figure 1. The four experiments. Red arrows indicate current, not magnetic field. In A the current circulates in the horizontal plane. In B-D all current flows in the axial direction. A dot inside a circle means a vector pointing towards the reader. The drawn prism is a schematic portion of an effectively infinite prism; its top and bottom faces do not carry the prescribed axial sheet current.

3.1 What is uniform, and what is a center?

3.1.1 Geometry inherited from the mass chapter

Let T be a nondegenerate triangle with position vectors A, B, C , counterclockwise in the plane. Its opposite side lengths are a, b, c , its area is S , its perimeter is $P = a + b + c$, and its interior angles are α, β, γ . The same letters A, B and C are often used for angles in triangle geometry; we use Greek letters to avoid confusing an angle with a position vector.

The centroid of uniform area mass and the centroid of uniform perimeter mass are, respectively,

$$G = X_2 = \frac{A + B + C}{3}, \quad W = X_{10} = \frac{(b + c)A + (c + a)B + (a + b)C}{2P}.$$

To recover the second formula, replace each uniform side by its mass at its midpoint. Side BC contributes $a(B + C)/2$, and similarly for the other sides. Dividing their sum by P gives W . This is the Spieker center, also the incenter of the medial triangle [1].

Our reference triangle will always be

$$A = (0, 0), \quad B = (4, 0), \quad C = (0, 3), \quad (a, b, c) = (5, 3, 4).$$

It has $S = 6$, $P = 12$, $X_1 = (1, 1)$, $G = (4/3, 1)$ and $W = (3/2, 1)$. Lengths can be in any consistent unit. The current and permeability multiply all field values, but do not alter the locations of the normalized centers discussed here.

3.1.2 Four different meanings of a uniform current

A steady current I in a single closed wire has the same current through every cross-section of that wire. In our ideal triangular loop the current is not divided into three parts: I passes successively along all three sides. Uniform wire thickness is needed when a finite-radius regularization is invoked, but not to write the filamentary Biot-Savart field.

In the three-prism experiments the currents are parallel rather than successive. In the vertex-wire case each vertical wire carries $I/3$. In the lateral-surface case the sheet current density is $K = I/P$, measured in amperes per unit transverse perimeter length. In the volume case the current density is $j = I/S$, measured in amperes per unit cross-sectional area. A side of length a therefore carries total axial current Ia/P in the sheet model; it does not generally carry $I/3$.

The prism is treated as infinite, or observed far from its ends on scales small compared with its length. The medium has constant permeability μ_0 . A real steady-current experiment must include a source and a return path. The local infinite-prism model neglects their field. Its full-plane limits are consequently mathematical limits of that idealization, not predictions independent of the return circuit.

3.1.3 Minimum, mean and center are different questions

For a nonnegative weight w on a region D , its centroid is

$$C_w(D) = \frac{\int_D p w(p) dA}{\int_D w(p) dA}.$$

The denominator must be finite and positive, and the first moment must exist. On a bounded triangle, a finite integral of w automatically gives a finite first moment. In the whole plane that implication fails. A signed weight can also have total weight zero, making a normalized centroid impossible.

We use the weights $B = |\mathbf{B}|$ and $B^2 = \mathbf{B} \cdot \mathbf{B}$. Magnetic energy density in a linear medium is $B^2/(2\mu_0)$; the constant cancels in the centroid. Thus an energy centroid is a physically meaningful mean. A minimum of B is a point of weak field, not the same physical quantity as a mean. Because squaring is increasing on nonnegative numbers, B and its square have the same extremum locations.

For the loop, the field in its plane is normal to the plane: $\mathbf{B} = B_z \hat{z}$, so $B = |B_z|$. For the axial currents, the field is transverse and $B_z = 0$. Carrying the notation B_z from the loop into a prism would therefore create a zero weight everywhere. In the prism sections B always means the transverse magnitude.

3.2 From a current element to an observable field

3.2.1 The Biot-Savart law and superposition

For a thin wire carrying a steady current I , a directed element $d\ell$ at q contributes

$$d\mathbf{B}(p) = \frac{\mu_0 I}{4\pi} \frac{d\ell \times (p - q)}{|p - q|^3}.$$

The cross product specifies the right-hand-rule direction. Its magnitude contributes a sine of the angle between the current element and the displacement to the observation point. Integrating adds the vectors from all source elements. The dependence on I is linear, but taking a magnitude or squaring the total field is nonlinear. In particular, the square of a sum contains cross terms; one must add fields before squaring [10, 11].

For planar vectors define the rotation $J(x, y) = (-y, x)$. Then $\hat{z} \times (x, y, 0)$ has transverse coordinates $J(x, y)$. This notation will make the prism formulas compact while preserving their direction.

3.2.2 What a contour diagram does and does not show

A contour map of B shows equal magnitudes. It does not show the vector direction. We therefore include both magnitude plots and streamline plots. Very large values near an ideal wire would overwhelm a color scale; some figures deliberately saturate their top few percent. Captions state this. Saturation never makes a singular field bounded.

A zero on a magnitude plot is a minimum of magnetic energy density. It is not by itself a proof of stable motion of a charged particle: the magnetic force is $q\mathbf{v} \times \mathbf{B}$, depends on velocity, and does no work on a point charge. Nor is a low-field minimum the same as the geometric attractivity of a triangle center under vertex motion, defined in Section 3.9.

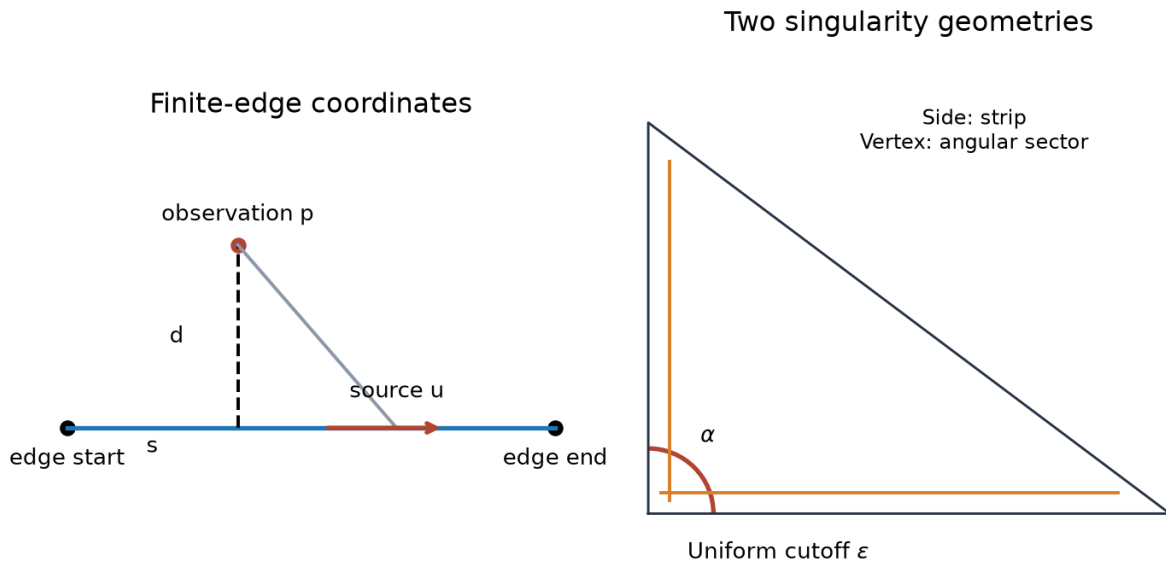


Figure 2. Left: coordinates used to integrate a straight side. The coordinate s is the projection of p along the side, and d is its perpendicular distance. Right: a boundary strip and a vertex sector lead to different divergent weights. This difference will explain why the loop produces X10 whereas vertex currents produce an angle-weighted center.

3.3 Case A: current circulating around the triangular perimeter

3.3.1 Integrating one side

Choose a side of length l , with unit tangent t in the current direction and inward unit normal $n = Jt$. Write an interior observation point as $p = q_0 + st + dn$, where $d > 0$. A source point on the side is $q(u) = q_0 + ut$, with $0 \leq u \leq l$. The current element is $t du$. Its cross product with $p - q(u)$ has z component d , independent of u . Thus

$$B_{z,e}(p) = \frac{\mu_0 I}{4\pi} \int_0^l \frac{d du}{((s-u)^2 + d^2)^{3/2}}.$$

An antiderivative is obtained by differentiating $v/\sqrt{v^2 + d^2}$. Evaluating at the two endpoints gives

$$B_{z,e}(p) = \frac{\mu_0 I}{4\pi d} \left(\frac{s}{\sqrt{s^2 + d^2}} + \frac{l-s}{\sqrt{(l-s)^2 + d^2}} \right).$$

This expression remains valid when the perpendicular projection falls beyond an endpoint; s need not lie between 0 and l . Summing the three sides gives the exact filamentary field. For counterclockwise current every element contributes positive z field at an interior point. Outside the triangle some contributions reverse sign.

For computation it is convenient to remove the dimensional constant and put $b = 4\pi B_z/(\mu_0 I)$. The function b has units of inverse length. We minimize b instead of B_z because they have the same minimizing point.

3.3.2 M1: why a unique minimum exists

Define M1 as the point minimizing B inside T . As a point approaches the middle of a side, the field behaves like $\mu_0 I/(2\pi d)$ and tends to infinity. The same conclusion holds on approach to a vertex from inside. The minimum is therefore attained away from the boundary.

A useful geometric representation establishes more than existence. Let $R_p(u)$ be the distance from p to the boundary along a unit direction u . Then

$$b(p) = \int_{\mathbb{R}^2 \setminus T} |q - p|^{-3} dA(q) = \int_0^{2\pi} \frac{d\theta}{R_p(u_\theta)}.$$

Here is why the formula holds. In the source plane the divergence of $(q - p)/|q - p|^3$ is $-|q - p|^{-3}$. Apply the divergence theorem to the exterior of the triangle, where p is absent. The outward normal of this exterior region points into T along the inner boundary. The resulting boundary integral is exactly the sum of the three wire expressions. The boundary contribution at infinity vanishes. Polar integration about p then gives the second expression, since the exterior begins at radius R_p in each direction and the radial integral is the integral of r^{-2} .

If $d_i(p)$ is the distance to side i and n_i its outward unit normal, the first side hit in direction u gives

$$\frac{1}{R_p(u)} = \max_i \frac{(n_i \cdot u)_+}{d_i(p)}, \quad x_+ = \max(x, 0).$$

Each positive numerator is constant as p varies, while $d_i(p)$ is positive affine. The reciprocal of a positive affine function is convex. Taking a maximum preserves convexity, and so does

integration. Strictness follows because every nonzero displacement changes the distance to at least one side; that side is the first side hit for an open interval of directions. Its reciprocal has a strictly positive second derivative in that displacement direction. Consequently b is strictly convex and $M1$ is unique.

The field formula uses rational operations and square roots only. Its stationary equations can be rewritten as polynomial equations by introducing positive auxiliary distances. $M1$ is therefore implicitly algebraic; a simple barycentric expression is not known here, but it should not be described as proved transcendental.

3.3.3 Computing and enclosing $M1$

One can minimize b in two barycentric coordinates while enforcing positive barycentric weights, then refine the equation $\nabla b = 0$ by Newton's method. Strict convexity removes ambiguity about which interior stationary point has been found. An implementation must still check its residual and avoid crossing the boundary.

There is also an inexpensive analytical location certificate. The exterior half-plane beyond side i is contained in the exterior of T . Its integral in the representation above equals $2/d_i$. Therefore, for any chosen interior trial point p_0 ,

$$d_i(M1) \geq \frac{2}{b(M1)} \geq \frac{2}{b(p_0)}.$$

This gives a closed inner triangular region containing $M1$. It is a true geometric bound, not a contour-plot impression. Its numerical value becomes a validated certificate if the trial-point field is enclosed using outward-rounded arithmetic. The earlier atlas supplies an interval-certified $M1$ counterexample to vertex attractivity; its certificate is retained in the chapter companion.

For the reference triangle, the stationary solution is approximately $(1.103976518, 0.998576185)$. It lies close to the incenter $(1, 1)$, but the two points are not equal. The field minimum and its square have the same location, so they do not deserve separate M indices.

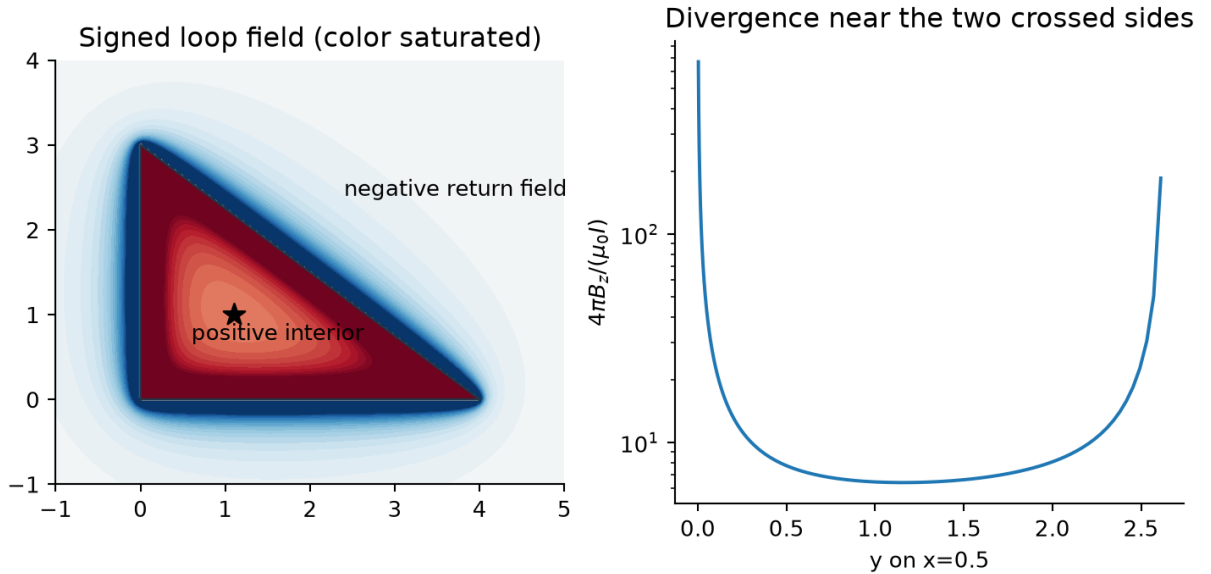


Figure 3. The loop's normal field is positive inside and has a negative return field outside. Colors are saturated to expose the sign structure; the black star is $M1$. The profile on $x=0.5$ diverges at the two sides it crosses. The plotting function is the exact finite-edge expression, not a finite-element approximation.

3.3.4 Interior means of B and B squared: both lead to X10

The unregularized mean of B does not exist: its total weight diverges logarithmically near every side. The square diverges more strongly. We therefore state a measurement protocol. Remove a strip of the same width ε along the entire boundary and call the remaining triangle T_ε . This is a uniform-distance cutoff, not a uniform barycentric cutoff.

Away from the vertices, put $k = \mu_0 I / (2\pi)$. Then $B \sim k/d$. Integrating perpendicular to a side gives a logarithmic weight for B and an inverse-cutoff weight for its square:

$$\int_{T_\varepsilon} B dA \sim kP \log(L/\varepsilon), \quad \int_{T_\varepsilon} B^2 dA \sim \frac{k^2 P}{\varepsilon}.$$

The leading first moments replace P by $\int_{\partial T} p ds$. Corner corrections are lower order. Dividing numerator by denominator therefore gives

$$\lim_{\varepsilon \rightarrow 0} C_B(T_\varepsilon) = \lim_{\varepsilon \rightarrow 0} C_{B^2}(T_\varepsilon) = W = X_{10}.$$

The normalized weight collapses onto the perimeter, uniformly per unit length. This is a new physical route to an old mass center. A fixed nonzero cutoff generally gives a different point. For the B mean the convergence is slow, of logarithmic order.

For the squared field, the uniformity assumption matters especially. A local cutoff $\varepsilon f(s)$ produces limiting line weight proportional to $1/f(s)$. A deliberately thicker excluded strip along one side changes the resulting mean. The regularization is part of the definition, rather than an inconvenience to be hidden.

3.3.5 The whole plane and a physically smooth regularization

The signed Bz centroid over the whole plane is not defined. On the filament plane it has positive and negative nonintegrable singularities. On a parallel plane at height h greater than zero the field is smooth, but its total signed flux is zero. Close a large disk by a hemisphere and use zero magnetic divergence: the hemisphere flux tends to zero because the loop's far field decays as the inverse cube of distance. The disk's total flux must therefore tend to zero too.

The square has no sign cancellation. Excluding a uniform tube around the wire gives two copies of the local strip divergence, one on each side. Its full-plane centroid tends to X10. There is no first-moment difficulty at infinity for the square, since $B_z^2 = O(r^{-6})$.

An alternative experiment measures Bz on a plane at uniform height h. Near a long straight portion of the loop,

$$B_z(d, h) \sim k \frac{d}{d^2 + h^2}, \quad \int_{-\infty}^{\infty} B_z(d, h)^2 dd = \frac{\pi k^2}{2h}.$$

Thus the squared normal-field centroid on that observation plane, or restricted to the projected triangle, tends to X10 as h tends to zero. At finite height it need not equal X10. Away from the wire plane Bz squared is only one component's contribution to the magnetic energy; it should not be silently equated with the full vector energy.

For completeness, the whole-plane absolute Bz weight at height h is integrable, but its absolute first moment diverges at infinity because the far field is of order r^{-3} . It consequently does not have an ordinary first-moment centroid either. This is a different obstruction from the zero denominator of the signed flux.

3.4 The common field equation for axial currents

3.4.1 Integrating along an infinite vertical wire

Place a wire through a transverse point q . At a transverse displacement $\rho = p - q$, an axial source element is $\hat{z} dz'$. Its cross product with $(\rho, -z')$ is simply $\hat{z} \times \rho$. Biot-Savart therefore gives

$$\mathbf{B}(p) = \frac{\mu_0 I}{4\pi} (\hat{z} \times \rho) \int_{-\infty}^{\infty} \frac{dz'}{(|\rho|^2 + z'^2)^{3/2}}.$$

The axial integral equals $2/|\rho|^2$, so

$$\mathbf{B}(p) = \frac{\mu_0 I}{2\pi} J \frac{p - q}{|p - q|^2}.$$

The magnitude is inversely proportional to the transverse distance; the vector is tangent to a circle centered on the wire. All three prism experiments are superpositions of this same kernel.

3.4.2 Three normalized source measures

Let $d\nu$ specify what fraction of the total axial current passes through a transverse source element. It has total mass one. The three choices are the sum of three equal point masses, uniform perimeter measure, and uniform area measure. Define

$$F(p) = \int \frac{p - q}{|p - q|^2} d\nu(q), \quad \mathbf{B}(p) = k J F(p), \quad k = \frac{\mu_0 I}{2\pi}.$$

The auxiliary field F is useful because it points in the radial rather than tangential direction. It is a gradient:

$$U(p) = \int \log(|p - q|/L) d\nu(q), \quad F = \nabla U.$$

The arbitrary reference length L adds a constant to U and has no effect on its gradient. This is the same logarithmic kernel as two-dimensional electrostatics, up to the sign and dimensional constant used for electric potential. It is not the inverse-distance Coulomb kernel of a triangular plate embedded in three dimensions. It also describes prescribed uniform source distributions, not the generally nonuniform equilibrium charge on a conductor.

The identity $\Delta \log |p - q| = 2\pi \delta_q$ implies $\Delta U = 0$ away from source points or source lines. In the uniform-volume case it gives $\Delta U = 2\pi/S$ inside T . Differentiating this constant shows that the two components of F , and hence of the magnetic field, are harmonic in the interior in all three cases.

3.4.3 A common maximum principle

If both components of a vector field are harmonic, differentiation of their squares gives

$$\Delta |\mathbf{B}|^2 = 2 \sum_{i,j} (\partial_i B_j)^2 \geq 0.$$

The maximum principle for such a subharmonic function says that an interior local maximum is impossible unless the function is constant in a neighborhood. Our triangle fields are not constant. Thus none of the axial models has an interior maximum of B or B squared. In the singular models the supremum is infinite at vertices. In the volume model the field extends continuously to the boundary, where its finite global maximum is attained.

This argument does not locate minima. In a source-free region the complex field is holomorphic, and the minimum-modulus principle says that an interior minimum of its magnitude must be a zero. In a region containing volume current that stronger conclusion requires additional work; one cannot use the source-free complex argument unchanged.

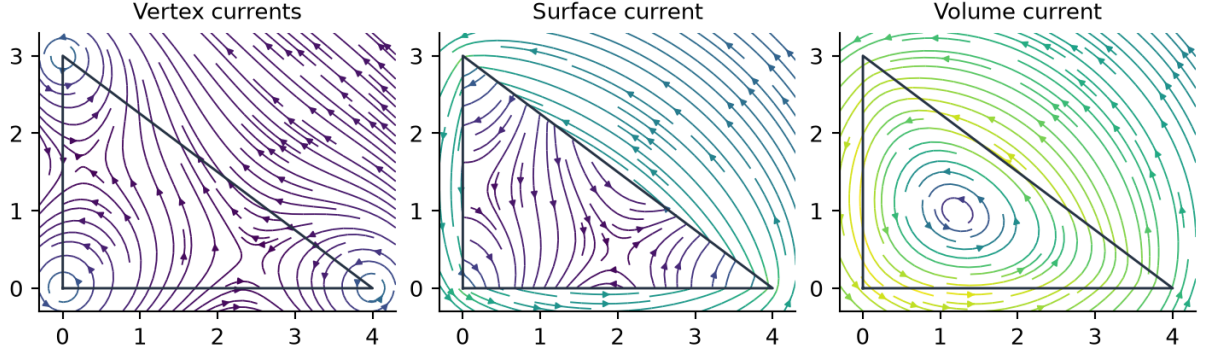


Figure 4. Field-line directions in and around the reference triangle for the three axial models. The arrows are magnetic directions; the auxiliary logarithmic field F is rotated through a right angle. All sources carry current towards the reader. The current sheet creates a tangential-field jump across a side; streamlines plotted across it are schematic near that discontinuity.

3.5 Case B: three equal axial edge currents

3.5.1 Exact field and its two zero-field minima

For three wires the source integral becomes a finite sum:

$$F(p) = \frac{1}{3} \left(\frac{p - A}{|p - A|^2} + \frac{p - B}{|p - B|^2} + \frac{p - C}{|p - C|^2} \right).$$

One may solve its two real equations directly. Complex notation reveals an exact geometric answer. Represent positions by complex numbers z, a, b, c . The complex conjugate of the radial field is proportional to the sum of the reciprocals $1/(z - a)$, $1/(z - b)$ and $1/(z - c)$. If $Q(z) = (z - a)(z - b)(z - c)$, this sum is $Q'(z)/Q(z)$. Away from the wires, the field vanishes precisely when $Q'(z) = 0$.

$$z_{\pm} = \frac{a + b + c \pm \sqrt{a^2 + b^2 + c^2 - ab - bc - ca}}{3}.$$

Marden's theorem identifies the two points as the foci of the Steiner inellipse, the ellipse tangent to all three side midpoints [12, 13]. Appendix A.2 derives its equation from the lamina's second moment and explains the identification. Both foci are strictly inside T . Their midpoint is G . They coincide in the equilateral triangle, where the inellipse is a circle.

There is an important lesson for a catalogue of centers. The physical minimum is generally a pair of points. Calling one of them the magnetic center without a selection rule would be

ambiguous, and a globally continuous choice through the circular case is problematic. We retain the unordered pair as a magnetic construction and assign M indices only to the single-valued means below. The ETC has a separate bicentric-pair catalogue, where these foci are P(118) and U(118) [1].

For the reference triangle, the two minima are approximately (0.255979477, 1.618800093) and (2.410687190, 0.381199907). Substitution in the explicit field gives zero, up to numerical precision. Their mean is $(4/3, 1)$ as required.

3.5.2 M2: the ordinary magnitude centroid

Near a vertex wire, $B \sim k/(3r)$. In polar coordinates the area element is $r dr d\theta$, so the apparent singularity cancels one radial factor. The local integral of B is finite. Thus

$$M2 = \frac{\int_T p |F(p)| dA}{\int_T |F(p)| dA}$$

is an ordinary centroid. It is unique, lies strictly inside T, and respects relabeling, rotations, reflections, translations and scaling of the triangle. These facts make it a legitimate triangle center even without a compact symbolic expression.

The magnitude must be taken after summing the three vector fields. Replacing the weight by a sum of their separate magnitudes would define a different center and erase the destructive interference responsible for the two zeros.

For the reference triangle, M2 is approximately (1.092932855, 0.961460421). The numerical algorithm resolves the corner singularity by splitting the triangle into six subtriangles and using a radial quadrature from each original vertex; Section 3.9 explains why this is more effective than an unadapted rectangular grid.

3.5.3 M3: the energy cutoff mean is exactly X360

Squaring changes integrability. Near A, $B^2 \sim k^2/(9r^2)$, and the available angular sector inside T has angle alpha. Excluding equal circular cores of radius epsilon at the three wires gives the leading divergent mass

$$\frac{k^2}{9} \alpha \log(L/\epsilon)$$

near A, with analogous terms involving beta and gamma. Cross terms with the fields of the other wires are locally integrable and do not alter this leading coefficient. Since the angles sum to pi, the normalized first moment has the exact limit

$$M3 = \frac{\alpha A + \beta B + \gamma C}{\pi}, \quad M3 = (\alpha : \beta : \gamma) = X_{360}.$$

The ETC name is the Hofstadter zero-point [1]. The database expression for its first barycentric coordinate is simply the angle A, and the independent numerical search recovers X360 to floating-point precision. Here M3 is a magnetic alias for a known center, not a claim of a new point. The earlier conversation had not identified its ETC index; this chapter supplies that identification.

The answer differs from the loop's X10 because the singular sources are now three points, not three lines in the observation plane. Each point receives weight proportional to the interior angle visible from it. In the reference triangle this gives approximately (0.819331059, 0.885501706).

3.5.4 An exact attractivity theorem for M3

Move A while holding B and C fixed, and denote the derivative matrix of M3 with respect to A by $D_A M3$. Let u and v be the unit vectors from B to A and from C to A. Differentiating the angles gives $\nabla_A \beta = Ju/|A - B|$ and $\nabla_A \gamma = -Jv/|A - C|$. Since alpha is pi minus the other two angles, differentiation of the angle-weighted mean yields

$$D_A M3 = \frac{1}{\pi} \left[\alpha I - u(Ju)^T + v(Jv)^T \right].$$

The two outer-product terms have opposite antisymmetric parts, so the matrix is symmetric. Their difference has trace zero and eigenvalues plus and minus sin alpha. Hence the eigenvalues of the full derivative are

$$\lambda_{\pm} = \frac{\alpha \pm \sin \alpha}{\pi} > 0 \quad (0 < \alpha < \pi).$$

This is an exact positivity result, not a numerical screen. Integrate the quadratic form along any straight displacement of A that remains in the same nondegenerate half-plane determined by BC. For every nonzero displacement h , its dot product with the resulting motion of M3 is positive. The same proof applies to the other vertices. M3 is therefore strictly vertex-attractive in this sense. Near degeneration the smaller eigenvalue can be extremely small, so numerical differentiation alone would give a much weaker conclusion.

3.6 Cases C and D: current on the surface and throughout the volume

3.6.1 Surface current: reducing the field to three edge integrals

A lateral strip of width ds carries axial current $K ds = I ds/P$. Superposition of infinite-wire fields gives

$$F_{\partial}(p) = \frac{1}{P} \int_{\partial T} \frac{p - q}{|p - q|^2} ds.$$

This looks similar to the loop integral, but it is not the same kernel or vector direction. The loop's source element lies along the side; the prism's source element is an entire vertical wire attached to a side point.

For a side of length l , write $p - q_0 = st + dn$ with signed distance d . Resolving the radial kernel along t and n reduces the side integral to

$$F_{e,t} = \frac{1}{2} \log \frac{s^2 + d^2}{(s - l)^2 + d^2},$$

$$F_{e,n} = \text{sgn}(d) \left[\arctan \frac{s}{|d|} - \arctan \frac{s - l}{|d|} \right].$$

Their vector sum is $F_{e,t}t + F_{e,n}n$, and dividing the sum over sides by P gives the normalized field. In software, `atan2` is used to preserve signs and handle limiting configurations. The field has a finite jump across a side away from its endpoints. At a vertex, integrating the tangent kernel along the two incident sides produces a logarithmic divergence. Both B and its square nevertheless have finite area integrals because the integrals of $r|\log r|$ and $r|\log r|^2$ near zero converge.

3.6.2 Surface minima, M4 and M5

The interior zero-field minima solve $F_{\partial}(p) = 0$. A root finder should start at multiple points and should not assume uniqueness. For the reference triangle two roots were found: approximately (0.084997098, 1.279266098) and (1.982619696, 0.457867825). They are not the Steiner foci of the vertex-current problem. The contour plots make the difference visible.

The source-free minimum-modulus principle excludes a positive interior local minimum. It does not supply a universal count of zeros, and this chapter does not claim one. In particular, no single-valued M index is assigned to a root selected merely by the initial guess of a numerical solver.

The two ordinary means are unambiguous:

$$M4 = \frac{\int_T p |F_{\partial}| dA}{\int_T |F_{\partial}| dA}, \quad M5 = \frac{\int_T p |F_{\partial}|^2 dA}{\int_T |F_{\partial}|^2 dA}.$$

For the reference triangle they are approximately (1.407774064, 1.072912382) and (1.581074930, 1.112103781). Neither equals the current-source centroid W. This is expected: the source centroid averages source positions, while M4 and M5 average observation positions weighted by an interfering vector field.

3.6.3 Volume current: why a singular area integral is finite

A small area element dA carries axial current $j dA = I dA/S$. Thus

$$F_T(p) = \frac{1}{S} \int_T \frac{p - q}{|p - q|^2} dA(q).$$

The kernel magnitude behaves like the inverse distance. Its area integral is locally finite, and the resulting field is continuous even on the boundary. For reliable computation we can remove the interior kernel singularity entirely. Since the p-gradient of log distance is minus its q-gradient, the planar divergence theorem gives

$$F_T(p) = -\frac{1}{S} \int_{\partial T} \log |p - q| n_{\text{out}}(q) ds.$$

The normal is constant on each straight side. Only a scalar log-distance integral remains. If $x = u - s$ and d is the perpendicular distance, an antiderivative is

$$H(x, d) = x \log \sqrt{x^2 + d^2} - x + |d| \arctan(x/|d|).$$

Use the continuous limiting values when d or x is zero. Evaluating H at the two side endpoints, multiplying by the outward normal, and summing gives the exact field in terms of elementary functions. No finite-element mesh is required for these uniform-source problems.

3.6.4 Volume minima and the need to separate existence from uniqueness

The zero equation is $F_T(p) = 0$. At least one interior zero exists. To see this, U grows like log distance at infinity, so it attains a global minimum. An exterior point cannot minimize U : projection onto the closed convex triangle decreases its distance to every source point. At a boundary point p , the negative gradient is a positive weighted average of vectors $q - p$ directed

into the triangle. It points strictly into the interior tangent cone and is nonzero, so a small step in that direction lowers U . The global minimum is therefore interior and satisfies zero gradient.

This proof establishes existence, not uniqueness of all stationary points. A numerical root near the reference triangle's centroid is approximately $(1.217335188, 0.990154002)$. We report it as a computed zero-field minimum. We do not promote it to a globally defined new triangle center before establishing uniqueness or a canonical rule for ties on every triangle.

The maximum of B on the closed reference triangle was found numerically at $(0, 0.746023187)$, with $B/k \approx 0.684555623$. It lies in the interior of a side. More generally the maximum-principle argument requires a boundary search, not a search over vertices alone. If the word inside excludes the boundary, the maximum is a supremum approached from inside, rather than an attained maximum.

3.6.5 M6 and M7: finite volume-field means

The two means are

$$M6 = \frac{\int_T p |F_T| dA}{\int_T |F_T| dA}, \quad M7 = \frac{\int_T p |F_T|^2 dA}{\int_T |F_T|^2 dA}.$$

Their weights are continuous, nonnegative and positive on open subsets. They define unique strict-interior points with all the required triangle symmetries. On the reference triangle they are approximately $(1.310681466, 1.006307117)$ and $(1.272070723, 1.011102413)$. The energy mean M7 is somewhat farther from G than the magnitude mean M6; squaring makes regions of stronger field more influential.

One might guess that uniform volume current must give G for every magnetic mean. That inference confuses two averages. The current itself has uniform source density and centroid G . The field created by that current is nonuniform, and its centroid inside the source region need not be G . A different, explicitly regularized full-plane experiment will recover G in the next section.

3.7 Whole-plane means and the role of the current centroid

3.7.1 Why the axial full-plane integrals diverge

All three axial models have total current I rather than zero net current. Far from the triangle they resemble a single axial wire. Thus $B \sim k/r$. Integrating B over a large annulus gives a contribution proportional to its radial width; integrating its square gives a logarithm. Neither total weight on the whole plane is finite. The absolute first moments also diverge.

There is therefore no ordinary whole-plane B or B -squared centroid for any of the three infinite-prism models. In the vertex-wire case the square additionally needs local regularization at the three wires. The loop is different: it is a localized closed current, whose far field decays as the inverse cube rather than inverse first power of distance.

3.7.2 A disk limit that recovers X2 or X10

Although the integrals diverge, a specified sequence of normalized finite measurements can converge. Let $D_R(O)$ be the disk of radius R centered at a fixed point O . Let m be the current-source centroid, namely $m = \int q d\nu(q)$. Measured relative to O , write an observation point as $p = O + re$, with e a unit direction. Expansion of the kernel gives

$$B(O + re) = \frac{k}{r} \left[1 + \frac{e \cdot (m - O)}{r} + O(r^{-2}) \right].$$

The leading radial term contains only total current; the next angular term contains its centroid. Integrating around a circle uses the identities that the integral of e is zero and the integral of the matrix ee^T is π times the identity. For B the denominator grows as $2\pi kR$, while its first moment relative to O grows as $\pi k(m - O)R$. For the square the respective coefficients are $2\pi k^2 \log R$ and $2\pi k^2(m - O) \log R$. Consequently,

$$\lim_{R \rightarrow \infty} C_B(D_R(O)) = \frac{O + m}{2}, \quad \lim_{R \rightarrow \infty} C_{B^2}(D_R(O)) = m.$$

The first answer depends on the arbitrary disk center. Choosing a preferred triangle center for O would create an additional convention, not a unique intrinsic whole-plane mean. The second disk limit is independent of O and yields the following exact aliases.

Axial source	Current centroid m	Disk-limit energy centroid
Three equal vertex currents	G	X2
Uniform perimeter sheet	W	X10
Uniform filled cross-section	G	X2

In the vertex case, hold small symmetric current cores fixed while the disk grows, then shrink the cores. This convention preserves the source centroid. An equivalent uniform-core local limit in a fixed disk assigns equal full angular weights to the three vertices and also gives G. Inside the triangle alone, the three angles are not equal; that is why the interior energy limit was X360 instead.

Circular exhaustion is part of these statements. An arbitrary asymmetric growing observation window need not have the same first-moment cancellation. Likewise, adding the return conductor changes the distant field and can make formerly divergent integrals convergent with a different answer. These limitations are physical information about the measurement, not merely technical exceptions.

3.7.3 What remains of the inertia tensor?

The first far-field correction finds the source centroid. Higher corrections involve the source's second moments, connecting this chapter to the preceding inertia chapter. For three equal vertex currents an even stronger connection occurs: their two interior zeros are determined exactly by the covariance tensor, through the axes and foci of the Steiner inellipse. Appendix A.2 derives this statement and separates it from the tensor of a uniform perimeter wire. The source measures have related centers but different second moments.

3.8 Which constructions deserve M indices?

We keep M1 for the established triangular-loop minimum and use consecutive indices for the six single-valued prism means. An alias is retained when the magnetic construction gives a known ETC center. The symbols identify physical selectors; they do not assert mathematical novelty or ETC nonmembership.

Index	Physical selector	Mathematical status
M1	Minimum of the loop field inside T	Unique; implicitly algebraic
M2	Interior magnitude mean, vertex currents	Finite integral-defined center
M3	Interior energy mean, equal-core limit, vertex currents	Exactly X360
M4	Interior magnitude mean, surface current	Finite integral-defined center
M5	Interior energy mean, surface current	Finite integral-defined center
M6	Interior magnitude mean, volume current	Finite integral-defined center
M7	Interior energy mean, volume current	Finite integral-defined center

The loop's cutoff means are X10; the disk-limit axial energy means are X2 or X10. Giving those additional M numbers would multiply labels for the same point without adding a new center. The Steiner focus pair and the potentially multiple surface-field zeros are recorded as sets. The volume zero-field selector remains a candidate for a future index once its global uniqueness question is settled.

3.8.1 A general location certificate for the finite means

Let λ_i be a barycentric coordinate. It is affine, nonnegative on T, and positive in its interior. For a finite nonnegative weight with positive mass on an interior open set,

$$\lambda_i(C_w) = \frac{\int_T \lambda_i(p) w(p) dA}{\int_T w(p) dA} > 0.$$

Thus M2 and M4-M7 are strictly interior. M3 has positive angle weights and is strictly interior too. This is a rigorous location certificate, though it gives no uniform positive distance from every side over all shapes. M1 has the stronger side-distance bound from Section 3.3. None of these observations alone places every center in an attractive-center region or proves an ETC identity.

3.8.2 What is meant by a barycentric center function?

Once a unique physical point is defined and transforms correctly under similarities and vertex permutations, its normalized barycentric coordinates provide a center function, even if they are evaluated by integrals. Lack of a short formula is not lack of a definition. Nor does the presence of an integral prove that the final function is transcendental. M1 illustrates this caution, and M3 illustrates how an apparently new singular-limit construction can coincide with an existing ETC entry involving angles.

3.9 Numerical methods, fingerprints and response to vertex motion

3.9.1 Quadrature adapted to a triangle's corners

The axial fields themselves use exact finite sums or analytical side integrals. The remaining numerical task for the means is two-dimensional integration. Divide T into six subtriangles, each with one original vertex, a neighboring side midpoint, and G . On such a subtriangle write

$$p(r, t) = V + r[(1 - t)e_1 + te_2], \quad 0 \leq r, t \leq 1.$$

The Jacobian is $r|\det(e_1, e_2)|$. For the vertex-current magnitude, the leading inverse- r field is canceled by this factor. Gauss-Legendre rules in r and t then integrate a much smoother function. Each subtriangle uses its actual Jacobian, so the weighting is geometrically correct. The six pieces are also treated symmetrically, avoiding a privileged Duffy-map vertex.

The reference means are evaluated at orders 24, 48, 96 and 192 per coordinate per subtriangle. The 100 general-shape samples and 40 independent holdouts use orders 24 and 48, retaining their displacement difference. These differences are resolution diagnostics, not rigorous continuum-error bounds. A magnitude has a cusp at a field zero, whereas its square is smoother; the corresponding quadrature convergence can therefore differ.

Root finding, minimum finding and quadrature have different failure modes. A small root residual does not prove that every root was found. A small integration change does not establish a rigorous enclosure. The companion preserves all four reference resolutions, the general-sample changes, and the root and field checks instead of presenting rounded numbers alone.

3.9.2 Five complementary shape tests

The atlas uses the same conventions as the larger project. Isosceles triangles have vertices $(-1, 0), (1, 0), (0, h)$; we plot the center height and its derivative. Right triangles have vertices $(0, 1), (0, 0), (h, 0)$; we plot both coordinates and their derivatives. The parameter h is not the height of the reference 3-4-5 drawing unless a similarity has first been applied.

Nearly collinear triangles have vertices $(-1, 0), (1, 0), (x, \varepsilon)$ with $0 \leq x \leq 1$. We record the horizontal coordinate and the vertical coordinate divided by epsilon. The endpoint $x=1$ can behave differently from a fixed interior x . The plotted smallest epsilon is 0.0125, accompanied by several larger values in the data; a thin-triangle plot is not itself an asymptotic proof.

Near the equilateral triangle we perturb the apex to $(x, \sqrt{3} + y)$. Symmetry makes the linear response diagonal at the symmetric shape. The atlas records its horizontal and vertical coefficients at two finite-difference steps and three quadrature orders. The sum is $2/3$ for a smooth similarity-equivariant symmetric center; the deviation provides a useful consistency check. Only the coefficients explicitly derived in the text are called exact.

Finally, 100 equal-area samples in the ordered-angle fundamental region, together with 40 holdouts, probe general scalene shapes. Their measure is uniform in angle-shape space, not in side lengths or Cartesian apex coordinates. These samples are important because distinct centers can have almost identical symmetry-family curves.

3.9.3 Searching the ETC without mistaking proximity for identity

The supplied WXF snapshot contains 72,807 entries. Its restricted numerical importer supplies 70,853 candidates finite on the common 100-shape grid. Unsupported expressions and singular

evaluations are excluded rather than silently filled in. For a candidate X and magnetic center M , the location score is

$$E(X, M) = \left[\frac{1}{N} \sum_{r=1}^N \frac{|X(T_r) - M(T_r)|^2}{D_r^2} \right]^{1/2},$$

where D_r is the diameter of the r th triangle. Each atlas entry reports the best location candidates and their independent holdout errors. The fingerprint panels compare the magnetic point with two shortlisted centers and the familiar X_1 , X_2 and X_{10} , with duplicate curves removed.

These are location rankings. They do not yet optimize a joint metric including derivative and attractivity similarity. The plotted derivative comparisons are additional evidence, not an unreported part of the score. Very small errors warrant symbolic investigation, not automatic identification. M_3 is an exception because its exact angle formula proves the X_{360} identity independently of the numerical match.

3.9.4 Geometric attractivity is a separate property

Fix two vertices and displace the third by h . A center is vertex-attractive if the dot product of h with its own displacement is nonnegative. At a differentiable configuration, the local criterion is that the symmetric part of the vertex derivative matrix be positive semidefinite:

$$h \cdot [M(T_i + h) - M(T)] \geq 0, \quad \frac{D_i M + (D_i M)^T}{2} \succeq 0.$$

The finite and infinitesimal statements are related by integration along admissible vertex paths. They have nothing to do with convergence of an iterative algorithm or with magnetic trapping of particles.

The original screen differentiates all three vertices on 100 triangles. The September 30 stress survey adds 284 shapes, then refines selected finite motions at orders 40, 80, 160 and 256 with two step sizes. M_6 receives an additional targeted search. Stable negative values are numerical counterexample evidence, not interval certificates. Absence of a negative value in a finite sample does not prove global attractivity.

M_1 has a previously validated negative finite-motion certificate. M_2 , M_4 , M_5 and M_7 now have stable negative numerical witnesses, refined to quadrature order 256 and checked with a halved displacement. M_6 remains unrefuted after the original screen, a 284-shape stress survey and a further 2,048-shape targeted search, with six low-margin cases refined. Its global status remains open. M_3 is strictly attractive by the exact derivative proof in Section 3.5.

3.10 Individual magnetic atlas entries

The following generated entries use the preceding definitions and reproducible computations. Each includes reference coordinates and barycentrics, location evidence, numerical resolution information, ETC neighbors, attractivity evidence, three-triangle field diagrams and the isosceles/right fingerprints. The common thin-family and equilateral tables follow them. The data are a chapter-scale extension of the earlier atlas, not a claim of complete classification.

3.10.1 M1. Loop-field minimum

On the 3-4-5 triangle, the computed Cartesian position is (1.103976518, 0.998576185), with normalized barycentrics (0.391147142, 0.275994129, 0.332858728). The change from reference quadrature order 96 to 192 is 0 times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the weakest normal magnetic field of a current circulating around the triangular wire. The finite-edge formula is minimized, then its stationary root is refined. Strict convexity proves uniqueness. The side-distance bound in Section 3.3 is stronger than a mere interiority statement. Its construction is implicitly algebraic.

Attractivity. Not globally attractive: the earlier interval-certified negative finite vertex motion remains the rigorous witness.

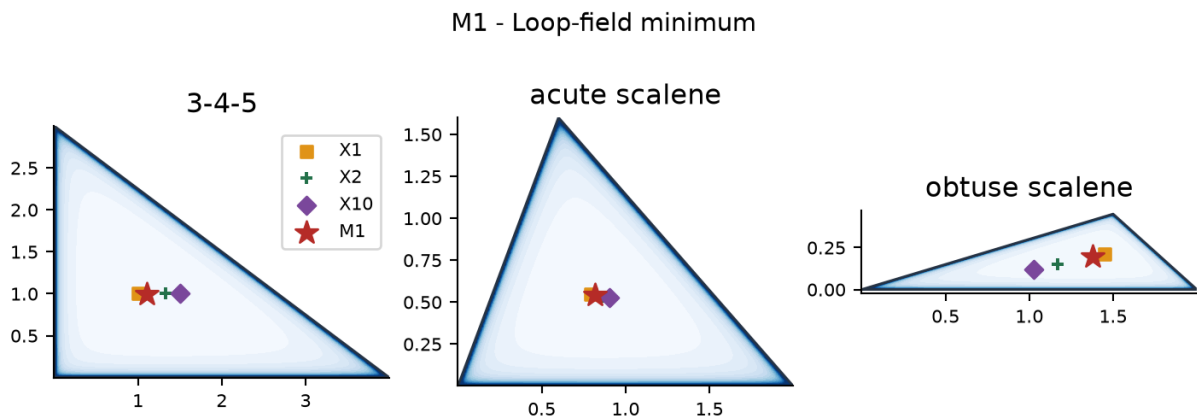
The original 100-shape screen minimum was -0.0602109. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

Resolution on general shapes: 100/100 grid shapes and 40/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were 0 and 0. All shapes, including less-resolved ones, are retained in the location ranking.

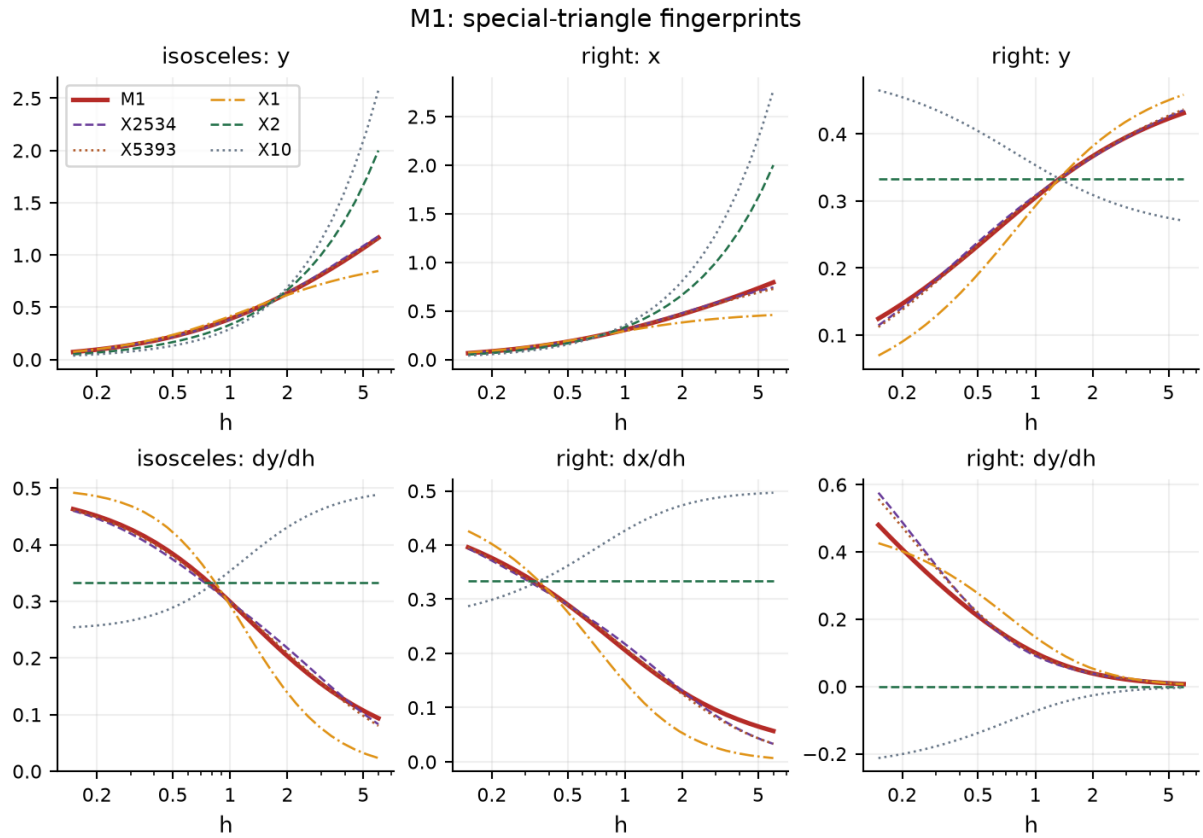
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X2534	0.00686286	0.00657577	0.0144098
X5393	0.00702362	0.00691894	0.015792
X53841	0.00710349	0.00668813	0.0138951
X29678	0.00741535	0.00636905	0.014007
X39621	0.00766273	0.00714979	0.0120833

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

For X2534, position-fingerprint RMS / D is 0.00386736, derivative-fingerprint RMS is 0.0275604, and response-margin RMS difference is 0.0243488. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.



Atlas figure 1a. M1 in the reference, acute and obtuse triangles. Red star: M1; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.



Atlas figure 1b. M1 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.2 M2. Vertex-current magnitude mean

On the 3-4-5 triangle, the computed Cartesian position is (1.092932855, 0.961460421), with normalized barycentrics (0.406279979, 0.273233214, 0.320486807). The change from reference quadrature order 96 to 192 is 5.6e-08 times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the mean observation position weighted by the magnitude of the sum of the fields of three equal axial vertex currents. The inverse-distance corner singularities are integrable and are handled by radial quadrature. A weighted average of source positions would be a different construction.

Location certificate: all normalized barycentric coordinates are strictly positive by the affine averaging argument in Section 3.8. No universal narrower region or rigorous numerical distance-to-side enclosure is asserted for this entry.

Attractivity. Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -1.61118595 and -1.61648215. No interval certificate is asserted.

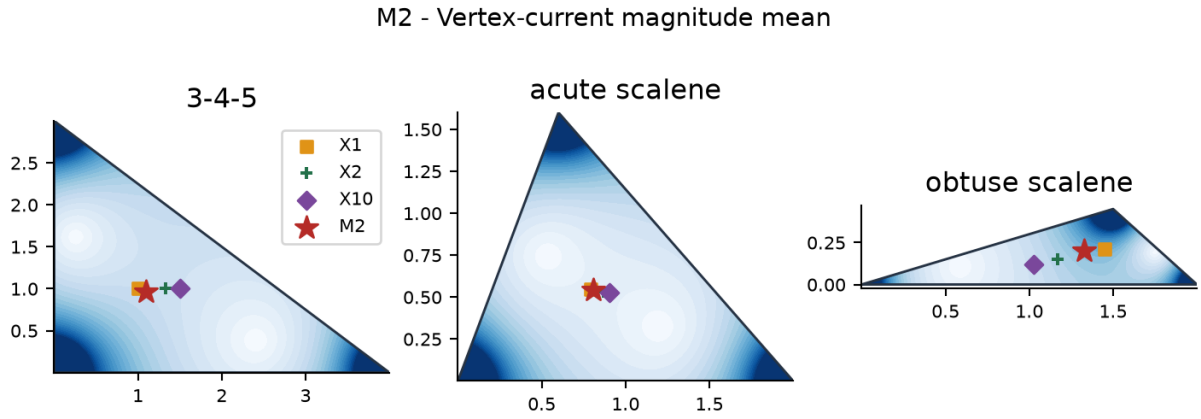
The original 100-shape screen minimum was -0.0461086. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

Resolution on general shapes: 100/100 grid shapes and 39/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were 4.81e-05 and 0.00189. All shapes, including less-resolved ones, are retained in the location ranking.

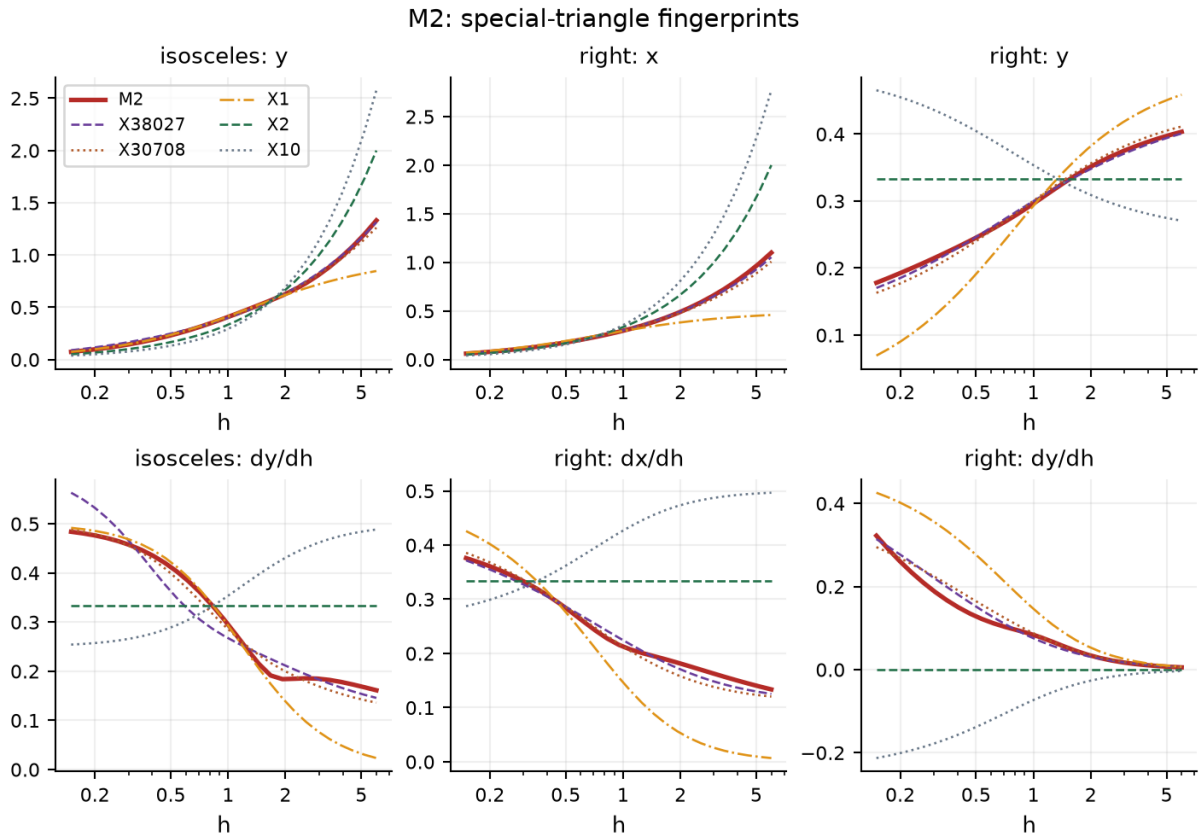
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X38027	0.00693143	0.0099179	0.0423536
X30708	0.0072812	0.0107841	0.0434288
X50293	0.00866959	0.0133356	0.055731
X29597	0.00909368	0.0121667	0.0434472
X3636	0.00939902	0.0138534	0.0557312

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

For X38027, position-fingerprint RMS / D is 0.00584481, derivative-fingerprint RMS is 0.0289519, and response-margin RMS difference is 0.0550474. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.



Atlas figure 2a. M2 in the reference, acute and obtuse triangles. Red star: M2; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.



Atlas figure 2b. M2 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.3 M3. Vertex-current energy limit: X360

On the 3-4-5 triangle, the computed Cartesian position is (0.819331059, 0.885501706), with normalized barycentrics (0.500000000, 0.204832765, 0.295167235). The change from reference quadrature order 96 to 192 is 0 times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the interior mean weighted by the squared vertex-current field, after excluding equal small circular wire cores. The exact answer is the angle-weighted barycentric point X360. This is a classical identification, not just a near numerical match. Its location certificate is positivity of all three angles.

Attractivity. Strictly globally vertex-attractive: $M3 = X360$ exactly, with symmetric vertex-derivative eigenvalues (α plus or minus $\sin \alpha$)/ π .

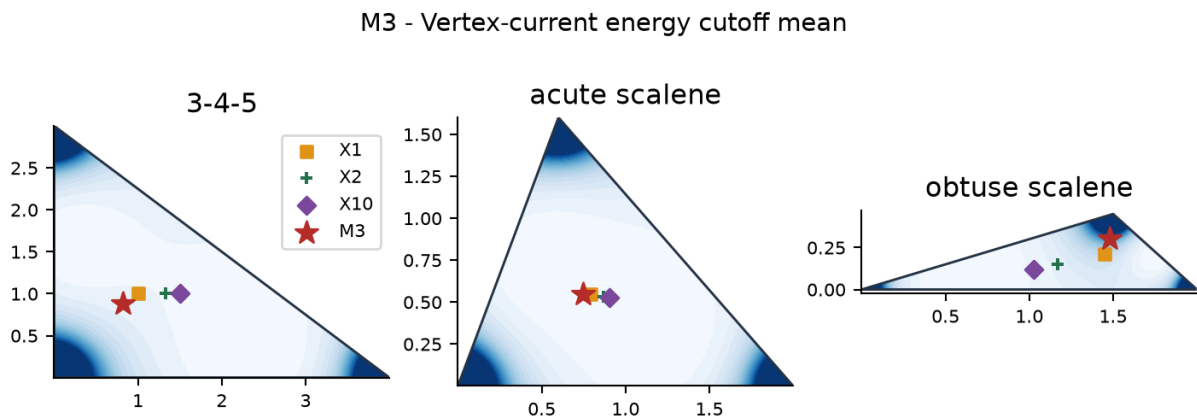
The original 100-shape screen minimum was 0.0000023. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

Resolution on general shapes: 100/100 grid shapes and 40/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were 0 and 0. All shapes, including less-resolved ones, are retained in the location ranking.

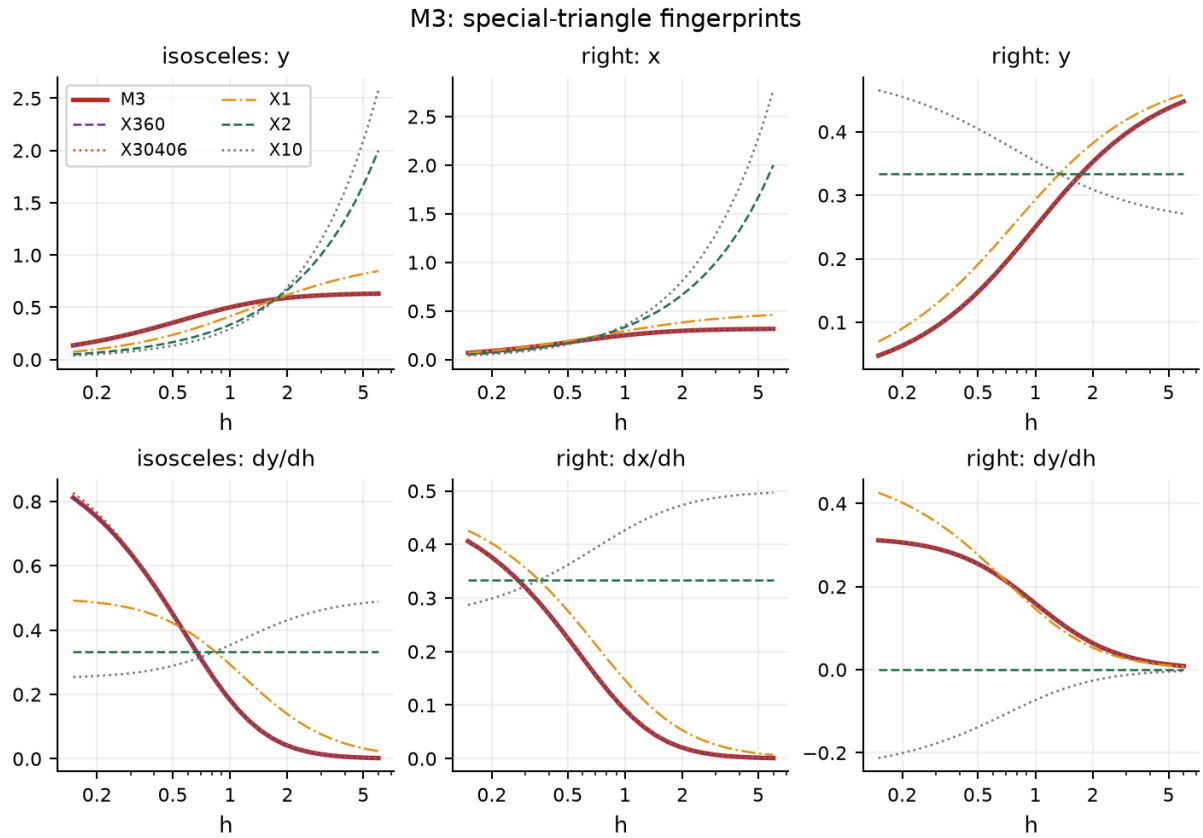
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X360	2.07319e-16	2.6653e-16	9.99219e-16
X30406	0.000861307	0.000797032	0.00182603
X53977	0.000918504	0.000924244	0.00139299
X30409	0.000955266	0.00094783	0.00162392
X8389	0.00114555	0.00114908	0.00172432

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

For X360, position-fingerprint RMS / D is 2.20496e-17, derivative-fingerprint RMS is 1.10773e-07, and response-margin RMS difference is 1.01909e-05. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.



Atlas figure 3a. M3 in the reference, acute and obtuse triangles. Red star: M3; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.



Atlas figure 3b. M3 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.4 M4. Surface-current magnitude mean

On the 3-4-5 triangle, the computed Cartesian position is (1.407774064, 1.072912382), with normalized barycentrics (0.290419023, 0.351943516, 0.357637461). The change from reference quadrature order 96 to 192 is 5.2e-09 times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the interior magnitude mean for a uniform axial surface current. Analytic edge fields are summed as vectors before their magnitude is integrated. The logarithmic corner divergence is integrable. The finite mean remains single-valued even when the field has multiple zero-field minima.

Location certificate: all normalized barycentric coordinates are strictly positive by the affine averaging argument in Section 3.8. No universal narrower region or rigorous numerical distance-to-side enclosure is asserted for this entry.

Attractivity. Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.11778058 and -0.11804977. No interval certificate is asserted.

The original 100-shape screen minimum was 0.1303961. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

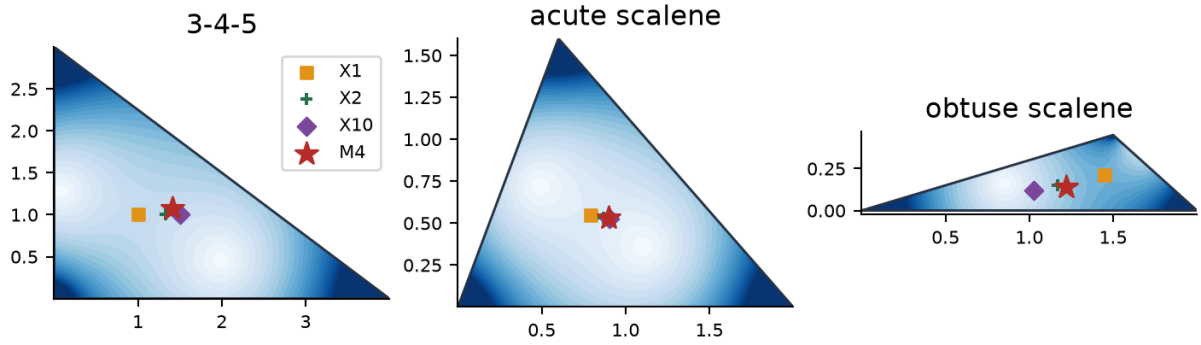
Resolution on general shapes: 100/100 grid shapes and 40/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were 6.23e-06 and 1.68e-05. All shapes, including less-resolved ones, are retained in the location ranking.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X17258	0.00735564	0.00699618	0.0144227
X24739	0.00869396	0.0101256	0.0310545
X33947	0.00966709	0.0104356	0.0194045
X24658	0.0101628	0.0116685	0.0316963
X24657	0.0103155	0.0117293	0.0316964

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

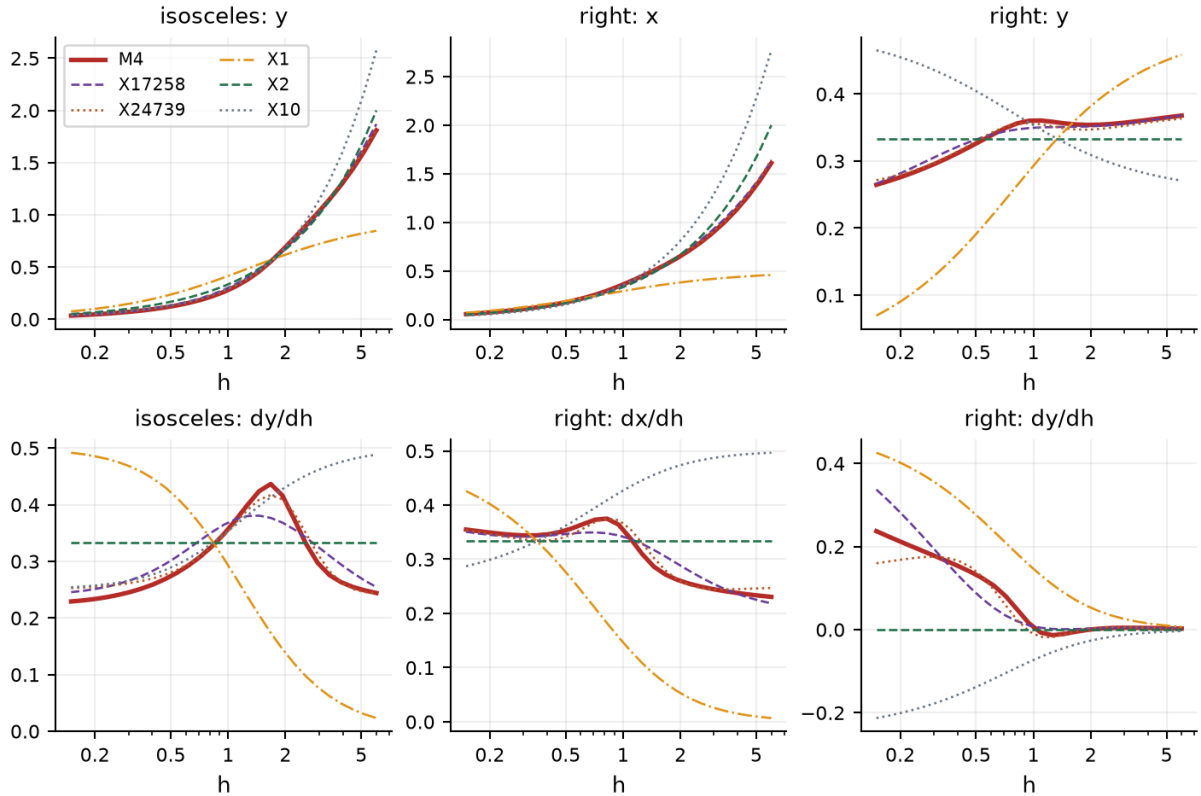
For X17258, position-fingerprint RMS / D is 0.00687787, derivative-fingerprint RMS is 0.0344165, and response-margin RMS difference is 0.0308281. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

M4 - Surface-current magnitude mean



Atlas figure 4a. M4 in the reference, acute and obtuse triangles. Red star: M4; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.

M4: special-triangle fingerprints



Atlas figure 4b. M4 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.5 M5. Surface-current energy mean

On the 3-4-5 triangle, the computed Cartesian position is (1.581074930, 1.112103781), with normalized barycentrics (0.234030007, 0.395268733, 0.370701260). The change from reference quadrature order 96 to 192 is 3.5e-09 times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the interior magnetic-energy mean for a uniform axial surface current. The squared total field includes interference terms between the three sides. Its corner logarithms are integrable after squaring. This differs from the circularly regularized full-plane mean, which is X10.

Location certificate: all normalized barycentric coordinates are strictly positive by the affine averaging argument in Section 3.8. No universal narrower region or rigorous numerical distance-to-side enclosure is asserted for this entry.

Attractivity. Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.72777353 and -0.72897714. No interval certificate is asserted.

The original 100-shape screen minimum was -0.0846799. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

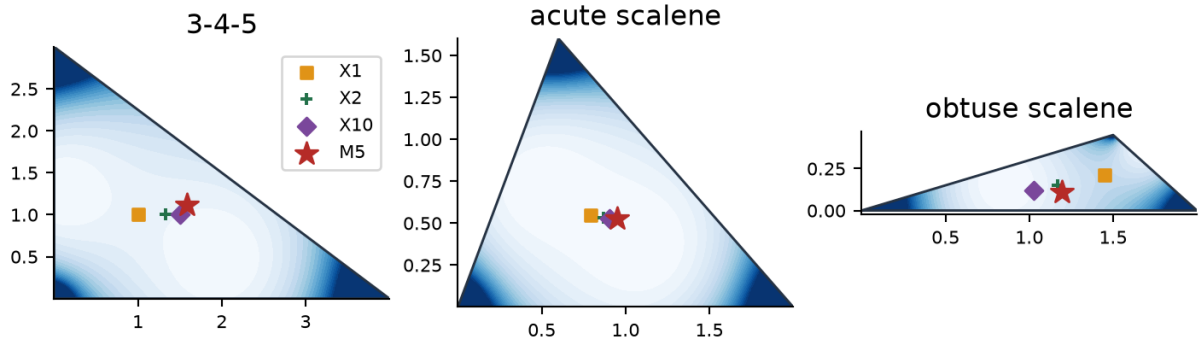
Resolution on general shapes: 100/100 grid shapes and 40/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were 2.68e-06 and 3.12e-05. All shapes, including less-resolved ones, are retained in the location ranking.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X4166	0.0198646	0.0214742	0.056023
X48629	0.0217717	0.0228648	0.0538496
X33947	0.0237758	0.0251878	0.055156
X60544	0.0252782	0.026135	0.0560113
X70526	0.0255545	0.0312075	0.109718

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

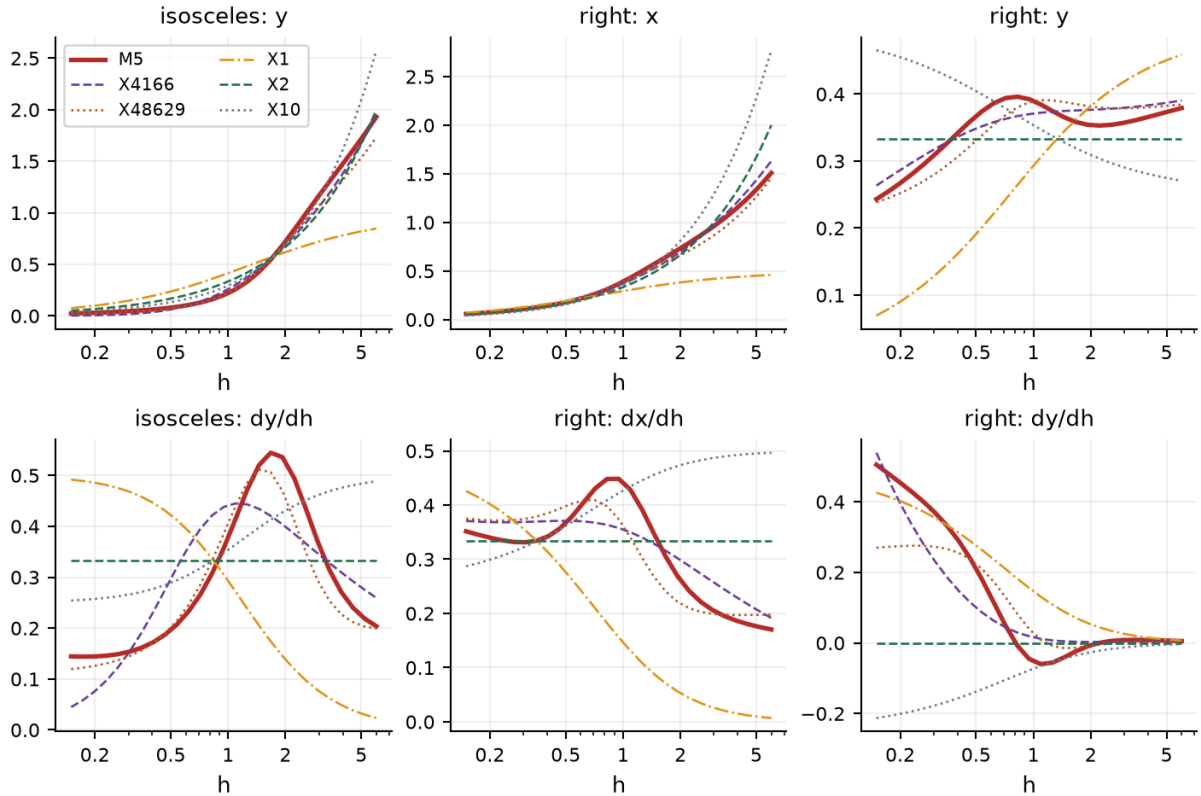
For X4166, position-fingerprint RMS / D is 0.0179651, derivative-fingerprint RMS is 0.0813805, and response-margin RMS difference is 0.098876. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

M5 - Surface-current energy mean



Atlas figure 5a. M5 in the reference, acute and obtuse triangles. Red star: M5; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.

M5: special-triangle fingerprints



Atlas figure 5b. M5 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.6 M6. Volume-current magnitude mean

On the 3-4-5 triangle, the computed Cartesian position is (1.310681466, 1.006307117), with normalized barycentrics (0.336893928, 0.327670366, 0.335435706). The change from reference quadrature order 96 to 192 is $9.1\text{e-}11$ times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the interior magnitude mean for a uniformly current-filled triangular prism. The area kernel is evaluated using a nonsingular boundary log-distance integral. This is a field-observation mean, while the source-current centroid itself is X2.

Location certificate: all normalized barycentric coordinates are strictly positive by the affine averaging argument in Section 3.8. No universal narrower region or rigorous numerical distance-to-side enclosure is asserted for this entry.

Attractivity. Global attractivity remains open. In addition to the earlier 100/284/2048-shape investigations, 512 new extreme shapes and 100 repeated shapes passed a complex-step response screen; ten difficult shapes retained positive signs through quadrature order 256. Initial subtraction-based negative values failed refinement and are not accepted as witnesses. No uniform positive lower bound or global theorem is claimed. See Appendix D.

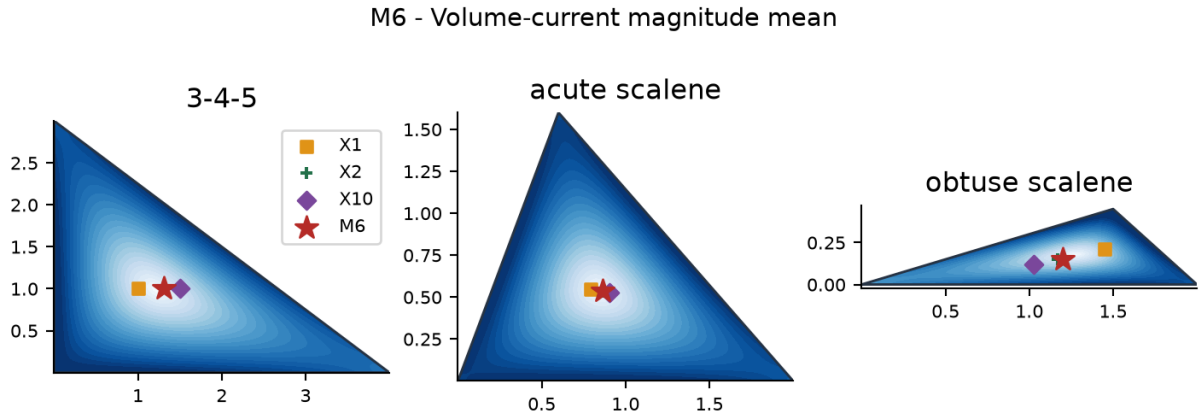
The original 100-shape screen minimum was 0.2659010. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

Resolution on general shapes: 100/100 grid shapes and 40/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were $2.96\text{e-}07$ and $8.04\text{e-}06$. All shapes, including less-resolved ones, are retained in the location ranking.

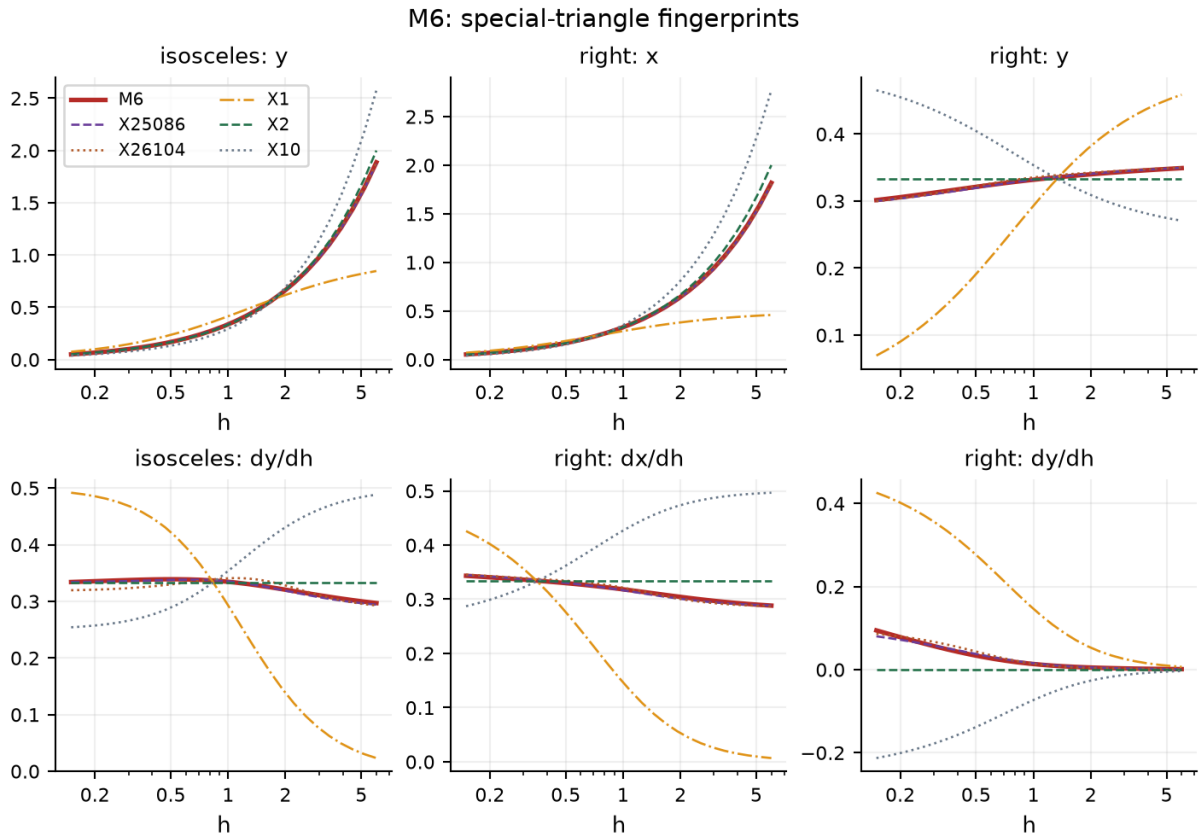
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X25086	0.00246473	0.00268194	0.00742478
X26104	0.00258123	0.00285751	0.00696481
X58859	0.00287241	0.0030829	0.00711097
X19935	0.00314996	0.00335865	0.00678586
X17325	0.00315741	0.00328295	0.00728227

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

For X25086, position-fingerprint RMS / D is 0.00125006, derivative-fingerprint RMS is 0.00350096, and response-margin RMS difference is 0.0122648. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.



Atlas figure 6a. M6 in the reference, acute and obtuse triangles. Red star: M6; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.



Atlas figure 6b. M6 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.7 M7. Volume-current energy mean

On the 3-4-5 triangle, the computed Cartesian position is (1.272070723, 1.011102413), with normalized barycentrics (0.344948182, 0.318017681, 0.337034138). The change from reference quadrature order 96 to 192 is $3.8\text{e-}14$ times the diameter. For M1 this compares the same stationary solver; for M3 the formula is exact and quadrature is not used.

Physical selector: the interior energy mean for uniform volume current. The weight is the square of the continuous total field. The location is a finite positive moment and therefore strictly interior. It is not generally the zero-field minimum, the area centroid, or the magnitude mean M6.

Location certificate: all normalized barycentric coordinates are strictly positive by the affine averaging argument in Section 3.8. No universal narrower region or rigorous numerical distance-to-side enclosure is asserted for this entry.

Attractivity. Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.10211640 and -0.10259846. No interval certificate is asserted.

The original 100-shape screen minimum was 0.1483760. The September 30 stress survey adds 284 shapes. Its retained negative finite motions are evaluated at orders 40, 80, 160 and 256, with the motion halved at each level. Complete vertices, directions, steps and values are in `chapter_atlas_update/refined_witnesses.json`. For M2, M4, M5 and M7 the reported refined witness is shape 117. The targeted M6 search and its six refinement records are separate. Positive sampled margins are not proofs.

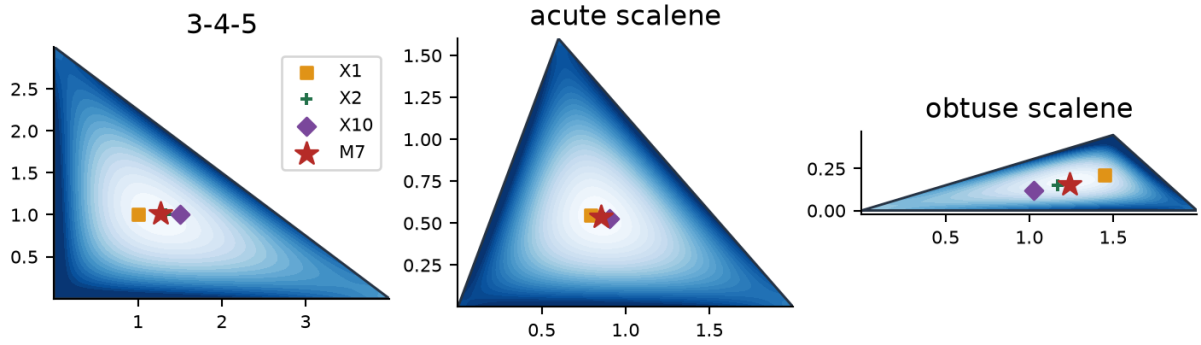
Resolution on general shapes: 100/100 grid shapes and 40/40 holdouts changed by less than 0.0001 diameter between orders 24 and 48. The largest observed changes were $3.33\text{e-}08$ and $2.28\text{e-}05$. All shapes, including less-resolved ones, are retained in the location ranking.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout maximum / D
X4687	0.00574687	0.00577726	0.0120352
X17324	0.00614127	0.00799723	0.0286794
X53034	0.00621419	0.00725259	0.0205268
X25581	0.00640863	0.00663014	0.0136762
X40197	0.0065313	0.00793585	0.0253556

The ranking uses location alone over the finite candidate set. It does not rank by vertex response or establish an identity. For M3, X360 is already proved equal from its barycentric formula.

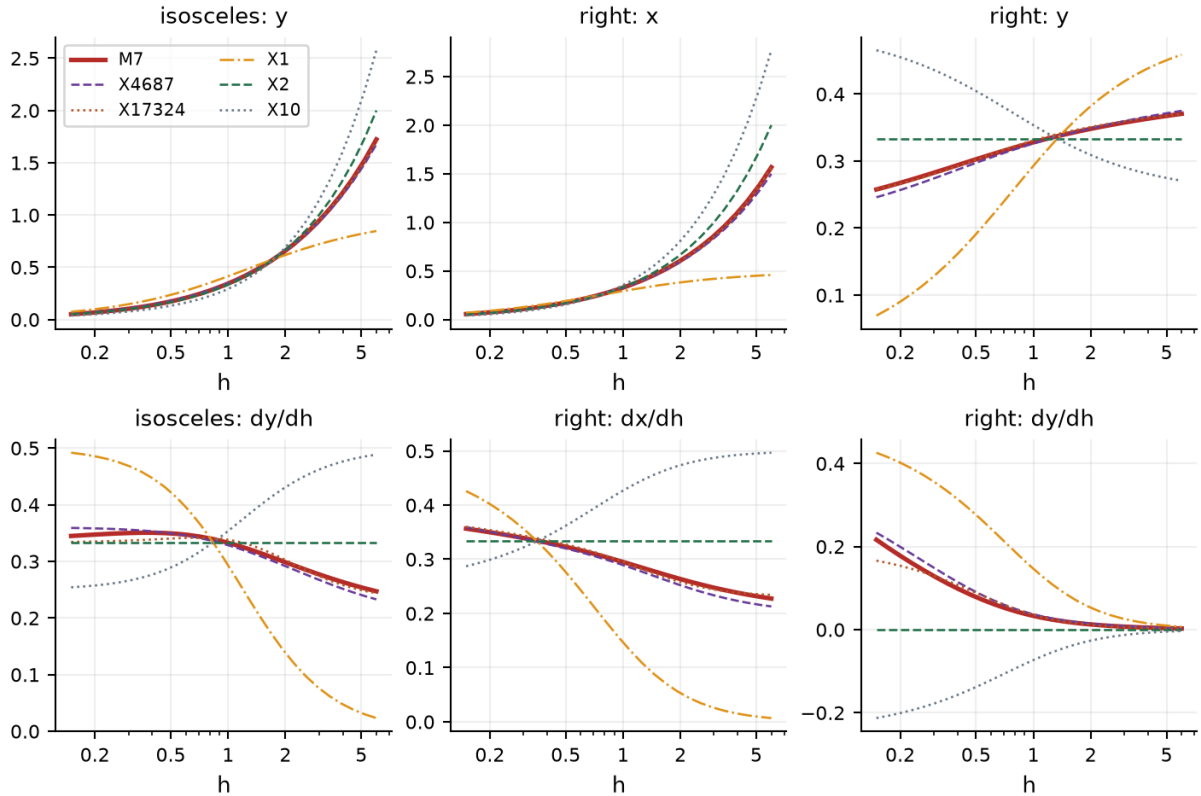
For X4687, position-fingerprint RMS / D is 0.00563704, derivative-fingerprint RMS is 0.0118469, and response-margin RMS difference is 0.0247311. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

M7 - Volume-current energy mean



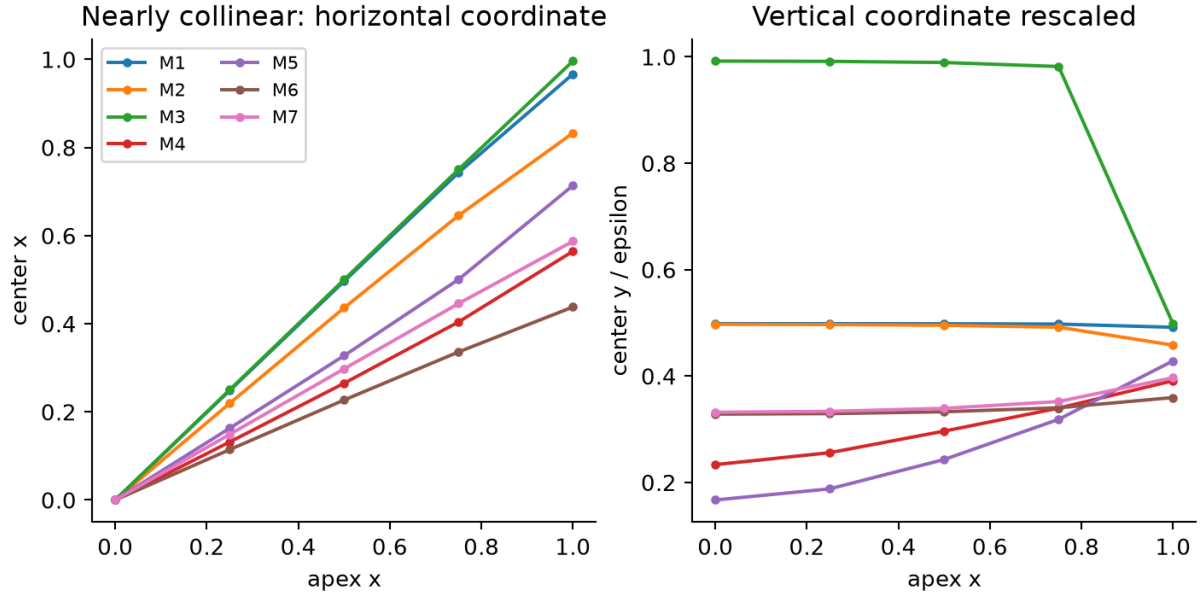
Atlas figure 7a. M7 in the reference, acute and obtuse triangles. Red star: M7; gold square: X1; green plus: X2; purple diamond: X10. Shading shows the defining field or squared field, clipped at its 95th percentile. For M3 the star marks a regularized limit rather than an ordinary squared-field centroid.

M7: special-triangle fingerprints



Atlas figure 7b. M7 in red, compared with the first two ETC location candidates and X1, X2, X10. The upper row contains position fingerprints; the lower row contains derivatives with respect to h . Identical candidates are drawn once. A candidate undefined on a symmetry locus can be absent there; removable singularities have not been reconstructed automatically.

3.10.8 Thin and near-equilateral comparisons



Atlas figure 8. All seven selectors on the nearly collinear family at $\epsilon=0.0125$. The endpoint $x=1$ is plotted separately from the fixed-interior regime. These finite-epsilon values are not certificates of limiting values for the integral-defined means. M3 has the exact endpoint distinction derived in Appendix A.2.

Center	Horizontal apex coefficient	Vertical apex coefficient	Evidence
M1	0.4444437	0.2222225	Exact limits: $4/9$ and $2/9$
M2	0.4790590	0.1876071	Numerical, order 96 and step 0.005
M3	0.6089961	0.0576695	Exact limits given in Appendix A.2
M4	0.2311511	0.4355160	Numerical, order 96 and step 0.005
M5	0.1210422	0.5456254	Numerical, order 96 and step 0.005
M6	0.3427775	0.3238891	Numerical, order 96 and step 0.005
M7	0.3605611	0.3061055	Numerical, order 96 and step 0.005

The table displays the computed coefficients even in the two cases with known exact limits, making finite-step error visible. Mixed coefficients are zero by reflection symmetry at the equilateral triangle. The two diagonal coefficients sum to $2/3$ in the exact smooth limit. Full order-and-step comparisons are retained in `equilateral_jets.json`.

Chapter 4

Membranes, hanging sheets and elastic laminae

4.1 What is being allowed to move?

A horizontal triangular frame does not, by itself, determine the lowest point of whatever spans it. A slack sheet can fold. A stretched membrane resists an increase of area. A thin solid plate resists bending even without initial tension. Heating a plate can create compression and then buckling. A probe applying a force at one point asks a different question from gravity acting everywhere. Each experiment must specify its material law, constraints, load and observable before it can define a triangle center.

We begin with the supplied manuscript A Variational Generalization of the Hanging-Chain Problem [16]. It contains two exact or limiting identifications: the incenter for a tetrahedral sheet, and the torsion maximum for a shallow sheet with prescribed excess area. It also gives an essential negative result: a fixed boundary and total sheet area alone do not provide a gravitational minimizer. The restriction that makes each model meaningful is part of its definition.

Our sign convention is downward displacement w positive. The projection of a lowest point is then a maximum of w over the horizontal triangle T . In the mass chapter, a centroid was a first moment of a specified density. Here a maximum is often the desired observation, but positive field densities also give natural first moments. We retain both when they answer different experiments. The same displacement field can therefore generate several centers.

For comparison we use the incenter $I=X1$, area centroid $G=X2$ and uniform-wire centroid $W=X10$. In the 3-4-5 triangle these are $(1,1)$, $(4/3,1)$ and $(3/2,1)$. Its area is 6, semiperimeter is 6 and inradius is 1. All reported numerical positions use these coordinates, rather than the 6-9-13 reference triangle of the older mechanical atlas.

4.1.1 A map of the models

Object and constraint	Quantity minimized or measured	Selected point
Sheet, boundary and total area only	Gravitational energy	No minimizer; no center
Three planar faces, fixed lateral area	Apex depth or sheet gravity	X1 exactly
Shallow sheet, fixed excess area	Leading gravitational energy	F1
Uniformly tensioned membrane	Deflection under uniform load	F1
Nonlinear slope-energy membrane	Maximum or mean deflection	Mem(p), MemMean(p)
Uniform clamped solid plate	Deflection under gravity	Pgrav
Compressed clamped plate	First unstable mode	Bmode, Bmean
Loaded plate below buckling	Maximum deflection	Pheat(tau)
Point probe or thermal fluctuations	Maximum or mean local compliance	Compliance, FluctMean
Loaded plate bending energy	Energy first moment	Energy

The last eight plate labels and the six nonlinear membrane instances already occur in the atlas. F1 and the ordinary membrane displacement mean are older aliases: Mem(2)=F1 and MemMean(2)=Abs(0). This chapter adds physical interpretations, clearer derivations, a new reference calculation and targeted refinements; it does not invent new names for the same functions.

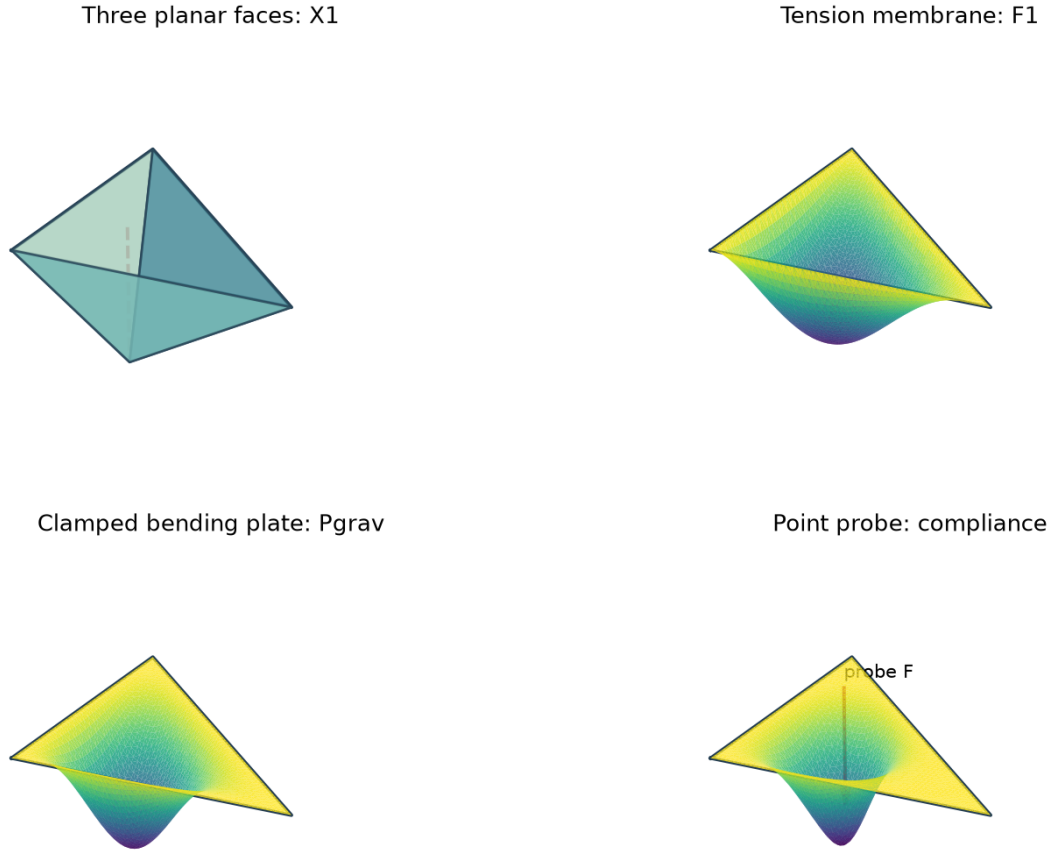


Figure 1. Four experiments on the same 3-4-5 footprint. Vertical scales are chosen for visibility. A three-face sheet has an exact incenter projection. A tension membrane and a bending plate obey different differential equations. The point probe measures a diagonal response, not the uniformly loaded deflection.

4.2 From a hanging chain to a hanging sheet

4.2.1 Why the one-dimensional analogy is useful

Suspend a chain between equal-height endpoints separated by d . Its linear density is constant and its length L exceeds d . With upward coordinate z , minimize the gravitational energy, proportional to the integral of z along the chain, at fixed length. The Lagrange multiplier for length acts as a tension scale. The Euler equation gives a catenary; symmetry places its minimum below the midpoint. Writing its scale as a , the length equation and sag are

$$L = 2a \sinh \frac{d}{2a}, \quad h = a \left(\cosh \frac{d}{2a} - 1 \right).$$

If instead only a heavy bead contributes appreciable mass, its accessible positions satisfy the sum of distances to the endpoints equal to L . The bead occupies the lowest point of this ellipse and has sag one half of the square root of L squared minus d squared. These statements presume that

the lengths on the two sides of the bead can adjust, for example by sliding; fixing the bead to a specified material point supplies two length constraints instead. Equal-height symmetry makes the projected location the same, but the two minimization problems and their depths differ.

In one dimension, extra length cannot carry a fixed positive amount of mass arbitrarily far down a needle of arbitrarily small width: the two descending strands themselves consume length. In two dimensions, the area of a thin cone depends on the product of width and depth. This changes the answer completely.

4.2.2 A fixed-area sheet can form an arbitrarily deep spike

Let the footprint have area K and prescribe actual sheet area $S=K+E$ with $E>0$. For this idealized model the density is constant per actual surface area. The only geometric constraints are the boundary and total area; no local material metric, bending penalty or bound on slope is prescribed.

Place a narrow circular cone of depth H inside the triangle and leave the rest of the sheet horizontal. If the cone has base radius r_0 , the area it adds beyond the disk it replaces is

$$E = \pi r_0 \sqrt{H^2 + r_0^2} - \pi r_0^2.$$

For sufficiently large H , the exact choice $r_0=(E/\pi)/\sqrt{H^2 - 2E/\pi}$ makes this added area E and fits the cone inside any prescribed interior neighborhood. Its lateral surface has mean downward depth $H/3$. Consequently its gravitational energy is minus ρg times $H/3$ times its lateral area, and tends to minus infinity as H grows. Rounding the cone tip and the circular join does not remove the construction: their areas can be made arbitrarily small and the radius adjusted to restore the constraint.

Thus a fixed total area is insufficient, even when the surface is a single-valued graph with zero boundary displacement. This is not a defect of a numerical optimizer; the unrestricted problem has no solution. A real sheet has additional structure, such as a fixed material metric, stretching or bending energy, finite thickness, contact, or a restricted range of shapes. Adding any of those changes the problem and can restore equilibrium.

The same warning applies to a practically massless sheet carrying an arbitrarily deep concentrated load. A total-area budget alone does not impose a positive lower width on a spike. The next restriction does.

4.3 The tetrahedral sheet: an exact incenter experiment

Join a point P at downward depth h to the three sides of T with three planar triangular faces. Let Q be its horizontal projection. The restriction to exactly these three faces is crucial. Denote side lengths by a, b, c and consistently signed inward distances from Q to the side lines by d_1, d_2, d_3 . A face has base a and slant altitude $\sqrt{h^2 + d_1^2}$, so the total lateral area is

$$S_{\text{lat}}(Q, h) = \frac{1}{2} \sum_{i=1}^3 a_i \sqrt{h^2 + d_i(Q)^2}.$$

Signed distances allow Q to range over the entire plane. The oriented area decomposition gives the affine identity $\sum a_i d_i(Q)$ equal to $2K$ for every Q . Divide by perimeter $2s$: the length-weighted mean of the three distances is $K/s=r$. The function t maps to $\sqrt{h^2 + t^2}$ plus t squared) is strictly convex when $h>0$. Jensen's inequality therefore gives

$$S_{\text{lat}}(Q, h) \geq s\sqrt{h^2 + r^2}.$$

Equality requires all three signed distances to equal r . Their unique common solution is $Q=I$, the incenter. Hence I minimizes required lateral area at each positive depth. Reversing the statement, it maximizes achievable depth at each prescribed lateral area $S>K$:

$$h_{\text{max}} = \sqrt{(S/s)^2 - r^2}.$$

If total tetrahedron area includes the horizontal base, replace S by total area minus K . At $S=K$ the only depth is zero and the strictness argument degenerates; we require positive excess area.

For a massive three-face sheet of constant surface density, the centroid of each face lies one third of the way down to its apex. Every face therefore has mean height $-h/3$, independently of its horizontal shape. At fixed S its gravitational energy is $-\rho g S h/3$. Minimizing this energy is exactly the same as maximizing h . A heavy apex on otherwise massless faces yields energy $-mgh$ and selects the same Q . These are two physical inputs for $X1$, with different distributions of mass.

In the 3-4-5 triangle, choosing lateral area $S=12$ gives maximum depth $\sqrt{3}$ below $I=(1,1)$. The boundary remains at height zero. This is an exact geometric result, not an approximation based on shallow slopes.

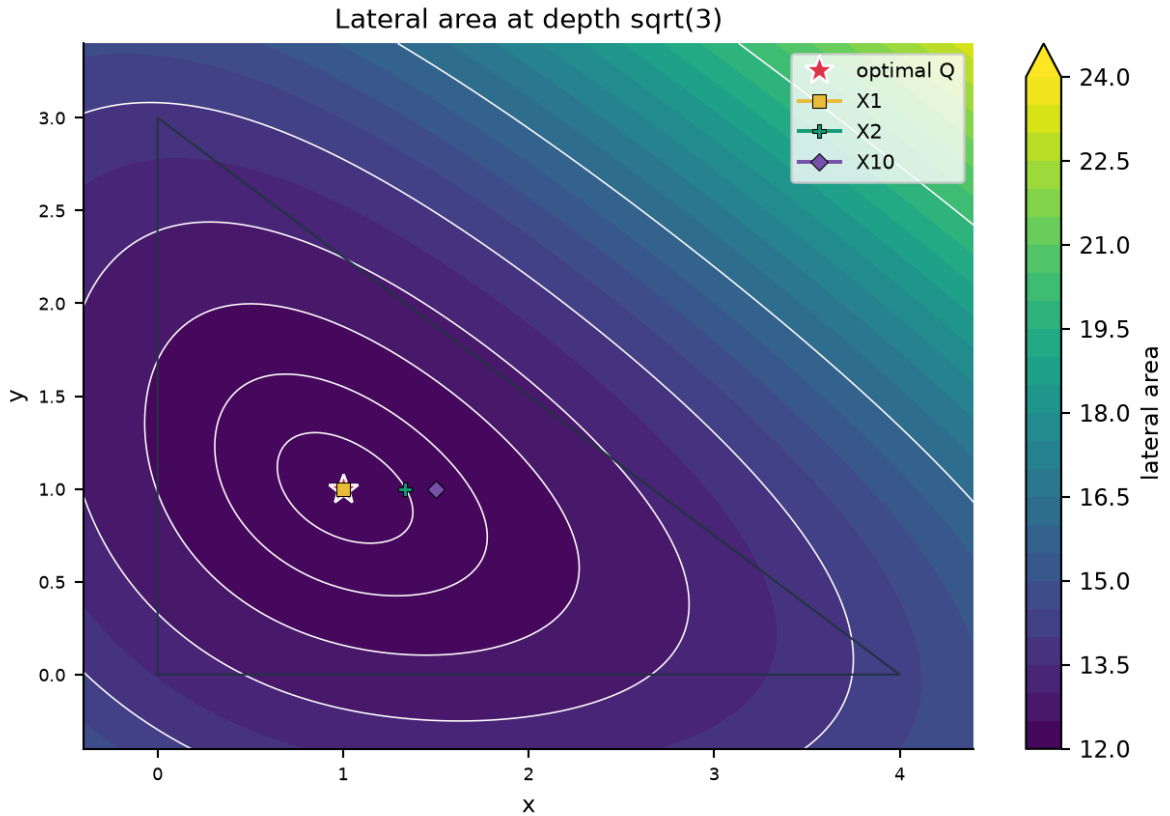


Figure 2. Required lateral area at fixed depth $\sqrt{3}$, shown over the base plane of the 3-4-5 triangle. The minimum lies at I and equals 12. The other marked classical centers require slightly more area at this same depth. The contour formula is exact; the colored grid merely samples it.

$X1$ is strictly vertex-attractive in the sense used throughout the atlas. One way to transfer the previous chapter's calculation is to use $I=3G-2W$. The symmetric vertex-derivative eigenvalues

of W lie strictly between zero and one half, so those of I are one minus twice these eigenvalues and are positive. The proof concerns motion of a vertex and of the projected center; it is separate from the sheet's gravitational stability.

4.4 The shallow fixed-area sheet: a precise torsion reduction

4.4.1 Expanding the two integrals

Let $w(x,y)$ be downward displacement and suppose its slope is uniformly small. The graph area element is $\sqrt{1+|\text{grad } w|^2}$ times the horizontal area element. To leading order, the excess area E and gravitational energy U are

$$E = \frac{1}{2} \int_T |\nabla w|^2 dA, \quad U = -\rho g \int_T w dA.$$

The omitted area terms start at fourth order in slope. The omitted mass-weight correction to gravity contains w times slope squared. Thus this is a controlled leading model in a shallow regime, not an exact treatment of arbitrary excess area. The material density of the unconstrained physical sheet and its local stretching law have not been supplied; only the stated asymptotic variational problem is solved.

We must maximize the integral of w at fixed Dirichlet energy E and zero boundary height. Define the torsion function u by

$$-\Delta u = 1 \quad \text{in } T, \quad u = 0 \quad \text{on } \partial T.$$

Integrate by parts. Because w is zero at the boundary, its integral equals the integral of $\text{grad } u \cdot \text{grad } w$. Cauchy-Schwarz bounds this by the product of their gradient norms. Also, testing the torsion equation against u gives the integral of $|\text{grad } u|^2$ equal to the integral of u . Equality in Cauchy-Schwarz occurs exactly when $\text{grad } w$ is a positive constant times $\text{grad } u$. Zero traces then give

$$w = Cu, \quad C = \sqrt{\frac{2E}{\int_T u dA}}.$$

This proves the optimizer of the quadratic constrained problem, not just its stationary equation. Its depth changes with E ; its projected lowest point is always $F1$, the unique maximum of u . The uniqueness of that maximum on a triangle follows from planar torsion theory. Section 4.6 states a theorem that includes both this case and the nonlinear extensions.

The Lagrange multiplier for excess area plays the mathematical role of a uniform tension. No internal elastic energy was assumed in this fixed-area argument. This explains why a sheet model with only gravitational energy can lead, after a small-slope restriction, to the same equation as a stretched membrane with real stored energy.

4.4.2 Why the full equation is not an existence theorem

Using upward displacement $z=-w$, the exact stationary functional in the supplied source is the integral of $(z+\lambda) \sqrt{1+|\text{grad } z|^2}$. Its Euler equation can be written as

$$\nabla \cdot \frac{\nabla z}{\sqrt{1+|\nabla z|^2}} = \frac{1}{(z+\lambda)\sqrt{1+|\nabla z|^2}}.$$

This follows by differentiating with respect to z and $\text{grad } z$ and cancelling the gradient-squared term. Smooth stationary solutions, where the denominator is meaningful, do not defeat the spike counterexample: they cannot be unrestricted global minimizers. A numerical branch of this equation would need additional admissibility and stability information before its deepest point could be assigned a new atlas center.

4.5 A stretched membrane: force balance, energy and observables

4.5.1 From small vertical forces to a Poisson equation

An isotropic membrane with uniform pretension T_0 exerts a tangential force per unit cut length. Consider a small rectangle in the horizontal plane. For a shallow graph, the vertical components of tension on opposite x faces differ by $T_0 w_{xx}$ times its area; the y faces contribute $T_0 w_{yy}$ times its area. Balancing an applied downward pressure q gives $-T_0 \Delta w = q$ with zero boundary displacement.

The same result follows from minimizing stored energy minus load work. Increasing the membrane's area by one unit costs T_0 . The leading area increase is one half of the integral of squared slope, while the load does work q times the integral of w . Therefore

$$\mathcal{E}(w) = \frac{T_0}{2} \int_T |\nabla w|^2 dA - q \int_T w dA.$$

Differentiate this expression at w in an arbitrary test direction v that vanishes at the boundary. The resulting equation is T_0 times the integral of $\text{grad } w \cdot \text{grad } v$ equal to q times the integral of v . Integration by parts gives the Poisson equation. Since the gradient quadratic form is positive on nonzero zero-boundary functions, the equilibrium field is unique and stable for this linear model.

The solution is $w = (q/T_0)u$. Thus its maximum is F_1 independently of q and T_0 . Here uniform self-weight means uniform load per projected area in the shallow approximation. Large slopes change force balance and the relation between actual-area mass and projected-area loading.

4.5.2 The volume below the membrane defines a different center

Fill the gap between the frame plane and the downward membrane with a fluid of uniform density, ignoring the fluid's feedback on the shape. A vertical column at position x has volume proportional to $w(x)$. Its horizontal center of mass is

$$C_{\text{vol}} = \frac{\int_T x u(x) dA}{\int_T u(x) dA}.$$

This is $\text{MemMean}(2)$, already stored as Abs0.0 in the reaction-diffusion atlas. It is not F_1 . Because u is positive in the interior, all three barycentric first moments are positive, giving a strict-interiority certificate for C_{vol} .

There is a useful second interpretation. Multiply $-\Delta u = 1$ by an affine coordinate function ℓ times u . Integration by parts gives the integral of $\ell |\text{grad } u|^2$ squared equal to the integral of ℓu ; the remaining term is one half the integral of $\text{grad } \ell \cdot \text{grad } (u^2)$, whose boundary contribution vanishes. It follows that the **tension-energy centroid equals the deflection-volume centroid**. This equality concerns the total and first moments. The two local densities are different and their maxima need not coincide.

The same torsion equation describes fully developed Poiseuille flow in a triangular duct and the mean exit time of Brownian motion with generator Delta. Those are additional physical realizations of F1 or the volume mean, depending on the measured observable. They are cross-references, not new membrane center labels.

4.5.3 Exact checks and the role of the Steiner inellipse

For an equilateral triangle of altitude H, let d1,d2,d3 be inward distances to its sides. Their sum is H and the dot product of any two inward unit normals is -1/2. Direct differentiation shows that $\Delta(d_1 d_2 d_3) = -H$. Therefore $u = d_1 d_2 d_3 / H$ solves the torsion problem exactly. Its unique maximum is the common center. For side length 2, the maximum value is 1/9. This exact polynomial is a useful solver check; it does not extend to an arbitrary triangle simply by applying an affine coordinate change.

For an ellipse $x^2/a^2 + y^2/b^2 \leq 1$, a quadratic function solves the same Poisson equation:

$$u(x, y) = \frac{1 - x^2/a^2 - y^2/b^2}{2(a^{-2} + b^{-2})}.$$

Its maximum and volume mean are both the ellipse center. Recall that the Steiner inellipse of T is centered at G, tangent at the side midpoints, and aligned with the covariance axes of uniform area mass. It supplies a natural comparison ellipse from the previous chapter. The elliptic membrane over that ellipse selects G exactly; the triangular membrane generally selects a different point F1.

The reason is mathematical as well as physical. A general affine map changes an isotropic Laplacian into an anisotropic divergence operator. It also changes a scalar tension into a tension tensor. Similarity maps preserve our experiment, but arbitrary affine maps do not. Hence covariance alignment or tangency of the Steiner ellipse cannot identify the triangular membrane maximum with G. Appendix A.3 makes the comparison explicit.

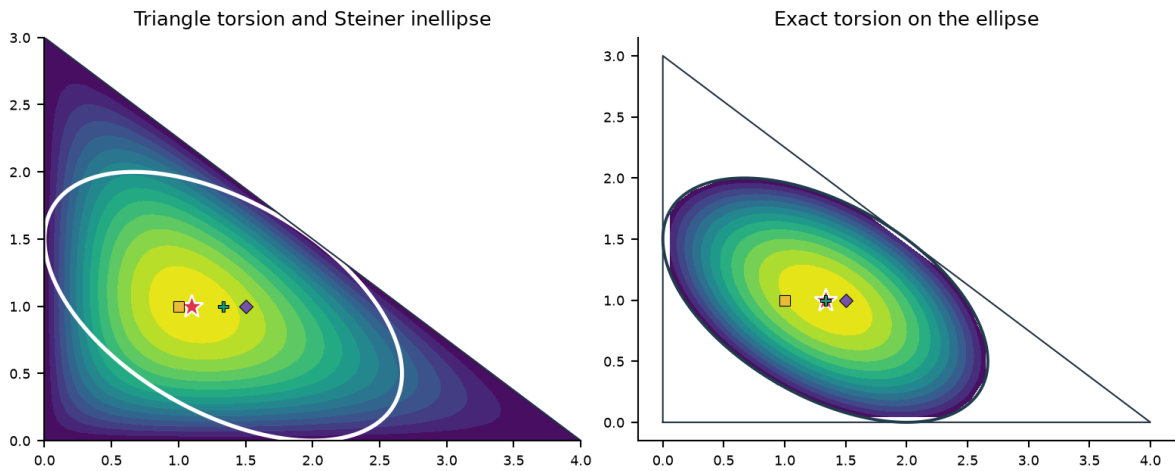


Figure 3. Triangle torsion contours, the Steiner inellipse, and the exact elliptic torsion field for the same 3-4-5 footprint. The ellipse is a covariance-based comparison domain, not an exact contour of the triangular solution.

4.6 Nonlinear tension: the p-torsion membrane family

4.6.1 A specified constitutive law

An ideal slope-energy law proportional to $|\text{grad } w|$ to the power p leads to the energy

$$\mathcal{E}_p(w) = \frac{k_p}{p} \int_T |\nabla w|^p dA - q \int_T w dA, \quad p > 1.$$

This is a phenomenological membrane law, not the generic large-deflection equation of an ordinary elastic metal sheet. Its coefficient k_p has the units needed for the chosen displacement scale. It provides a clean isotropic nonlinear family because multiplying the load changes the amplitude by a power and does not change the selected location.

Differentiate the power of slope. The first variation contains $|\text{grad } w|^{p-2} \text{grad } w \cdot \text{grad } v$. After integration by parts and removal of the amplitude factor, the dimensionless field solves

$$-\nabla \cdot (|\nabla u_p|^{p-2} \nabla u_p) = 1, \quad u_p|_{\partial T} = 0.$$

Strict convexity of the energy for $p > 1$ proves uniqueness of the weak field. Define $\text{Mem}(p)$ as its maximum and $\text{MemMean}(p)$ as its first moment weighted by u_p . The positive field gives interiority of both selectors. Uniqueness of a field does not alone imply uniqueness of its maximum; a separate result is needed.

Borrelli, Mosconi and Squassina [2, Theorem 1.1 and Remarks 1.2, 1.4] prove uniqueness of the maximum for the planar convex p-torsion problem. Their assumptions allow flat sides and corners. The p-torsion field is power-concave and satisfies the theorem with constant positive right-hand side. Thus $\text{Mem}(p)$ is a single point on every nondegenerate triangle, including the atlas exponents 1.5, 3 and 4. This updates the earlier catalogue's cautious maximum-uniqueness wording. It does not prove smooth shape dependence, numerical accuracy, or vertex attractivity.

4.6.2 Discretization and the difficulty at zero slope

Use functions vanishing on all three edges, form a finite coefficient vector, and minimize the strictly convex discrete energy. Numerical integration supplies both energy and its gradient. The existing solver uses a boundary bubble times Bernstein polynomials and a small declared regularization, adding 10^{-18} to squared slope before raising it to a fractional power. This avoids undefined differentiated expressions at an exactly zero slope.

That convenience does not remove approximation error near the maximum. For $p > 2$ even the radially symmetric p-torsion field in a disk has a nonquadratic peak; for $p < 2$ it can be unusually flat. A tiny field error can move its computed maximum more than it moves a first moment. The inherited $\text{Mem}(1.5)$ and $\text{Mem}(3)$ permutation and quadrature checks reveal this sensitivity. Their fitted equilateral second-order coefficients are therefore omitted as unresolved, and unstable negative responses are not accepted as counterexamples.

4.6.3 Thin triangles and the large-exponent limit

Across a narrow strip of width H , integrate the one-dimensional equation once. On either side of the strip midpoint the flux is linear, so the slope magnitude is distance from the midpoint to the power $1/(p-1)$. Integrating again gives a displacement scale $H^{p/(p-1)}$. Integrating displacement across the strip adds one power of H . Thus a slow triangular width profile contributes longitudinal mass proportional to H^k with $k = 1 + p/(p-1)$.

For vertices $(-1,0), (1,0), (x, \epsilon)$, direct integration of this piecewise-linear width profile raised to the power k gives the limiting longitudinal mean $kx/(k+2)$. Consequently

$$\text{MemMean}(p)_x \longrightarrow \frac{2p-1}{4p-3}x.$$

At $p=2$ this is $3x/5$. It differs from the uniform-area mean $x/3$. These are leading strip asymptotics at fixed p and fixed interior x , without a certified finite-epsilon remainder. Near the endpoint $x=1$, or while p grows with thinness, the orders of limits require separate analysis. The large- p distance-function limit places the maximum at the incenter. It does not make every finite- p maximum equal to X_1 .

4.7 A solid elastic lamina under gravity

4.7.1 From three-dimensional strain to bending energy

Now replace the membrane by a thin homogeneous isotropic solid of thickness t , Young modulus E_Y and Poisson ratio ν . In the Kirchhoff-Love approximation, normals to the middle surface remain normal during bending. A layer at signed distance z from the middle plane has in-plane bending strain proportional to $-z$ times the Hessian of w . The elastic energy is quadratic in that strain. Integrating z squared from $-t/2$ to $t/2$ supplies the factor t cubed/12.

The resulting bending rigidity and energy are

$$D_b = \frac{E_Y t^3}{12(1-\nu^2)}, \quad U_b = \frac{D_b}{2} \int_T \left((1-\nu)|D^2 w|^2 + \nu(\Delta w)^2 \right) dA.$$

Here $|D$ squared $w|$ squared means the sum of the squares of all Hessian entries, so the mixed derivative is counted twice. This positive quadratic form applies in the stable isotropic material range. Small strain, small transverse slopes and thickness much smaller than the horizontal lengths are part of the approximation. The derivation and boundary quantities are described in the elasticity notes [18]; the calculation here fixes the particular isotropic homogeneous experiment.

For uniform downward load q , subtract q times the integral of w from U_b . Fixing both height and slope at the boundary gives the clamped conditions $w=0$ and normal derivative of $w=0$. Varying the energy and integrating twice by parts gives

$$D_b \Delta^2 w = q, \quad w = \partial_n w = 0 \quad \text{on } \partial T.$$

Uniform self-weight has $q=\rho t g$. Amplitude is proportional to q/D_b , but its maximum position is independent of these uniform constants. The atlas center **Pgrav** is therefore the maximum of the normalized clamped solution $\Delta w=1$.

4.7.2 Why boundary conditions and positivity matter

A frame that fixes vertical position but allows rotation is not fully clamped. The physical simply supported condition is zero normal bending moment together with zero height; along a straight edge it reduces, under the appropriate smooth trace interpretation, to the Navier condition $\Delta w=0$. Free edges carry different moment and shear conditions, including corner balances. Holding only the three vertices requires a separate support model and may introduce size-dependent singularities. None of these can be substituted silently for **Pgrav**.

The clamped weak problem has a unique field because bending energy is coercive on functions with both clamped traces. A fourth-order equation has no general second-order maximum principle. We therefore do not infer pointwise positivity, uniqueness of its global maximum, or a universal shape-response sign merely from the PDE order and a positive load. The atlas's plate maxima remain conditional point selectors where the global maximum is unique; its numerical searches retain competing stationary candidates.

There is nevertheless a simple interior location statement for a global maximum of the uniformly loaded field. Testing against w gives a positive bending quadratic form equal to the integral of w . The integral is positive, so w is positive somewhere. Boundary values are zero. A positive global maximum must consequently occur in the interior. This proves location of every maximizing point, even without a theorem that there is only one.

4.7.3 The Ritz solution explained

Write normalized barycentric coordinates as $\lambda_A, \lambda_B, \lambda_C$. Their product b is zero on every side. A trial function b squared times a polynomial has both zero value and zero normal derivative on every side. Choose a basis ϕ_i of these functions and write w as a sum of $c_i \phi_i$. Substitution in the energy gives a finite quadratic function of the coefficients:

$$\mathcal{E}(c) = \frac{1}{2} c^T K c - f^T c, \quad K_{ij} = \int_T \Delta \phi_i \Delta \phi_j dA, \quad f_i = \int_T \phi_i dA.$$

Its derivative is $Kc=f$, hence solve $Kc=f$. For clamped functions the determinant term in the bending energy has zero integral, so this simpler matrix gives the same field for uniform isotropic material. We use Bernstein polynomials of degree d , a squared boundary bubble, and QR rescaling to improve conditioning. The actual trial polynomial degree is $d+6$. Increasing d nests the spaces, but point locations need not converge monotonically. Record both the field diagnostics and changes of the extracted center.

Distributed-load deflection is relatively smooth. A point-probe Green function is less smooth at its source and converges more slowly in the same global polynomial space. This distinction will be visible in the resolution table; using one displayed degree for all plate observables does not give them equal accuracy.

4.8 Uniform heating, compression and buckling

4.8.1 Why a uniformly heated flat sheet has no density maximum

Suppose all in-plane boundary displacement is restrained and the initial plate is flat and homogeneous. Uniform temperature rise produces a uniform thermal eigenstrain $\alpha_T \Delta T$ times the identity. The identically zero in-plane displacement satisfies equilibrium: its stress is uniform compression and its divergence vanishes. In the plane-stress constitutive model the compression resultant per unit length is $E_Y \alpha_T \Delta T / (1-\nu)$.

The flat configuration is spatially uniform in this approximation. It has no distinguished point of extremal density. Uniform thickness expansion or contraction is likewise uniform. Once buckling occurs, density per actual area depends on the in-plane displacement and the surface-area Jacobian. Computing it requires nonlinear elasticity and a selected branch. A point of greatest downward buckling displacement is not automatically a point of greatest material density.

4.8.2 The first unstable mode

An isotropic compressive resultant N releases energy when the middle surface slopes. To quadratic order, subtract one half N times the integral of $|\text{grad } w|$ squared from bending energy. At zero transverse load the flat state is stable while the resulting quadratic form is positive. The first critical value satisfies

$$\lambda_b = \min_{v \neq 0} \frac{\int_T (\Delta v)^2 dA}{\int_T |\nabla v|^2 dA}, \quad N_{\text{cr}} = D_b \lambda_b.$$

The minimum is over clamped nonzero trial functions. Variation gives $\Delta^2 \psi = -\lambda_b \Delta \psi$. The Ritz version is the generalized eigenproblem $Kc = \lambda_b S c$, where S_{ij} is the integral of $\text{grad } \phi_i \cdot \text{grad } \phi_j$. This is a buckling eigenproblem, not the freely vibrating plate equation, whose denominator is a mass integral of v squared.

The sign of a buckling mode is arbitrary: up and down have equal linear energy. The atlas defines B_{mode} as the maximum of ψ squared and B_{mean} as its normalized first moment. B_{mean} is strictly interior because its density is nonnegative and has positive mass in the interior. A simple eigenvalue determines the mode up to scale and sign, but a repeated eigenvalue needs an eigenspace convention. A natural mean uses the sum of squared modes in an L^2 -orthonormal basis of the entire critical eigenspace; selecting one vector arbitrarily would violate symmetry. The inherited calculations track spectral gaps rather than presuming universal simplicity.

4.8.3 A loaded plate below the critical compression

If uniform gravity is present before the instability, the equation is $D_b \Delta^2 w + N \Delta w = q$. To compare triangle shapes as centers, the atlas fixes $\tau = N/N_{\text{cr}}$ for each triangle, with $0 \leq \tau < 1$. After normalization,

$$(\Delta^2 + \tau \lambda_b \Delta) w = 1.$$

The selected maximum is $\text{Pheat}(\tau)$; the stored instances are $\tau=0.5$ and 0.9 . At $\tau=0$ it equals Pgrav . The defining energy remains positive below threshold, so the field is unique. Its maximum still requires a point-uniqueness qualification. Fixing τ means adjusting compression with shape. A fixed physical temperature experiment would follow a different path and has different vertex derivatives.

If the uniform load has nonzero projection on a simple first buckling mode, that mode dominates as τ approaches one. The linear displacement then grows without bound; eventually small-deflection theory breaks down. This asymptotic observation explains a curve approaching a buckling-mode selector. It is not a post-buckling solution, a prediction of the final amplitude, or a branch-selection rule.

4.9 Point probes and thermal fluctuations

4.9.1 The compliance maximum

Apply a small force F at an interior point p and measure downward displacement there. In the linear plate model, $w(x) = F G_b(x, p)$, where the clamped Green function responds to a point source. Displacement at the force location per unit force is $G_b(p, p)$, the local compliance. Define Compliance as the point where this diagonal is largest.

The diagonal is finite in two dimensions for a bending plate: point evaluation is a continuous functional in the clamped H2 energy space. If a_p is the vector of basis values at p , the discrete response is $a_p^T K^{-1} a_p$. It is positive in the interior and zero on the clamped boundary, so every maximum is interior. Uniqueness of the maximizer is not supplied by matrix positivity; the atlas leaves that question open for arbitrary triangles.

A classroom demonstration is a small force probe or calibrated bead moved across the plate, with a laser displacement sensor at the same location. The force must stay in the linear regime and the footprint must be small relative to the plate. A finite contact disk defines a regularized family, which approaches the ideal point-probe measurement only in the appropriate limit.

4.9.2 The fluctuation interpretation

Let the plate exchange energy with a heat bath at temperature Θ while remaining in the harmonic bending regime. For discrete coefficient vector c the energy is one half $c^T K c$. The Boltzmann distribution is Gaussian, so its coefficient covariance is $k_B \Theta K^{-1}$. Evaluating the field at p gives

$$\text{Var } w(p) = k_B \Theta a_p^T K^{-1} a_p.$$

Thus the location of largest displacement variance is exactly the compliance maximum in this model. This is an equilibrium fluctuation identity derived from the same quadratic energy, not a separate center. Define `FluctMean` as the first moment of the positive variance diagonal. Temperature and uniform stiffness rescale the variance but cancel from both selectors.

`FluctMean` is not the center of mass of the plate and not the mean instantaneous position of an atom. It is the horizontal mean of a spatial map of displacement variance. A long sequence of small-amplitude surface measurements can estimate that map. Its continuum plate definition is finite, but high spatial frequencies contribute appreciably, explaining slow Ritz convergence.

4.9.3 The membrane counterpart returns to E1

A tension membrane has a logarithmic Green singularity in two dimensions. An ideal point force and the continuum pointwise fluctuation variance therefore have divergent diagonals. They do not directly define the finite plate compliance observable. Average the force and displacement over a common small circular footprint of radius ϵ . Away from the boundary the regularized diagonal has a universal logarithmic term plus a finite position-dependent part proportional to $\log$ of conformal radius.

Removing the common divergence, or comparing equally sized tiny probes, selects the largest conformal radius: `E1` from the preceding chapter. This provides a membrane interpretation of an already catalogued center. The limiting location uses a common footprint convention; finite probes form a size-dependent family. It also makes a useful conceptual point: a stable membrane can give a center that fails vertex attractivity, since `E1` now has an interval-certified negative vertex-motion witness.

4.10 Bending energy and the deflection-volume center

The local bending-energy density, apart from a common positive factor, is

$$e_\nu = (1 - \nu)|D^2 w|^2 + \nu(\Delta w)^2 = (\Delta w)^2 - 2(1 - \nu) \det D^2 w.$$

Define its first moment as Energy. The density changes with Poisson ratio, but its center does not for a fully clamped homogeneous plate. The reason is an affine-moment identity, which is stronger than cancellation of the total Gaussian-curvature term alone.

For every affine function ℓ and clamped w , integration by parts gives the integral of $\ell \det(D \text{ squared } w)$ equal to zero. One useful identity writes the determinant as the x derivative of $w_x w_{yy}$ minus the y derivative of $w_x w_{xy}$. Boundary terms vanish because the entire gradient trace vanishes; the remaining affine-derivative terms vanish after another integration by parts. Appendix C.2 supplies the moment calculation.

Next suppose $\Delta w = q$ is constant. Test the equation against ℓw and integrate twice. Expanding $\Delta(\ell w)$ leaves $\ell \Delta w$ plus twice $\text{grad } \ell \cdot \text{grad } w$. The integral of Δw times any fixed directional derivative of w is zero for clamped w . Hence

$$\int_T \ell (\Delta w)^2 dA = q \int_T \ell w dA.$$

It follows that Energy equals the deflection-volume first moment and is independent of ν . This is exact in the continuum model. It does not assert equality of the local densities, nor does it identify their maxima. Since the energy density is nonnegative for a stable material, its first moment is strictly interior. This argument remains valid without assuming the clamped Green function is everywhere positive.

The stored name `Energy0.3` records the display choice $\nu=0.3$. `Energy0.0` and `Energy0.49` are identity checks, not additional centers. At finite Ritz degree, ℓw can lie outside the trial space, so the energy and volume means need not agree exactly. Their discrepancy is a useful convergence check.

In a narrow clamped strip the uniform-load deflection is proportional to y squared times $(H-y)$ squared. Its integral across the strip is proportional to H to the fifth power. The same triangular-profile integration used for the membrane predicts `Energy_x` tending to $5x/7$ in the nearly collinear family. The maximum instead approaches the widest part, x . These different limiting fingerprints make the distinction between maximum and first moment experimentally visible.

4.11 The 3-4-5 reference calculations and numerical methods

Selector	x	y	Last change / D
F1	1.094705	0.998017	1.47e-06
Abs0.0	1.217077	1.000677	6.71e-09
Mem1.5	1.113005	0.995299	0.000832
Mem3.0	1.067726	1.001787	0.0011
Mem4.0	1.053668	1.002724	0.00169
MemMean1.5	1.208870	1.001201	2.01e-06
MemMean3.0	1.226974	1.000220	1.79e-05
MemMean4.0	1.232600	1.000030	4.14e-05
Pgrav	1.075307	0.999027	1.3e-06
Bmode	1.063986	0.999776	9.39e-08
Bmean	1.075133	1.002212	2.77e-10
Compliance	1.053417	1.000141	4.82e-05
FluctMean	1.198115	1.000251	0.000624
Energy0.3	1.155748	1.000923	7.65e-09
Pheat0.5	1.073768	0.999005	6.98e-07
Pheat0.9	1.068063	0.999287	2.12e-07

The last displayed digits identify a reproducible computation, not certified coordinate accuracy. For nonlinear membranes the last comparison is degree 11 to 17; for the other entries it is degree 17 to 23. The independent field, optimization and spectral-gap diagnostics are retained in reference.json.

The plate calculations use nested polynomial degrees 11, 17 and 23 for the reference table. The nonlinear membrane reference uses degrees 11 and 17, with declared regularization and residuals retained. These are new calculations on the chapter's 3-4-5 triangle. The general-shape ETC rankings and fingerprints remain the existing atlas experiments; changing one reference triangle is not a reason to silently rerank a different training set.

Figure colors are normalized within each panel. They show the defining field or density, not equal physical deflections across materials. Reference fields use degree 14 for the plate and nonlinear membrane, and a separate degree-20 torsion solution. Two additional scalene-triangle panels reuse the original atlas's illustrative fields. Center coordinates in the tables come from the stated refined calculations, so small visible differences from coarse field markers can occur.

4.11.1 What the ETC comparison does and does not establish

The atlas traversed all 72,807 entries of the supplied X1-X72807 snapshot; 70,853 were evaluable and finite on all 100 generic training shapes. The remaining 1,954 were excluded for unsupported expressions or singular evaluations. This is a whole-snapshot numerical survey with explicit exclusions, not a proof that a candidate is absent from the ETC.

The training shapes uniformly sample the ordered-angle fundamental region through the documented equal-area construction, with 40 independent held-out shapes. Location error is Euclidean distance divided by triangle diameter. Top-ten candidates also have fingerprint-derivative and vertex-response discrepancies. These different metrics are reported separately. No hidden weighting makes a poor derivative match into a good one, and a small RMS location error is never an identity proof.

Isosceles fingerprints use $A=(-1,0)$, $B=(1,0)$, $C=(0,h)$. Right-triangle fingerprints use $A=(0,1)$, $B=(0,0)$, $C=(h,0)$. The thin family uses $C=(x,\epsilon)$ over base $(-1,0),(1,0)$, and almost-equilateral tests perturb $C=(0,\sqrt{3})$. The atlas retains positions, derivatives, resolution changes and omitted/undefined candidate values. Its polynomial approximations use positive barycentric weights and are empirical surrogates, not algebraic representations of the exact PDE centers.

4.11.2 Attractivity and stability must be tested separately

The physical stability of a field means the energy rises under an admissible perturbation of that field on a fixed triangle. Geometric attractivity asks how a selected point moves when a vertex of the domain is displaced. Write J_A for the center's derivative with respect to A . Its symmetric part must be positive semidefinite if every small displacement has a nonnegative dot product with the center displacement. A finite negative dot product refutes global attractivity only when the computed center error is sufficiently controlled.

The atlas's coarse screens contain negative plate and nonlinear membrane responses. Several change sign with degree or quadrature refinement. We retain those failures of numerical convergence as evidence about the method, rather than calling them established counterexamples. The new targeted plate checks use degrees 20, 26 and 32 and both the original and halved displacement. Their results accompany the individual entries. F1 has refined negative finite-element evidence from the earlier atlas, but no interval certificate. E1's membrane-probe limit has the separate certified counterexample from Chapter 2.

4.11.3 Positive symbolic surrogates

The companion also supplies six-coefficient shape approximations for the mechanical entries. Let ell_i be the opposite side length divided by the perimeter, let $u_i = \text{ell}_i - 1/3$, and let $v_i = \text{ell}_i^2 - \text{mean}(\text{ell}_i^2)$. A dimensionless shape factor is $z = 3 \sqrt{3} K/s^2$, where K is area and s is semiperimeter; $z=1$ at equilateral and tends to zero at degeneration. The stored convention uses three powers of either z or $z^{2/3}$, with coefficients multiplying u_i and v_i at each power.

Start with positive barycentric baseline weights b_i , such as incenter weights ell_i or a declared convex mixture of incenter and centroid weights. Form a correction f_i from those six coefficients and normalize exponentially:

$$w_i = \frac{b_i \exp(f_i)}{\sum_j b_j \exp(f_j)}, \quad C_{\text{fit}} = \sum_i w_i V_i.$$

The result has positive barycentrics, respects permutation and similarity symmetries, and returns the common center at equilateral. Appropriate baseline mixtures encode the leading longitudinal thin mean. The exact power convention and coefficients are stored in `mech_opt_fit.py` and `symbolic_models.json`. These are useful compact approximations, but positivity of their weights proves only interiority, not geometric attractivity or equality with the physical center. Their validation errors also inherit uncertainty in the numerical training fields.

4.12 Individual atlas entries

4.12.1 F1: Fluid torsion / linear membrane maximum

Measure maximum axial velocity in a triangular Poiseuille duct, maximum deflection of a uniformly tensioned loaded membrane, or maximal Brownian mean exit time.

$$-\Delta u = 1, \quad u|_{\partial T} = 0, \quad F1 = \arg \max_T u$$

Solve the quadratic FEM Poisson system with exact load integration and maximize the piecewise quadratic field.

On the 3-4-5 triangle the refined position is (1.094705, 0.998017), with normalized barycentric coordinates (0.393652, 0.273676, 0.332672).

Location and definition. The positive torsion field has a unique interior maximum by the planar convex torsion theorem, also included in [17]. This location statement concerns the maximum itself, not a first moment of the field.

Attractivity. Negative numerical evidence, with unresolved refinement sensitivity. The inherited finite-element witness is negative at subdivisions 128, 192 and 256. Independent Ritz degrees 20 and 32 retain both negative steps, but degree 44 changes the half-step sign. The data do not supply a converged, validated continuum counterexample; global nonattractivity is not certified.

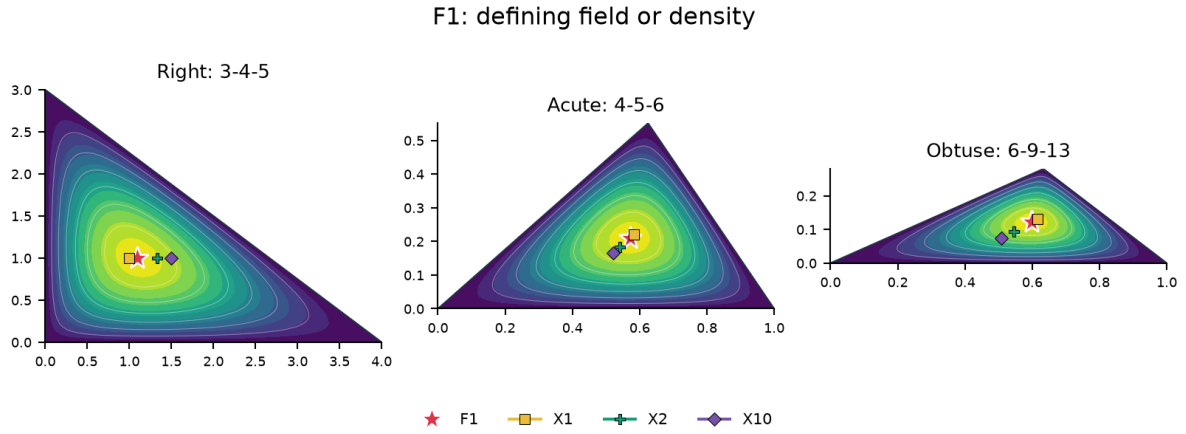
F1 retains the earlier quadratic finite-element location and fingerprint experiment. Its largest last training-grid change was 0.000657934 D. Maximum locations can be more sensitive than mean-field errors.

Independent Ritz degree	Original-step ratio	Half-step ratio
20	-0.07240683	-0.06769307
32	-0.06703628	-0.06659708
44	-0.08868240	0.00121947

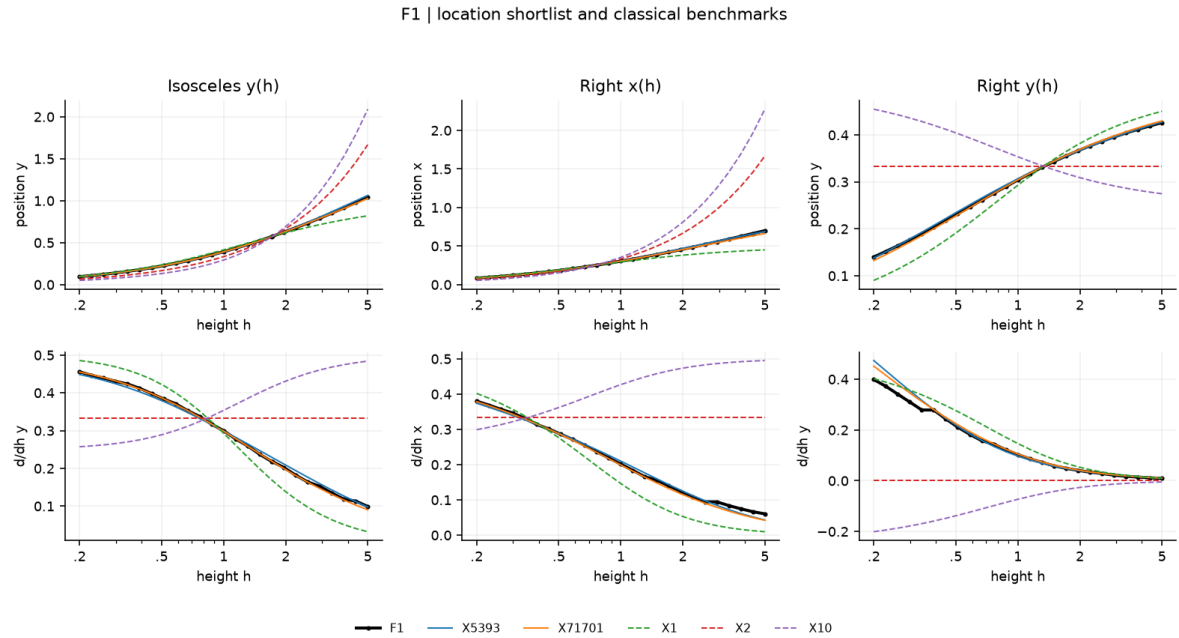
The Ritz solver is an independent spatial discretization of the inherited FEM witness. Degrees 20 and 32 agree with its negative sign, but the degree-44 half-step becomes positive. This loss of stability is retained explicitly. It prevents upgrading the old numerical claim to a settled continuum result; field and maximum-location error control are still required.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5393	0.00422012	0.00405025	0.00826325
X71701	0.00506842	0.00513996	0.0109041
X2534	0.00546374	0.00513597	0.00846997

For the leading candidate X5393, the stored derivative-fingerprint RMS discrepancy is 0.0174586, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0493412. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.



Atlas figure 1a. F1: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 1b. F1: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.2 Abs0.0: MemMean(2): tension-membrane volume and energy centroid

Measure the first moment of the gap beneath a uniformly tensioned membrane under constant transverse load, or the first moment of its tension-energy density. The inherited atlas key is Abs0.0.

$$-\Delta u = 1, \quad u|_{\partial T} = 0, \quad C = \int_T xu / \int_T u$$

Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual.

On the 3-4-5 triangle the refined position is (1.217077, 1.000677), with normalized barycentric coordinates (0.362172, 0.304269, 0.333559).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate.

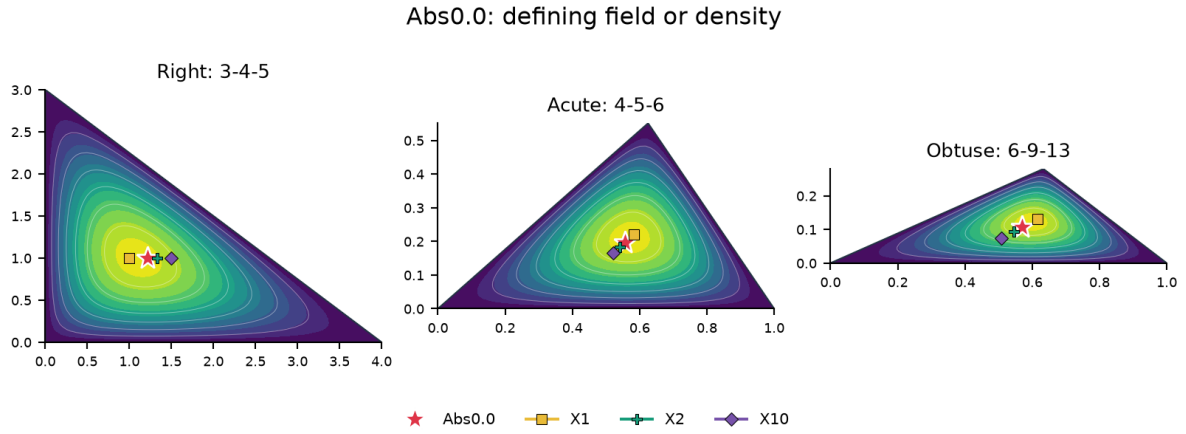
Attractivity. Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large mu, and mu=10000 has independent negative Ritz/FEM evidence. That does not refute this entry. See Appendix D.

Inherited resolution record: 100/100 training triangles and 40/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 7.66774e-05 D. The coarse minimum response was 0.201143, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

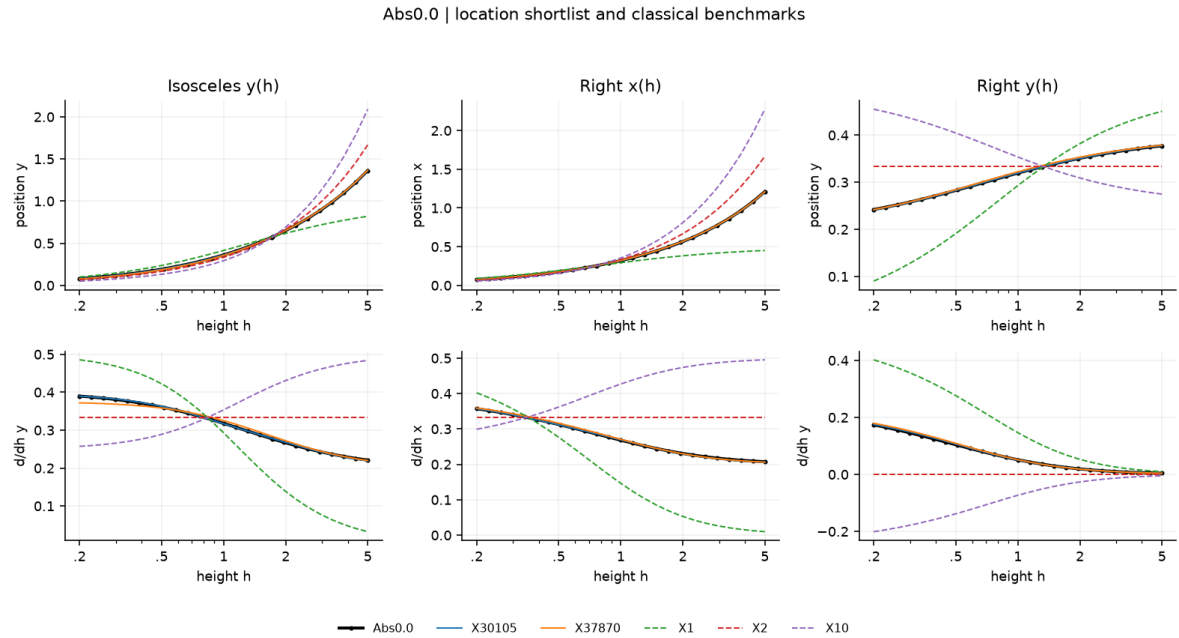
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X30105	0.000752051	0.000724443	0.0021002
X37870	0.00214214	0.00213078	0.00370082
X16831	0.00252882	0.00237092	0.00553802

For the leading candidate X30105, the stored derivative-fingerprint RMS discrepancy is 0.00151126, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.00661625. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.000259866 D. Its baseline is 0.4 I + 0.6 G; q=z. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 2a. Abs0.0: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 2b. Abs0.0: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.3 Mem1.5: Nonlinear membrane maximum, exponent 1.5

Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Locate the greatest transverse deflection.

$$-\operatorname{div}(|\nabla u|^{-0.5}\nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \arg \max u$$

Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree.

On the 3-4-5 triangle the refined position is (1.113005, 0.995299), with normalized barycentric coordinates (0.389983, 0.278251, 0.331766).

Location and definition. The positive field has a unique interior maximum by the planar convex p-torsion theorem [17]. This is a well-defined center, although its numerical shape derivatives may be unresolved.

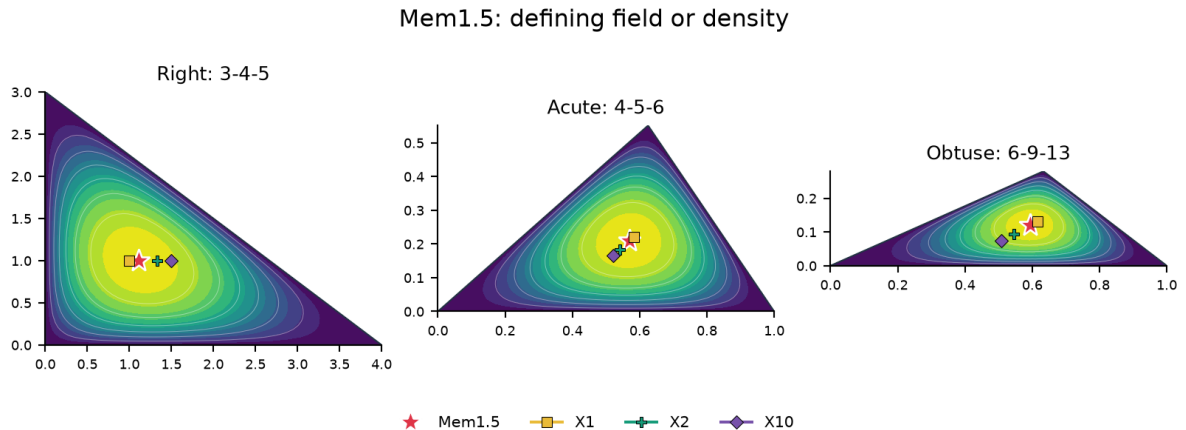
Attractivity. Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

Inherited resolution record: 89/100 training triangles and 37/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.0109493 D. The coarse minimum response was -0.0958127, with 5 negative sampled triangles. A negative count is not a count of established counterexamples.

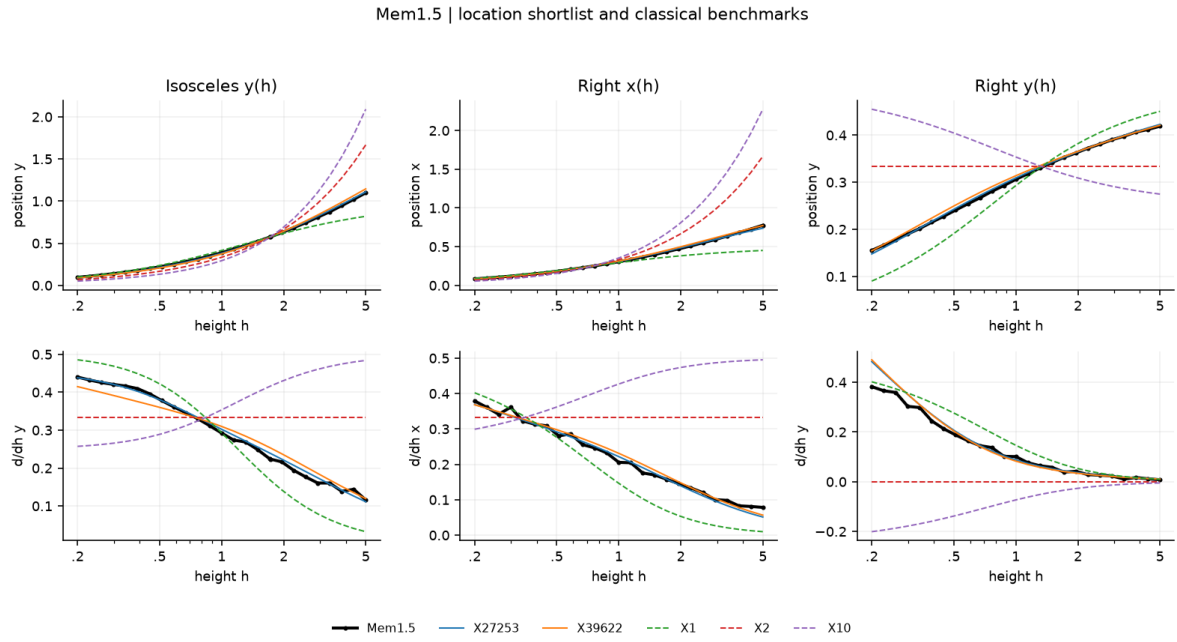
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X27253	0.00787725	0.00670068	0.0158216
X39622	0.0084687	0.0078793	0.0161139
X24936	0.00872403	0.00727812	0.0189091

For the leading candidate X27253, the stored derivative-fingerprint RMS discrepancy is 0.0227549, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0480147. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00760439 D. Its baseline is incenter; $q=\hat{z}(2/3)$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 3a. Mem1.5: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 3b. Mem1.5: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.4 Mem3.0: Nonlinear membrane maximum, exponent 3.0

Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Locate the greatest transverse deflection.

$$-\operatorname{div}(|\nabla u|^1 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \arg \max u$$

Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree.

On the 3-4-5 triangle the refined position is (1.067726, 1.001787), with normalized barycentric coordinates (0.399140, 0.266931, 0.333929).

Location and definition. The positive field has a unique interior maximum by the planar convex p-torsion theorem [17]. This is a well-defined center, although its numerical shape derivatives may be unresolved.

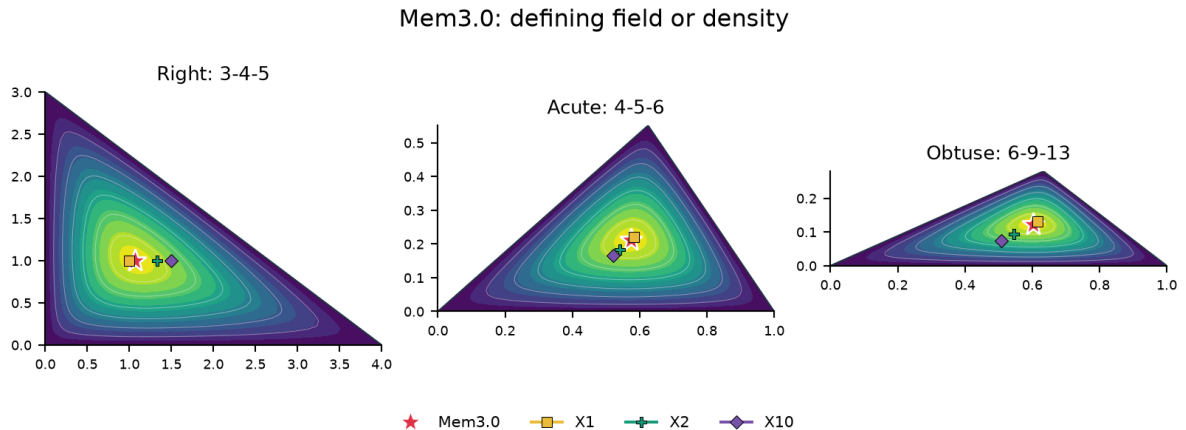
Attractivity. Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

Inherited resolution record: 85/100 training triangles and 32/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00551773 D. The coarse minimum response was -0.246157, with 6 negative sampled triangles. A negative count is not a count of established counterexamples.

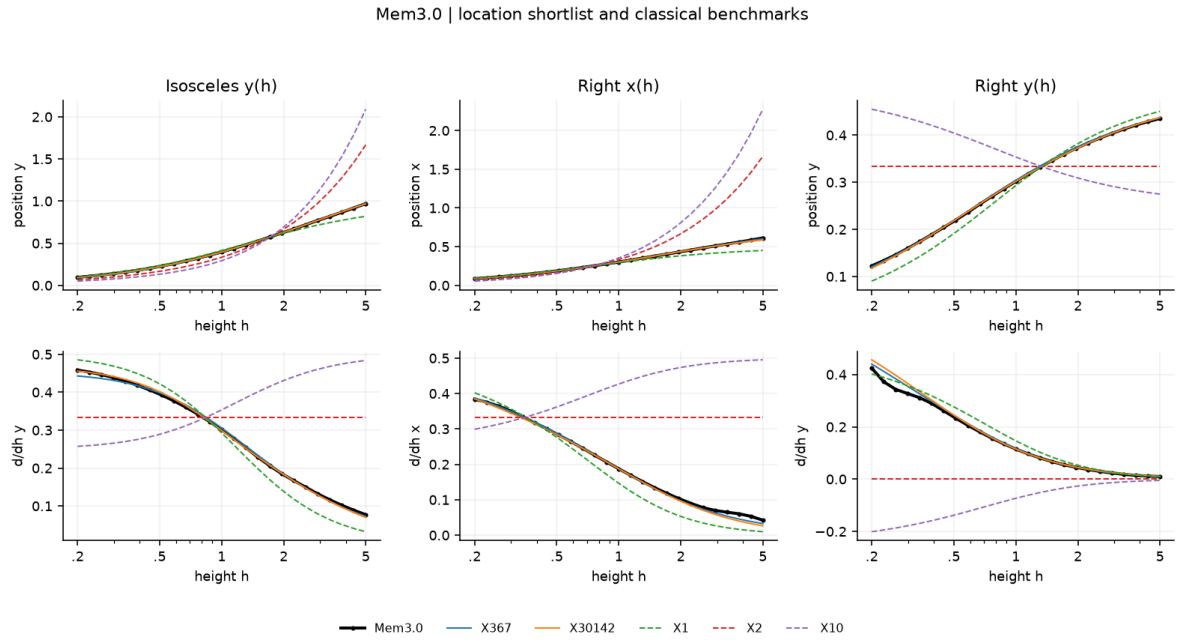
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X367	0.00333057	0.0100715	0.0610012
X30142	0.00400582	0.0105042	0.062678
X29624	0.00406333	0.0106432	0.0626914

For the leading candidate X367, the stored derivative-fingerprint RMS discrepancy is 0.0112436, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0518432. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.0106619 D. Its baseline is incenter; $q = \hat{z}^{(2/3)}$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 4a. Mem3.0: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 4b. Mem3.0: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.5 Mem4.0: Nonlinear membrane maximum, exponent 4.0

Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Locate the greatest transverse deflection.

$$-\operatorname{div}(|\nabla u|^2 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \arg \max u$$

Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree.

On the 3-4-5 triangle the refined position is (1.053668, 1.002724), with normalized barycentric coordinates (0.402342, 0.263417, 0.334241).

Location and definition. The positive field has a unique interior maximum by the planar convex p-torsion theorem [17]. This is a well-defined center, although its numerical shape derivatives may be unresolved.

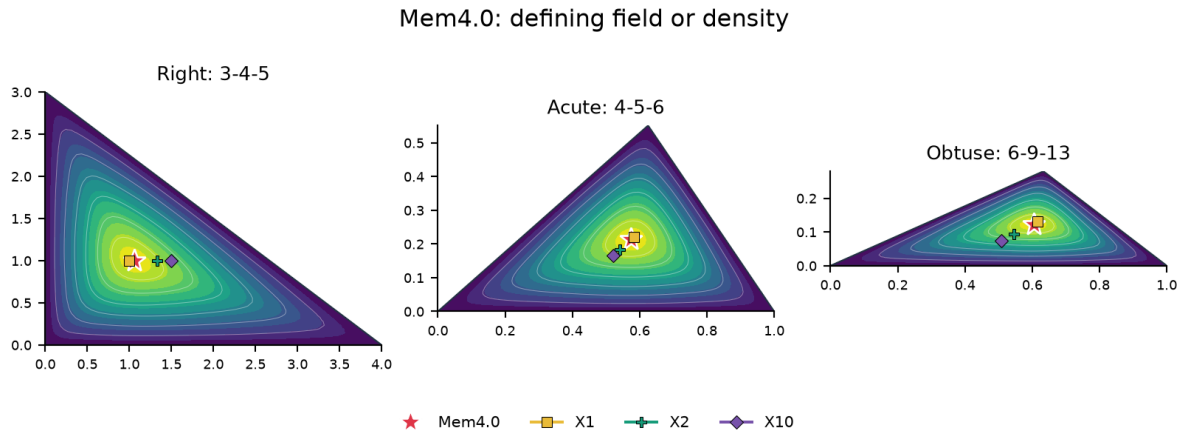
Attractivity. Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

Inherited resolution record: 76/100 training triangles and 28/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00640818 D. The coarse minimum response was -0.301827, with 7 negative sampled triangles. A negative count is not a count of established counterexamples.

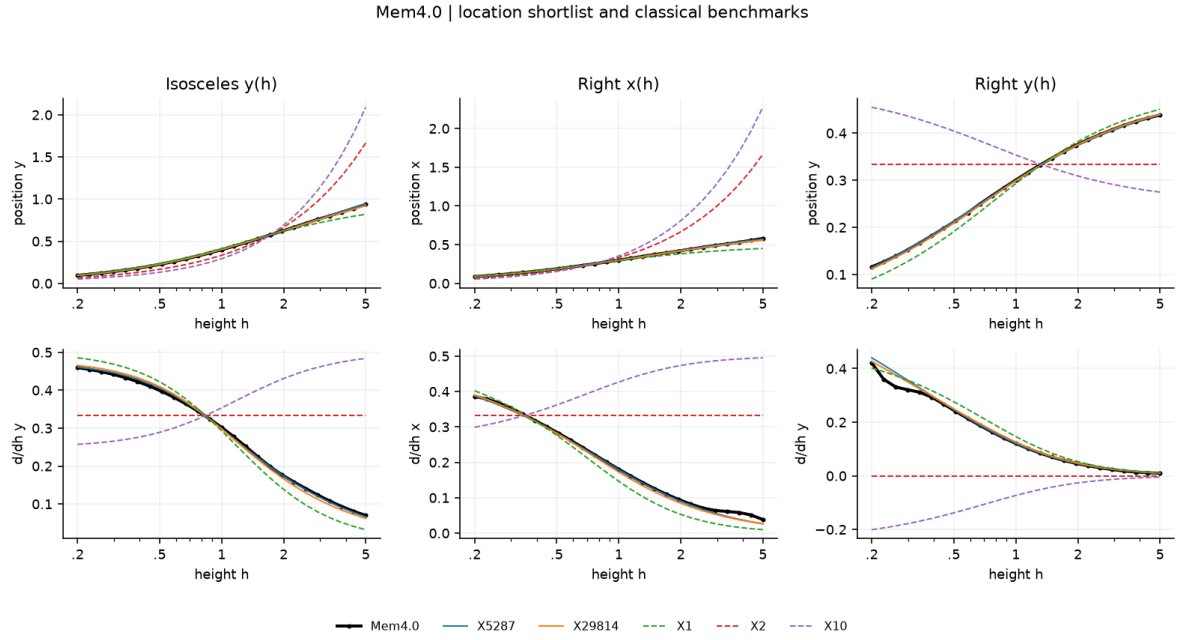
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5287	0.00325486	0.010334	0.0632229
X29814	0.00330034	0.0104446	0.0632489
X29624	0.00355909	0.0104102	0.0632163

For the leading candidate X5287, the stored derivative-fingerprint RMS discrepancy is 0.0135001, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0557729. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.0106155 D. Its baseline is incenter; $q = \hat{z}^{(2/3)}$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 5a. Mem4.0: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 5b. Mem4.0: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.6 MemMean1.5: Nonlinear membrane density mean, exponent 1.5

Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Measure the deflection-volume centroid.

$$-\operatorname{div}(|\nabla u|^{-0.5}\nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \int pu / \int u$$

Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree.

On the 3-4-5 triangle the refined position is (1.208870, 1.001201), with normalized barycentric coordinates (0.364049, 0.302218, 0.333734).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate.

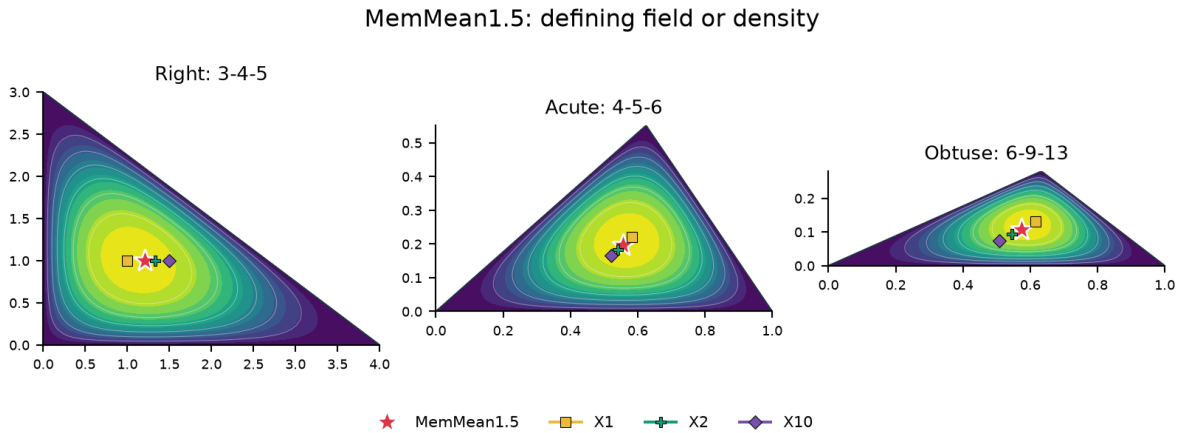
Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Inherited resolution record: 100/100 training triangles and 39/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.000145485 D. The coarse minimum response was 0.168377, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X10197	0.00122722	0.001177	0.00230877
X41312	0.0036405	0.00376844	0.00604839
X10180	0.00365199	0.00376097	0.00706274

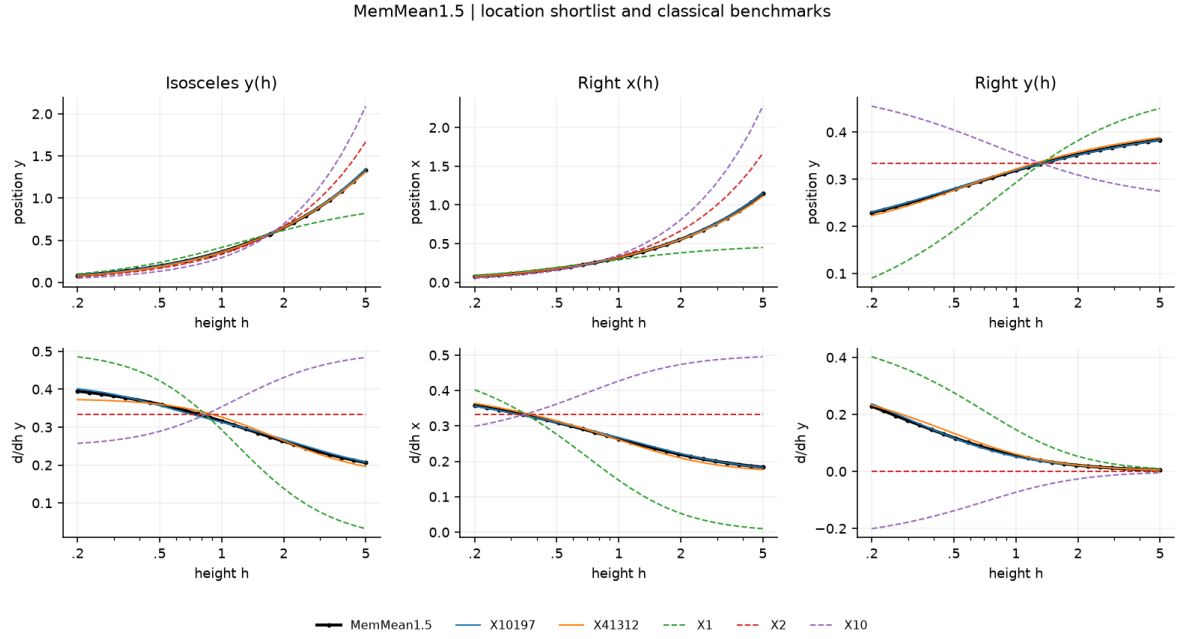
For the leading candidate X10197, the stored derivative-fingerprint RMS discrepancy is 0.00403896, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.00751452. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.000530781 D. Its baseline is 0.5 I + 0.5 G; q=z. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 6a. MemMean1.5: reference 3-4-5 triangle and two further shapes. Red star marks

the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 6b. MemMean1.5: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.7 MemMean3.0: Nonlinear membrane density mean, exponent 3.0

Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Measure the deflection-volume centroid.

$$-\operatorname{div}(|\nabla u|^1 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \int p u / \int u$$

Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree.

On the 3-4-5 triangle the refined position is (1.226974, 1.000220), with normalized barycentric coordinates (0.359850, 0.306744, 0.333407).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate.

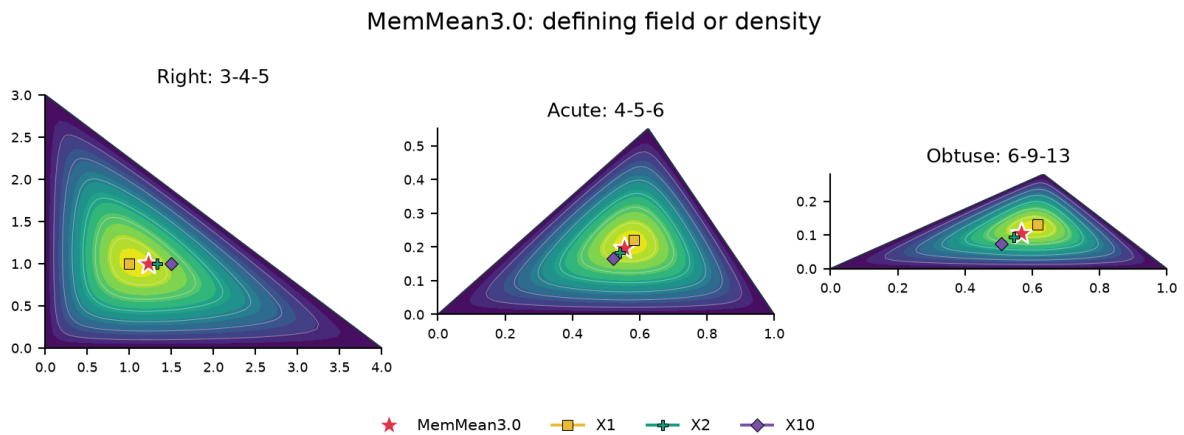
Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Inherited resolution record: 100/100 training triangles and 39/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.000143505 D. The coarse minimum response was 0.223209, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X25055	0.00106607	0.00107558	0.00186137
X28641	0.00123534	0.00124878	0.00178708
X20153	0.00181509	0.00188658	0.00494619

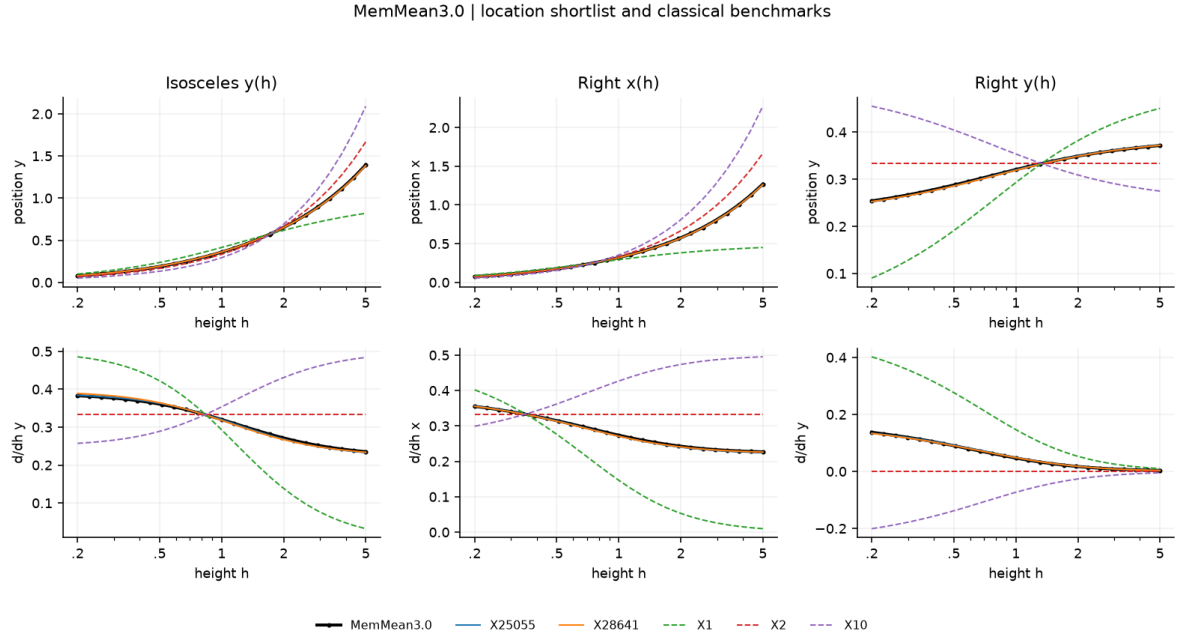
For the leading candidate X25055, the stored derivative-fingerprint RMS discrepancy is 0.00229699, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.00625614. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.000134396 D. Its baseline is 0.3333333333333333 I + 0.6666666666666667 G; q=z. Coefficients and the precise feature formula are in `symbolic_models.json` and `mech_opt_fit.py`; fitting positions does not establish an attractivity theorem.



Atlas figure 7a. MemMean3.0: reference 3-4-5 triangle and two further shapes. Red star marks

the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 7b. MemMean3.0: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.8 MemMean4.0: Nonlinear membrane density mean, exponent 4.0

Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Measure the deflection-volume centroid.

$$-\operatorname{div}(|\nabla u|^2 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \int p u / \int u$$

Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree.

On the 3-4-5 triangle the refined position is (1.232600, 1.000030), with normalized barycentric coordinates (0.358507, 0.308150, 0.333343).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate.

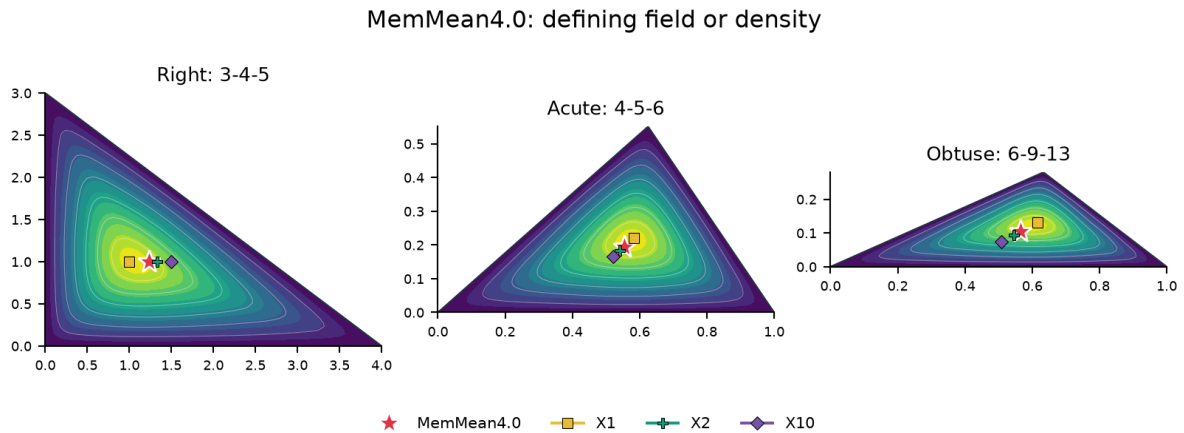
Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Inherited resolution record: 100/100 training triangles and 39/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.000201592 D. The coarse minimum response was 0.231737, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X46934	0.000469201	0.000490445	0.000796369
X51109	0.000958451	0.00100726	0.00197283
X17397	0.00249824	0.00244893	0.00383414

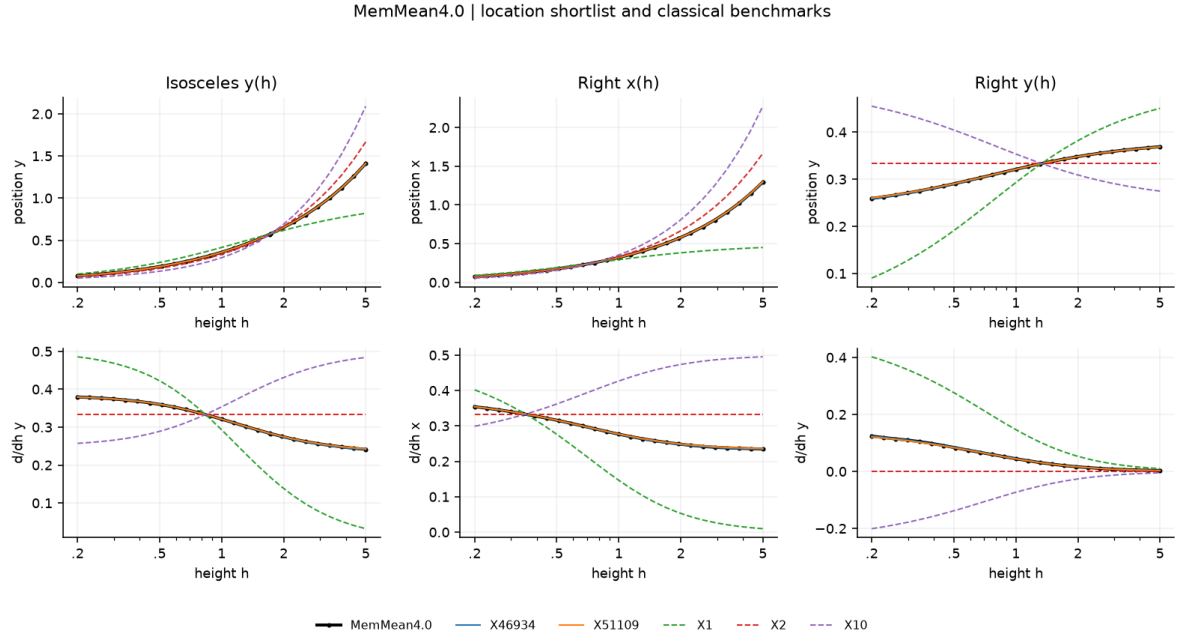
For the leading candidate X46934, the stored derivative-fingerprint RMS discrepancy is 0.000931885, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0075691. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.000141859 D. Its baseline is 0.30769230769230765 I + 0.6923076923076923 G; q=z. Coefficients and the precise feature formula are in `symbolic_models.json` and `mech_opt_fit.py`; fitting positions does not establish an attractivity theorem.



Atlas figure 8a. MemMean4.0: reference 3-4-5 triangle and two further shapes. Red star marks

the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 8b. MemMean4.0: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.9 Pgrav: Gravity-loaded clamped plate maximum

Uniform gravity or pressure deflects a homogeneous thin plate whose boundary height and slope are fixed.

$$\Delta^2 w = 1, \quad w = \partial_n w = 0, \quad C = \arg \max w$$

Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field.

On the 3-4-5 triangle the refined position is (1.075307, 0.999027), with normalized barycentric coordinates (0.398164, 0.268827, 0.333009).

Location and definition. Every selected global maximum is interior in the stated model. The numerical point selector remains conditional on a unique maximum; the atlas supplies no universal plate-maximizer theorem. For Bmode a simple critical eigenspace is also needed unless a full-projector convention is adopted.

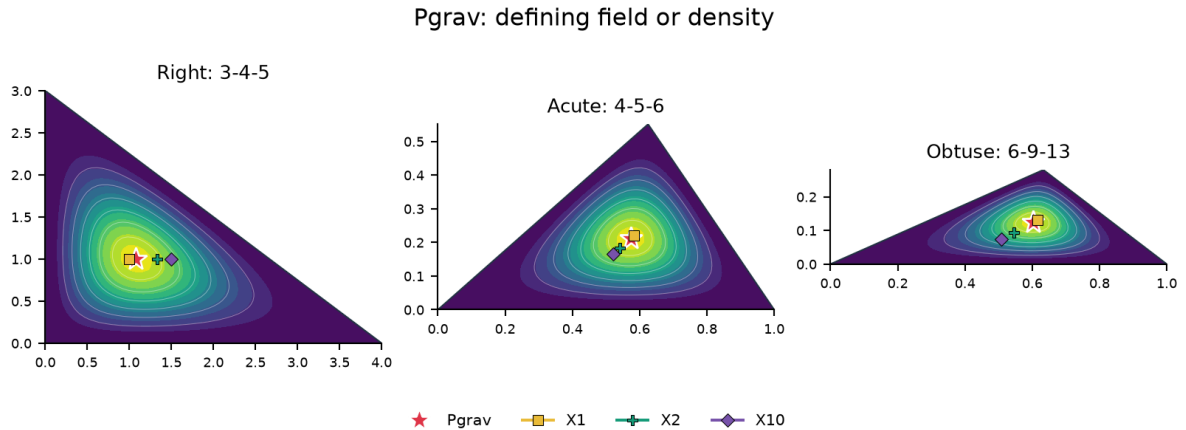
Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Inherited resolution record: 92/100 training triangles and 33/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.0025991 D. The coarse minimum response was 0.0126308, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

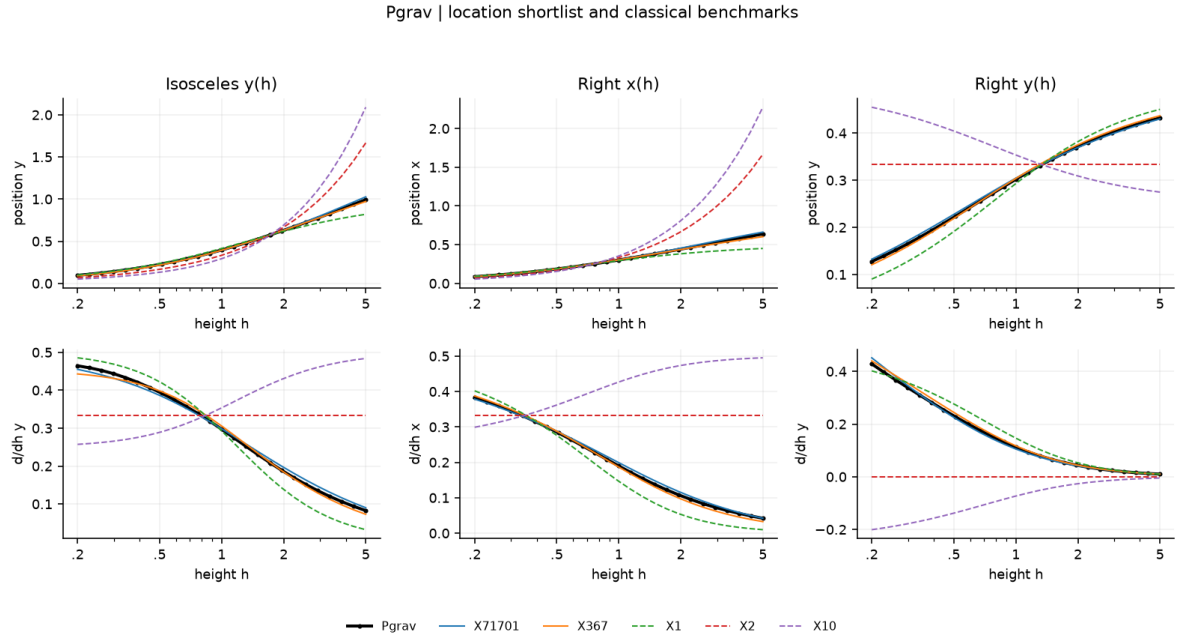
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X71701	0.00357896	0.00596688	0.0296268
X367	0.00433759	0.00651723	0.0303618
X53588	0.00446225	0.00692264	0.0316805

For the leading candidate X71701, the stored derivative-fingerprint RMS discrepancy is 0.00963835, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0818316. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00661363 D. Its baseline is incenter; $q=z^{(2/3)}$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 9a. Pgrav: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 9b. Pgrav: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.10 Bmode: First buckling-mode maximum

Increase uniform compression to its first linear instability and locate the largest squared amplitude of the critical mode.

$$\Delta^2\psi = -\lambda_b\Delta\psi, \quad C = \arg \max \psi^2$$

Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field.

On the 3-4-5 triangle the refined position is (1.063986, 0.999776), with normalized barycentric coordinates (0.400745, 0.265996, 0.333259).

Location and definition. Every selected global maximum is interior in the stated model. The numerical point selector remains conditional on a unique maximum; the atlas supplies no universal plate-maximizer theorem. For Bmode a simple critical eigenspace is also needed unless a full-projector convention is adopted.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Inherited resolution record: 88/100 training triangles and 34/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00303264 D. The coarse minimum response was -0.226962, with 5 negative sampled triangles. A negative count is not a count of established counterexamples.

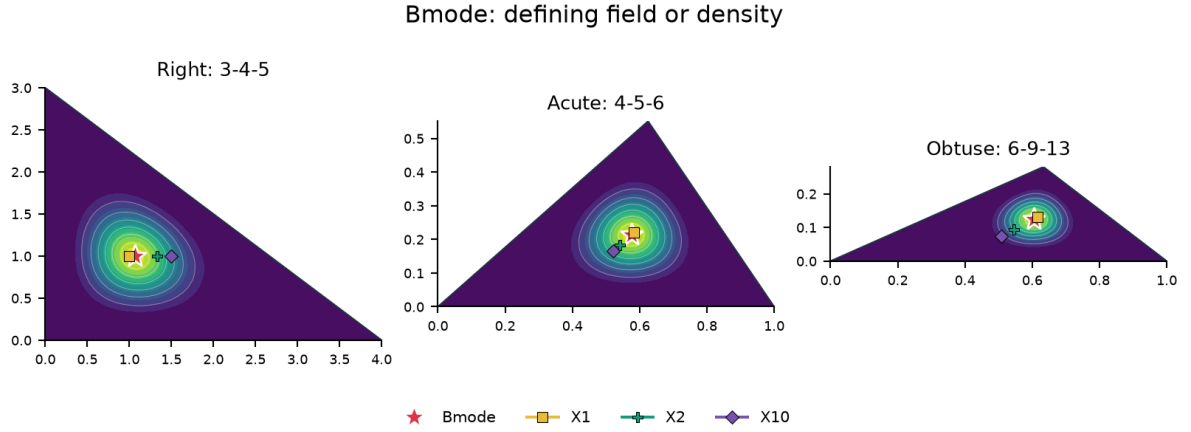
Ritz degree	Original-step ratio	Half-step ratio
20	0.08178634	0.08329733
26	0.10315393	0.10166499
32	0.10361293	0.10211685

These values use the exact retained witness vertices and direction. All inputs and mode gaps are stored in `plate_witness_refinement.json`. They test that particular motion; they are not lower bounds over triangle space.

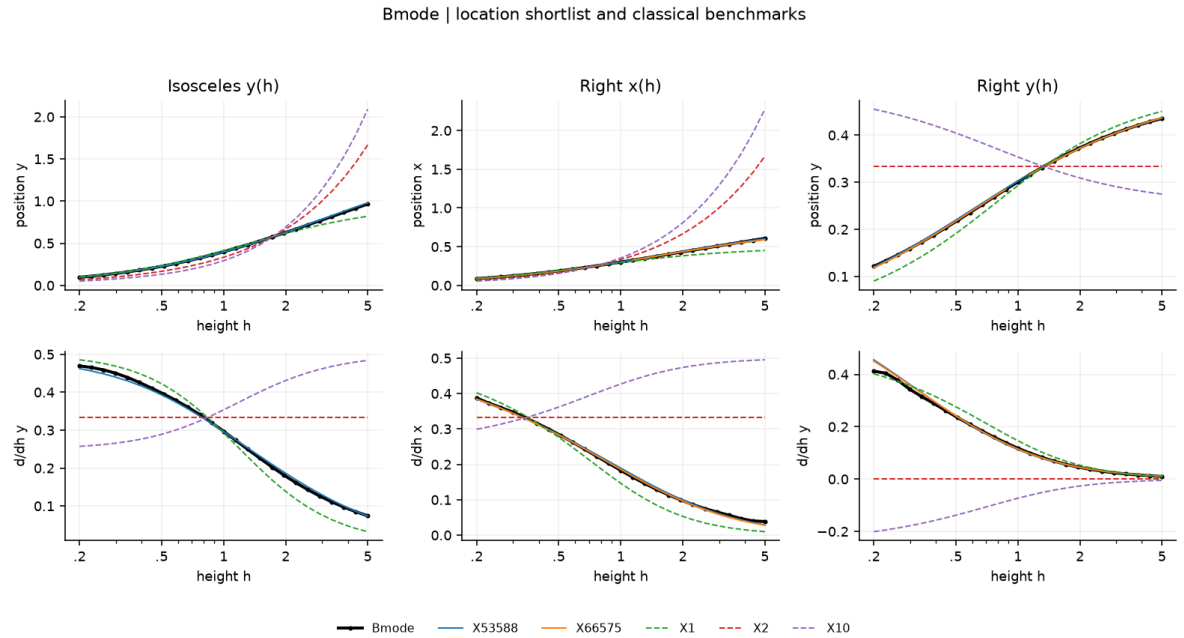
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X53588	0.00267737	0.0196489	0.0970227
X66575	0.00282533	0.0197031	0.0970315
X29624	0.00352851	0.0194358	0.0938258

For the leading candidate X53588, the stored derivative-fingerprint RMS discrepancy is 0.00972212, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0723683. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.0199121 D. Its baseline is incenter; $q=z^{2/3}$. Coefficients and the precise feature formula are in `symbolic_models.json` and `mech_opt_fit.py`; fitting positions does not establish an attractivity theorem.



Atlas figure 10a. Bmode: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 10b. Bmode: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.11 Bmean: First buckling-mode expectation

Use the spatial first moment of the squared critical buckling mode, avoiding its arbitrary sign.

$$\Delta^2\psi = -\lambda_b\Delta\psi, \quad C = \int p\psi^2 / \int \psi^2$$

Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field.

On the 3-4-5 triangle the refined position is (1.075133, 1.002212), with normalized barycentric coordinates (0.397146, 0.268783, 0.334071).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate. The individual-mode definition requires a simple critical eigenspace or the explicitly stated projector convention.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Inherited resolution record: 89/100 training triangles and 33/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00299849 D. The coarse minimum response was -0.2036, with 4 negative sampled triangles. A negative count is not a count of established counterexamples.

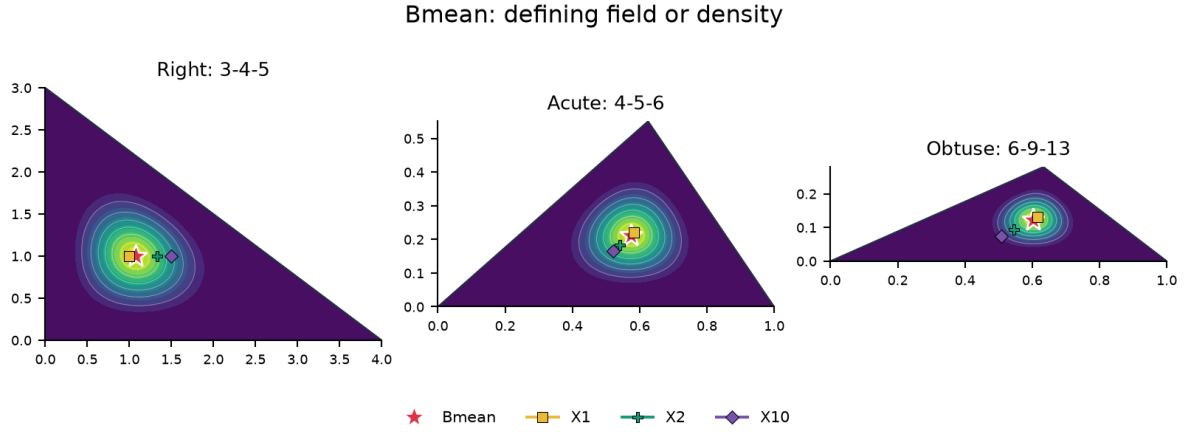
Ritz degree	Original-step ratio	Half-step ratio
20	0.09583589	0.09698384
26	0.10527426	0.10373161
32	0.10566495	0.10418017

These values use the exact retained witness vertices and direction. All inputs and mode gaps are stored in `plate_witness_refinement.json`. They test that particular motion; they are not lower bounds over triangle space.

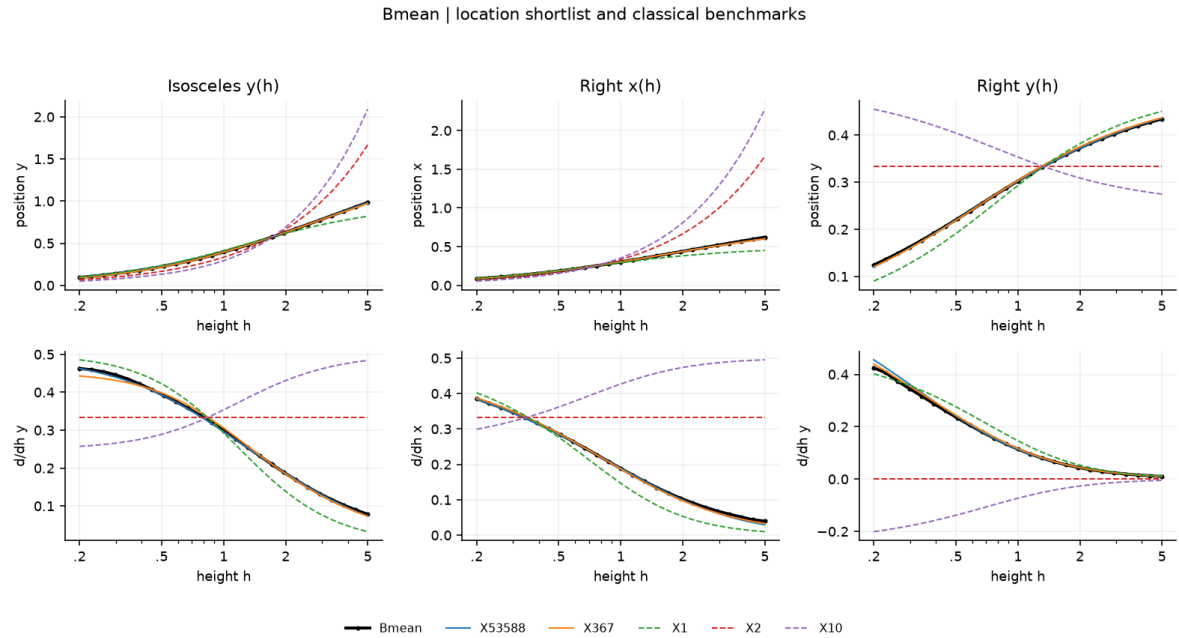
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X53588	0.00338569	0.00394908	0.0130349
X367	0.00365808	0.00379262	0.00970521
X66560	0.00394934	0.0044619	0.0130431

For the leading candidate X53588, the stored derivative-fingerprint RMS discrepancy is 0.00759843, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0722444. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00417451 D. Its baseline is incenter; $q=z^{(2/3)}$. Coefficients and the precise feature formula are in `symbolic_models.json` and `mech_opt_fit.py`; fitting positions does not establish an attractivity theorem.



Atlas figure 11a. Bmean: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 11b. Bmean: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.12 Compliance: Maximum point compliance / thermal variance

Scan a point load and measure displacement at the load point, or locate the greatest equilibrium thermal displacement variance.

$$C = \arg \max_p G_b(p, p)$$

Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field.

On the 3-4-5 triangle the refined position is (1.053417, 1.000141), with normalized barycentric coordinates (0.403265, 0.263354, 0.333380).

Location and definition. Every selected global maximum is interior in the stated model. The numerical point selector remains conditional on a unique maximum; the atlas supplies no universal plate-maximizer theorem. For Bmode a simple critical eigenspace is also needed unless a full-projector convention is adopted.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Inherited resolution record: 80/100 training triangles and 34/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.0026068 D. The coarse minimum response was -0.725125, with 8 negative sampled triangles. A negative count is not a count of established counterexamples.

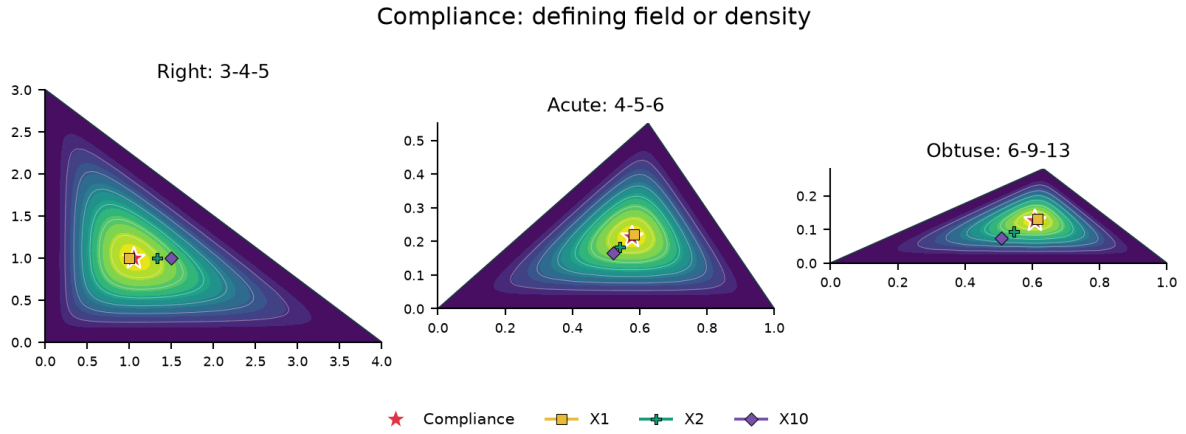
Ritz degree	Original-step ratio	Half-step ratio
20	0.23066628	0.22751602
26	0.17044776	0.16524098
32	0.11832031	0.11428299

These values use the exact retained witness vertices and direction. All inputs and mode gaps are stored in plate_witness_refinement.json. They test that particular motion; they are not lower bounds over triangle space.

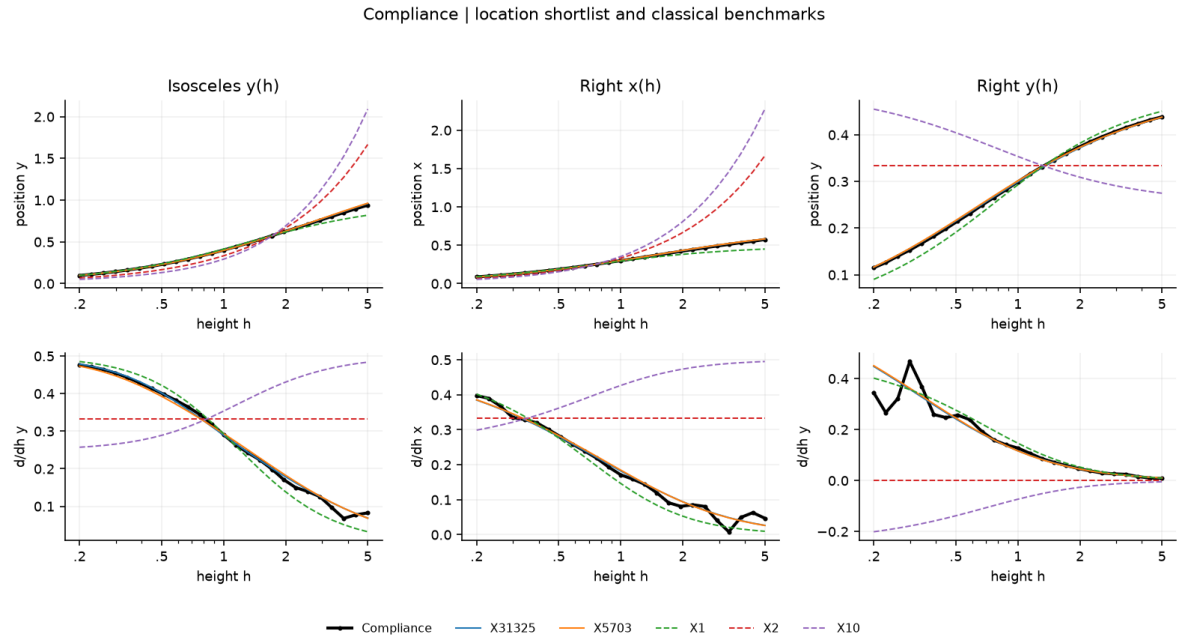
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X31325	0.00253157	0.00266588	0.0084842
X5703	0.00264369	0.0028636	0.00848288
X16015	0.00278309	0.00303297	0.00801462

For the leading candidate X31325, the stored derivative-fingerprint RMS discrepancy is 0.0349092, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.107681. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00267817 D. Its baseline is incenter; $q=z^{(2/3)}$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 12a. Compliance: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 12b. Compliance: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.13 FluctMean: Thermal fluctuation-density centroid

Average position using the local plate displacement variance as a positive density.

$$C = \int_T p G_b(p, p) dp / \int_T G_b(p, p) dp$$

Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field.

On the 3-4-5 triangle the refined position is (1.198115, 1.000251), with normalized barycentric coordinates (0.367054, 0.299529, 0.333417).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate.

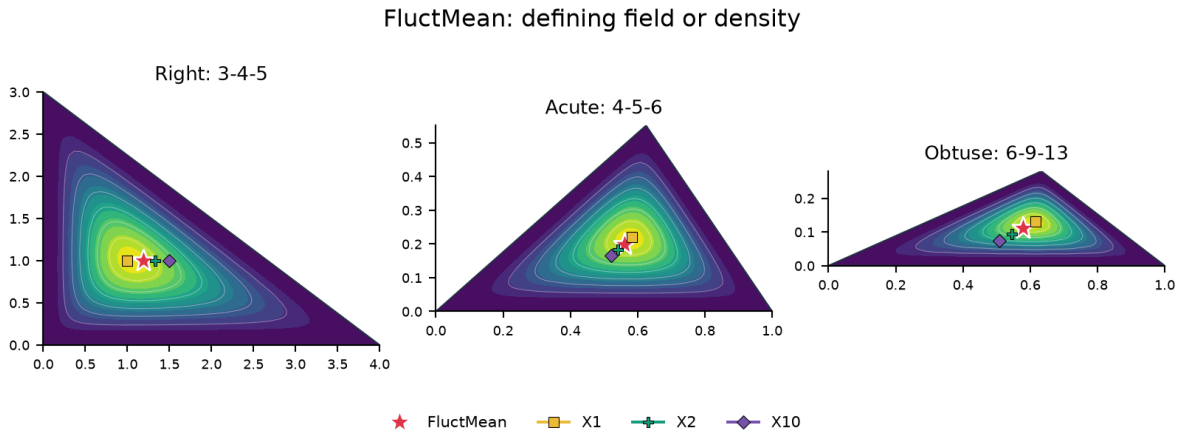
Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Inherited resolution record: 46/100 training triangles and 18/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00537727 D. The coarse minimum response was 0.141952, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X31319	0.0071399	0.00885627	0.0211029
X27268	0.00730219	0.00859276	0.0223509
X39738	0.00829442	0.00988825	0.0206975

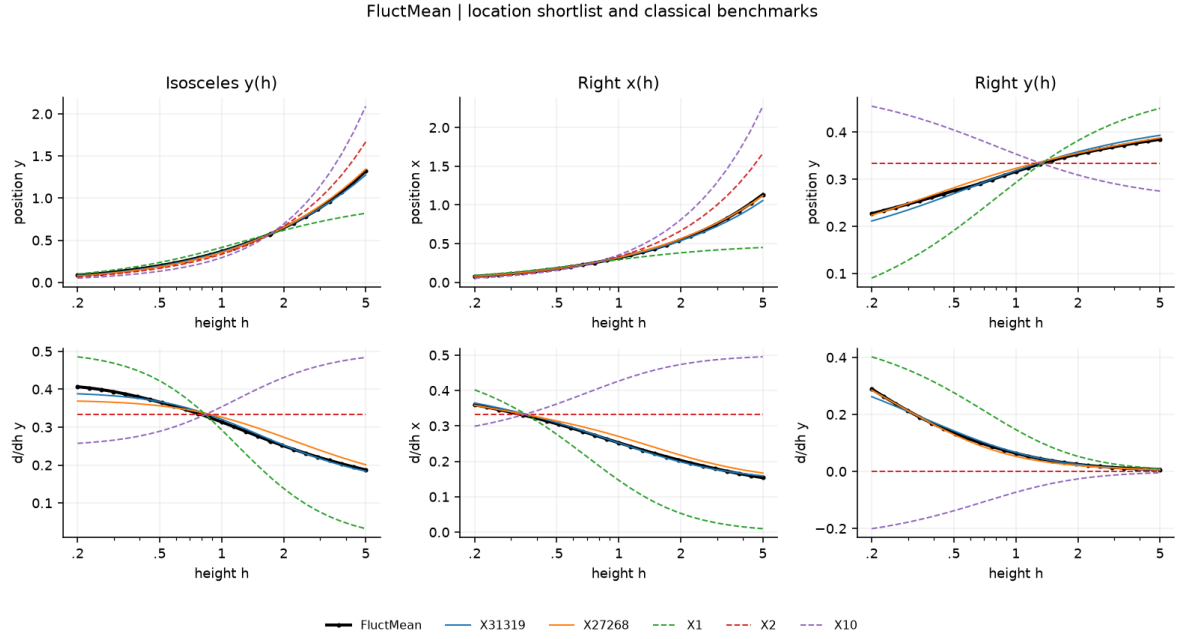
For the leading candidate X31319, the stored derivative-fingerprint RMS discrepancy is 0.00772528, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0339903. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.0164652 D. Its baseline is 0.4 I + 0.6 G; q=z. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 13a. FluctMean: reference 3-4-5 triangle and two further shapes. Red star marks the

selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 13b. FluctMean: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.14 Energy0.3: Bending-energy centroid

Use local bending energy under a uniform load as a spatial density; the plotted representative has Poisson ratio 0.3.

$$C = \int_T p e_\nu / \int_T e_\nu, \quad e_\nu = (1 - \nu)|D^2 w|^2 + \nu(\Delta w)^2$$

Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field.

On the 3-4-5 triangle the refined position is (1.155748, 1.000923), with normalized barycentric coordinates (0.377422, 0.288937, 0.333641).

Location and definition. The defining nonnegative density has positive mass in the interior. Its normalized barycentric first moments are strictly positive, giving a strict-interiority certificate.

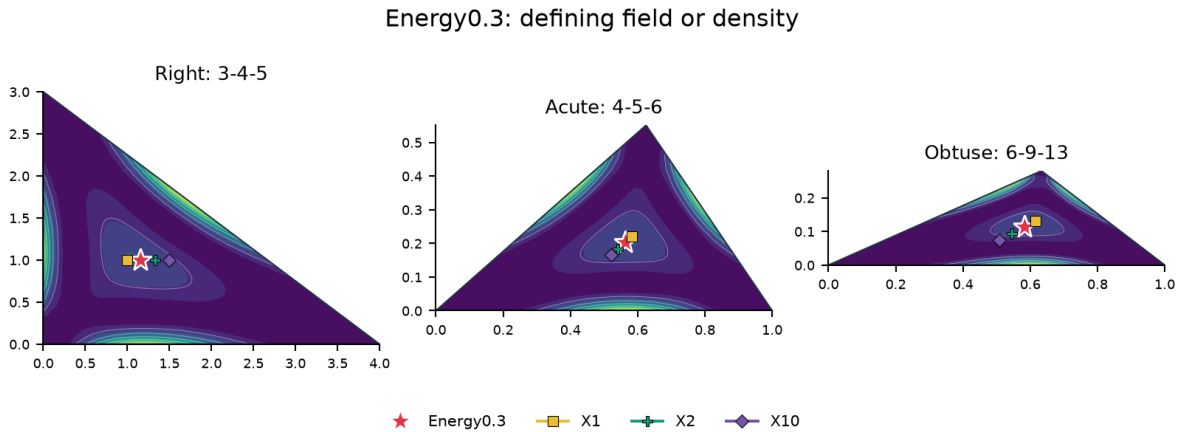
Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Inherited resolution record: 100/100 training triangles and 38/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.000222445 D. The coarse minimum response was 0.148725, with 0 negative sampled triangles. A negative count is not a count of established counterexamples.

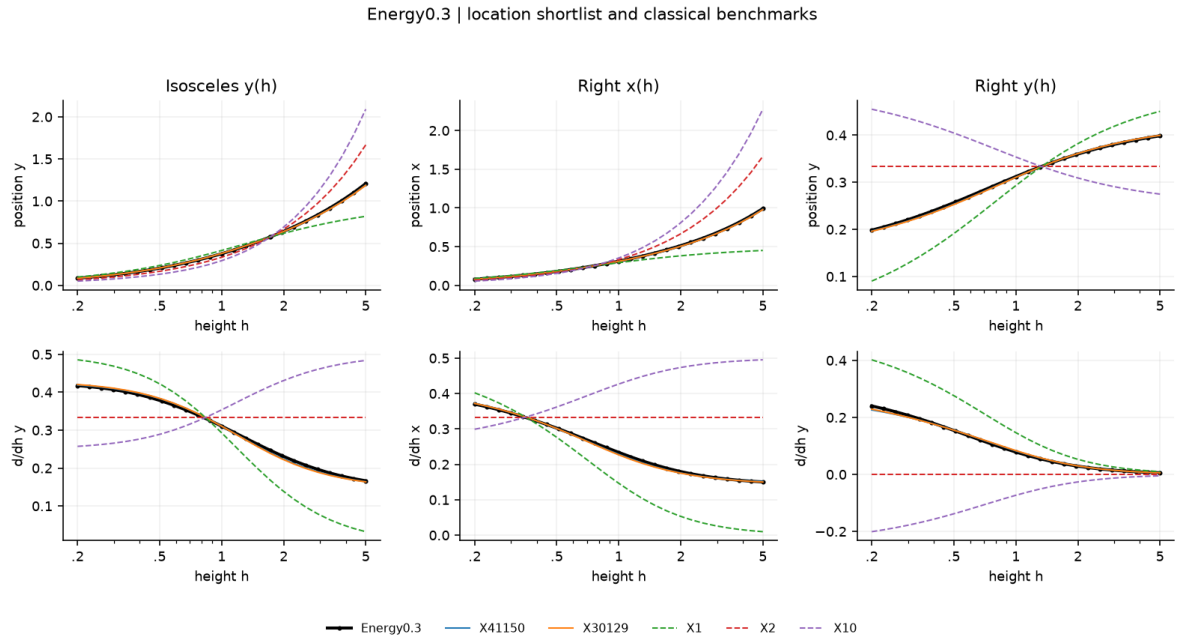
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X41150	0.00173009	0.00171295	0.00306125
X30129	0.00207949	0.00204529	0.00369216
X29823	0.00221246	0.00203805	0.00363572

For the leading candidate X41150, the stored derivative-fingerprint RMS discrepancy is 0.00517696, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0256887. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00028163 D. Its baseline is (4/7) I + (3/7) G; q=z. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 14a. Energy0.3: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 14b. Energy0.3: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.15 Pheat0.5: Compressed, gravity-biased plate at fraction 0.5

Uniform heating produces compression; a weak downward uniform load selects a pre-buckling deflection. Fix the stated fraction of the shape-dependent critical compression.

$$(\Delta^2 + 0.5\lambda_b\Delta)w = 1, \quad w = \partial_n w = 0, \quad C = \arg \max w$$

Compute the first generalized buckling eigenvalue, solve the shifted bending system, and refine the maximum. Include the eigenvalue change in vertex derivatives.

On the 3-4-5 triangle the refined position is (1.073768, 0.999005), with normalized barycentric coordinates (0.398556, 0.268442, 0.333002).

Location and definition. Every selected global maximum is interior in the stated model. The numerical point selector remains conditional on a unique maximum; the atlas supplies no universal plate-maximizer theorem. For Bmode a simple critical eigenspace is also needed unless a full-projector convention is adopted.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Inherited resolution record: 90/100 training triangles and 33/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00300986 D. The coarse minimum response was -0.021344, with 2 negative sampled triangles. A negative count is not a count of established counterexamples.

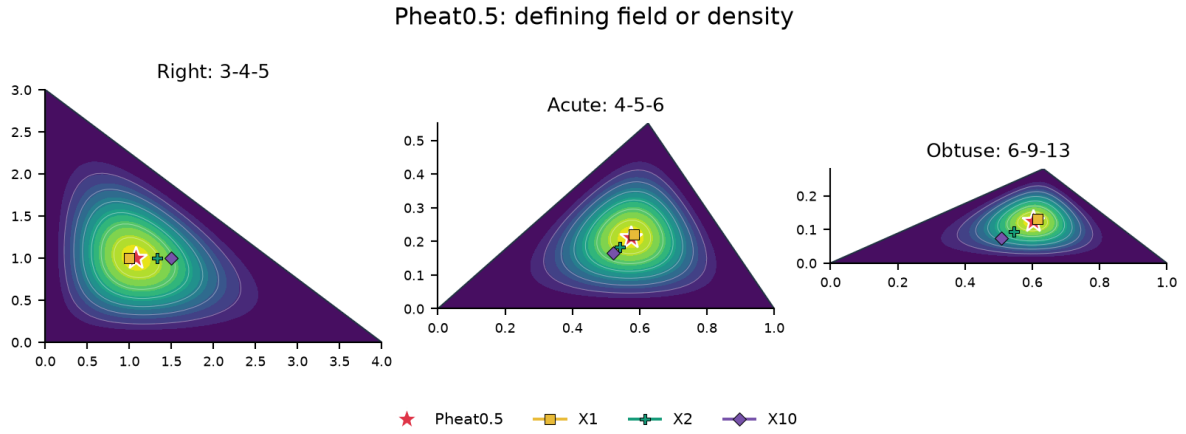
Ritz degree	Original-step ratio	Half-step ratio
20	0.05828073	0.05833194
26	0.06855832	0.06780801
32	0.07194423	0.07117039

These values use the exact retained witness vertices and direction. All inputs and mode gaps are stored in plate_witness_refinement.json. They test that particular motion; they are not lower bounds over triangle space.

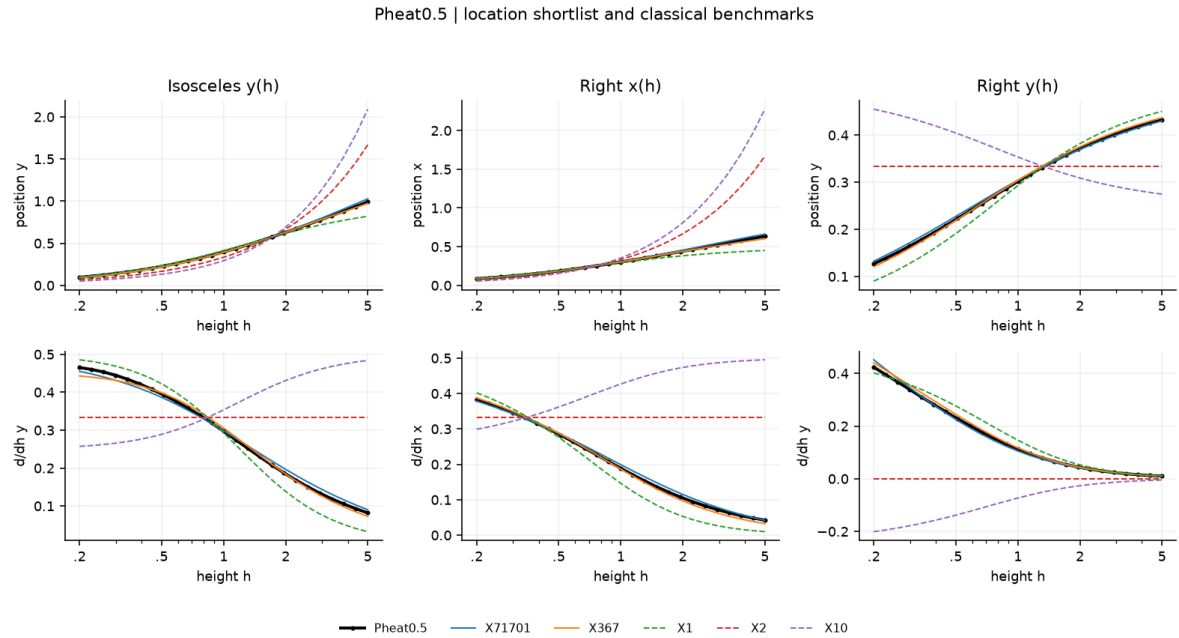
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X71701	0.00391058	0.00412491	0.0103653
X367	0.00415574	0.00431233	0.0088127
X53588	0.00418174	0.00464599	0.0108283

For the leading candidate X71701, the stored derivative-fingerprint RMS discrepancy is 0.0106385, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0805007. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00414258 D. Its baseline is incenter; $q=\hat{z}(2/3)$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.



Atlas figure 15a. Pheat0.5: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.



Atlas figure 15b. Pheat0.5: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.16 Pheat0.9: Compressed, gravity-biased plate at fraction 0.9

Uniform heating produces compression; a weak downward uniform load selects a pre-buckling deflection. Fix the stated fraction of the shape-dependent critical compression.

$$(\Delta^2 + 0.9\lambda_b\Delta)w = 1, \quad w = \partial_n w = 0, \quad C = \arg \max w$$

Compute the first generalized buckling eigenvalue, solve the shifted bending system, and refine the maximum. Include the eigenvalue change in vertex derivatives.

On the 3-4-5 triangle the refined position is (1.068063, 0.999287), with normalized barycentric coordinates (0.399889, 0.267016, 0.333096).

Location and definition. Every selected global maximum is interior in the stated model. The numerical point selector remains conditional on a unique maximum; the atlas supplies no universal plate-maximizer theorem. For Bmode a simple critical eigenspace is also needed unless a full-projector convention is adopted.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Inherited resolution record: 90/100 training triangles and 33/40 holdouts meet the entry's declared successive-change tolerance. The largest training change was 0.00322378 D. The coarse minimum response was -0.130021, with 4 negative sampled triangles. A negative count is not a count of established counterexamples.

Ritz degree	Original-step ratio	Half-step ratio
20	0.07650946	0.07641517
26	0.09945462	0.09815169
32	0.10162926	0.10023592

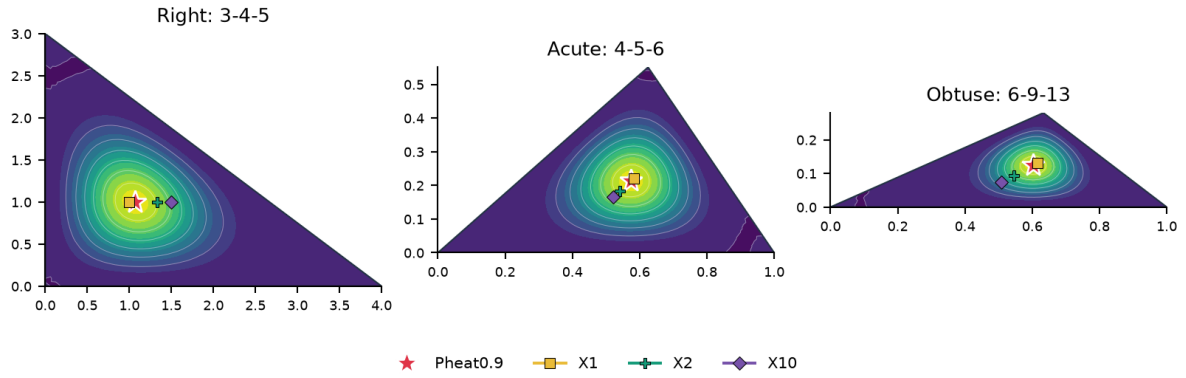
These values use the exact retained witness vertices and direction. All inputs and mode gaps are stored in plate_witness_refinement.json. They test that particular motion; they are not lower bounds over triangle space.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X53588	0.00378373	0.00430395	0.0126643
X367	0.00389252	0.00399021	0.0086326
X30142	0.00439795	0.00454679	0.0100278

For the leading candidate X53588, the stored derivative-fingerprint RMS discrepancy is 0.0108633, based on 51/51 evaluable family samples. The response-margin RMS discrepancy is 0.0677979. These diagnostics use the inherited numerical derivatives and their documented resolution limitations. The location ranking supplies a useful neighbor, not an exact identity.

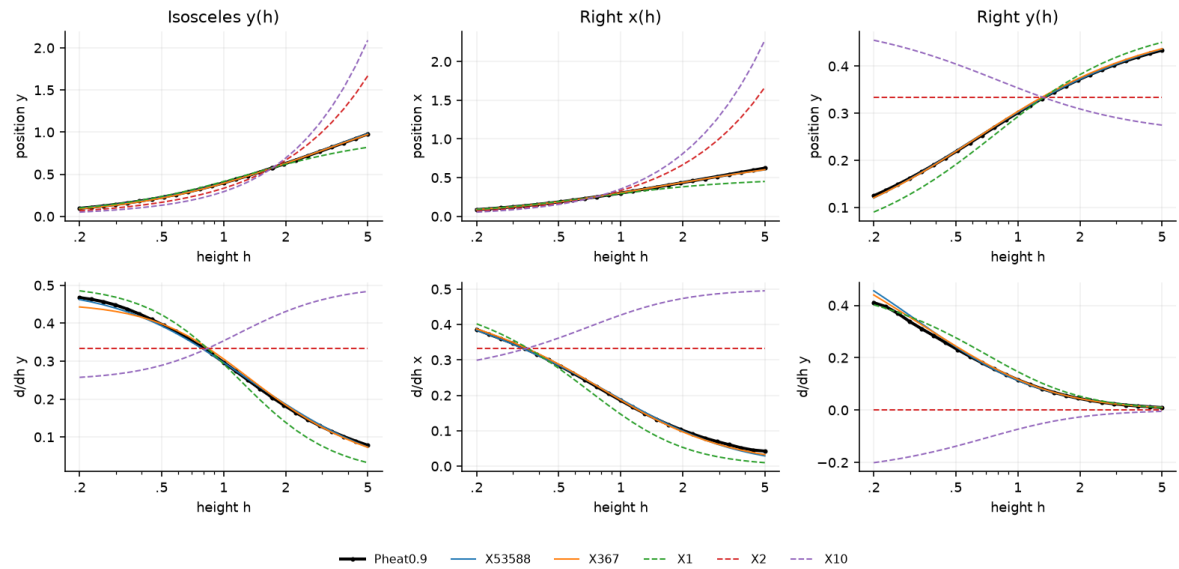
The six-coefficient positive-barycentric surrogate has inherited holdout RMS error 0.00416349 D. Its baseline is incenter; $q=\hat{z}(2/3)$. Coefficients and the precise feature formula are in symbolic_models.json and mech_opt_fit.py; fitting positions does not establish an attractivity theorem.

Pheat0.9: defining field or density



Atlas figure 16a. Pheat0.9: reference 3-4-5 triangle and two further shapes. Red star marks the selected point; comparison markers are X1, X2 and X10. Each panel is normalized separately. Field plots are illustrative discretizations, not enclosures.

Pheat0.9 | location shortlist and classical benchmarks



Atlas figure 16b. Pheat0.9: inherited isosceles and right-triangle positions and their height derivatives. Curves include two ETC neighbors and X1, X2, X10. Gaps indicate undefined or omitted values. For the nonlinear maxima, derivative curves are exploratory where degree or quadrature sensitivity remains.

4.12.17 Thin and almost-equilateral fingerprints

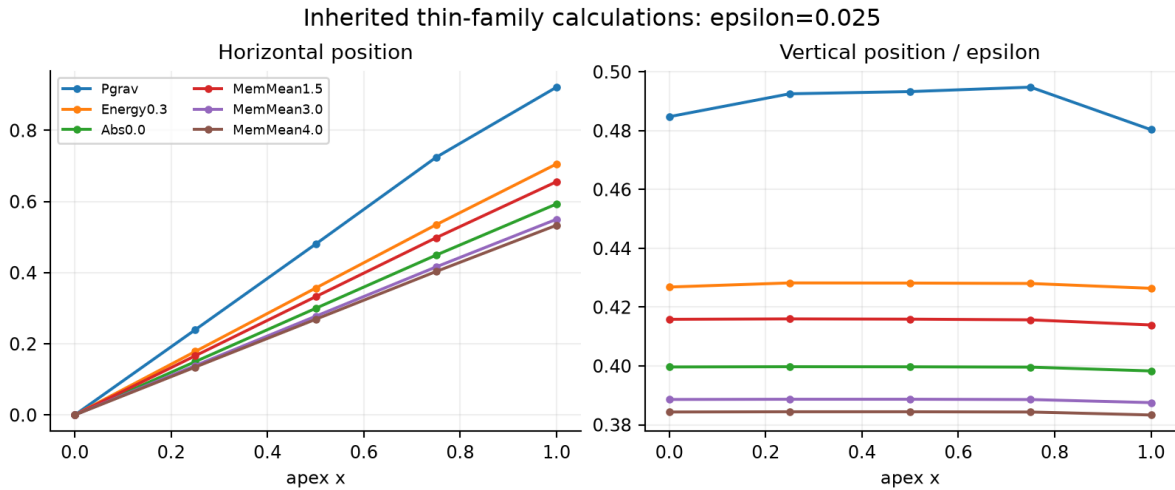


Figure. Inherited finite-height thin-triangle comparisons for maxima and means. The plotted points retain their original resolution flags in thin.json; the curves are not certified limiting functions.

For $C=(x, \sqrt{3}+y)$ over the normalized symmetric base, write $\text{center_x}=\alpha x+\beta xy+\dots$ and $\text{center_y}=1/\sqrt{3}+\gamma y+\delta x^2+\eta y^2+\dots$. Reflection fixes the parity, and similarity equivariance gives $\alpha+\gamma=2/3$. The following coefficients are numerical fits. The problematic Mem(1.5) and Mem(3) fits are omitted rather than presented as Taylor constants.

Entry	alpha	gamma	Trace defect
Abs0.0	0.389978	0.276689	9.4e-07
Mem4.0	0.470072	0.196593	-1.1e-06
MemMean1.5	0.393061	0.273609	2.7e-06
MemMean3.0	0.386122	0.280539	-6.2e-06
MemMean4.0	0.383709	0.282958	8.4e-07
Pgrav	0.459413	0.207249	-4.9e-06
Bmode	0.465376	0.201295	4.9e-06
Bmean	0.458639	0.208030	2.1e-06
Compliance	0.465574	0.201088	-4.8e-06
FluctMean	0.402090	0.264578	1.1e-06
Energy0.3	0.420654	0.246014	1.5e-06
Pheat0.5	0.460337	0.206331	1.5e-06
Pheat0.9	0.463428	0.203243	4.5e-06

The remaining quadratic coefficients, step comparisons and symmetry checks are in equilateral_jets.json. In particular a plausible trace does not certify each coefficient independently. A symbolic approximation should respect these symmetry constraints, positivity for means, and the applicable thin asymptotics before additional fitted terms are introduced.

4.13 Demonstrations and directions for further study

A three-face construction can use three flexible triangular panels joined at an apex and constrained to remain planar. Measure the lateral area and search for maximum apex depth. The

predicted projection is I ; for a genuinely floppy continuous sheet the planar-face restriction is lost and the experiment no longer tests that theorem.

A uniformly tensioned film stretched over a rigid triangular frame can be loaded by a small pressure difference. Surface-height contours reveal $F1$, while integrating the height map estimates $\text{MemMean}(2)$. To compare with a plate, use a thin solid sheet clamped between two triangular frames and a small uniform pressure or its own weight. Calibrate boundary rotation: an edge that slips or hinges can change the selected point more than numerical approximation does.

For the thermal-compression experiment, record the first instability as restrained temperature rises. The squared spatial mode estimates B_{mode} and B_{mean} , but the observed post-buckling shape is affected by imperfections and nonlinear mechanics. The mathematical critical-mode calculation predicts an onset pattern, not a unique final thermal shape.

For a compliance map, apply the same small force with the same contact footprint at each location and measure displacement at the load point. For a fluctuation map, estimate the variance over time after removing rigid motion and drift. Their proportionality tests the linear equilibrium model and its boundary conditions. Finite bandwidth and probe size are experimental parameters, not details to suppress.

The next rigorous tasks are validated shape responses for $F1$, reliable nonlinear membrane derivatives near their maxima, and point-uniqueness results for the clamped plate selectors. Slow convergence of a Green diagonal calls for local singular enrichment or an independent finite-element formulation, not just a larger global polynomial degree. Bounded support patches, anisotropic materials and a combined bending-plus-tension operator would define useful parameter families, but their additional parameters must be scaled with triangle size before they define similarity-equivariant triangle centers.

Chapter 5

Quantum confinement, billiards and Brownian motion

5.1 Three particles in the same triangle

A classical particle flies along straight lines and reflects from a wall. A Brownian particle wanders randomly and may be killed when it reaches the boundary. A quantum particle has a probability amplitude constrained by the wall. These are three different experiments. Their centers agree in some limits because they share geometric symmetries or the same differential operator. Their agreement should be derived, rather than inferred from the word particle.

We use T for the open triangular region, K for its area, D for its diameter, and L squared for the mean of the three squared side lengths. Position is denoted by p . In the reference triangle, $K=6$, $D=5$, $L^2=50/3$, the centroid is $G=(4/3,1)$, the incenter is $I=(1,1)$, and the perimeter centroid is $X_{10}=(3/2,1)$. Using the same vertices throughout lets the reader compare the figures directly with the first three chapters.

A center must commute with rigid motion, scaling and relabelling of vertices. A parameter such as time or temperature must therefore be specified in dimensionless form. A probability mean is a single point whenever its density is defined. A maximum requires an additional uniqueness argument. An energy level needs an eigenspace convention when its multiplicity exceeds one. These distinctions will determine which proposed constructions belong in an atlas of single-valued triangle centers.

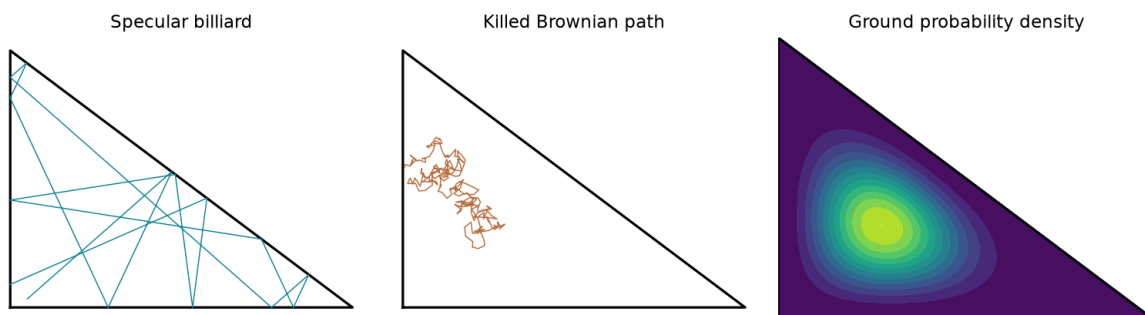


Figure 1. Specular flights, killed random motion, and a ground-state probability density in the

same 3-4-5 triangle. The Brownian trace is an illustrative discrete path; the calculations below use the differential equation, not this drawing's time step.

5.2 From the infinite quantum well to geometric eigenfunctions

5.2.1 The physical assumptions

Consider a nonrelativistic particle of mass m with no force inside a triangle. An ideal infinitely high surrounding potential confines it to the triangle. The stationary Schrodinger equation inside is the kinetic-energy equation, and the infinite wall imposes zero wavefunction on the boundary:

$$-\frac{\hbar^2}{2m}\Delta\psi = E\psi, \quad \psi|_{\partial T} = 0.$$

Divide by the constant in front of the Laplacian and write A for the positive Dirichlet Laplacian. We count from $j=0$, including multiplicity. Normalize every eigenfunction to have total probability one:

$$A\psi_j = \lambda_j\psi_j, \quad A = -\Delta_D, \quad \int_T |\psi_j|^2 dp = 1.$$

The energies are $E_j = \hbar^2 \lambda_j$ divided by $2m$. Thus changing the particle mass changes the energies, but does not change these spatial distributions. Magnifying the triangle by s multiplies eigenvalues by s to the power minus two and scales every mean and maximum by s . No material constant survives in the geometric center.

5.2.2 Mean and most likely point

Born's rule assigns probability $|\psi_j|^2$ times dp to a small area. The mean of many repeated position measurements, each on a newly prepared particle in the same state, is

$$Q_j = \int_T p |\psi_j(p)|^2 dp, \quad j = 0, 1, 2, 3, 4.$$

The most likely location in the ground state is a different observable:

$$Q_{0m} = \operatorname{argmax}_{p \in T} \psi_0(p) = \operatorname{argmax}_{p \in T} |\psi_0(p)|^2.$$

Choose the ground eigenfunction positive. It is simple: multiplication by a constant phase changes no probability. Its mean samples the entire density; its maximum uses only the top of the largest peak. In a scalene triangle there is no reason for those two operations to agree. A measurement of one particle is neither of these points in general.

Log-concavity supplies the missing uniqueness argument for Q_{0m} . The positive ground eigenfunction of a convex Dirichlet domain is log-concave; this follows also by taking the long-time limit of the log-concavity-preserving heat flow [22]. If two distinct points were maxima, log-concavity would make the whole segment between them a maximum. The eigenfunction is analytic in the interior, so its restriction to the line would be constant along the entire interior chord. Continuity and its zero boundary values would then contradict its positive maximum. Thus a triangle has a unique ground-state maximum. This argument proves a location selector exists; it says nothing about its response to a moving vertex.

5.2.3 Repeated energies and a necessary repair

For a simple excited eigenvalue, sign or phase changes do not change Q_j . For a repeated eigenvalue, two independent eigenfunctions can have different position expectations. A numerical eigensolver chooses some orthonormal basis in this space; its choice is not a geometric rule.

At the equilateral triangle the first excited level is the Q1/Q2 doublet, and the level containing Q4 also contains Q5. The archived almost-equilateral calculations approach different individual-state means along different deformation directions. Assigning G at the exactly equilateral triangle would not remove the discontinuity. The first excited eigenvalue is simple for non-equilateral triangles in the cited simplicity work [23]; higher levels still require gap monitoring.

There is a canonical replacement for a complete eigenspace E:

$$\rho_E(p) = \frac{1}{d_E} \sum_{j \in E} |\psi_j(p)|^2, \quad Q_E = \int_T p \rho_E(p) dp.$$

Here d_E is the dimension of E. A unitary change of orthonormal basis leaves the sum of squared amplitudes unchanged. This is the position mean of an equal incoherent mixture over the whole level. One must keep the entire cluster: cutting a doublet in half reinstates the ambiguity. If a cluster remains spectrally separated as the shape changes, its projector gives a smooth family even when its internal levels cross. The individual atlas entries below retain the original requested Q1, Q2 and Q4, with their domain-of-definition qualifications; the projector construction is recorded as an extension, not silently substituted.

5.3 Numerical solution, explained from the weak equation

Split the triangle into small triangles. On each use a quadratic polynomial, matched continuously to its neighbors and set to zero on the boundary. Write a trial wavefunction as a sum of these basis functions ϕ_k with unknown coefficients c_k . Multiply the differential equation by ϕ_l and integrate. Integration by parts removes second derivatives; its boundary term vanishes. The resulting matrix equation is

$$Kc = \lambda Mc, \quad K_{lk} = \int_T \nabla \phi_l \cdot \nabla \phi_k, \quad M_{lk} = \int_T \phi_l \phi_k.$$

K measures gradient energy; M measures probability normalization. Both are assembled by polynomial quadrature. Normalize c by $c^T M c = 1$. Coordinate-weighted mass matrices M_x and M_y , formed by inserting x or y in the mass integral, give the expectation without approximating the density by nodal point masses:

$$Q_j = (c_j^T M_x c_j, c_j^T M_y c_j).$$

For Q0m, inspect the quadratic polynomial on every element. Its stationary point solves a two-by-two linear system. Check that it lies inside the element, and also inspect edge stationary points and nodes. Select the largest value and repeat on a finer mesh. An eigenvalue may converge well while the maximum location is still moving; the two error questions must not be merged.

The new common example uses 24, 40 and 56 subdivisions along each original side, with 48, 80 and 120 eigenpairs. The first six selectors need only the first few modes, while heat and thermal averages need more. The archive records changes, adjacent relative gaps and half-spectrum tests.

These are convergence diagnostics, not outward-rounded error bounds. The inherited all-shape survey used additional adaptive refinement for difficult narrow triangles.

5.3.1 The equivalent conformal problem has a weight

A conformal map F from a disk or half-plane onto T gives another useful formulation. If $\phi(z) = \psi(F(z))$, the Laplacian transformation is $\Delta_z \phi = |F'(z)|^2 (\Delta_p \psi)(F(z))$. Thus the eigenproblem on the simpler domain becomes a weighted eigenproblem:

$$-\Delta_z \phi = \lambda |\phi|^2 |F'|^2 \phi, \quad \int |\phi|^2 |F'|^2 dz = 1.$$

The physical mean is integral $F(z) |\phi(z)|^2 |F'(z)|^2 dz$. Conformal mapping simplifies the boundary, but it does not turn an arbitrary triangular well into the ordinary unweighted disk well. The varying area weight carries the shape information and is singular near mapped corners. Harmonic exit probabilities, used later, are conformally invariant without this eigenvalue weight because their Laplacian is zero.

Selector	x	y	Last change / D
Q0m	1.0954507	0.9975085	0.000116
Q0	1.1265611	1.0015308	2.06e-07
Q1	1.4133624	0.9877257	8.28e-07
Q2	0.9581951	1.0509560	2.37e-07
Q3	1.5264102	0.9056026	3.02e-06
Q4	1.1919157	1.1275766	1.25e-06
E0	0.9306884	0.9606674	not enclosed

All values use $A=(0,0)$, $B=(4,0)$, $C=(0,3)$. A last-change value compares two discretizations; it is neither an error bound nor the number of reliable printed digits. Exact and numerical location statements are identified separately in each entry.

5.4 Where can a quantum mean lie?

5.4.1 A convex-hull certificate

Barycentric coordinates $w_i(p)$ are affine functions of p , nonnegative in the closed triangle. Therefore a normalized positive density ρ satisfies

$$w_i \left(\int_T p \rho(p) dp \right) = \int_T w_i(p) \rho(p) dp.$$

Every quantum density has positive mass away from each side, so every barycentric mean is strictly positive. This proves strict interiority of Q_j and of projector and thermal means. It is stronger than simply observing the red marker inside a numerical plot, but it supplies no uniform distance from the boundary for arbitrary energy.

5.4.2 An explicit smaller triangle

Let $d_i(p)$ be the perpendicular distance to side i , and r the inradius. Apply the one-dimensional Hardy inequality along lines perpendicular to the supporting side, extending the Dirichlet eigenfunction by zero outside T . Integrating the resulting inequalities along the side gives

$$\int_T \frac{|\psi_j|^2}{d_i^2} dp \leq 4 \int_T |\nabla \psi_j|^2 dp = 4\lambda_j.$$

Convexity of t to the power minus two and the affinity of d_i give a lower bound on the side distance of the mean. The incircle is contained in T ; the min-max principle bounds each triangle eigenvalue above by the corresponding disk eigenvalue. If ζ_j is the j th positive unit-disk wave number counted with multiplicity, these two steps yield

$$d_i(Q_j) \geq \frac{1}{2\sqrt{\lambda_j}} \geq \frac{r}{2\zeta_j}, \quad i = 1, 2, 3.$$

For $j=0$ through 4, the disk numbers are approximately 2.40483, 3.83171, 3.83171, 5.13562, 5.13562. The three inequalities describe an inner parallel triangle. This is an analytic location certificate for expectations, including any state in a repeated level. It is not a location bound for Q_0 . The numerical eigenvalue approximation should not be inserted as though it were a validated exact eigenvalue; the explicit disk bound needs no numerical triangle solve.

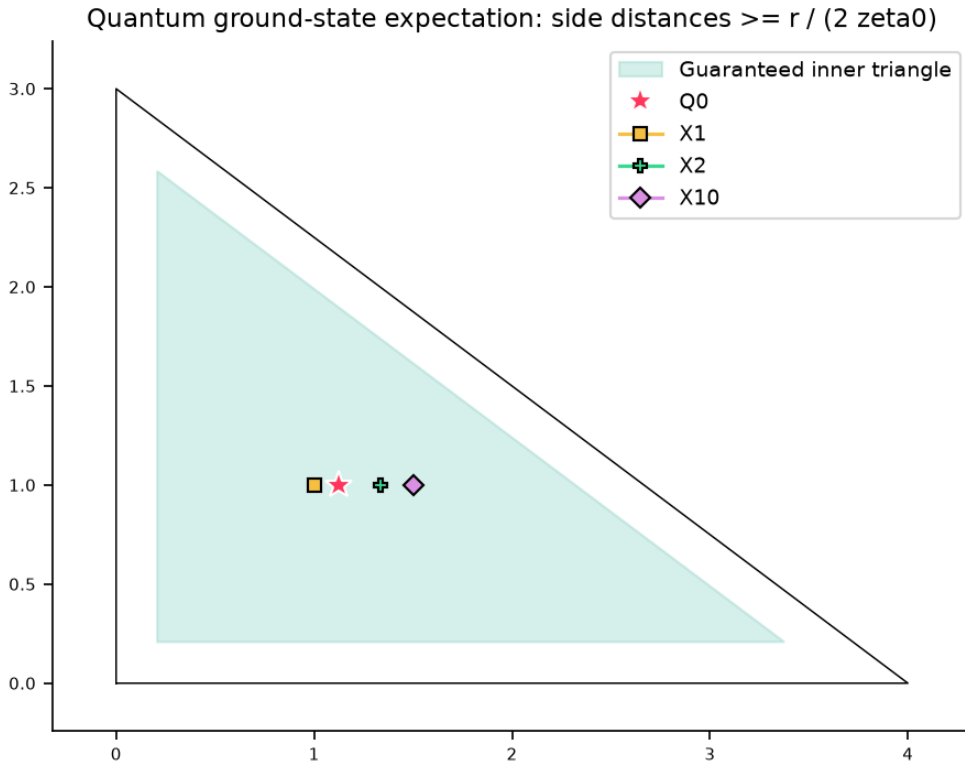


Figure 2. The ground-state expectation and the disk-based inner parallel triangle for the 3-4-5 example. The shaded region follows from an inequality, independently of the numerical position.

5.5 Fingerprints near symmetry and degeneration

5.5.1 The four shape families

Besides generic shapes, the atlas studies isosceles triangles with vertices $(-1,0)$, $(1,0)$, $(0,h)$, and right triangles with vertices $(0,1)$, $(0,0)$, $(h,0)$. Position curves and their h derivatives give reproducible low-dimensional fingerprints. A degenerating family has third vertex (x,ϵ) ,

with $0 \leq x \leq 1$, while an almost-equilateral family has third vertex $(x, \sqrt{3}+y)$. Their limiting behaviors detect distinctions that a few generic positions can conceal.

For a simple center differentiable at the equilateral triangle, reflection implies a diagonal apex Jacobian. Translation, rotation and scale equivariance imply that its diagonal entries add to $2/3$. This trace identity is a useful numerical consistency check; neither a correct trace nor positive derivatives along one family proves attractivity under all vertex motions.

The independent perturbation calculation differentiates the eigenproblem. In a fixed reference triangle its simple-state derivative solves a linear equation on the mass-orthogonal complement of the eigenfunction. The normalization equation removes the arbitrary component along the state. In a repeated eigenspace, first diagonalize the perturbation restricted to that eigenspace. The latter matrix depends on the direction (x,y) , explaining why no single Taylor expansion describes the individual doublet branches.

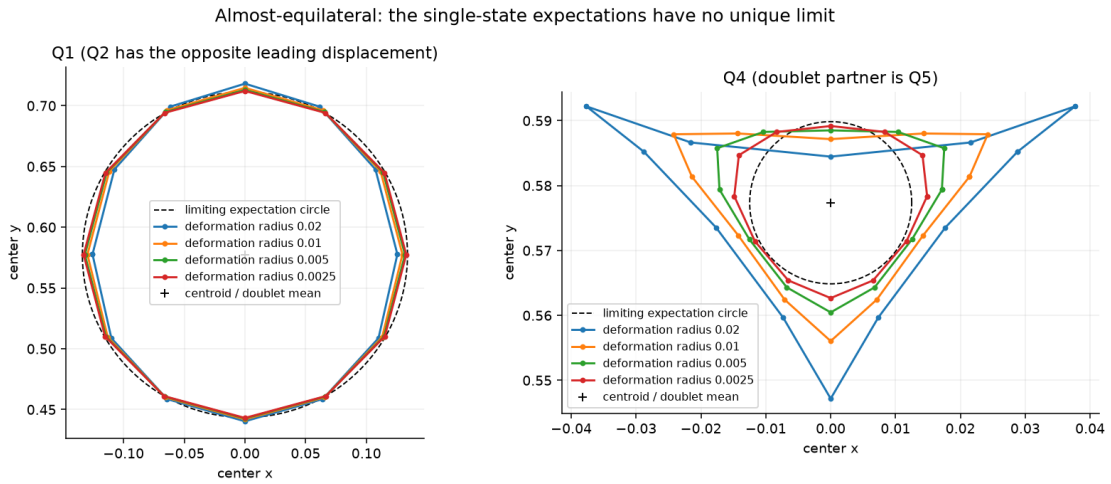


Figure 3. Archived deformation-direction limits of the excited-state branches. A smooth surrogate forced to give a single value at equilateral cannot reproduce these individual-state limits.

5.5.2 Why the thin quantum center follows the widest section

At a longitudinal coordinate s in a thin triangle, a transverse Dirichlet strip of height $H(s)$ has first energy approximately π^2 divided by $H(s)^2$. The height is greatest beneath the apex. A low state lowers its energy by concentrating near that widest section rather than spreading uniformly over the limiting segment. This explains why the first fixed number of quantum means tend horizontally toward x , not generally toward the midpoint of $[-1,1]$.

The change of transverse energy near the widest section is linear in longitudinal displacement and has size ϵ to the power minus two. If localization has width ℓ , longitudinal kinetic energy has size ℓ to the power minus two. Balancing ℓ to the power minus two with ℓ divided by ϵ^2 gives ℓ proportional to $\epsilon^{2/3}$. The effective one-dimensional equation is of Airy type. This noninteger power is a useful structural ingredient in a fit. The endpoint $x=1$ has a one-sided profile and must be treated separately; a fixed-interior expansion is not automatically uniform up to it.

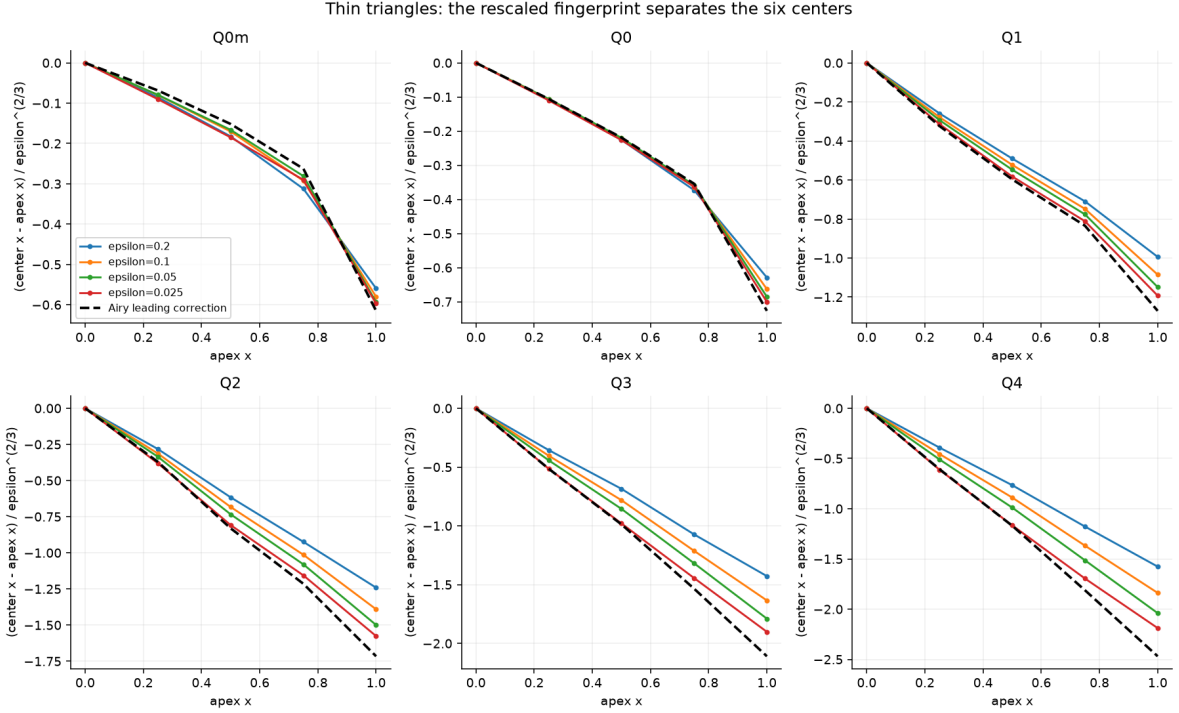


Figure 4. Inherited finite-height quantum calculations, with their archived resolution flags. A plot approaching a limit is numerical evidence for an asymptotic description, not a certified remainder estimate.

5.6 Is the classical centroid the limit of high quantum states?

There are three different limiting questions. Averaging all energy levels below a growing cutoff gives a basis-independent density if complete eigenspaces are included. The local Weyl law leads to uniform area density and hence

$$\frac{1}{N(\Lambda)} \sum_{\lambda_j \leq \Lambda} Q_j \longrightarrow G.$$

A thermal mixture is another canonical average. With dimensionless inverse temperature τ , its density is the diagonal heat kernel divided by its trace:

$$Q_{\text{th}}(\tau) = \frac{\sum_j e^{-\tau L^2 \lambda_j} Q_j}{\sum_j e^{-\tau L^2 \lambda_j}}.$$

At low temperature, τ tending to infinity, this tends to Q_0 . At high temperature, τ tending to zero, it tends to G . Low τ requires many eigenpairs; a fixed finite truncation cannot demonstrate the high-temperature limit. The reference calculations compare full and half spectra rather than hiding that limitation.

The assertion that every individual Q_j tends to G is stronger. For rational polygons, Marklof and Rudnick prove spatial equidistribution along a density-one subsequence [24]. Thus almost all indices, in the asymptotic counting sense, have means tending to G . This does not exclude exceptional subsequences. It does not establish an all-state result for an arbitrary triangle. The side lengths 3,4,5 do not make our reference triangle a rational polygon: rationality here means that every angle is a rational multiple of π .

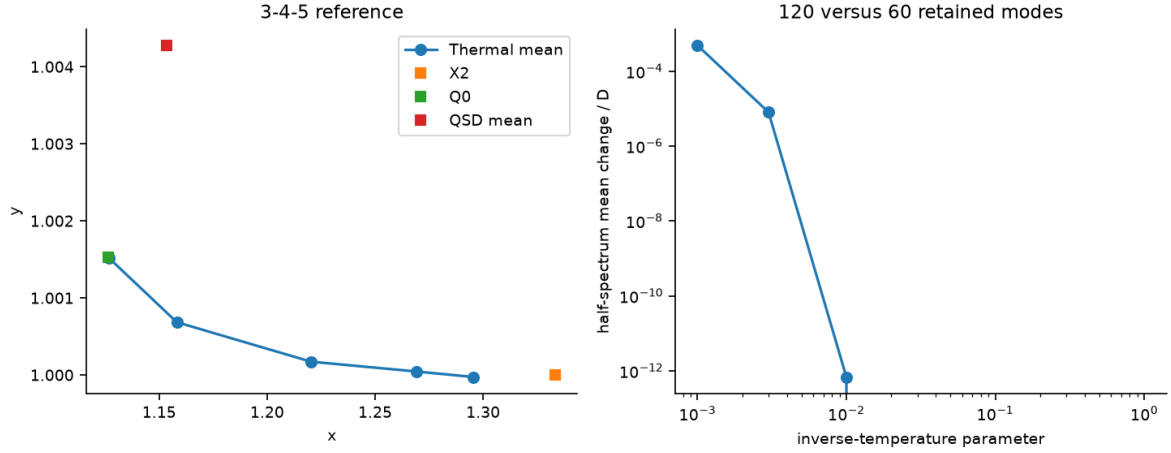


Figure 5. The thermal quantum mean in the reference triangle, compared with G, Q0 and the quasi-stationary Brownian mean. The smallest inverse temperatures carry explicit finite-spectrum diagnostics in the data.

5.7 Billiards: what exactly is being averaged?

5.7.1 Reflection and exact flight integration

Let the particle have speed one. Between collisions it follows $p(s)=p_0+sv$. At a straight wall with unit normal n , reverse the normal velocity component and leave the tangential one unchanged:

$$v_{\text{after}} = v_{\text{before}} - 2(v_{\text{before}} \cdot n)n.$$

This preserves speed and gives the usual equal-angle law. A vertex collision has no unique wall normal and is excluded; it has probability zero in the ideal continuous launch ensemble. Floating-point near-vertex events are counted in the diagnostics.

For a single launch z , the observation is its time-averaged position, not its current position and not its average collision point:

$$C_z(t) = \frac{1}{t} \int_0^t X_z(s) ds.$$

Each flight lasting h contributes $p_0 h$ plus $v h$ squared divided by two to the integral. Partial flights are integrated in the same way when an observation time falls before the next collision. Consequently no time-stepping approximation is needed between collisions. Equal weighting of collision points would overrepresent short flights and can give another answer.

The requested random boundary launch is uniform in arclength, with an inward angle uniform between minus and plus $\pi/2$ measured from the inward normal. We also compare the cosine-flux law. The latter is the invariant collision-section law associated with uniform position and direction in the interior; a boundary-started sample at time zero is still not itself a stationary interior sample.

5.7.2 Distance, spread and what tends to zero

For independent launches define distance $D_z(t)=|C_z(t)-G|$. Its ensemble mean μ_D , standard deviation σ_D and root mean square R answer different questions:

$$R(t)^2 = \mathbb{E}D_z(t)^2 = \sigma_D(t)^2 + \mu_D(t)^2.$$

The standard deviation can be small even if every orbit mean is the same positive distance from G. Therefore the figures show all three quantities, together with upper quantiles and the norm of the ensemble mean error. Instantaneous positions continue to fill the triangle; their RMS distance from G does not tend to zero. For the uniform area distribution, that RMS is determined by the mass covariance from Chapter 2.

The new reference experiment uses 1,024 launches per law, speed one and diameter normalized to one, through time 30,000. The following table uses uniform inward angles:

Time / D	Mean distance / D	SD / D	RMS / D
1	0.137200	0.068470	0.153337
10	0.043429	0.030733	0.053203
100	0.027545	0.022803	0.035759
1000	0.021271	0.017599	0.027608
3000	0.019313	0.015263	0.024616
10000	0.017419	0.013865	0.022263
30000	0.015829	0.012895	0.020416

For cosine-flux launches, the final mean distance, SD and RMS are 0.015850, 0.012770, 0.020354 diameter. The time-average cloud still has appreciable width at the last recorded time. This experiment demonstrates slow finite-time behavior; it does not establish a zero asymptotic limit for the 3-4-5 triangle. A small ensemble-mean bias would not substitute for these trajectory-wise statistics. Bootstrap intervals, seeds, collision checks and trajectories are saved.

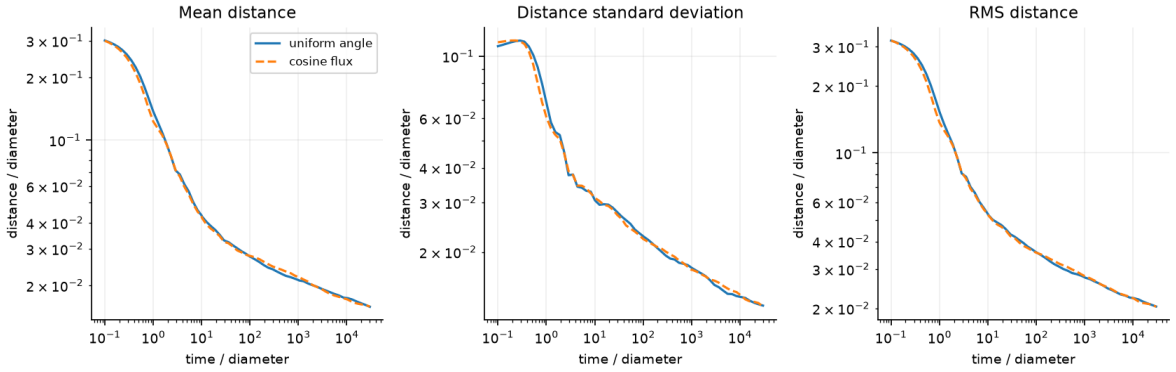


Figure 6. Mean distance, its standard deviation, and RMS of the 3-4-5 trajectory means. Solid and dashed curves correspond to the two launch laws. Neither curve is fitted to a universal decay rate.

For rational triangles the directional unique-ergodicity theorem applies to almost every direction [25]. Projecting the directional invariant area measure back into T gives the centroid as the time mean. The bounded distance $D_z(t) \leq D$ then permits dominated convergence, giving μ_D , σ_D and R tending to zero for absolutely continuous nonsingular launch laws. Periodic directions are exceptional. The archived three-bounce orbit in an acute scalene triangle has a non-centroid time mean; this is a useful demonstration of why the qualification almost every matters.

No universal t to the power minus one-half law follows. For a stationary ensemble, direct expansion of the squared integral expresses the RMS through the position autocorrelation $R_X(s)$:

$$\mathbb{E}|C_z(t) - G|^2 = \frac{2}{t^2} \int_0^t (t-s) R_X(s) ds.$$

An integrable correlation with suitable control would suggest a square-root rate, but polygonal billiards need not have that mixing behavior. Neither the rational theorem nor our finite simulation proves convergence for every irrational triangle. The book's billiard center is G when uniform-area averaging or an applicable ergodic limit is specified; an individual exceptional orbit defines an orbit-dependent quantity, not a universal triangle center.

5.8 Brownian motion separates several familiar centers

5.8.1 A short-time balance gives the generator

Use Brownian increments with covariance $2D_B dt$ times the identity. Taylor expansion and averaging cancel the linear term and retain D_B times the Laplacian in the quadratic term. This is the generator $D_B \Delta$. Stopping at the first boundary hit gives an absorbing process; reflecting at the boundary gives a different process with a uniform stationary density.

Let τ be the first exit time. The expected remaining lifetime $m(p)$ satisfies $m(p) = dt + \mathbb{E}_p[m(X(dt))]$ before exit. Subtract $m(p)$, divide by dt and pass to the limit:

$$-D_B \Delta m = 1, \quad m|_{\partial T} = 0.$$

Thus $m=u/D_B$ where u is the torsion function. Its maximum is F_1 , the same point that is deepest in the uniformly tensioned membrane and fastest in the pressure-driven duct. Uniformly releasing particles and accumulating their pre-exit occupation density instead gives a density proportional to u , whose mean is $Abs(0)$. Expected lifetime maximum and accumulated occupation mean are distinct measurements.

5.8.2 Equal exit through the three sides: E_0

Let $h_i(p)$ be the probability of first exit through side i . The same short-time averaging argument has no accumulated time term, so $\Delta h_i=0$, with boundary value one on side i and zero on the other sides. Values at vertices do not affect the probability. We seek the point where all three probabilities are one third. This is Iannaccone's equal-exit center E_0 [26].

Map the upper half-plane conformally onto T , sending $0,1,\infty$ to A,B,C . Write α, β, γ for the three angles divided by π . The Schwarz-Christoffel derivative has powers $\alpha-1$ and $\beta-1$; integrating and fixing the base length gives

$$F(z) = A + (B - A) \frac{B_z(\alpha, \beta)}{B(\alpha, \beta)}.$$

Here complex notation identifies the plane with the complex numbers, and the upper-half-plane branches are chosen consistently. Harmonic functions compose with a conformal map in two dimensions. Brownian paths are preserved after a change of clock, so exit probabilities are unchanged. In the half-plane, harmonic measure of a real interval is its subtended angle divided by π . Equal one-third probabilities for the intervals ending at 0 and 1 put the starting point at $z=1/2+i\sqrt{3}/2$. Therefore $E_0=F(z)$.

There is a simple barycentric certificate. The affine coordinate w_i is harmonic, and the bounded stopped martingale gives $w_i(E_0)=E[w_i(X_\tau)]$. On the opposite side $w_i=0$; on the two

adjacent sides it lies strictly between zero and one except at vertices of zero exit probability. Each side has exit probability one third. Hence $0 < w_i(E_0) < 2/3$. The intersection of these three strips is a central hexagon, a narrower guaranteed region than the whole triangle.

5.8.3 Survival, quasi-stationarity and conditioning forever

The probability $S(p, t) = P_p(\tau > t)$ solves the heat equation with initial value one and absorbing boundary. Its most favorable starting point at a prescribed dimensionless time is $H(\Delta)$, developed in Chapter 6. Integrating S over time returns m . Maximizing the time integral is not the same operation as maximizing at one time.

There are two further long-lived Brownian distributions. The killed transition kernel has leading term $\exp(-D_B \lambda_0 t) \psi_0(p) \psi_0(q)$. Conditional on survival to a distant final time, the endpoint density in q therefore approaches $\psi_0(q)$ divided by its integral. Its mean is a new, well-defined extension called the quasi-stationary mean in this chapter; it is not Q_0 .

Conditioning to survive indefinitely changes the process itself. The ground-state Doob transform adds drift $2D_B \text{grad} \log \psi_0$. Its invariant density is proportional to ψ_0 squared: substituting this density in the zero-flux equation verifies the claim. Its long-time occupation mean is Q_0 . This gives a clean probabilistic realization of the quantum ground-state expectation without confusing it with either ordinary reflecting motion, whose center is G , or the quasi-stationary endpoint law.

Experiment	Density or objective	Selected center
Reflecting Brownian equilibrium	Uniform area	X_2
Maximum mean time before absorption	Torsion maximum	F_1
Uniform injection, accumulated occupation	Torsion density mean	$Abs(0)$
Equal first-exit side probabilities	Three harmonic measures	E_0
Maximum survival to a fixed time	Heat maximum	$H(\Delta)$
Endpoint conditioned on long survival	Density proportional to ψ_0	Quasi-stationary mean
Occupation conditioned to survive forever	Density proportional to ψ_0 squared	Q_0

5.9 Comparing with the ETC and testing attractivity

All comparisons use a frozen ETC snapshot rather than a presumed complete classification of physically defined centers. The main score is RMS Euclidean location error divided by the triangle diameter, over 100 shapes distributed by equal-area sampling in the ordered angle region $\theta_1 \leq \theta_2 \leq \theta_3$. Forty separate shapes are held out for validation. Uniformity refers to area in angle space, equivalently distances to the sides of an equilateral angle diagram; it is not uniformity in side lengths or Cartesian apex coordinates.

The top candidates are then compared by derivatives along isosceles and right-triangle families and by their vertex-response margins. A fingerprint derivative RMS is dimensionless in the stated normalized families; a location RMS is normalized by diameter. They are listed separately, not combined into an unexplained weighted score. A leading location neighbor need not be the leading response neighbor, and proximity to an ETC center is not an identity theorem.

For a center C and a vertex V_i , write J_i for the two-by-two derivative of C with respect to that vertex. An infinitesimal displacement d has nonnegative dot product with the induced center displacement precisely when $d^T J_i d$ is nonnegative. Only the symmetric part contributes. Thus local vertex attractivity requires the smaller eigenvalue of $(J_i + J_i^T)/2$ to be nonnegative for all three vertices. A finite witness instead reports the ratio $d \cdot [C(T+d) - C(T)] / |d|^2$. A strictly negative rigorously enclosed ratio disproves the global property; an unvalidated negative floating-point ratio is numerical evidence requiring resolution checks.

Maximum-location errors, spectral crossings, quadrature errors and cancellation can all corrupt finite differences. Passing sampled triangles establishes no universal theorem. Missing symmetry-locus expressions remain missing; the evaluator does not invent values for a potentially removable singularity. The tables retain evaluated counts, and the figures compare two leading location candidates with X1, X2 and X10. No claim of transcendence follows merely because an elementary center function is unavailable.

The quantum location survey visits the full supplied X1–X72807 snapshot, with 70,853 expressions finite on all 100 generic shapes. The discarded 1,954 expressions are not classified. Almost-equilateral doublet branches and narrow triangles are additional stress tests, not substitutes for a generic sample. Derivative comparison numbers added here use the archived family curves and the independent vertex-response screens. Numerical model error can exceed differences between two nearby ETC candidates.

The six quantum entries and E0 follow. F1 is treated fully in Chapter 6 as well as Chapter 4; the Brownian interpretation above adds an exact physical alias rather than a duplicate numerical center. Projector, thermal and quasi-stationary means are defined here and have reference demonstrations, but no new complete ETC or attractivity survey is claimed for those extension families.

5.10 Individual atlas entries

5.10.1 Q0m: Quantum ground-state maximum

Confine a quantum particle to an infinite triangular well. Select the most probable location in the positive ground state.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_{0m} = \arg \max \psi_0$$

Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Search stationary points within quadratic elements for the maximum, then refine its location.

Reference calculation. In the 3-4-5 triangle: (1.0954507, 0.9975085); normalized barycentrics (0.3936345, 0.2738627, 0.3325028). Method: FEM subdivisions 40 to 56. Last position change is 0.000116 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Unique interior maximum: log-concavity, interior analyticity and zero boundary trace give the line-restriction argument in Section 5.2. The Hardy/disk bound applies to expectations, not this maximum.

Attractivity. A candidate negative finite response appears only after substantial refinement; its small magnitude and maximum-location sensitivity leave the rigorous conclusion open.

The six-selector adaptive experiment met its joint refinement targets on 99/100 grid shapes and 39/40 holdouts. This entry's largest last changes were 0.000493343 and 0.000861405 diameter, respectively. Unresolved shapes remain identified in the archived rankings.

FEM subdivisions	Finite-motion ratio
40	0.215027508
64	0.033841049
96	0.000704255
144	0.001683553
192	-0.015603443
256	-0.010309579
320	-0.005782037
384	-0.008483760

The retained witness uses vertex 1 (zero-based), step 0.005 in the archived normalized triangle. Vertices, direction and both center positions are in quantum/response_witnesses.json. Its earlier positive signs and later small negative signs remain part of the evidence; this maximum is not treated as a certified counterexample.

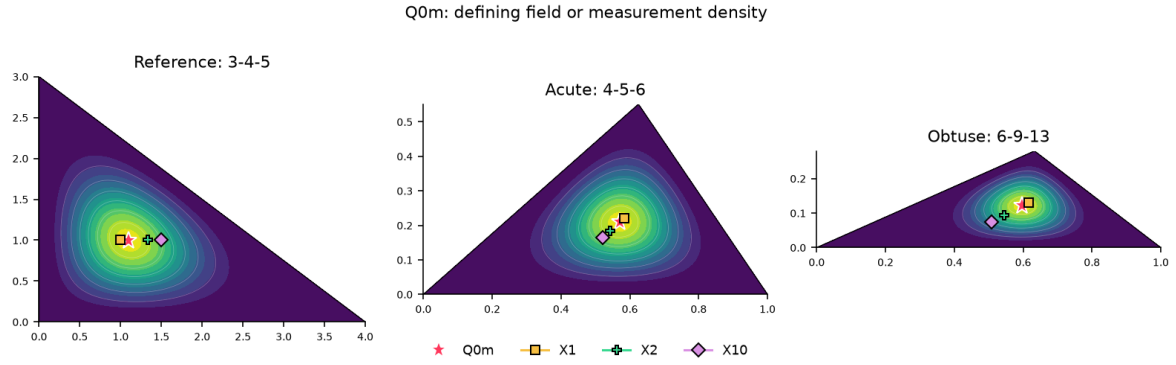
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5393	0.00698967	0.00689852	0.0151762
X5308	0.00720699	0.00614069	0.0138465
X29678	0.00722491	0.00612803	0.0125855
X38825	0.00723628	0.00637187	0.0132697
X17022	0.0072723	0.00634256	0.0146304

For location winner X5393, the separately computed derivative-fingerprint RMS discrepancy is 0.0180176, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0399643, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.0696081. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

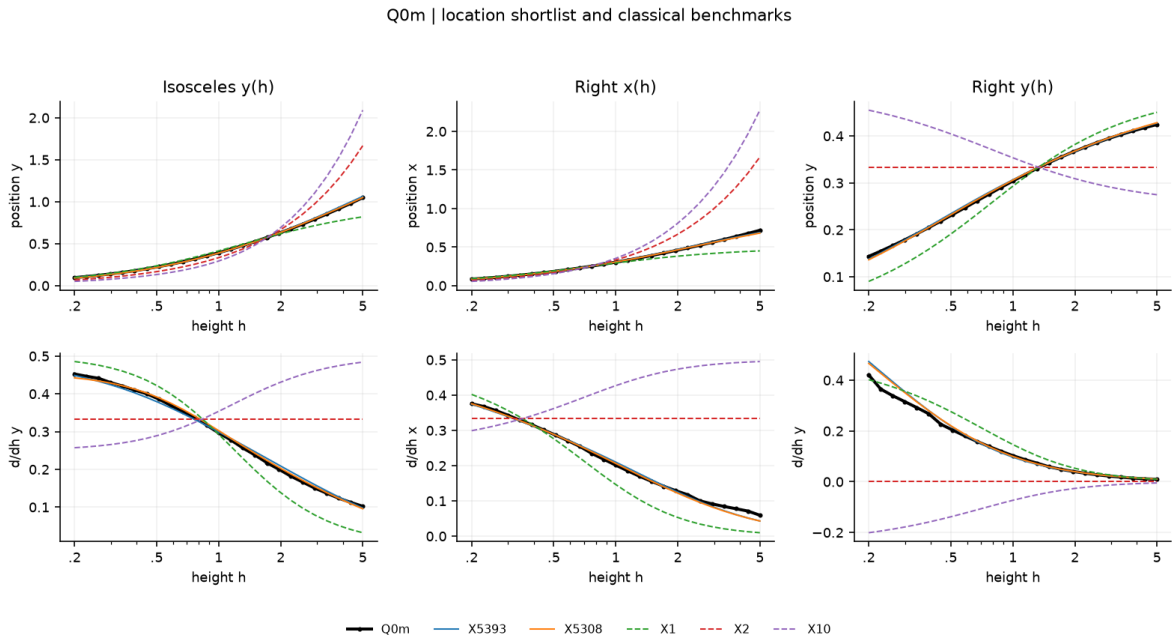
Symbolic approximation. The selected archived fit uses degree 3 in its declared shape-power features, with branch terms set to False. Held-out RMS error is 0.000502907 diameter and

maximum error 0.00107412. Minimum predicted barycentric weight on that holdout is 0.0145191. This linear barycentric fit does not enforce positivity globally. The branch terms and their fixed equilateral constants are retained in quantum/symbolic_models.json; they cannot repair an intrinsically basis-dependent individual eigenstate.

Further fingerprint information. At equilateral the center is unambiguous. The thin correction has the ground-mode Airy scale.



Atlas figure 1a. Q0m in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 1b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.10.2 Q0: Quantum ground-state expectation

Confine a quantum particle to an infinite triangular well. Prepare energy level 0 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_0 = \int_T p|\psi_0(p)|^2 dp$$

Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps.

Reference calculation. In the 3-4-5 triangle: (1.1265611, 1.0015308); normalized barycentrics (0.3845161, 0.2816403, 0.3338436). Method: FEM subdivisions 40 to 56. Last position change is 2.06e-07 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

The reference state's closest retained relative spectral gap is 0.487928. A gap measured at this triangle does not prove a gap on every triangle.

The six-selector adaptive experiment met its joint refinement targets on 99/100 grid shapes and 39/40 holdouts. This entry's largest last changes were 3.52967e-05 and 0.000563522 diameter, respectively. Unresolved shapes remain identified in the archived rankings.

FEM subdivisions	Finite-motion ratio
40	-0.097625273
64	-0.101129032
96	-0.102015658
144	-0.102199452
192	-0.102220696
256	-0.102221881

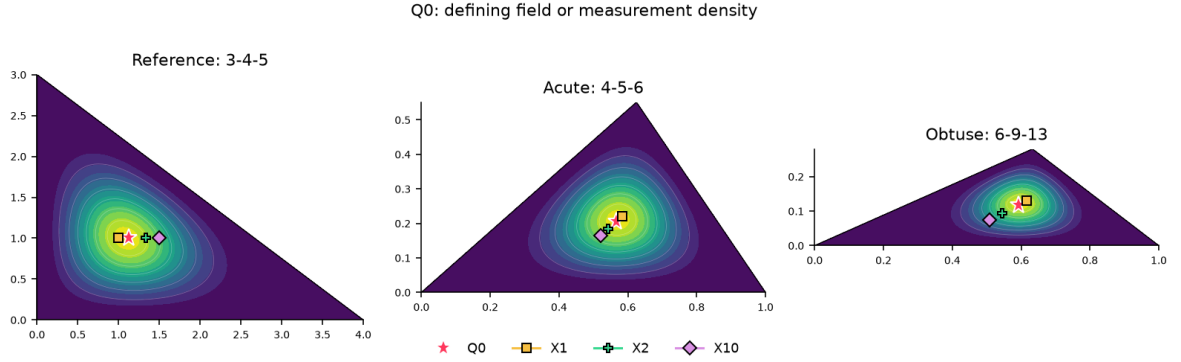
The retained witness uses vertex 1 (zero-based), step 0.005 in the archived normalized triangle. Vertices, direction and both center positions are in quantum/response_witnesses.json. The convergent negative finite-response values are numerical evidence, not a validated continuum certificate.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X24936	0.00771298	0.00626347	0.0147166
X39622	0.0077708	0.00756458	0.0163788
X27253	0.00785032	0.00704734	0.0173955
X39619	0.00835912	0.00778243	0.0126674
X42310	0.00839134	0.00743397	0.016455

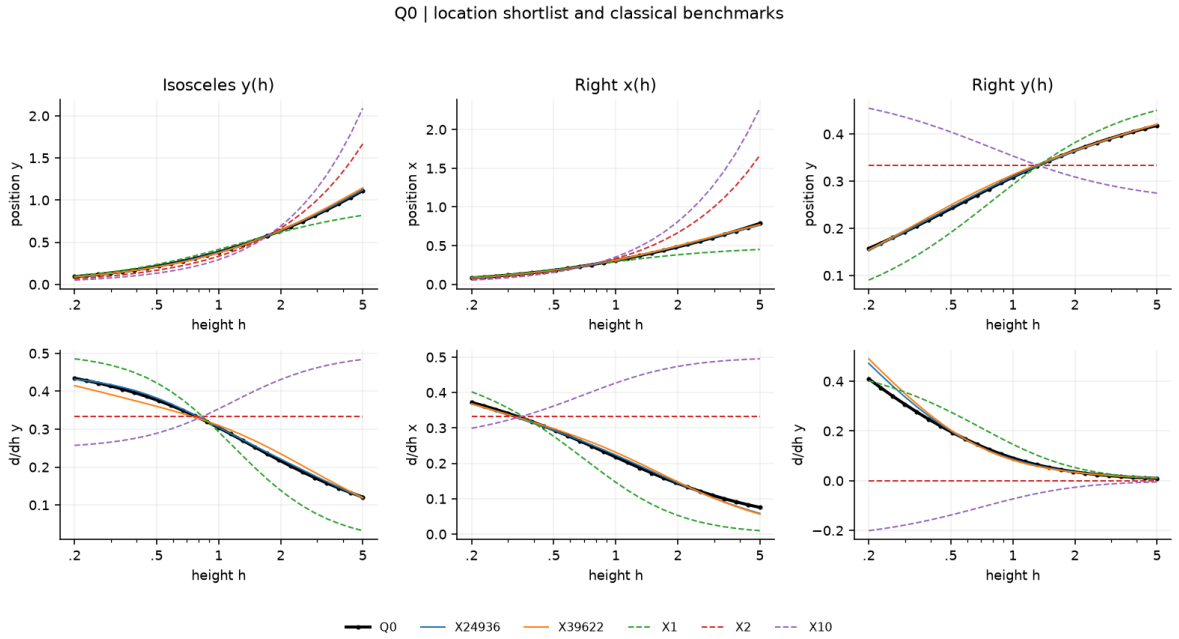
For location winner X24936, the separately computed derivative-fingerprint RMS discrepancy is 0.0149248, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0451799, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.305743. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Symbolic approximation. The selected archived fit uses degree 3 in its declared shape-power features, with branch terms set to False. Held-out RMS error is 0.000607437 diameter and maximum error 0.00120214. Minimum predicted barycentric weight on that holdout is 0.0160365. This linear barycentric fit does not enforce positivity globally. The branch terms and their fixed equilateral constants are retained in `quantum/symbolic_models.json`; they cannot repair an intrinsically basis-dependent individual eigenstate.

Further fingerprint information. At equilateral the center is unambiguous. The thin correction has the ground-mode Airy scale.



Atlas figure 2a. Q0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 2b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.10.3 Q1: Quantum first excited-state expectation

Confine a quantum particle to an infinite triangular well. Prepare energy level 1 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_1 = \int_T p|\psi_1(p)|^2 dp$$

Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps.

Reference calculation. In the 3-4-5 triangle: (1.4133624, 0.9877257); normalized barycentrics (0.3174175, 0.3533406, 0.3292419). Method: FEM subdivisions 40 to 56. Last position change is 8.28e-07 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

The reference state's closest retained relative spectral gap is 0.261677. A gap measured at this triangle does not prove a gap on every triangle.

The six-selector adaptive experiment met its joint refinement targets on 99/100 grid shapes and 39/40 holdouts. This entry's largest last changes were 6.54227e-05 and 0.0011214 diameter, respectively. Unresolved shapes remain identified in the archived rankings.

FEM subdivisions	Finite-motion ratio
40	-0.401893637
64	-0.401891425
96	-0.401891105

The retained witness uses vertex 2 (zero-based), step 0.005 in the archived normalized triangle. Vertices, direction and both center positions are in quantum/response_witnesses.json. The convergent negative finite-response values are numerical evidence, not a validated continuum certificate.

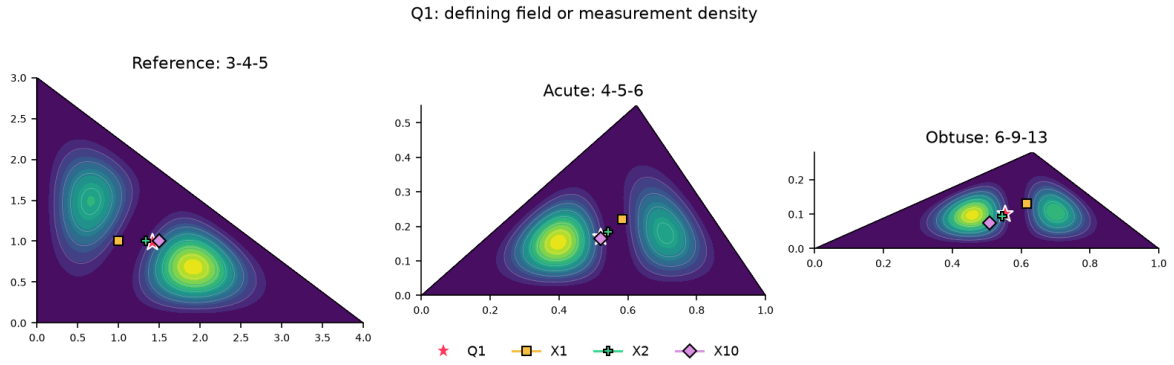
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X7324	0.0170086	0.0142091	0.0268953
X21939	0.0201476	0.0160438	0.0345568
X25061	0.02083	0.0171202	0.0368156
X71237	0.021199	0.0171212	0.0338498
X69285	0.0215016	0.0177873	0.0268722

For location winner X7324, the separately computed derivative-fingerprint RMS discrepancy is 0.0959936, using 50/51 finite family values. The vertex-response margin RMS discrepancy is 0.109865, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.509204. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

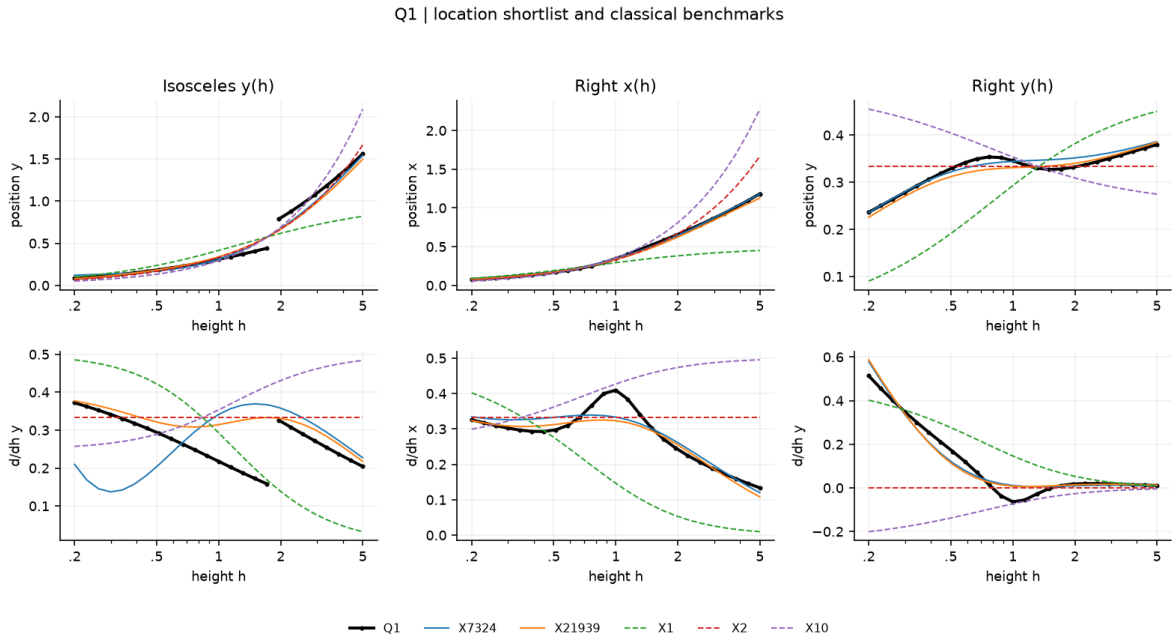
Symbolic approximation. The selected archived fit uses degree 3 in its declared shape-power features, with branch terms set to True. Held-out RMS error is 0.000669469 diameter and maximum error 0.00140063. Minimum predicted barycentric weight on that holdout is 0.023329.

This linear barycentric fit does not enforce positivity globally. The branch terms and their fixed equilateral constants are retained in `quantum/symbolic_models.json`; they cannot repair an intrinsically basis-dependent individual eigenstate.

Further fingerprint information. At equilateral this state belongs to a doublet. Individual-state means are branch dependent; use a full eigenspace projector for a canonical alternative. Missing curve values are intentional.



Atlas figure 3a. Q1 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 3b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.10.4 Q2: Quantum second excited-state expectation

Confine a quantum particle to an infinite triangular well. Prepare energy level 2 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_2 = \int_T p|\psi_2(p)|^2 dp$$

Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps.

Reference calculation. In the 3-4-5 triangle: (0.9581951, 1.0509560); normalized barycentrics (0.4101326, 0.2395488, 0.3503187). Method: FEM subdivisions 40 to 56. Last position change is 2.37e-07 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

The reference state's closest retained relative spectral gap is 0.184798. A gap measured at this triangle does not prove a gap on every triangle.

The six-selector adaptive experiment met its joint refinement targets on 99/100 grid shapes and 39/40 holdouts. This entry's largest last changes were 0.000145266 and 0.00181594 diameter, respectively. Unresolved shapes remain identified in the archived rankings.

FEM subdivisions	Finite-motion ratio
40	-1.834868708
64	-1.835330089
96	-1.835397056

The retained witness uses vertex 2 (zero-based), step 0.005 in the archived normalized triangle. Vertices, direction and both center positions are in quantum/response_witnesses.json. The convergent negative finite-response values are numerical evidence, not a validated continuum certificate.

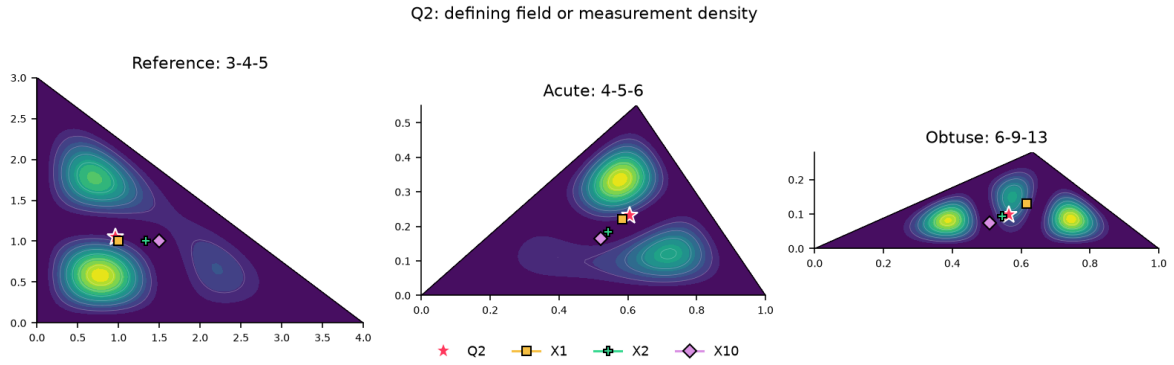
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X43441	0.0390798	0.0328104	0.10824
X43469	0.0393271	0.0329801	0.109933
X72043	0.0413763	0.0363413	0.0996381
X43467	0.0430361	0.0376426	0.104753
X43443	0.0431425	0.0380032	0.101653

For location winner X43441, the separately computed derivative-fingerprint RMS discrepancy is 0.250999, using 50/51 finite family values. The vertex-response margin RMS discrepancy is 0.267567, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.702093. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

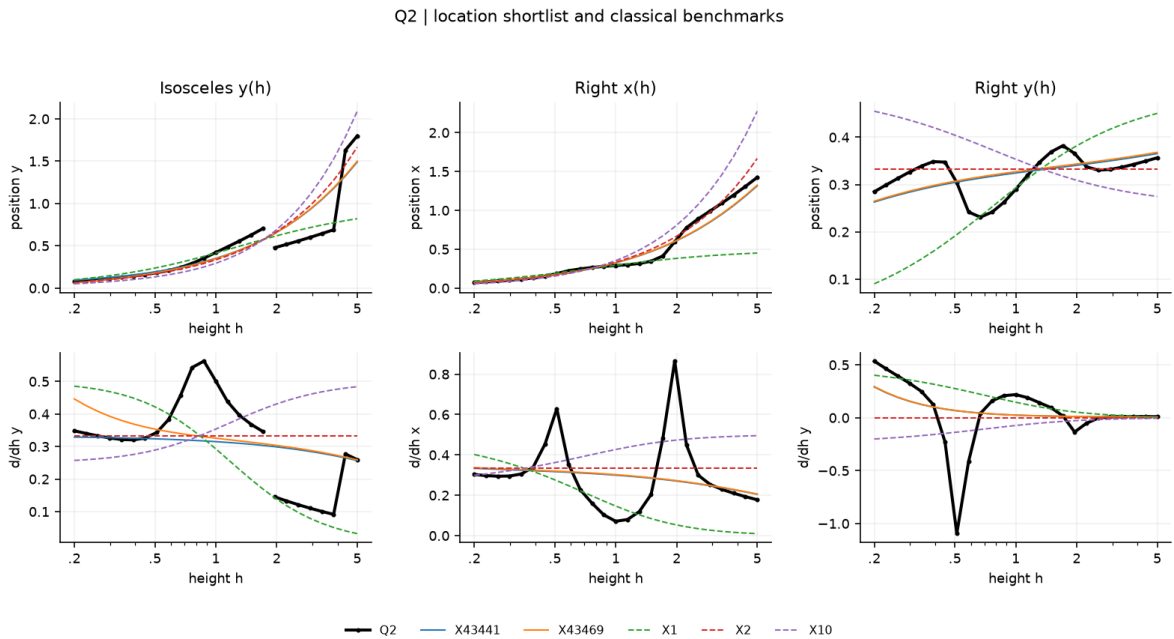
Symbolic approximation. The selected archived fit uses degree 3 in its declared shape-power features, with branch terms set to True. Held-out RMS error is 0.0135787 diameter and maximum error 0.0439658. Minimum predicted barycentric weight on that holdout is 0.031323. This linear

barycentric fit does not enforce positivity globally. The branch terms and their fixed equilateral constants are retained in `quantum/symbolic_models.json`; they cannot repair an intrinsically basis-dependent individual eigenstate.

Further fingerprint information. At equilateral this state belongs to a doublet. Individual-state means are branch dependent; use a full eigenspace projector for a canonical alternative. Missing curve values are intentional.



Atlas figure 4a. Q2 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 4b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.10.5 Q3: Quantum third excited-state expectation

Confine a quantum particle to an infinite triangular well. Prepare energy level 3 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_3 = \int_T p|\psi_3(p)|^2 dp$$

Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps.

Reference calculation. In the 3-4-5 triangle: (1.5264102, 0.9056026); normalized barycentrics (0.3165299, 0.3816025, 0.3018675). Method: FEM subdivisions 40 to 56. Last position change is 3.02e-06 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

The reference state's closest retained relative spectral gap is 0.184798. A gap measured at this triangle does not prove a gap on every triangle.

The six-selector adaptive experiment met its joint refinement targets on 99/100 grid shapes and 39/40 holdouts. This entry's largest last changes were 0.000192402 and 0.00256896 diameter, respectively. Unresolved shapes remain identified in the archived rankings.

FEM subdivisions	Finite-motion ratio
40	-1.673008024
64	-1.672835911
96	-1.672810899

The retained witness uses vertex 0 (zero-based), step 0.005 in the archived normalized triangle. Vertices, direction and both center positions are in quantum/response_witnesses.json. The convergent negative finite-response values are numerical evidence, not a validated continuum certificate.

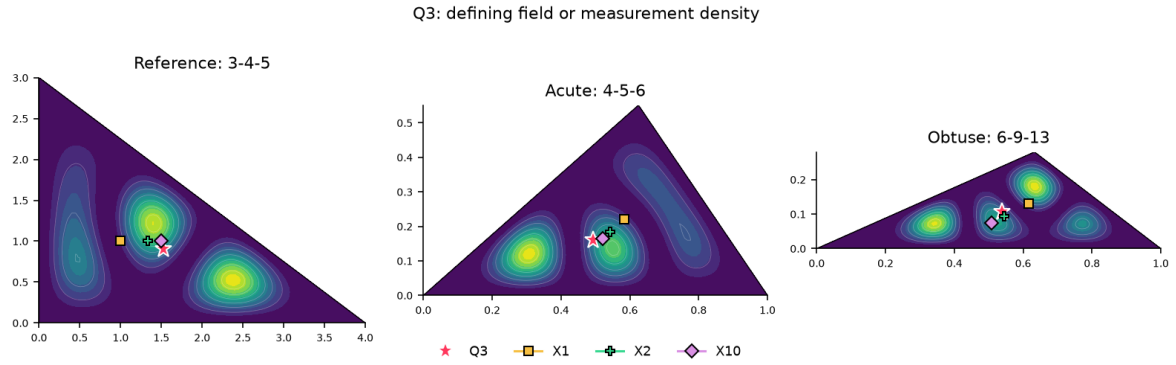
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X24762	0.0490889	0.0553898	0.13434
X43469	0.0493322	0.0470226	0.134015
X4179	0.0493579	0.0484279	0.143663
X364	0.0495083	0.0488672	0.133652
X25077	0.0495864	0.0472736	0.145666

For location winner X24762, the separately computed derivative-fingerprint RMS discrepancy is 0.455026, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.296383, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.0246518. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

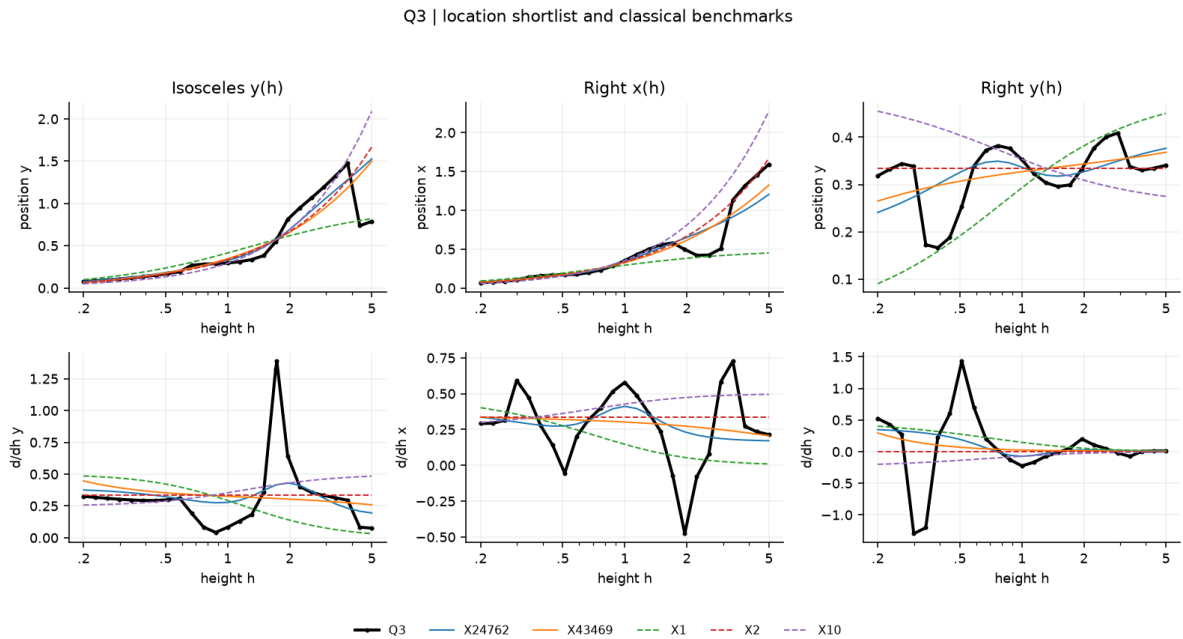
Symbolic approximation. The selected archived fit uses degree 3 in its declared shape-power features, with branch terms set to False. Held-out RMS error is 0.0367428 diameter and maximum error 0.105337. Minimum predicted barycentric weight on that holdout is 0.0194342. This linear

barycentric fit does not enforce positivity globally. The branch terms and their fixed equilateral constants are retained in `quantum/symbolic_models.json`; they cannot repair an intrinsically basis-dependent individual eigenstate.

Further fingerprint information. Q3 is simple at equilateral and has a numerically negative horizontal response there (about -0.7206).



Atlas figure 5a. Q3 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 5b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.10.6 Q4: Quantum fourth excited-state expectation

Confine a quantum particle to an infinite triangular well. Prepare energy level 4 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_4 = \int_T p|\psi_4(p)|^2 dp$$

Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps.

Reference calculation. In the 3-4-5 triangle: (1.1919157, 1.1275766); normalized barycentrics (0.3261622, 0.2979789, 0.3758589). Method: FEM subdivisions 40 to 56. Last position change is 1.25e-06 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

The reference state's closest retained relative spectral gap is 0.16479. A gap measured at this triangle does not prove a gap on every triangle.

The six-selector adaptive experiment met its joint refinement targets on 99/100 grid shapes and 39/40 holdouts. This entry's largest last changes were 0.000254007 and 0.00333695 diameter, respectively. Unresolved shapes remain identified in the archived rankings.

FEM subdivisions	Finite-motion ratio
40	-1.396463936
64	-1.398527778
96	-1.398828411

The retained witness uses vertex 2 (zero-based), step 0.005 in the archived normalized triangle. Vertices, direction and both center positions are in quantum/response_witnesses.json. The convergent negative finite-response values are numerical evidence, not a validated continuum certificate.

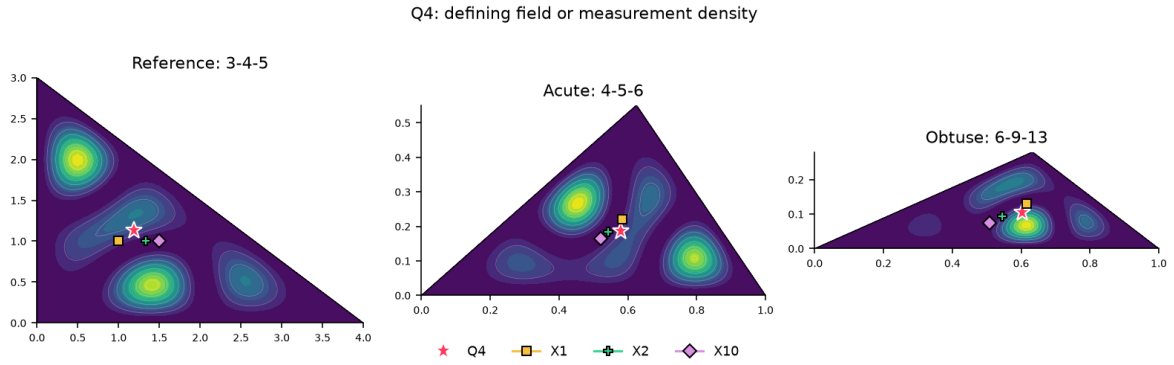
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X4179	0.0468775	0.0485523	0.152281
X50094	0.0468846	0.053908	0.144509
X31323	0.0469881	0.0525907	0.141231
X49737	0.0471213	0.0542953	0.137712
X364	0.0476006	0.0496503	0.13949

For location winner X4179, the separately computed derivative-fingerprint RMS discrepancy is 2.80847, using 50/51 finite family values. The vertex-response margin RMS discrepancy is 0.311196, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.717127. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

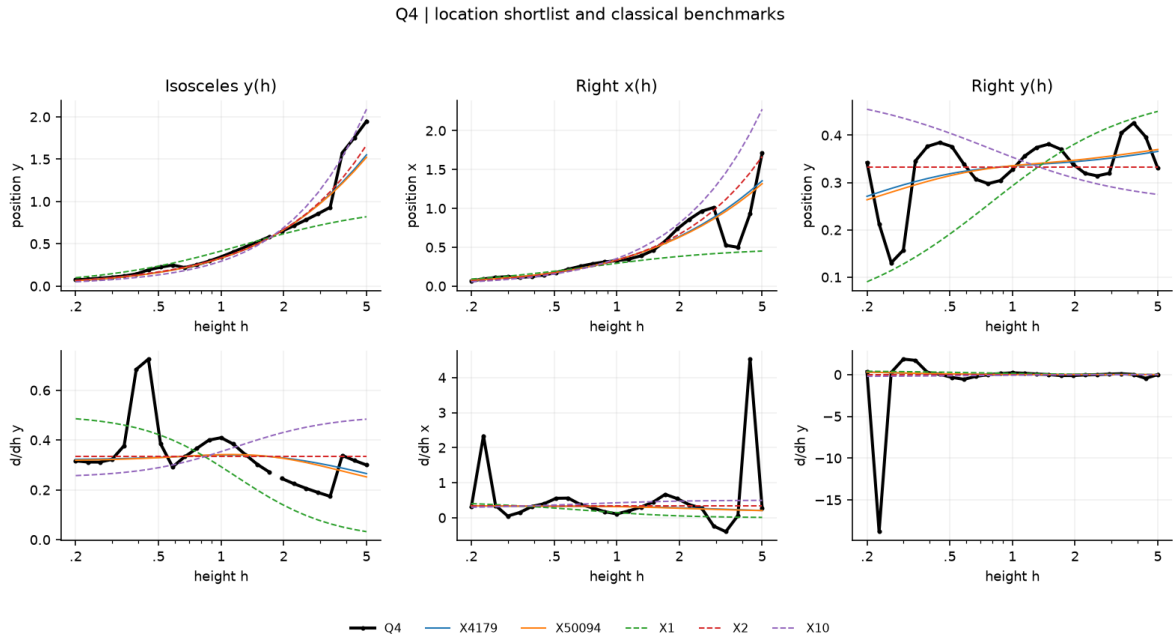
Symbolic approximation. The selected archived fit uses degree 3 in its declared shape-power features, with branch terms set to True. Held-out RMS error is 0.0438093 diameter and maximum error 0.15138. Minimum predicted barycentric weight on that holdout is 0.0473106. This linear

barycentric fit does not enforce positivity globally. The branch terms and their fixed equilateral constants are retained in `quantum/symbolic_models.json`; they cannot repair an intrinsically basis-dependent individual eigenstate.

Further fingerprint information. At equilateral this state belongs to a doublet. Individual-state means are branch dependent; use a full eigenspace projector for a canonical alternative. Missing curve values are intentional.



Atlas figure 6a. Q4 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 6b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.10.7 E0: Equal-exit conformal center

Release Brownian particles from a common point. Choose that point so each side has first-exit probability one third.

$$E0 = F\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right), \quad h_1 = h_2 = h_3 = 1/3$$

Compute triangle angles and the normalized incomplete-beta map in high precision. The contour plots show squared harmonic-measure mismatch.

Reference calculation. In the 3-4-5 triangle: (0.9306884, 0.9606674); normalized barycentrics (0.4471054, 0.2326721, 0.3202225). Method: high-precision conformal map. No independent coordinate error enclosure is asserted.

Location and well-posedness. An exact certificate is $0 < w_i(E0) < 2/3$ for all three normalized barycentric coordinates.

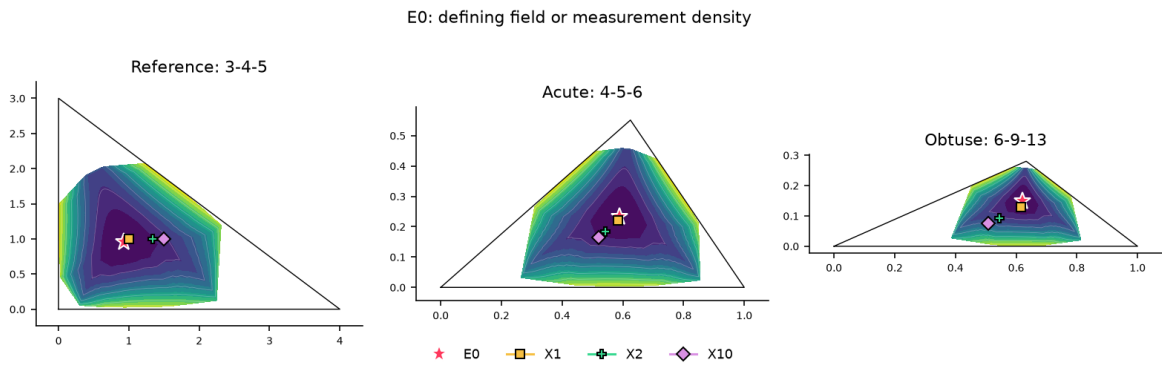
Attractivity. Not globally vertex-attractive: a new outward-rounded interval certificate encloses a finite-motion ratio near $-3.13100322614\text{e-}7$. This supersedes the previous 100-shape no-failure screen. See output/attractivity_candidates/E0_interval_certificate.json and its proof note. See Appendix D.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

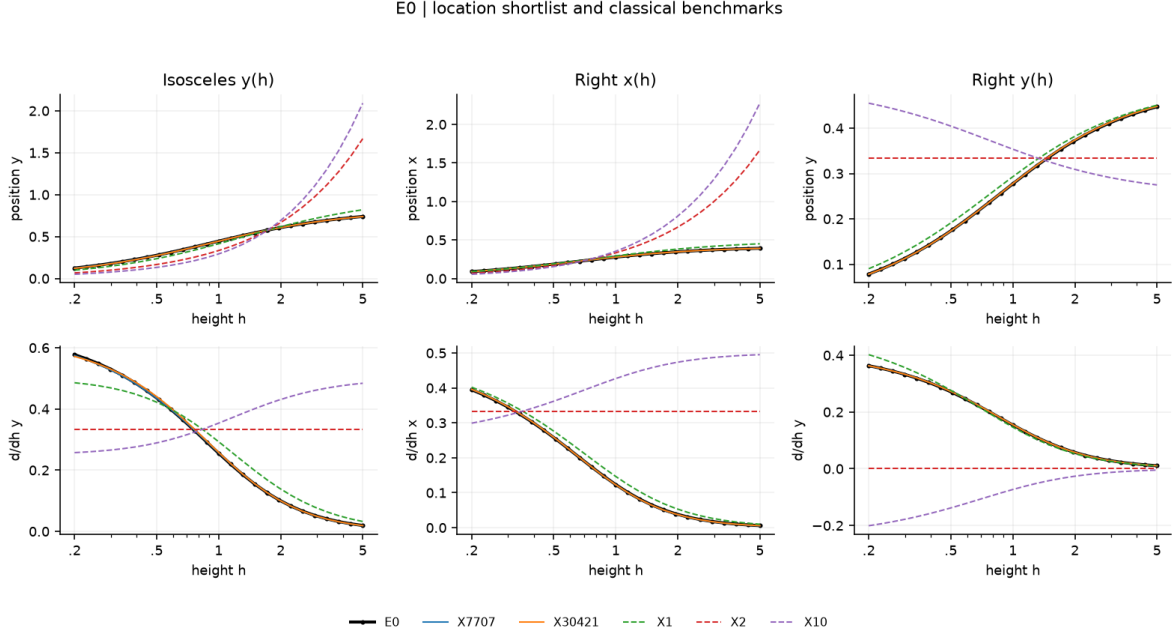
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X7707	0.000419467	0.000419947	0.000591908
X30421	0.00084768	0.000810156	0.00173164
X483	0.000898195	0.000900583	0.00136078
X11191	0.000981506	0.000973867	0.00142833
X1507	0.000983828	0.000984845	0.00155013

For location winner X7707, the separately computed derivative-fingerprint RMS discrepancy is 0.00133705, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00132048, using 300/300 finite vertex margins. The candidate's minimum screened margin is $9.90334\text{e-}05$. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. Thin limit: $C_x \rightarrow x$, $C_y/\epsilon \rightarrow 2/3$ for fixed $x < 1$; the endpoint $x = 1$ instead has vertical limit $1/2$.



Atlas figure 7a. E0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures. White regions are outside the sampled conformal patch, not zero-mismatch regions.



Atlas figure 7b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

5.11 Demonstrations and a reproducibility guide

A triangular microwave or acoustic cavity with the appropriate scalar Dirichlet boundary condition can display analogues of the quantum mode patterns. It reproduces the scalar eigenproblem only when the boundary and field component match the model; a general elastic plate has a biharmonic equation instead. Squaring a calibrated mode amplitude and integrating its image distinguishes the expectation from the brightest point.

A billiard simulation should draw the integrated mean as a second moving marker while the particle continues to bounce. Showing both makes the distinction between an instantaneous position and an average immediately visible. Repeat with the exact archived periodic orbit. For Brownian exit, color each trajectory by its first-hit side; moving the launch point toward E0 equalizes the three proportions. A finite-step simulation must control missed boundary crossings before being used as a quantitative probability solver.

The package contains the source chapters, the full reference records, scripts, archived 100+40 samples, witness meshes and plot data. The new reference solver is `final_chapters_compute.py`. Every numerical coordinate is accompanied by its discretization; displayed digits identify a reproducible computation, not a certified number of correct digits. The proof of the inner parallel triangle, the E0 hexagon and the probabilistic identities does not depend on those coordinates.

5.11.1 The explicit quantum surrogate convention

For reproducibility, let $\ell_i = a_i/(a+b+c)$, $u_i = \ell_i - 1/3$, and $v_i = \ell_i^2$ minus the mean of the three ℓ squared values. Put $q = [12 \sqrt{3} K / (a+b+c)^2]$ to the power $2/3$, equal to one at equilateral. The smooth fitted barycentric weights have the form

$$w_i = \ell_i + \sum_{k=0}^2 q^{k+1} (A_k u_i + B_k v_i), \quad C_{\text{fit}} = \sum_i w_i V_i.$$

The weights sum to one since each feature sums to zero, but need not be positive on every triangle. The doublet fits add directional terms divided by the norm of u , with the prescribed branch constants listed in the data. Such terms are intentionally undefined at $u=0$, the equilateral triangle. The selected coefficients are in `quantum/symbolic_models.json`; its training and independent holdout errors and minimum sampled weights distinguish an empirical approximation from a proven center formula. The immutable mathematical definitions remain the eigenfunction integrals above.

Chapter 6

Fluids, heat diffusion and reaction centers

6.1 A center depends on what the instrument measures

Pour a viscous fluid through a triangular duct, cool a triangular sheet at its edges, or supply a dissolved substance uniformly to a triangular habitat. Each experiment creates a scalar field. One instrument locates its maximum; another integrates its spatial first moment. The two readings can differ even when the underlying field is identical.

This chapter starts with conservation and the constitutive law appropriate to each experiment. It then derives the field equation and the chosen center. We assume homogeneous isotropic material, straight stationary walls and the stated boundary condition. Changing a no-slip wall to a slipping wall, an absorbing boundary to an insulating boundary, or a dimensionless parameter to a dimensional one changes the problem and can change its center.

Write T for the triangle, K for area, D for diameter, and $L^2 = (a^2 + b^2 + c^2)/3$. The reference triangle has $K=6$, $D=5$ and $L^2=50/3$. Its familiar points are $X_1=(1,1)$, $X_2=(4/3,1)$ and $X_{10}=(3/2,1)$. In figures a maximum is marked at its location even if most of the field's mass lies elsewhere.

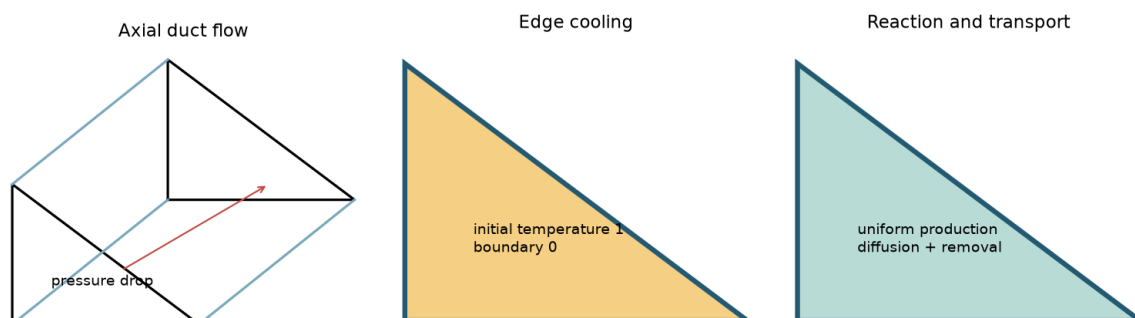


Figure 1. A triangular duct driven along its axis, edge cooling of an initially uniform sheet, and uniform production with diffusion and removal. These drawings specify the forcing and boundaries before a center is selected.

6.2 The fastest point in a pressure-driven triangular duct

6.2.1 Axial momentum balance

Take a long straight prism with cross-section T and axial coordinate z . Suppose an incompressible Newtonian fluid has a fully developed velocity $U(x,y)$ in the z direction, with no transverse velocity. Fully developed means U is independent of z . The viscous force per unit volume in the axial direction is $\eta \Delta U$, and the pressure force is $-\frac{dp}{dz}$. With no acceleration the forces balance:

$$0 = -\frac{dp}{dz} + \eta \Delta U.$$

For this ansatz the convective acceleration is zero even before making a small-Reynolds-number approximation. The physical experiment should nevertheless remain in a regime where the steady laminar profile is realized and entrance effects are negligible. At a stationary no-slip wall, $U=0$. If $g=-dp/dz>0$, put $U=(g/\eta)u$. The geometric problem is

$$-\Delta u = 1 \quad \text{in } T, \quad u = 0 \quad \text{on } \partial T.$$

The maximum speed occurs at $F1=\operatorname{argmax} u$. Viscosity and pressure gradient multiply the field by a positive constant and therefore disappear from its maximizing point. This is exactly the torsion or Laplace center previously encountered in the membrane chapter and as the maximum Brownian mean exit time.

6.2.2 Existence, positivity and uniqueness of the maximum

Multiplying the equation by a zero-boundary test function v and integrating by parts gives the weak equation $\int_T \operatorname{grad} u \cdot \operatorname{grad} v = \int_T v$. Equivalently u minimizes

$$\mathcal{E}(u) = \frac{1}{2} \int_T |\nabla u|^2 dp - \int_T u dp.$$

The gradient term is strictly convex on the Dirichlet space, which gives a unique field. The maximum principle gives $u>0$ in the interior. Uniqueness of the field does not by itself prove uniqueness of its largest point. For convex planar domains the p -torsion uniqueness theorem, including $p=2$ and polygonal triangles, supplies that result [17]. The $F1$ selector is therefore well-defined independently of a numerical maximum search.

The finite-element method replaces u by a sum of continuous quadratic basis functions and solves $Kc=f$, where K is the same stiffness matrix as in the quantum chapter and f contains the integrals of the basis functions. Each element polynomial can be maximized exactly within that element; mesh refinement tests its location. Polynomial Ritz spaces offer an independent discretization and were used in the preceding chapter's witness audit.

6.2.3 A second fluid center: the flow centroid

The volume flux through a small cross-sectional patch is $U(p)dp$. A flow meter measuring where the transported volume crosses the section therefore reads

$$C_{\text{flux}} = \frac{\int_T pU(p) dp}{\int_T U(p) dp} = \frac{\int_T pu(p) dp}{\int_T u(p) dp}.$$

This is $\text{Abs}(0)$, also called $\text{MemMean}(2)$ in Chapter 4. It is not the fastest point $F1$. A camera measuring passive tracer flux should weight its image by axial speed; a camera taking an instantaneous image of a uniformly mixed tracer concentration would instead read the ordinary area centroid G .

There is a useful exact energy identity. With any affine scalar function $\text{ell}(p)$, integration by parts gives $\int \text{ell} |\text{grad } u|^2 = \int \text{ell } u$. The extra term $\int u \text{grad } \text{ell} \cdot \text{grad } u$ vanishes because it is one half of a boundary integral of u squared and u is zero there. Consequently the spatial centroid of viscous dissipation $\int |\text{grad } U|^2$ is also $\text{Abs}(0)$. This is the fluid counterpart of the tension-energy identity in the membrane chapter. The location of maximum dissipation, which depends on $|\text{grad } u|$, is another observable and is not asserted to be $F1$.

6.2.4 Elliptical ducts and the Steiner connection

For an ellipse centered at c with semiaxes a_e, b_e along orthogonal coordinates, try $u = C(1 - X^2/a_e^2 - Y^2/b_e^2)$. Its Laplacian is constant, its boundary value is zero, and choosing $C = 1/[2(a_e^2 + b_e^2)]$ solves the equation. Both the fastest point and flux centroid are c . Applied to the Steiner inellipse, this center is G .

This comparison is instructive but does not solve the triangular duct. The ellipse has a different boundary. An arbitrary affine transformation of the Laplacian also produces an anisotropic operator, so affine equivalence alone cannot move an isotropic circular solution into an isotropic triangular one. The inertia tensor and Steiner axes from Chapter 2 organize an elliptical comparison experiment; they do not force the triangle's $F1$ to be its centroid.

6.3 Cooling a sheet with its boundary held at zero

6.3.1 Conservation plus Fourier's law

Let θ be temperature above the boundary temperature, c_v the heat capacity per unit volume and k_t the conductivity. Assume a homogeneous thin sheet with negligible through-thickness variation and no heat loss from its faces. Fourier's law is heat flux $j = -k_t \text{grad } \theta$. Conservation in any small interior region says $c_v \partial_t \theta = -\text{div } j$. Hence

$$\partial_t \theta = \kappa \Delta \theta, \quad \kappa = k_t / c_v.$$

Start at temperature one and clamp the whole boundary to zero for positive times. The corner incompatibility at time zero is part of an ideal sudden-cooling experiment; the positive-time solution is well-defined. If the edges were instead insulated, uniform initial temperature would stay uniform and no hottest-point center would be selected.

To compare similar triangles, hold $\Delta t = \kappa t / L^2$ fixed. The heat center is

$$H(\Delta) = \arg\max_{p \in T} \left(e^{-\Delta L^2 A_1} \right) (p), \quad A = -\Delta_D.$$

Holding dimensional time fixed while scaling the triangle would describe different stages of cooling. In every derivative with respect to a vertex, L^2 must change with the triangle. Omitting that change silently differentiates a different family.

6.3.2 Why there is one hottest point

The positive-time temperature is positive in the interior and zero at the boundary. Preservation of log-concavity by the Dirichlet heat flow in a convex domain applies to the initial constant density extended by zero [22]. If there were two maxima, log-concavity would make a segment of maxima. Interior spatial analyticity at positive time extends its constant value along the interior chord, contradicting zero temperature at its endpoints. Thus $H(\Delta)$ is a unique interior point for each positive Δ . This establishes a center; it does not give an attractivity theorem.

6.3.3 A spectral solution and two independent truncations

Expand the initial constant function in Dirichlet eigenfunctions. Orthogonality gives coefficient $b_j = \int_T \psi_j$. Each mode decays at its own rate:

$$\theta(p, t) = \sum_{j=0}^{\infty} b_j e^{-\kappa t \lambda_j} \psi_j(p), \quad b_j = \int_T \psi_j(q) dq.$$

This uses signed eigenfunctions with signed coefficients. It differs from the thermal quantum density, which is a sum of squared eigenfunctions. At late times we factor out $\exp(-\kappa t \lambda_0)$; the remaining field has the same maximum and avoids numerical underflow. At early times, many modes are needed to reproduce a nearly flat interior and a thin boundary layer.

There are two numerical limits: spatial mesh size tending to zero and number of retained modes tending to infinity. Comparing a full spectrum with its first half tests only one of them. Moreover a small field difference need not bound a maximum-location error when the field is flat. The reference records therefore retain both mesh changes and half-spectrum changes. The smallest sampled Δ here is 0.01; an older 6-9-13 experiment includes an unresolved 0.001 result, which remains flagged and is not advertised as a converged center.

6.3.4 Short time, long time and integrated time

For very short time, the heat lost at an interior point is dominated exponentially by the closest boundary. Its leading exponential scale is distance to the boundary squared divided by $4\kappa t$. Maximizing surviving heat therefore tends toward the point maximizing the nearest-side distance: the incenter. At time zero itself every interior point has temperature one, so this is a limiting center, not a unique zero-time maximum.

At long time the positive ground-state term dominates and H tends to Q_0 . Integrating the heat equation in time instead gives the torsion field:

$$\int_0^{\infty} e^{-sA} 1 ds = A^{-1} 1 = u.$$

The fastest-flow point F_1 maximizes this integrated field. In general taking a maximum does not commute with time integration; F_1 need not equal H at any prescribed time. A thermal-energy first moment would use $\int p \theta$ divided by $\int \theta$ and is yet another center. At long time that moment tends to the mean of ψ_0 , the Brownian quasi-stationary mean from Chapter 5, rather than Q_0 .

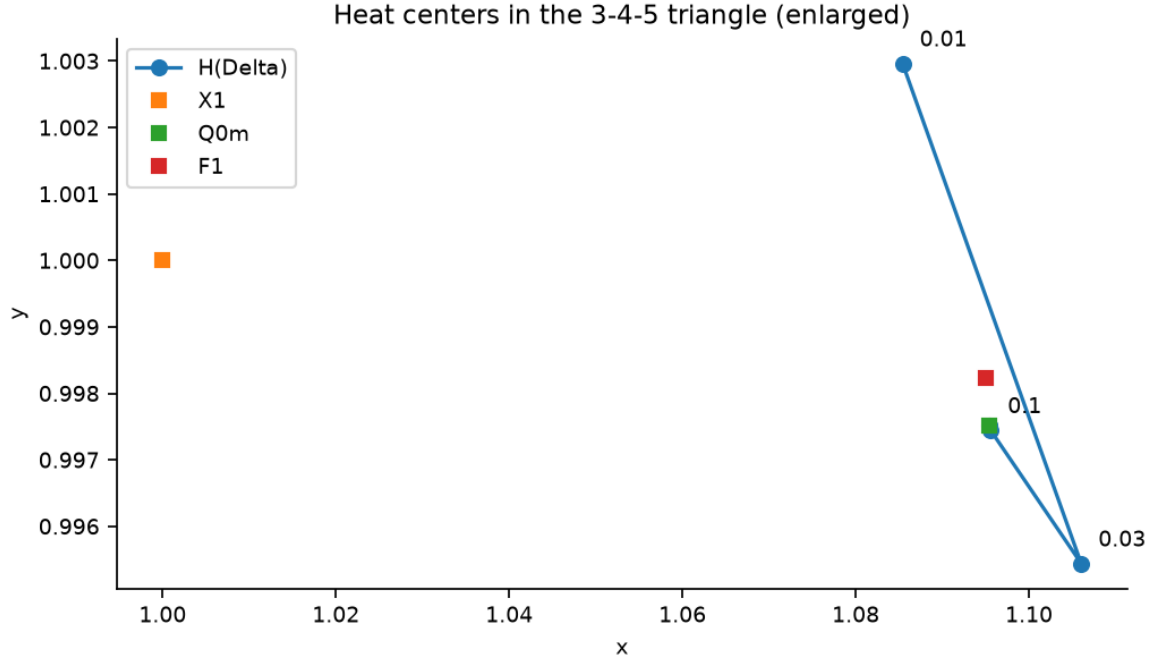


Figure 2. New 3-4-5 heat-center samples and comparison points. The curve connects computed positive times; it is not a proof of monotonic motion, and its visual closeness to Q0m at late time does not imply exact finite-time equality.

6.4 Uniform production, diffusion and bulk removal

6.4.1 From particle balance to a screened Poisson equation

Let $c(p)$ be the steady concentration of a dissolved species. It is produced uniformly at rate s , diffuses with coefficient D_c , disappears in the bulk at rate $k c$, and is removed immediately at the boundary. Fick's flux is $j = -D_c \text{grad } c$. In a stationary region, production minus removal minus outward flux must vanish, giving

$$-D_c \Delta c + k c = s, \quad c|_{\partial T} = 0.$$

Scale concentration by s/D_c and set $\mu = kL^2/D_c$. The geometric equation and the center are

$$-\Delta c + \frac{\mu}{L^2} c = 1, \quad \text{Abs}(\mu) = \frac{\int_T p c \, dp}{\int_T c \, dp}.$$

The name Abs is inherited from the atlas. For $\mu > 0$, bulk removal events have density proportional to $k c$, so their mean equals this concentration mean. At $\mu = 0$ there are no bulk removal events; the same formula remains meaningful as concentration, residence density or transported flux. It should not be described as an event centroid for nonexistent events.

The maximum principle makes c positive. The linear matrix system is $(K + \mu M/L^2)c = f$. Its positive mass shift improves coercivity, but a small algebraic residual only establishes that the discrete equations were solved; polynomial degree or mesh changes still test approximation error. The new reference uses boundary-bubble Ritz degrees 11,17,23, retaining the original field residuals and sampled positivity checks.

6.4.2 A probabilistic form and a limit toward G

The same field is an exponentially discounted Brownian lifetime, using a Brownian generator Delta:

$$c(p) = \mathbb{E}_p \int_0^\tau e^{-\mu s/L^2} ds = \int_0^\infty e^{-\mu s/L^2} S(p, s) ds.$$

Thus the absorption family continuously changes the weight given to late survival. At $\mu=0$ it is the torsion field and its mean is $\text{Abs}(0)$. For large μ , away from a boundary layer, removal balances production locally and c is approximately L^2/μ . The normalized density becomes uniform in the area limit and $\text{Abs}(\mu)$ tends to G . The boundary layer thickness has order $L/\sqrt{\mu}$. A finite μ , even 100, is not the infinite-removal endpoint.

Every finite member is a positive-density mean, hence strictly interior by the affine barycentric argument. The finite-element or Ritz values are not needed for that certificate. The atlas tests $\mu=0,1,10,100$. Their no-failure response screens leave global vertex attractivity open.

6.5 Reaction and diffusion in a triangular habitat

6.5.1 Why extinction must be excluded explicitly

Take diffusion coefficient D_c , intrinsic growth rate r and carrying capacity N_* . After scaling population density by N_* and dividing by D_c , the stationary logistic equation is

$$-\Delta n = an(1 - n), \quad n|_{\partial T} = 0.$$

The zero field solves it for every a . It cannot be normalized to a spatial mean. The intended experiment selects the positive persistent equilibrium. Testing against the positive first Dirichlet eigenfunction gives

$$\lambda_0 \int_T n\psi_0 = a \int_T n\psi_0 - a \int_T n^2\psi_0.$$

For $n>0$ the last integral is positive, so a must exceed λ_0 . Conversely a small multiple of ψ_0 is a subsolution when $a>\lambda_0$, while the constant carrying-capacity value is a supersolution; the usual monotone construction produces a positive solution between them. The ratio of the reaction to n is $a(1-n)$, strictly decreasing in n . A comparison or Picone-identity argument gives uniqueness of the positive equilibrium. Its maximum lies below one by the maximum principle. These are continuum statements about the branch, not consequences of an optimizer reporting success.

Set $a=\kappa\lambda_0(T)$, where $\kappa>1$ is fixed, and define

$$\text{Bio}(\kappa) = \frac{\int_T pn(p) dp}{\int_T n(p) dp}.$$

The parameter is a fixed factor above each shape's persistence threshold. Therefore the eigenvalue λ_0 changes when the triangle changes. Holding physical growth rate fixed is a different family and is not a similarity-equivariant center without another scale convention.

6.5.2 Computation of the positive branch

In the reference Ritz method, first compute the lowest eigenpair of the discrete stiffness and mass forms. Use a small positive ground-state profile as a branch-aware starting value. Minimize the energy consisting of gradient energy minus a n squared/2 plus a $|n|$ cubed/3. The absolute value gives a coercive extension for intermediate iterates. After solving, check the residual of the original n squared equation, not just the derivative of the extended objective. A negative or near-zero mass, large negative field undershoot, or large original residual disqualifies the result as a positive-branch solve.

The atlas uses $\kappa=1.1, 2, 10$. Near threshold the positive solution is a small multiple of ψ_0 , so its normalized mean tends to $\int p \psi_0$ divided by $\int \psi_0$. The square is absent. Far above threshold the density is nearly one away from its boundary layer and the mean tends to G . Those endpoints connect population dynamics with the Brownian and quantum constructions, without identifying their finite parameters.

An important conceptual separation appears here. The positive equilibrium can be dynamically stable against perturbations of the population density while its center moves against a vertex displacement. $\text{Bio}(1.1)$ has an independently reproduced negative finite vertex response. Physical stability and geometric attractivity answer different questions.

6.6 Common reference values and conservation checks

Selector	x	y	Last change / D
F1	1.0949876	0.9982276	0.000231
H0.01	1.0854107	1.0029634	0.000111
H0.03	1.1060295	0.9954395	0.000167
H0.1	1.0955679	0.9974450	0.000117
H0.3	1.0954507	0.9975085	0.000116
H1.0	1.0954507	0.9975085	0.000116
Abs0.0	1.2170772	1.0006773	6.71e-09
Abs1.0	1.2175718	1.0006652	6.8e-09
Abs10.0	1.2217632	1.0005695	7.61e-09
Abs100.0	1.2481443	1.0001988	1.54e-08
Bio1.1	1.1566856	1.0041136	1.74e-10
Bio2.0	1.1834208	1.0028060	7.04e-11
Bio10.0	1.2633595	1.0003868	1.42e-09

All values use $A=(0,0)$, $B=(4,0)$, $C=(0,3)$. A last-change value compares two discretizations; it is neither an error bound nor the number of reliable printed digits. Exact and numerical location statements are identified separately in each entry.

The load, flux and energy identities offer more informative checks than agreement of two printed coordinates. The integral of $|\text{grad } u|$ squared must equal the integral of u . Their affine first moments must agree as well. Heat fields integrated over all time solve the Poisson problem. Removing the production term or changing the boundary law would break these particular identities. The package keeps numerical identity checks separate from rigorous statements about the continuum model.

6.6.1 Equivalent conformal formulations

Let F map a disk or half-plane onto T and let a tilde denote composition with F . The planar Laplacian transformation gives the following equivalent equations on the mapped domain:

$$-\Delta_z \tilde{u} = |F'|^2, \quad -\Delta_z \tilde{c} + \frac{\mu}{L^2} |F'|^2 \tilde{c} = |F'|^2.$$

For heat, partial_t tilde theta=kappa |F'| to the power minus two Delta_z tilde theta; the logistic equation becomes -Delta_z tilde n=kappa lambda0 |F'| squared tilde n(1-tilde n). Each has zero boundary trace. The physical first moment must also include the area factor |F'| squared and the physical coordinate F(z). These are genuine equivalent problems, but the nonconstant coefficient means ordinary disk solutions cannot simply be transplanted into a triangle. Only the zero-right-hand-side harmonic equation loses the conformal factor entirely.

6.7 Thin triangles, almost-equilateral triangles and fitted formulas

6.7.1 A strip calculation for means

Put the vertices at (-1,0),(1,0),(x,epsilon), with fixed $0 \leq x < 1$. The transverse width is epsilon times a tent function $h_x(s)$, linear up to $s=x$ and linear down afterwards. In a strip of width H, solving $-u_{yy}=1$ with zero endpoint values gives $u(y)=y(H-y)/2$. Its integral across the strip is $H^3/12$. Thus the longitudinal mass of the torsion density is proportional to h_x cubed.

An elementary two-piece integration gives the horizontal mean for a density proportional to h_x to the power k:

$$\frac{\int_{-1}^1 s h_x(s)^k ds}{\int_{-1}^1 h_x(s)^k ds} = \frac{k}{k+2} x.$$

At $k=3$ this yields $\text{Abs}(0)_x$ tending to $3x/5$. For fixed finite μ , transverse diffusion dominates the bulk term as epsilon tends to zero, giving the same leading strip prediction for $\text{Abs}(\mu)$. Holding μ proportional to epsilon to the power minus two would give a different joint limit. The heat and quantum maxima instead concentrate near the widest section in the fixed-positive-time regime. This is why a mean and a maximum should have different thin fingerprints even when they share a PDE.

6.7.2 Symmetry constraints on local fits

For $C=(x, \sqrt{3}+y)$, reflection forces the horizontal center coordinate to be odd in x and the vertical coordinate to be even. Write $\text{center}_x = \alpha x + \beta xy + \dots$, $\text{center}_y = 1/\sqrt{3} + \gamma y + \delta x^2 + \eta y^2 + \dots$. A differentiable similarity-equivariant center has $\alpha + \gamma = 2/3$. The inherited data include numerical jets and their step comparisons. The trace is a consistency condition, not an independent enclosure of each coefficient.

Positive symbolic surrogates can be constructed in barycentric form. Begin with a symmetric baseline, such as uniform or incenter weights. Multiply each weight by the exponential of a cyclically permuted feature polynomial, then normalize by the sum. Positivity and relabelling equivariance are built in. The atlas's six-coefficient models use the normalized shape measure $z=3 \sqrt{3} K/s^2$, where s is semiperimeter, and powers chosen to reflect the expected thin scale. All coefficients and held-out errors are retained. Such a formula is an approximation to a center function; it neither proves an ETC identity nor transfers attractivity from a nearby classical center.

More explicitly, put $\ell_i = a_i/(a+b+c)$, $u_i = \ell_i - 1/3$ and $v_i = \ell_i^2$ minus the mean of the three ℓ_i^2 values. Form P_i as a sum of three powers of q , each multiplying A_k

$u_i + B_k v_i$. Then $w_i = b_i \exp(P_i)$ divided by the sum over all three vertices, where b_i are the baseline barycentrics. The exact q , starting exponent, baseline and six coefficients are stored with each fit. This prescription makes the numerical approximation reproducible without an ETC identification.

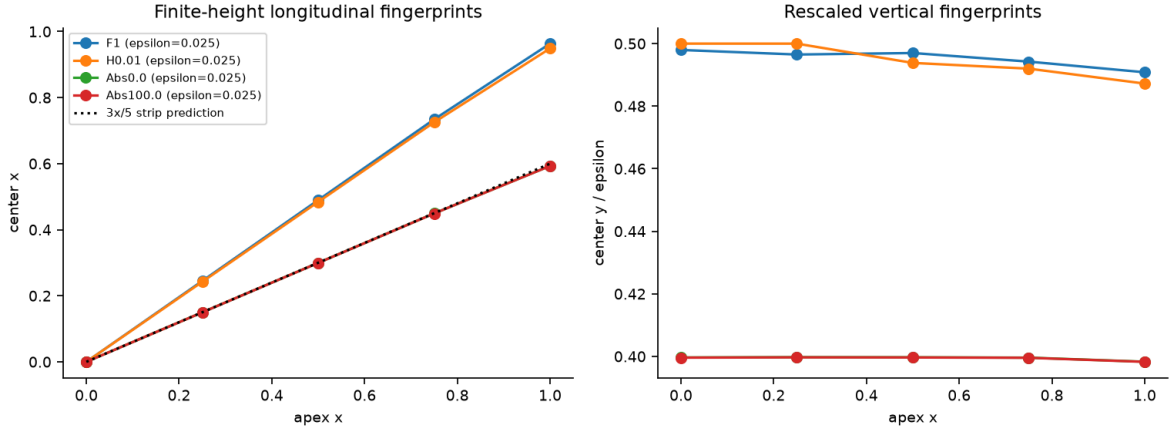


Figure 3. Archived thin-family examples for F1, H and absorption means. Finite-height values and their asymptotic interpretations must be distinguished, especially near the endpoint $x=1$.

6.8 ETC comparison and the meaning of an attractivity result

All comparisons use a frozen ETC snapshot rather than a presumed complete classification of physically defined centers. The main score is RMS Euclidean location error divided by the triangle diameter, over 100 shapes distributed by equal-area sampling in the ordered angle region $\theta_1 \leq \theta_2 \leq \theta_3$. Forty separate shapes are held out for validation. Uniformity refers to area in angle space, equivalently distances to the sides of an equilateral angle diagram; it is not uniformity in side lengths or Cartesian apex coordinates.

The top candidates are then compared by derivatives along isosceles and right-triangle families and by their vertex-response margins. A fingerprint derivative RMS is dimensionless in the stated normalized families; a location RMS is normalized by diameter. They are listed separately, not combined into an unexplained weighted score. A leading location neighbor need not be the leading response neighbor, and proximity to an ETC center is not an identity theorem.

For a center C and a vertex V_i , write J_i for the two-by-two derivative of C with respect to that vertex. An infinitesimal displacement d has nonnegative dot product with the induced center displacement precisely when $d^T J_i d$ is nonnegative. Only the symmetric part contributes. Thus local vertex attractivity requires the smaller eigenvalue of $(J_i + J_i^T)/2$ to be nonnegative for all three vertices. A finite witness instead reports the ratio $d \cdot [C(T+d) - C(T)] / |d|^2$ squared. A strictly negative rigorously enclosed ratio disproves the global property; an unvalidated negative floating-point ratio is numerical evidence requiring resolution checks.

Maximum-location errors, spectral crossings, quadrature errors and cancellation can all corrupt finite differences. Passing sampled triangles establishes no universal theorem. Missing symmetry-locus expressions remain missing; the evaluator does not invent values for a potentially removable singularity. The tables retain evaluated counts, and the figures compare two leading location candidates with X_1 , X_2 and X_{10} . No claim of transcendence follows merely because an elementary center function is unavailable.

The fluid and heat families reuse the electro-heat atlas; the concentration and population means reuse the mechanical-optical atlas. Both visited the supplied full ETC snapshot. Their numerical PDE resolutions differ, so ranking discrepancies should not be interpreted more accurately than the underlying fields. The new 3-4-5 examples do not replace or silently resample the inherited 100-shape ranking grid.

For F1, the negative FEM witness was re-examined in Chapter 4 using independent high-degree Ritz calculations. Its finest half-step changed sign, leaving the continuum conclusion unresolved. This chapter carries that corrected status forward. The heat family has retained negative responses at three fine FEM resolutions, but maximum-location validation remains absent. For Bio(1.1), an independent quadratic FEM calculation reproduces the negative response of the Ritz method; the other population parameters remain screening results. Passing a screen is never called proof of global attraction.

6.9 Individual atlas entries

6.9.1 F1: Fluid torsion / linear membrane maximum

Measure maximum axial velocity in a triangular Poiseuille duct, maximum deflection of a uniformly tensioned loaded membrane, or maximal Brownian mean exit time.

$$-\Delta u = 1, \quad u|_{\partial T} = 0, \quad F1 = \arg \max_T u$$

Solve the quadratic FEM Poisson system with exact load integration and maximize the piecewise quadratic field.

Reference calculation. In the 3-4-5 triangle: (1.0949876, 0.9982276); normalized barycentrics (0.3935106, 0.2737469, 0.3327425). Method: FEM subdivisions 40 to 56. Last position change is 0.000231 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive Dirichlet field has its maximum in the interior.

Attractivity. Negative numerical evidence, with unresolved refinement sensitivity. The inherited finite-element witness is negative at subdivisions 128, 192 and 256. Independent Ritz degrees 20 and 32 retain both negative steps, but degree 44 changes the half-step sign. The data do not supply a converged, validated continuum counterexample; global nonattractivity is not certified.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

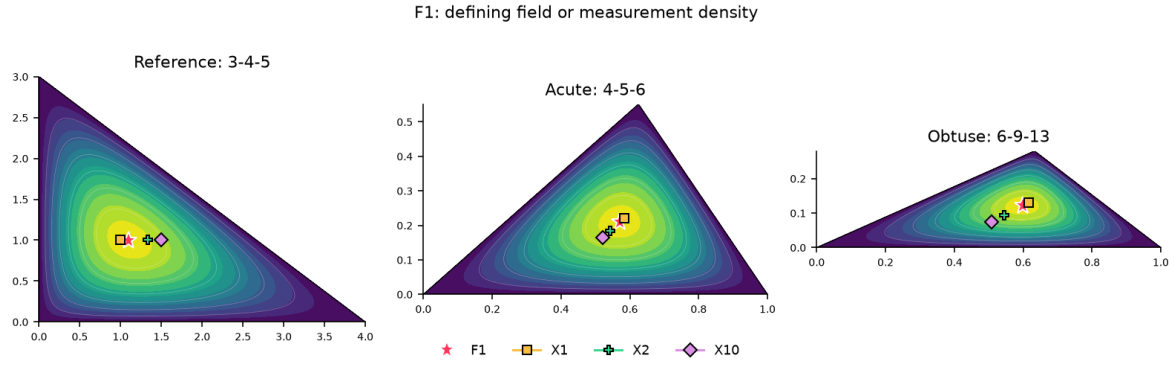
Ritz degree	Original ratio	Half-step ratio
20	-0.07240683	-0.06769307
32	-0.06703628	-0.06659708
44	-0.08868240	0.00121947

The changed finest half-step sign prevents promoting the old negative witness to a settled continuum result. This table carries forward the corrected Chapter 4 assessment.

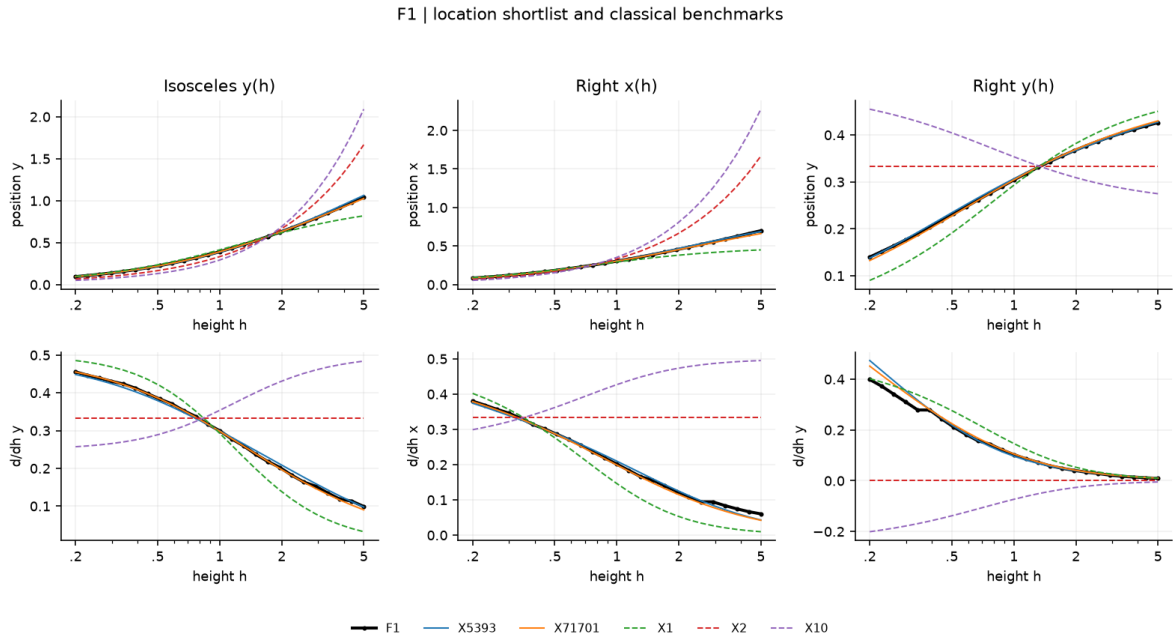
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5393	0.00422012	0.00405025	0.00826325
X71701	0.00506842	0.00513996	0.0109041
X2534	0.00546374	0.00513597	0.00846997
X33993	0.00629982	0.0063368	0.0137487
X13411	0.00635668	0.00638337	0.013735

For location winner X5393, the separately computed derivative-fingerprint RMS discrepancy is 0.0192409, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0493853, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.0696081. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. This is a maximum, whereas Abs(0) is the mean of the same field. In thin triangles the maximum approaches the widest section. The heat integral identity relates fields but does not identify their maxima. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum.



Atlas figure 1a. F1 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 1b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.2 H0.01: Heat maximum at dimensionless time 0.01

Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.01) = \arg \max_T e^{-0.01L^2 A} 1, \quad A = -\Delta_D$$

Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time.

Reference calculation. In the 3-4-5 triangle: (1.0854107, 1.0029634); normalized barycentrics (0.3943262, 0.2713527, 0.3343211). Method: FEM subdivisions 40 to 56. Last position change is 0.000111 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Unique interior maximum at every positive time, by preservation of log-concavity and the analytic line-restriction argument. This supplies existence and uniqueness, not a quantitative distance-to-side bound.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

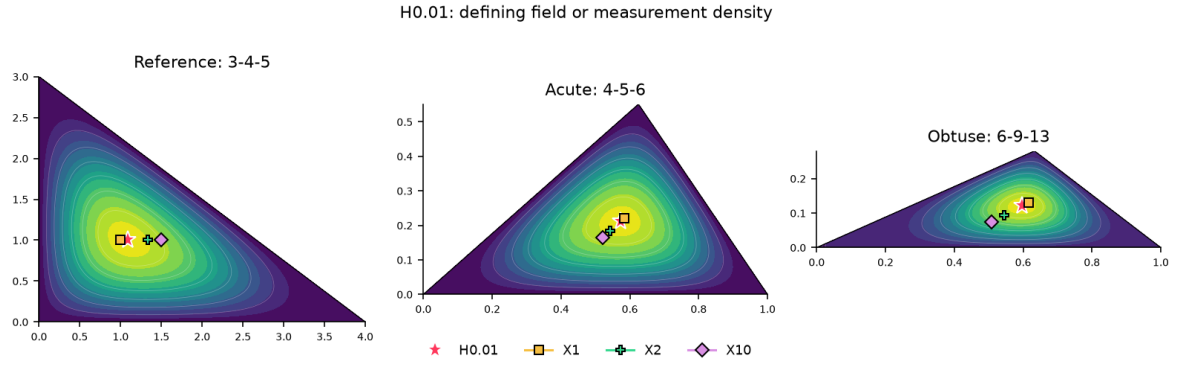
FEM subdivisions	Finite-motion ratio
128	-0.050688688
192	-0.082006085
256	-0.068188518

These are the shared retained thin-triangle heat witness, evaluated at the stated time for each shape. Higher modes are negligible in this example at late dimensionless times; that does not bound the spatial error in the ground-state maximum.

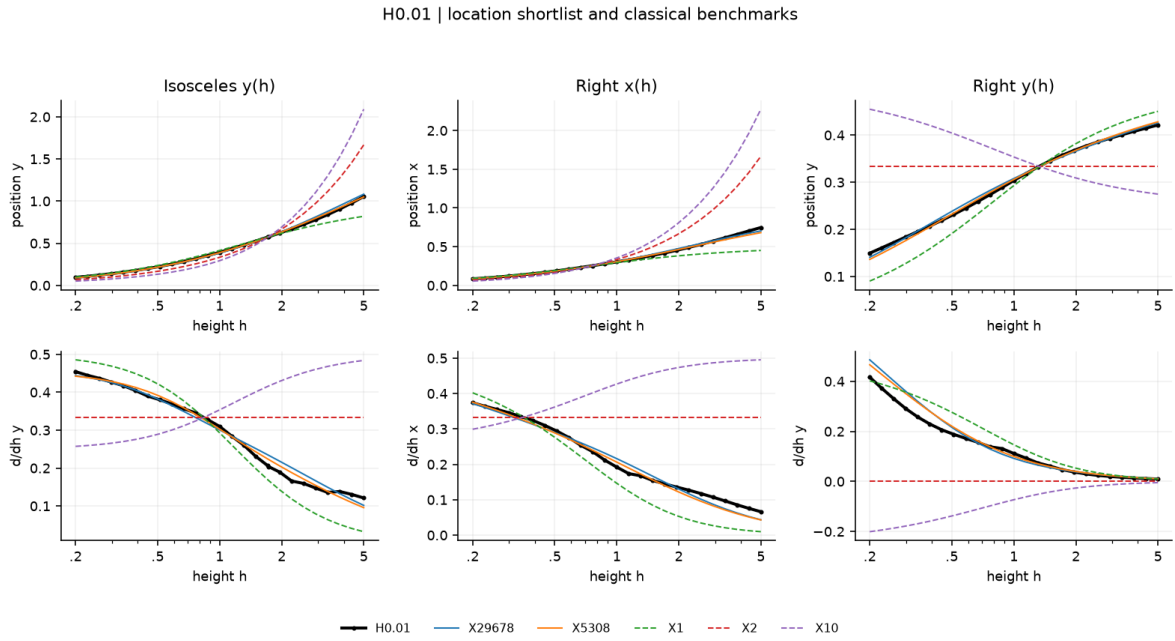
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X29678	0.00771364	0.00698506	0.0136716
X5308	0.00790053	0.00737719	0.0159203
X56032	0.00795127	0.00729019	0.0154675
X38825	0.00799185	0.00763133	0.0153434
X27577	0.008001	0.00723969	0.0145988

For location winner X29678, the separately computed derivative-fingerprint RMS discrepancy is 0.0286536, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0451534, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.197374. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.



Atlas figure 2a. H0.01 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 2b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.3 H0.03: Heat maximum at dimensionless time 0.03

Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.03) = \arg \max_T e^{-0.03L^2 A} 1, \quad A = -\Delta_D$$

Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time.

Reference calculation. In the 3-4-5 triangle: (1.1060295, 0.9954395); normalized barycentrics (0.3916795, 0.2765074, 0.3318132). Method: FEM subdivisions 40 to 56. Last position change is 0.000167 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Unique interior maximum at every positive time, by preservation of log-concavity and the analytic line-restriction argument. This supplies existence and uniqueness, not a quantitative distance-to-side bound.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

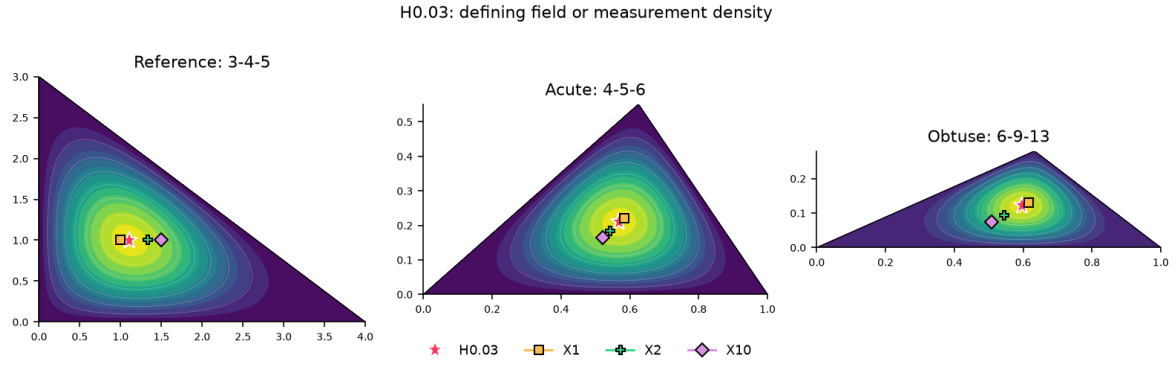
FEM subdivisions	Finite-motion ratio
128	-0.050493940
192	-0.081849800
256	-0.068014129

These are the shared retained thin-triangle heat witness, evaluated at the stated time for each shape. Higher modes are negligible in this example at late dimensionless times; that does not bound the spatial error in the ground-state maximum.

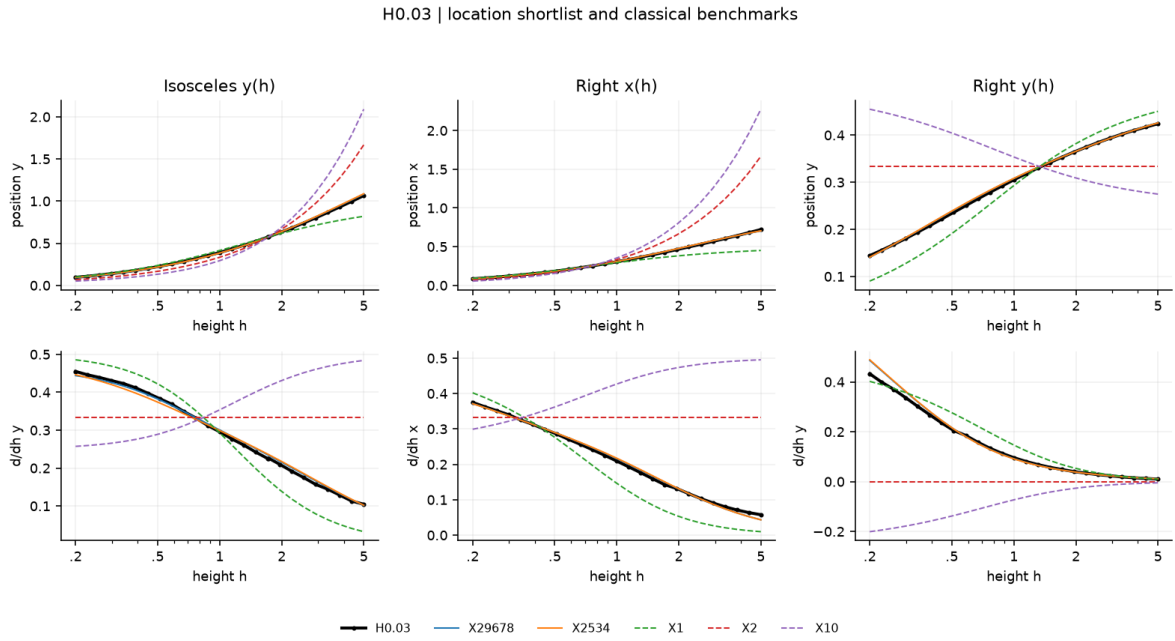
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X29678	0.00655751	0.00546278	0.0127374
X2534	0.00672261	0.00662839	0.0149772
X5393	0.00684258	0.00686945	0.0151762
X53841	0.0069209	0.00671024	0.0144934
X27577	0.00712187	0.00589488	0.0127534

For location winner X29678, the separately computed derivative-fingerprint RMS discrepancy is 0.0134405, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0440766, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.197374. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.



Atlas figure 3a. H0.03 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 3b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.4 H0.1: Heat maximum at dimensionless time 0.1

Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.1) = \arg \max_T e^{-0.1L^2 A} 1, \quad A = -\Delta_D$$

Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time.

Reference calculation. In the 3-4-5 triangle: (1.0955679, 0.9974450); normalized barycentrics (0.3936264, 0.2738920, 0.3324817). Method: FEM subdivisions 40 to 56. Last position change is 0.000117 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Unique interior maximum at every positive time, by preservation of log-concavity and the analytic line-restriction argument. This supplies existence and uniqueness, not a quantitative distance-to-side bound.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

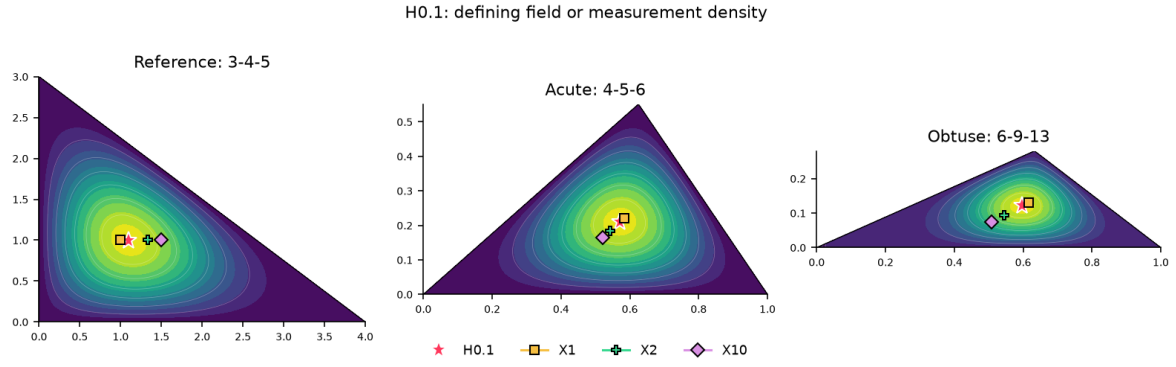
FEM subdivisions	Finite-motion ratio
128	-0.050493940
192	-0.081849800
256	-0.068014129

These are the shared retained thin-triangle heat witness, evaluated at the stated time for each shape. Higher modes are negligible in this example at late dimensionless times; that does not bound the spatial error in the ground-state maximum.

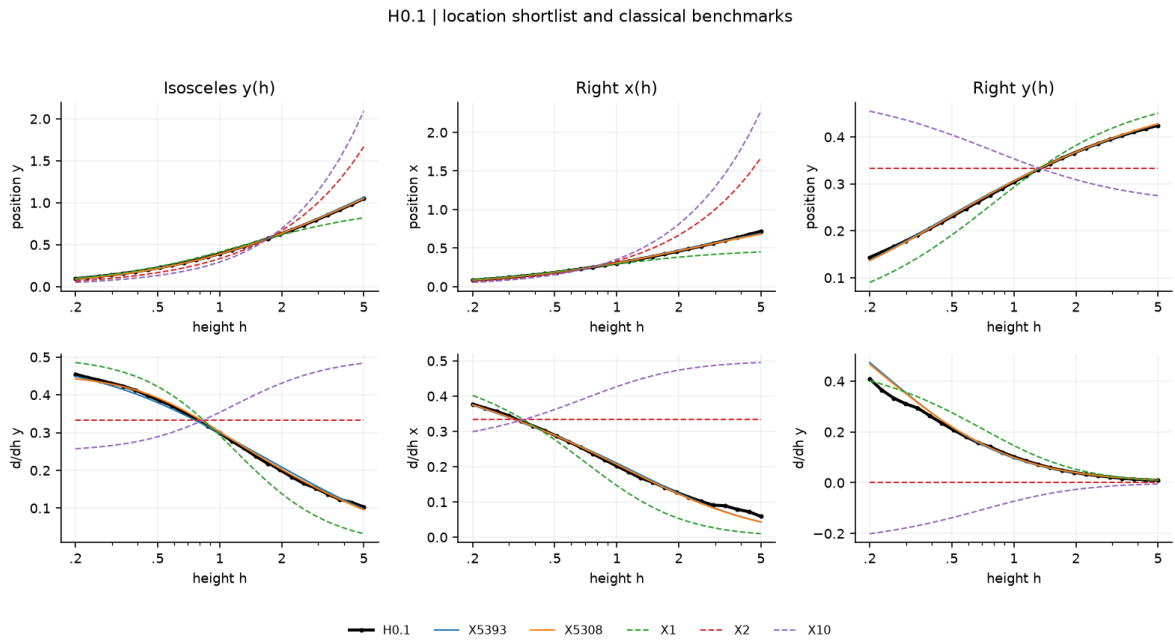
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5393	0.00698709	0.00699028	0.0151762
X5308	0.00719351	0.00614742	0.0137549
X29678	0.00720552	0.00613381	0.0127374
X38825	0.00722733	0.00638409	0.0131781
X17022	0.0072619	0.00635158	0.0145388

For location winner X5393, the separately computed derivative-fingerprint RMS discrepancy is 0.0191678, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0330423, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.0696081. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.



Atlas figure 4a. H0.1 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 4b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.5 H0.3: Heat maximum at dimensionless time 0.3

Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.3) = \arg \max_T e^{-0.3L^2 A} 1, \quad A = -\Delta_D$$

Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time.

Reference calculation. In the 3-4-5 triangle: (1.0954507, 0.9975085); normalized barycentrics (0.3936345, 0.2738627, 0.3325028). Method: FEM subdivisions 40 to 56. Last position change is 0.000116 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Unique interior maximum at every positive time, by preservation of log-concavity and the analytic line-restriction argument. This supplies existence and uniqueness, not a quantitative distance-to-side bound.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

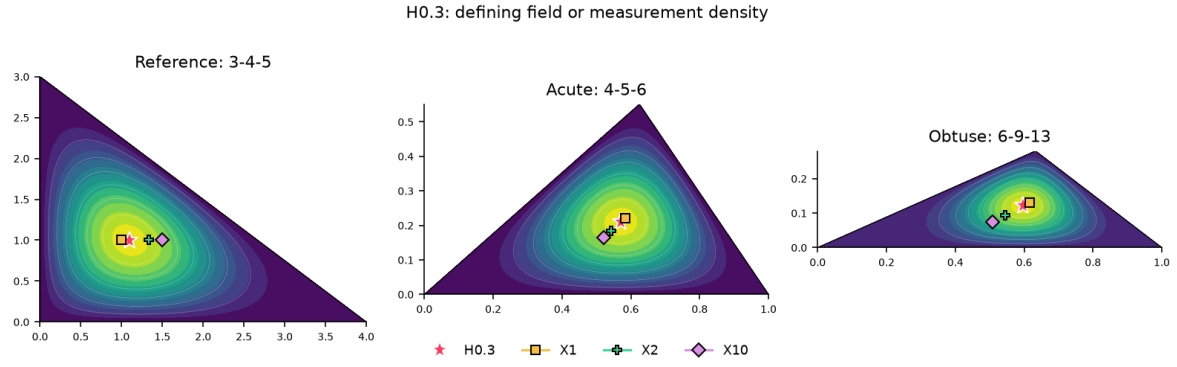
FEM subdivisions	Finite-motion ratio
128	-0.050493940
192	-0.081849800
256	-0.068014129

These are the shared retained thin-triangle heat witness, evaluated at the stated time for each shape. Higher modes are negligible in this example at late dimensionless times; that does not bound the spatial error in the ground-state maximum.

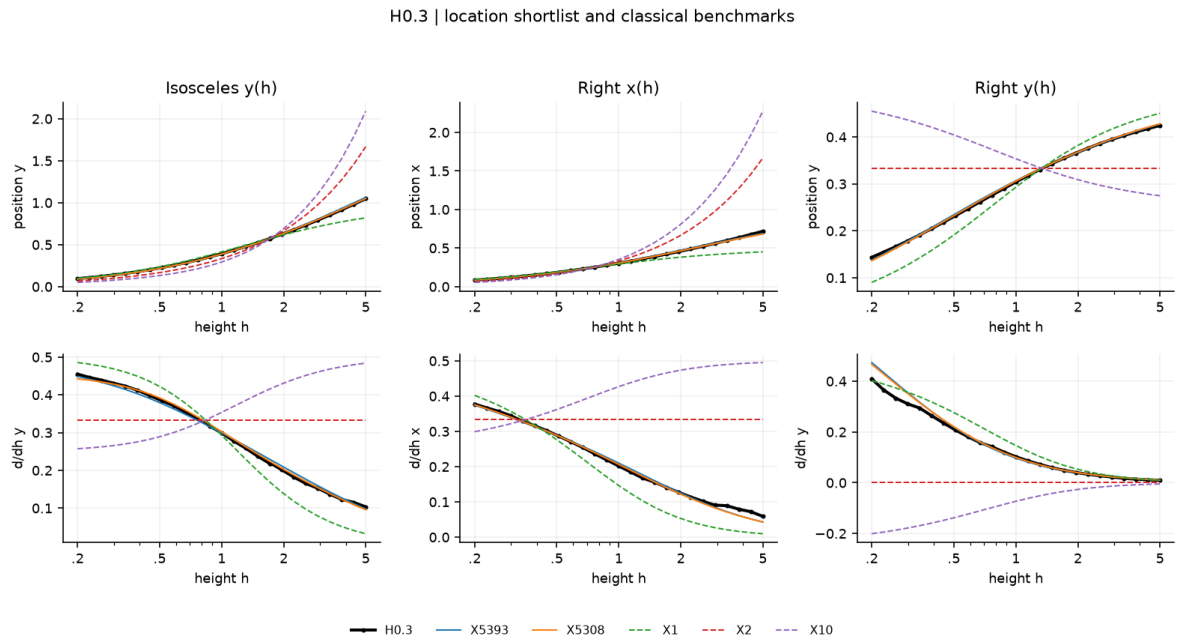
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5393	0.00699033	0.00699275	0.0151762
X5308	0.00719503	0.00614886	0.0137549
X29678	0.00721136	0.00613925	0.0127374
X38825	0.0072272	0.00638408	0.0131781
X17022	0.00726172	0.00635144	0.0145388

For location winner X5393, the separately computed derivative-fingerprint RMS discrepancy is 0.0191963, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0330478, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.0696081. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.



Atlas figure 5a. H0.3 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 5b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.6 H1.0: Heat maximum at dimensionless time 1.0

Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(1.0) = \arg \max_T e^{-1.0L^2 A} 1, \quad A = -\Delta_D$$

Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time.

Reference calculation. In the 3-4-5 triangle: (1.0954507, 0.9975085); normalized barycentrics (0.3936345, 0.2738627, 0.3325028). Method: FEM subdivisions 40 to 56. Last position change is 0.000116 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Unique interior maximum at every positive time, by preservation of log-concavity and the analytic line-restriction argument. This supplies existence and uniqueness, not a quantitative distance-to-side bound.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

The inherited electro-heat experiment met all of its joint targets on 94/100 grid shapes and 37/40 holdouts. Those joint counts include other fields; they are not a separate certification of this selector. Exact conformal evaluations and finite-element maxima have different error sources.

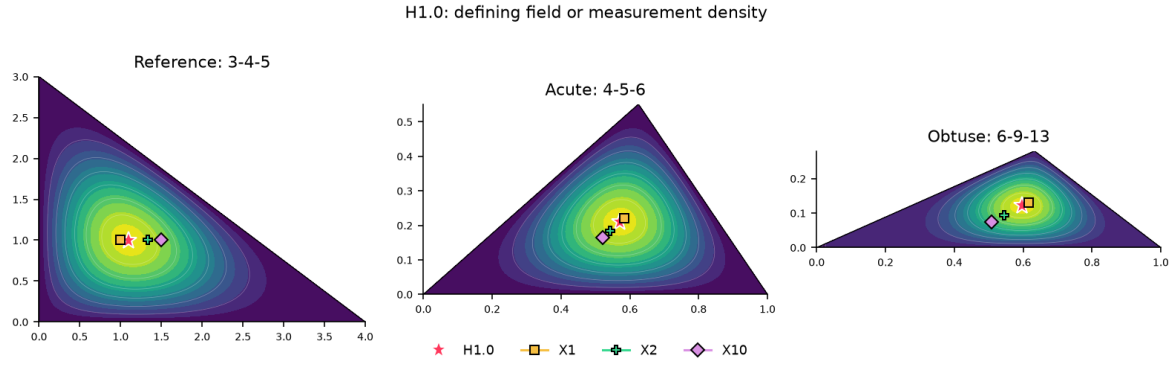
FEM subdivisions	Finite-motion ratio
128	-0.050493940
192	-0.081849800
256	-0.068014129

These are the shared retained thin-triangle heat witness, evaluated at the stated time for each shape. Higher modes are negligible in this example at late dimensionless times; that does not bound the spatial error in the ground-state maximum.

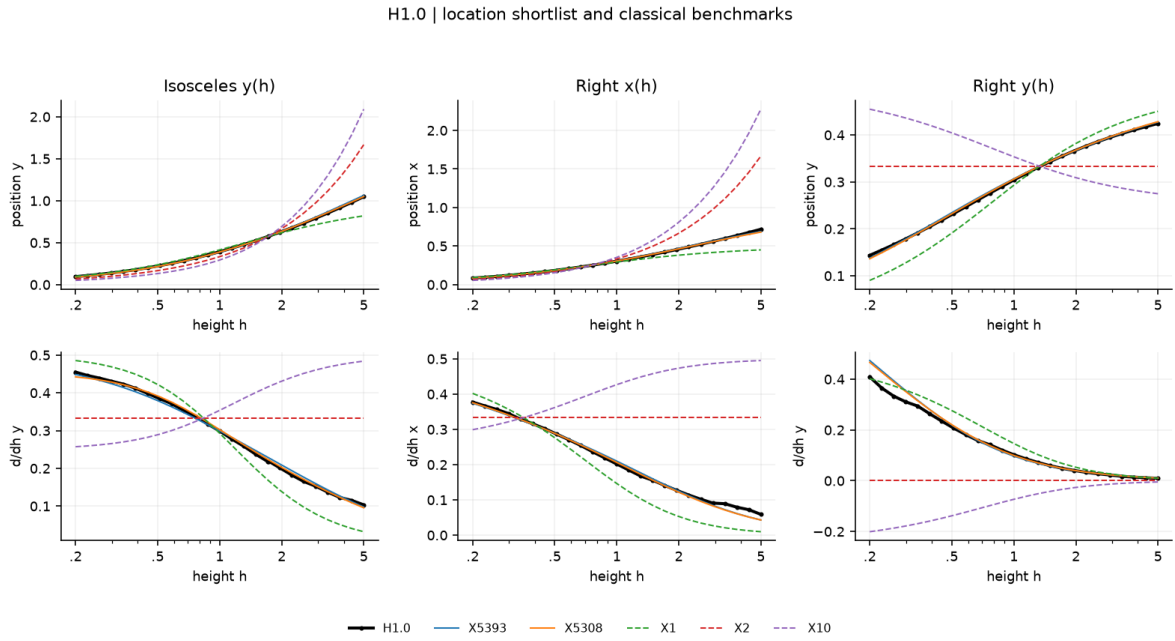
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5393	0.00699033	0.00699275	0.0151762
X5308	0.00719503	0.00614886	0.0137549
X29678	0.00721136	0.00613925	0.0127374
X38825	0.0072272	0.00638408	0.0131781
X17022	0.00726172	0.00635144	0.0145388

For location winner X5393, the separately computed derivative-fingerprint RMS discrepancy is 0.0191963, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0330478, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.0696081. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.



Atlas figure 6a. H1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 6b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.7 Abs0.0: Production–removal concentration mean, $\mu=0$

Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 0c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual.

Reference calculation. In the 3-4-5 triangle: (1.2170772, 1.0006773); normalized barycentrics (0.3621716, 0.3042693, 0.3335591). Method: Ritz degree 17 to 23. Last position change is 6.71e-09 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry. See Appendix D.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 7.66774e-05 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

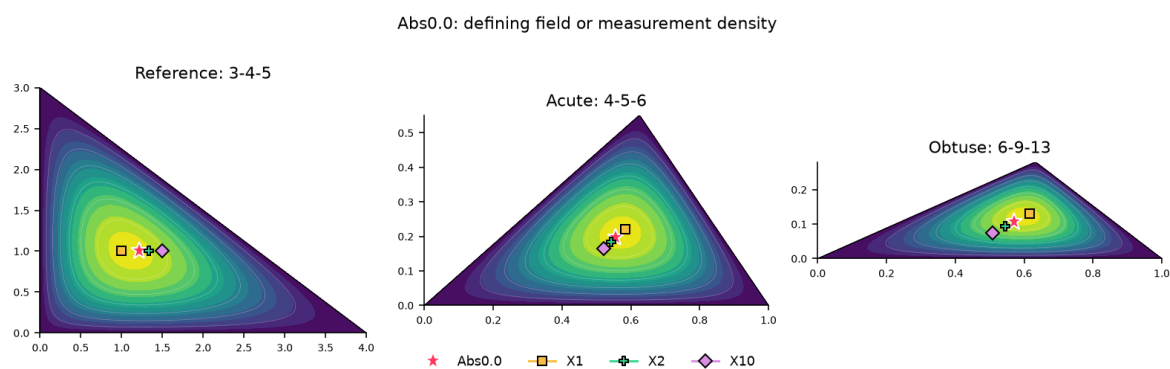
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X30105	0.000752051	0.000724443	0.0021002
X37870	0.00214214	0.00213078	0.00370082
X16831	0.00252882	0.00237092	0.00553802
X20135	0.00261556	0.00261313	0.00398413
X51108	0.0026821	0.00295312	0.00751632

For location winner X30105, the separately computed derivative-fingerprint RMS discrepancy is 0.00151123, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00662568, using 300/300 finite vertex margins. The candidate’s minimum screened margin is 0.189351. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

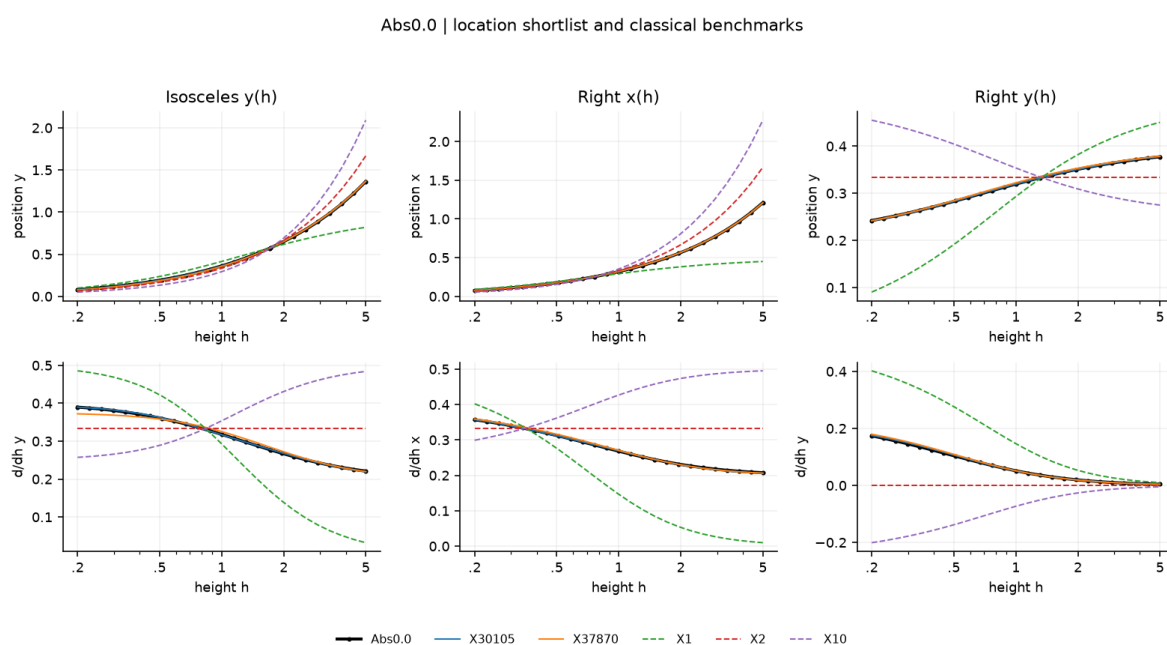
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000259866 diameter and maximum error 0.000520116. Baseline: 0.4 I + 0.6 G; $q=z$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. At zero removal this is the torsion-density mean, not the torsion maximum; there are no bulk removal events at this endpoint.

The archived numerical almost-equilateral coefficients are $\alpha=0.3899782$, $\gamma=0.2766894$, $\beta=-0.0743709$, $\delta=-0.0118697$, $\eta=-0.0208339$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 7a. Abs0.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 7b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.8 Abs1.0: Production–removal concentration mean, $\mu=1$

Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 1c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual.

Reference calculation. In the 3-4-5 triangle: (1.2175718, 1.0006652); normalized barycentrics (0.3620520, 0.3043929, 0.3335551). Method: Ritz degree 17 to 23. Last position change is 6.8e-09 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry. See Appendix D.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 7.66669e-05 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

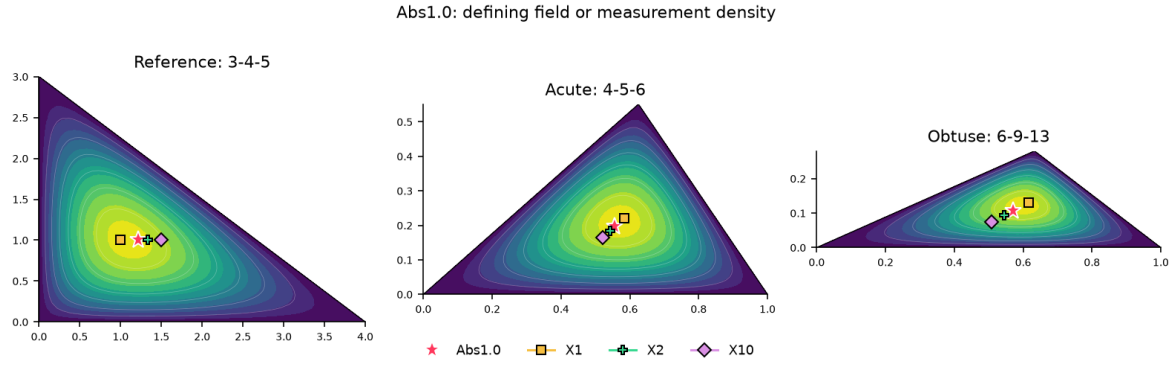
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X30105	0.000753835	0.000735388	0.00208656
X37870	0.00210025	0.00209181	0.00367286
X16831	0.00247744	0.0023214	0.00551937
X20135	0.00266955	0.0026683	0.00403759
X51108	0.00270994	0.00297559	0.00751629

For location winner X30105, the separately computed derivative-fingerprint RMS discrepancy is 0.00147015, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00660725, using 300/300 finite vertex margins. The candidate’s minimum screened margin is 0.189351. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

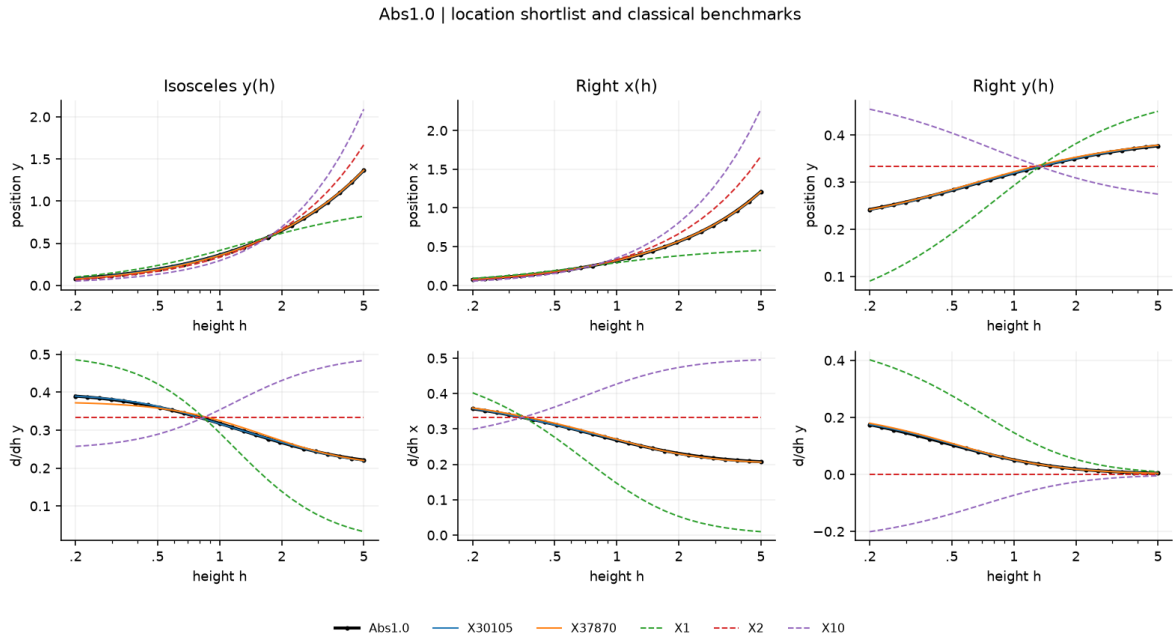
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000258901 diameter and maximum error 0.000519689. Baseline: 0.4 I + 0.6 G; $q=z$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. For positive removal the event-density centroid is the same. The large- μ limit is X(2); fixed- μ thin strip asymptotics give horizontal limit $3x/5$.

The archived numerical almost-equilateral coefficients are $\alpha=0.3896694$, $\gamma=0.2769982$, $\beta=-0.0739514$, $\delta=-0.0118121$, $\eta=-0.0207132$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 8a. Abs1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 8b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.9 Abs10.0: Production–removal concentration mean, $\mu=10$

Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 10c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual.

Reference calculation. In the 3-4-5 triangle: (1.2217632, 1.0005695); normalized barycentrics (0.3610360, 0.3054408, 0.3335232). Method: Ritz degree 17 to 23. Last position change is 7.61e-09 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry. See Appendix D.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 7.65717e-05 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

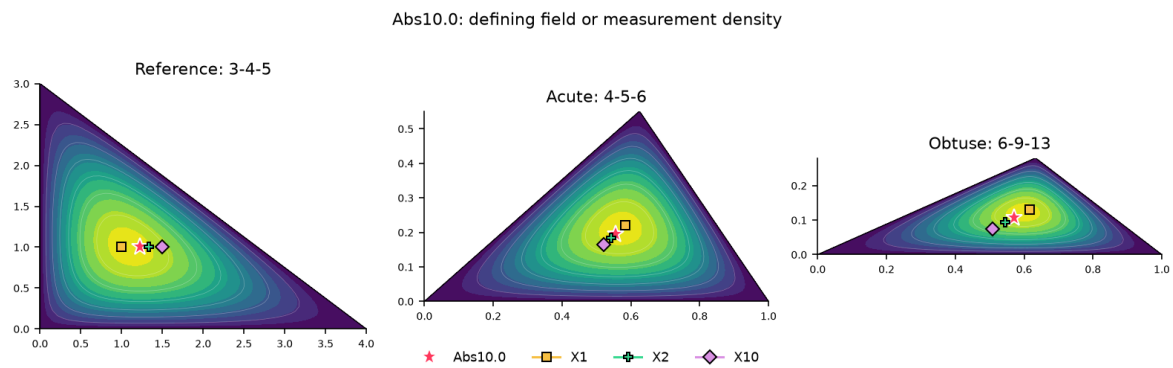
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X30105	0.00110312	0.00112483	0.00196418
X37870	0.00187236	0.00188269	0.00343048
X16831	0.00212494	0.00198556	0.00535213
X17321	0.00247063	0.00249083	0.0046329
X51108	0.00303952	0.00325017	0.00751601

For location winner X30105, the separately computed derivative-fingerprint RMS discrepancy is 0.0025271, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00668602, using 300/300 finite vertex margins. The candidate’s minimum screened margin is 0.189351. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

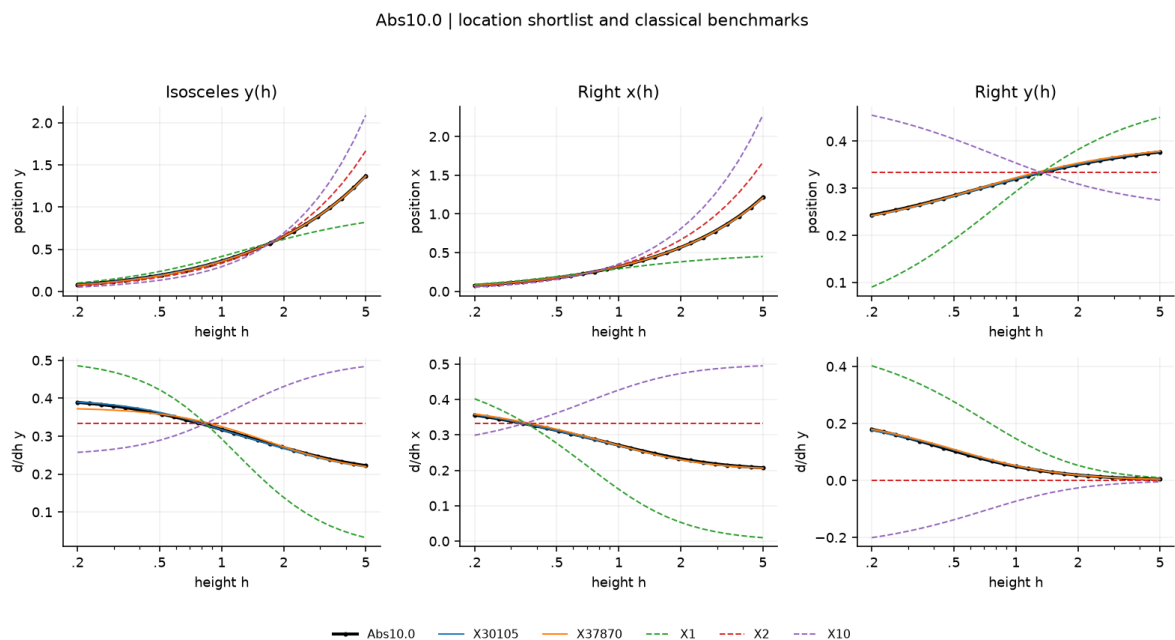
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000250835 diameter and maximum error 0.000515894. Baseline: 0.4 I + 0.6 G; $q=z$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. For positive removal the event-density centroid is the same. The large- μ limit is X(2); fixed- μ thin strip asymptotics give horizontal limit $3x/5$.

The archived numerical almost-equilateral coefficients are $\alpha=0.3870899$, $\gamma=0.2795777$, $\beta=-0.0704604$, $\delta=-0.0113237$, $\epsilon=-0.0197123$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 9a. Abs10.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 9b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.10 Abs100.0: Production–removal concentration mean, $\mu=100$

Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 100c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual.

Reference calculation. In the 3-4-5 triangle: (1.2481443, 1.0001988); normalized barycentrics (0.3545643, 0.3120361, 0.3333996). Method: Ritz degree 17 to 23. Last position change is 1.54e-08 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry. See Appendix D.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 7.56361e-05 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

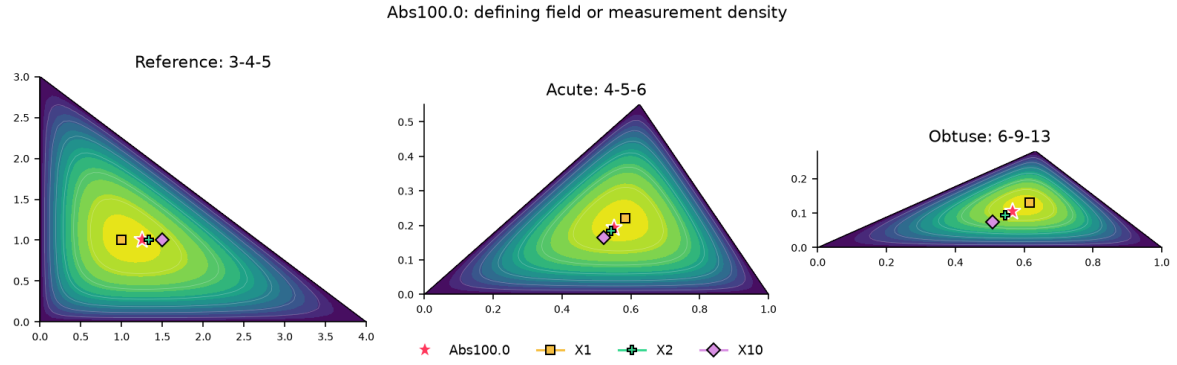
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X32009	0.00321593	0.00312739	0.00755951
X70563	0.00401207	0.00396792	0.00670264
X31336	0.0040506	0.00395864	0.0103054
X51154	0.00438534	0.00452585	0.00986297
X16830	0.00463343	0.00443055	0.0115578

For location winner X32009, the separately computed derivative-fingerprint RMS discrepancy is 0.00860812, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0145301, using 300/300 finite vertex margins. The candidate’s minimum screened margin is 0.148684. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

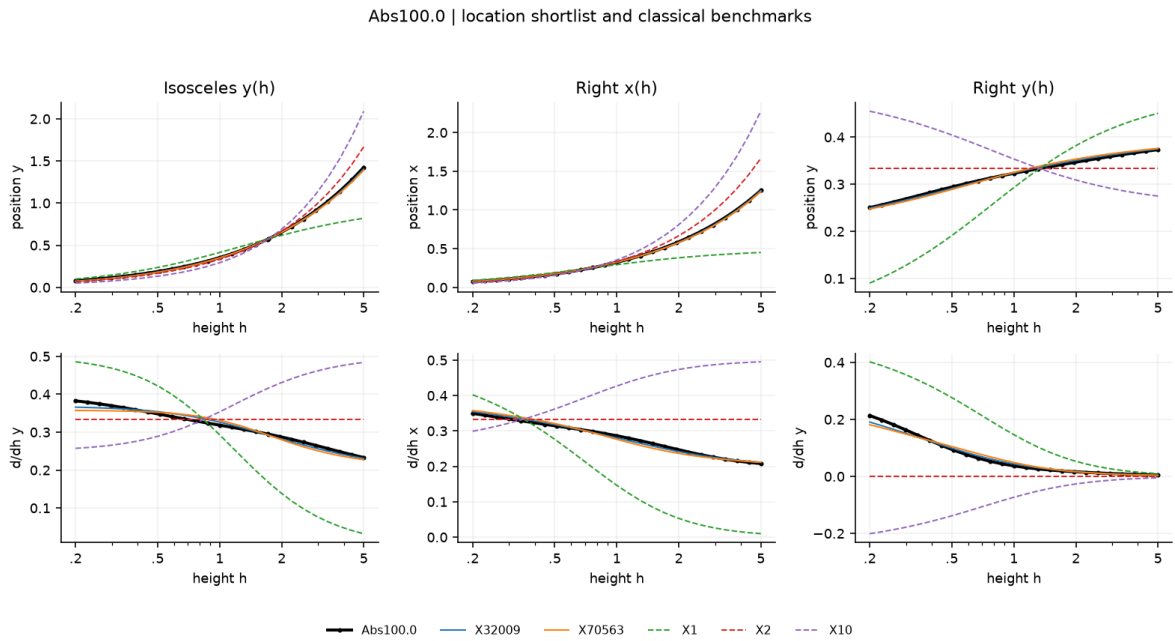
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000221468 diameter and maximum error 0.000526772. Baseline: 0.4 I + 0.6 G; $q=z$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. For positive removal the event-density centroid is the same. The large- μ limit is X(2); fixed- μ thin strip asymptotics give horizontal limit $3x/5$.

The archived numerical almost-equilateral coefficients are $\alpha=0.3722268$, $\gamma=0.2944405$, $\beta=-0.0507057$, $\delta=-0.0083295$, $\epsilon=-0.0141251$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 10a. Abs100.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 10b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.11 Bio1.1: Logistic population mean, kappa=1.1

Let a population diffuse, grow logistically and die at the triangular boundary. Choose the positive persistent equilibrium, not extinction.

$$-\Delta n = 1.1\lambda_0 n(1-n), \quad n|_{\partial T} = 0, \quad C = \int pn / \int n$$

Compute the first Dirichlet eigenvalue, minimize the positive-branch energy, then check the original logistic residual and integrate the density.

Reference calculation. In the 3-4-5 triangle: (1.1566856, 1.0041136); normalized barycentrics (0.3761241, 0.2891714, 0.3347045). Method: Ritz degree 17 to 23. Last position change is 1.74e-10 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The unique positive continuum branch has a strictly interior mean.

Attractivity. A strong finite numerical counterexample is independently reproduced by quadratic FEM; no interval certificate is supplied.

Inherited resolution record: 98/100 training shapes and 39/40 holdouts meet the declared successive-change tolerance. Largest training change: 0.00143393 diameter. Coarse screen: 8 sampled triangles with a negative response; those counts are not established continuum counterexamples.

Independent FEM subdivisions	Finite-motion ratio
48	-0.109269055
80	-0.109350445
128	-0.109219537

The original logistic residual and field extrema accompany both endpoints in `biomass_independent_witness.json`. This independent discretization strengthens the negative numerical evidence without supplying an interval certificate.

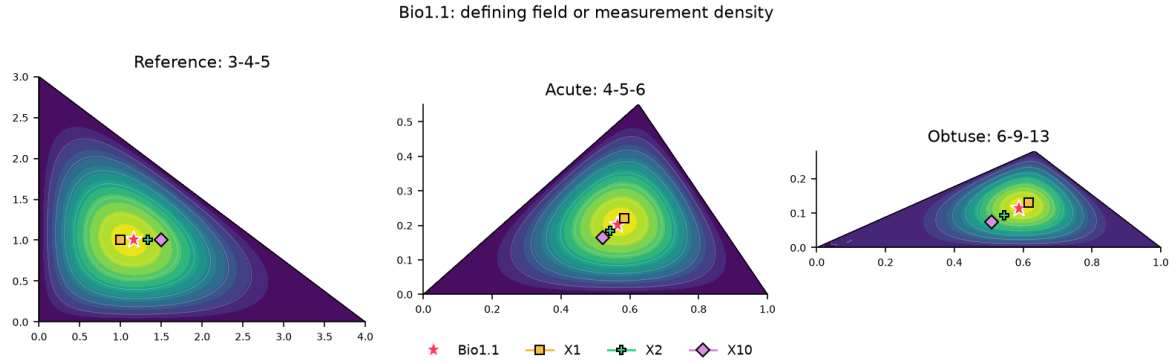
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X37884	0.00536317	0.00629449	0.0210963
X29640	0.00872738	0.00763495	0.0239452
X63628	0.00878533	0.00771972	0.023898
X25422	0.0094939	0.0085477	0.0262081
X30571	0.00957107	0.00865033	0.0239936

For location winner X37884, the separately computed derivative-fingerprint RMS discrepancy is 0.00670977, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0563817, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.243833. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

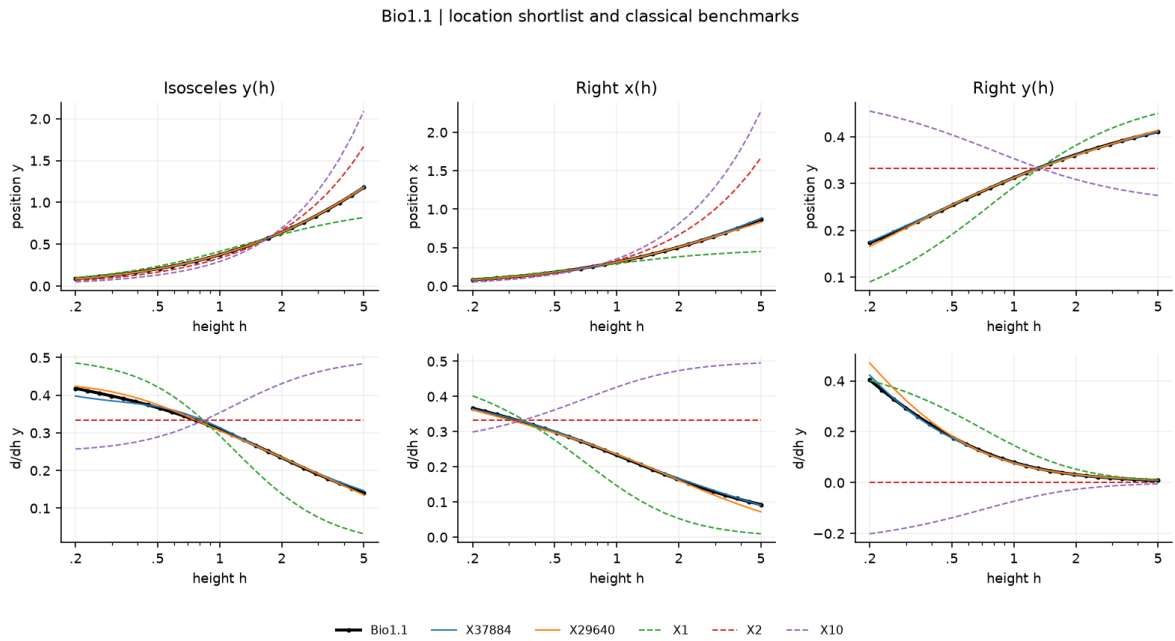
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.00573209 diameter and maximum error 0.0260166. Baseline: G; powers start at $z \hat{=} 0$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Near threshold the mean tends to the first moment of ψ_0 , not ψ_0^2 . Large growth tends to X(2). Fixed κ includes the shape dependence of λ_0 .

The archived numerical almost-equilateral coefficients are $\alpha=0.4165254$, $\gamma=0.2501427$, $\beta=-0.1125212$, $\delta=-0.0157900$, $\epsilon=-0.0322322$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 11a. Bio1.1 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 11b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.12 Bio2.0: Logistic population mean, kappa=2

Let a population diffuse, grow logistically and die at the triangular boundary. Choose the positive persistent equilibrium, not extinction.

$$-\Delta n = 2\lambda_0 n(1 - n), \quad n|_{\partial T} = 0, \quad C = \int pn / \int n$$

Compute the first Dirichlet eigenvalue, minimize the positive-branch energy, then check the original logistic residual and integrate the density.

Reference calculation. In the 3-4-5 triangle: (1.1834208, 1.0028060); normalized barycentrics (0.3698761, 0.2958552, 0.3342687). Method: Ritz degree 17 to 23. Last position change is 7.04e-11 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The unique positive continuum branch has a strictly interior mean.

Attractivity. Global attractivity is unresolved; numerical screening is not a theorem.

Inherited resolution record: 98/100 training shapes and 37/40 holdouts meet the declared successive-change tolerance. Largest training change: 0.000703176 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

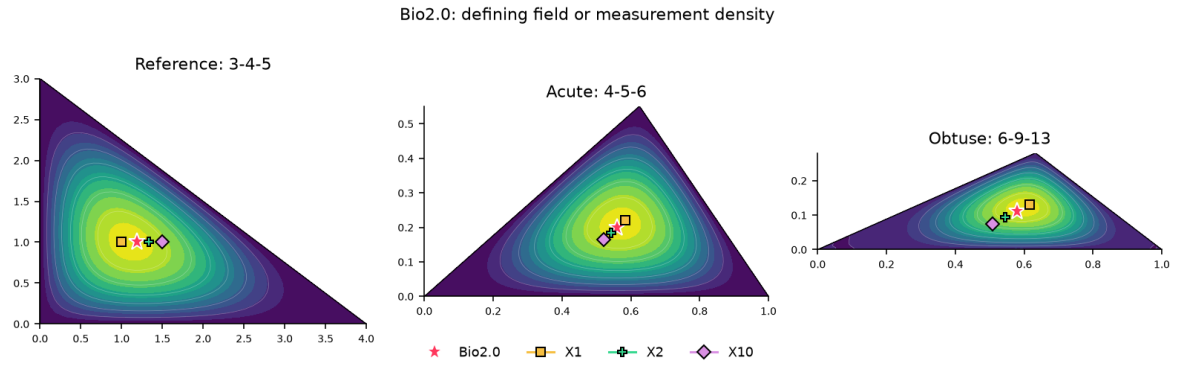
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X58387	0.00532099	0.00637545	0.0192593
X32090	0.00646407	0.00617431	0.0148107
X24857	0.00761763	0.00891416	0.0259895
X51488	0.00805171	0.00765564	0.0171067
X10587	0.00853929	0.0105171	0.034623

For location winner X58387, the separately computed derivative-fingerprint RMS discrepancy is 0.00959455, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0242982, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.0757506. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

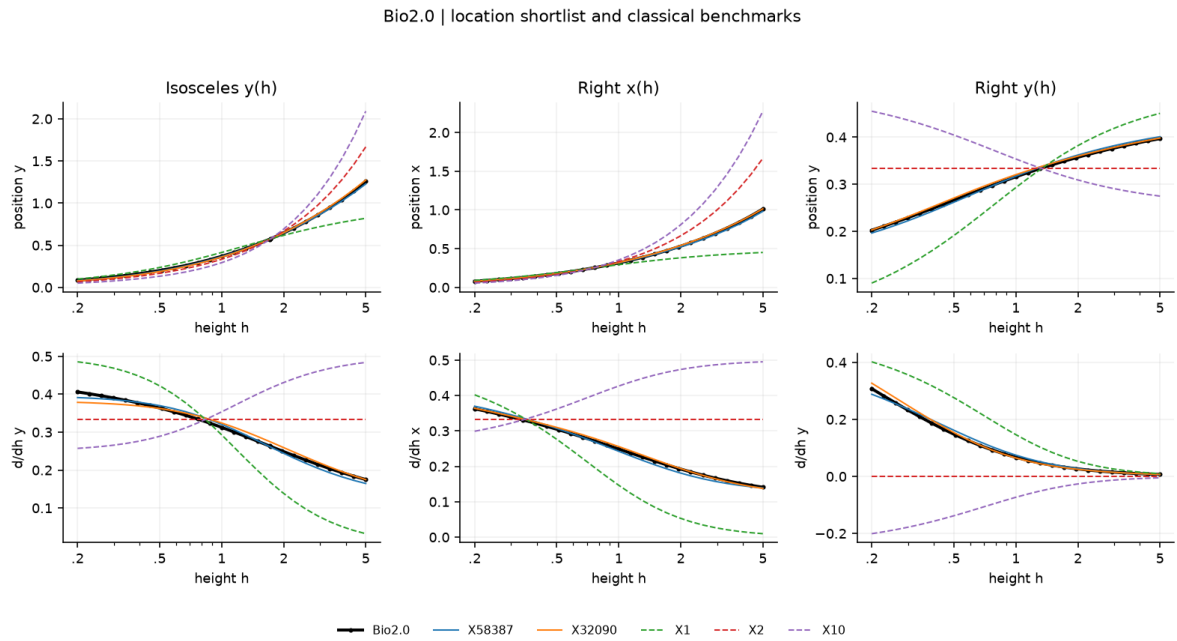
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.00114208 diameter and maximum error 0.00472535. Baseline: G; powers start at z^0 . Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Near threshold the mean tends to the first moment of ψ_0 , not ψ_0^2 . Large growth tends to X(2). Fixed κ includes the shape dependence of λ_0 .

The archived numerical almost-equilateral coefficients are $\alpha=0.4044965$, $\gamma=0.2621714$, $\beta=-0.0955115$, $\delta=-0.0138724$, $\epsilon=-0.0272140$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 12a. Bio2.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 12b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.9.13 Bio10.0: Logistic population mean, kappa=10

Let a population diffuse, grow logistically and die at the triangular boundary. Choose the positive persistent equilibrium, not extinction.

$$-\Delta n = 10\lambda_0 n(1 - n), \quad n|_{\partial T} = 0, \quad C = \int pn / \int n$$

Compute the first Dirichlet eigenvalue, minimize the positive-branch energy, then check the original logistic residual and integrate the density.

Reference calculation. In the 3-4-5 triangle: (1.2633595, 1.0003868); normalized barycentrics (0.3506978, 0.3158399, 0.3334623). Method: Ritz degree 17 to 23. Last position change is 1.42e-09 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The unique positive continuum branch has a strictly interior mean.

Attractivity. Global attractivity is unresolved; numerical screening is not a theorem.

Inherited resolution record: 91/100 training shapes and 36/40 holdouts meet the declared successive-change tolerance. Largest training change: 0.000552309 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

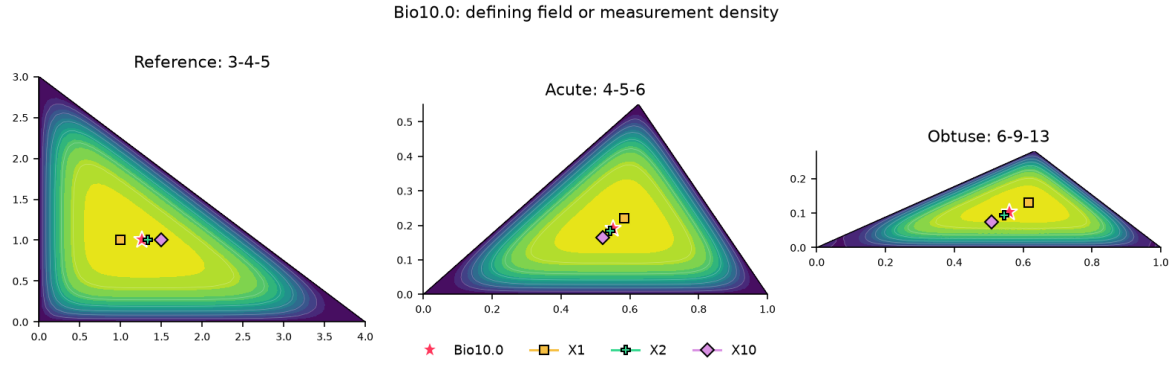
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X29648	0.00256368	0.00311174	0.0100531
X4657	0.00265298	0.00318012	0.00980612
X62648	0.00273426	0.00327087	0.0102929
X29826	0.00281212	0.00332008	0.0102989
X71484	0.00281681	0.00334577	0.00980612

For location winner X29648, the separately computed derivative-fingerprint RMS discrepancy is 0.00486924, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00769402, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.244877. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

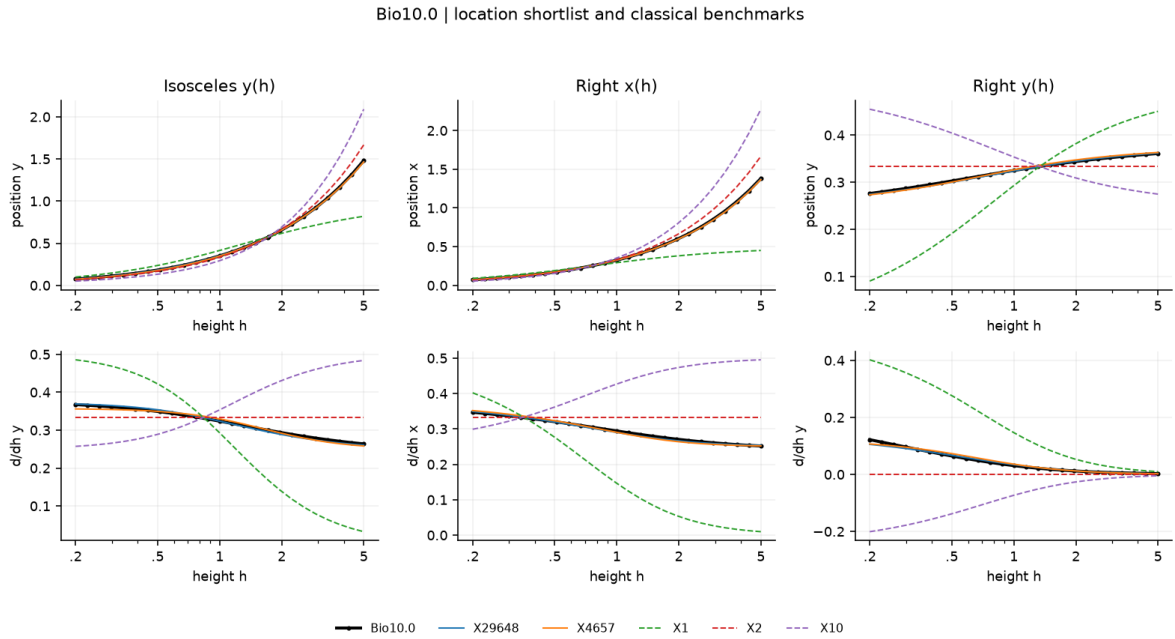
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.00021369 diameter and maximum error 0.00100796. Baseline: G; powers start at z^0 . Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Near threshold the mean tends to the first moment of ψ_0 , not ψ_0^2 . Large growth tends to X(2). Fixed κ includes the shape dependence of λ_0 .

The archived numerical almost-equilateral coefficients are $\alpha=0.3672410$, $\gamma=0.2994263$, $\beta=-0.0444836$, $\delta=-0.0071212$, $\epsilon=-0.0124550$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 13a. Bio10.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 13b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

6.10 Further physical realizations and controlled extensions

A pressure-driven microchannel can reveal $F1$ through the greatest tracked axial speed and $Abs(0)$ through the first moment of tracer flux. Edge-cooled sheets can be imaged thermally at equal values of $\kappa t/L^2$. A uniformly fed diffusive reactor with tunable decay measures $Abs(\mu)$. The same mathematics also models steady heat with uniform internal heating and a linear bulk cooling term; this is a legitimate thermodynamic realization of the absorption family.

Non-Newtonian power-law duct flow gives a useful connection to the previous chapter. If the shear stress is proportional to $|\text{grad } U|$ to the power $p-2$ times $\text{grad } U$, axial balance gives the p -torsion equation. Its maximum and flux mean are exactly $Mem(p)$ and $MemMean(p)$ under

the same normalization. This adds physical interpretations to existing entries rather than new independent centers. The Newtonian case $p=2$ is F1 and Abs(0).

A reflecting rather than absorbing thermal or concentration boundary makes uniform forcing and uniform coefficients spatially uniform at equilibrium, so its first moment is X2 and a maximum may fail to select a point. Robin boundaries introduce a dimensionless transfer coefficient and a new family. Advection introduces a direction and usually breaks the unoriented triangle-center definition unless the flow direction is itself selected equivariantly. These extensions require explicit parameter conventions and new surveys; the present chapter does not claim their numerical investigation is complete.

The chapter's PDE solvers, 3-4-5 data, cross-family identity checks and figures are in the handoff. The individual entries identify the inherited evidence file and separate proven interiority, unique-field or unique-maximum results, stable numerical negative witnesses, unresolved candidates and no-failure screens.

Chapter 7

Optical and illumination centers

7.1 An optical experiment must specify a scale and a measurement

An illuminated triangular aperture produces a field on a screen, but not every feature of that field is an intrinsic point of the aperture. The location may depend on wavelength, screen distance, polarization, incident phase and detector response. A bright far-field direction is a direction, not a point in the triangular plane. An intensity minimum can be nonunique or occur outside the projected aperture.

We seek reproducible geometric experiments. The aperture is uniformly illuminated with constant phase. A declared dimensionless propagation distance or blur width is held fixed while the triangle is moved, reflected or scaled. We then specify whether the detector reads a maximum, a minimum, or a weighted first moment. The resulting families may be natural and clean without being parameter-free. Their chosen parameter values are conventions, not newly discovered constants of nature.

The reference triangle has vertices $(0,0),(4,0),(0,3)$, area $K=6$, diameter $D=5$ and $L^2=(a^2+b^2+c^2)/3=50/3$. The plotted comparison points are $X_1=(1,1)$, $X_2=(4/3,1)$ and $X_{10}=(3/2,1)$. All area integrals are over the aperture unless another domain is explicitly named.

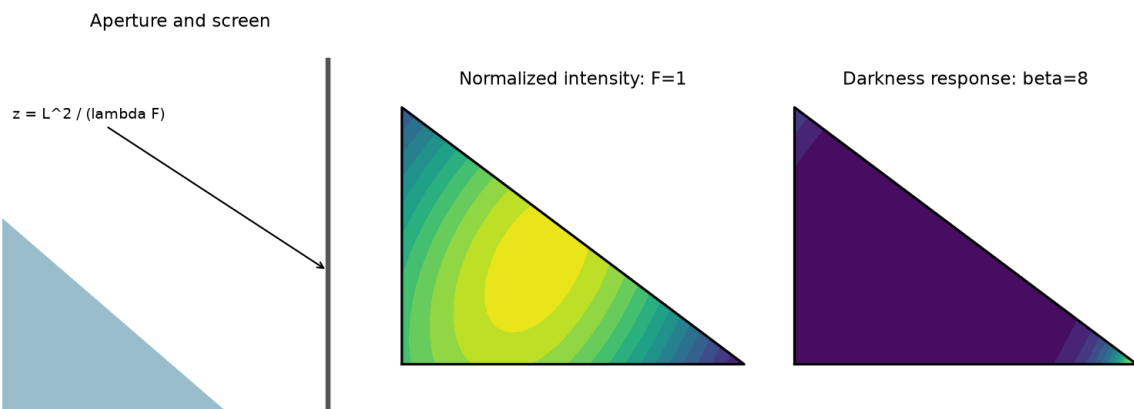


Figure 1. Left: uniform triangular aperture and screen at fixed Fresnel number. Middle: normalized intensity over the projected triangle at $F=1$. Right: its exponential darkness response at $\beta=8$. The detector response changes the spatial weighting used to define the center.

7.2 Deriving the Fresnel field

7.2.1 From propagation distance to a quadratic phase

In scalar diffraction, each aperture element contributes an oscillatory wave to the observation point. If the screen is parallel to the aperture at distance z , the path from source q to transverse screen coordinate p has length $\sqrt{z^2 + |p-q|^2}$. For transverse separation small relative to z , expand it as $z + |p-q|^2/(2z)$. The common phase $\exp(ikz)$, the constant distance factor and any nonzero uniform illumination amplitude cannot move an intensity-normalized center. The remaining paraxial amplitude is proportional to

$$A_F(p) = \frac{1}{K} \int_T \exp\left(\frac{i\pi F |p-q|^2}{L^2}\right) dq, \quad F = \frac{L^2}{\lambda z}.$$

Its intensity is $I_F = |A_F|^2$. The scalar paraxial approximation neglects vector polarization and higher path-length terms. It is most credible when the aperture dimensions and transverse coordinates are small compared with z ; fixing F alone does not guarantee that condition. For a laboratory realization choose wavelength small relative to L and then set z accordingly. The introductory diffraction source [28] explains the underlying Fresnel approximation.

7.2.2 What wavelength independence means here

Fix F as part of the protocol. If wavelength changes, move the screen to $z = L^2/(\lambda F)$. The normalized integral then depends only on triangle shape and the dimensionless coordinate p/L . That is the intended wavelength-independent geometric construction. At a physically fixed screen, changing wavelength changes F and generally changes the center; the apparatus is not achromatic in that sense.

Under a similarity $p \rightarrow sp+t$ and $q \rightarrow sq+t$, area scales by s^2 and L^2 scales by s^2 . The normalized field at corresponding coordinates is unchanged, so all the moment constructions below scale correctly. Relabelling the triangle changes no integral. Uniform incident phase and isotropy are essential: a fixed laboratory phase gradient or preferred polarization can introduce a direction not determined by the triangle.

7.3 Why there is no exact interior Fresnel dark point

The first natural proposal is the point where the amplitude vanishes. In this particular model it fails throughout the closed projected triangle, at every finite positive Fresnel number. The reason is geometric and can be shown without numerical root searches.

Take p inside the triangle. In polar coordinates centered at p , each ray stays in the triangle for $0 \leq r \leq R_p(\theta)$. Put $c = \pi F/L^2$. Integrate the radial factor first:

$$\int_T e^{ic|q-p|^2} dq = \frac{1}{2ic} \int_0^{2\pi} \left(e^{icR_p(\theta)^2} - 1 \right) d\theta.$$

The imaginary part is

$$\text{Im} \int_T e^{ic|q-p|^2} dq = \frac{1}{2c} \int_0^{2\pi} \left[1 - \cos(cR_p(\theta)^2) \right] d\theta.$$

Every integrand is nonnegative. On an open range of directions hitting one straight side, the ray length varies continuously and is not constant. It cannot take only the discrete radii whose squared phase is an integer multiple of 2π . Thus the integral is strictly positive. If p lies on an edge or vertex, integrate only over its inward angular sector; an open set of rays still ends on a varying straight-side distance, so the same strictness argument applies.

Multiplying the amplitude by a global phase changes its imaginary part but cannot create a zero. Consequently the theorem is about nonvanishing, not about the special choice of phase convention. A circular aperture is not a counterexample: at its center R is constant, so the strictness step fails and exact cancellation can occur at selected parameters.

This proof excludes an exact dark-point center in the projected triangle for a uniformly phased triangular aperture under scalar Fresnel propagation. It does not exclude exterior zeros, an opaque obstacle's complementary field, a deliberately phase-structured aperture or a different electromagnetic model. It also does not make numerical intensity minima uninteresting. It tells us to name and interpret them correctly.

7.3.1 A minimum is not automatically a triangle center

The continuous intensity has a minimum on the compact projected triangle. That minimum can lie at the boundary, and different points can tie. A root finder returning one minimum does not certify that it found the global one. Averaging several discovered minima need not produce another minimum and can depend on which minima the search found. The inherited DarkMin diagnostics are therefore retained as optimization diagnostics, not promoted to single-valued canonical triangle-center entries.

The following alternative uses the entire intensity image and is always a unique point. It preserves the intuitive preference for darkness without an arbitrary tie-breaker.

7.4 Darkness-weighted centers

7.4.1 A calibrated detector response

First divide intensity by its mean over the triangular observation window. Then apply a strictly positive response that decreases with brightness:

$$J_F(p) = \frac{|A_F(p)|^2}{K^{-1} \int_T |A_F|^2 dp}, \quad w_{F,\beta}(p) = e^{-\beta J_F(p)}.$$

The darkness center is its first moment:

$$C_{F,\beta} = \frac{\int_T p w_{F,\beta}(p) dp}{\int_T w_{F,\beta}(p) dp}, \quad \beta > 0.$$

A camera can implement this by calibrating and normalizing its intensity image, applying the exponential response in software or an equivalent detector calibration, and calculating the centroid. This is an operational definition. The exponential is a declared measurement rule, not a claim that the raw optical field naturally possesses that centroid.

The denominator is positive, and every barycentric coordinate has a positive weighted integral. Thus every finite-parameter darkness center is unique as a moment and strictly inside the triangle. No uniqueness theorem for intensity minima is needed. The atlas studies $F=0.5, 1, 2, 4$ with $\beta=1$

and 8. The weaker response samples a broad area; the stronger response emphasizes darker patches.

At $\beta=0$ the response is constant and the center is G. At F tending to zero the amplitude becomes uniformly constant, so for any fixed finite β the center also tends to G. These are exact limiting identifications with X2. At β tending to infinity the density can concentrate on multiple or boundary minima. The positive finite- β location theorem does not justify a smooth, single-valued limiting minimum selector.

7.4.2 Persistence over a range of propagation parameters

An alternative avoids choosing one Fresnel number. Normalize each intensity at its own F , average uniformly in $\log F$ over $[1/2, 2]$, and then apply the same response:

$$\bar{J}(p) = \frac{1}{2 \log 2} \int_{1/2}^2 J_F(p) \frac{dF}{F}, \quad C_{\text{persist}, \beta} = \frac{\int_T p e^{-\beta \bar{J}(p)} dp}{\int_T e^{-\beta \bar{J}(p)} dp}.$$

The logarithmic interval gives equal weight to multiplicative scales above and below $F=1$. It is an explicit convention. Averaging center positions would be a different construction; averaging exponential responses before normalizing would be another. The present order of operations is part of the definition and must be reproduced by both numerical code and experiment. The atlas includes $\beta=1$ and 8.

7.4.3 Numerical quadrature and attractivity

Map the triangle affinely to a reference triangle. Gaussian product quadrature with the appropriate triangular Jacobian approximates its area integrals. For each observation point, evaluate the complex amplitude as a sum over source nodes, then normalize intensity over observation nodes and form the weighted mean. Increase the source and observation order together, and independently increase the log-frequency order for persistence.

At large F , the integrand oscillates rapidly; low-order quadrature can manufacture a nearly zero amplitude or move a narrow dark region. At large β , small intensity errors are magnified by the exponential. Therefore both phase resolution and response resolution matter. The new 3-4-5 reference compares spatial orders 22,34,46 and log-frequency orders 10,16,24. The reported changes measure successive numerical results, not rigorous upper bounds on omitted error.

For equilateral deformations one can differentiate the fixed-domain integral itself. This gives smoother response estimates than subtracting independently optimized extrema. In the archived calculations, strong darkness at $F=1$ and $F=2$ has a negative horizontal apex derivative. The corresponding approximate diagonal Jacobians are $(-0.22453347, 0.89120014)$ and $(-1.57953177, 2.24619843)$. Their traces equal $2/3$ as required by similarity equivariance. Multiple quadrature orders preserve the negative signs. These are strong numerical nonattractivity results, not interval-enclosed continuum certificates.

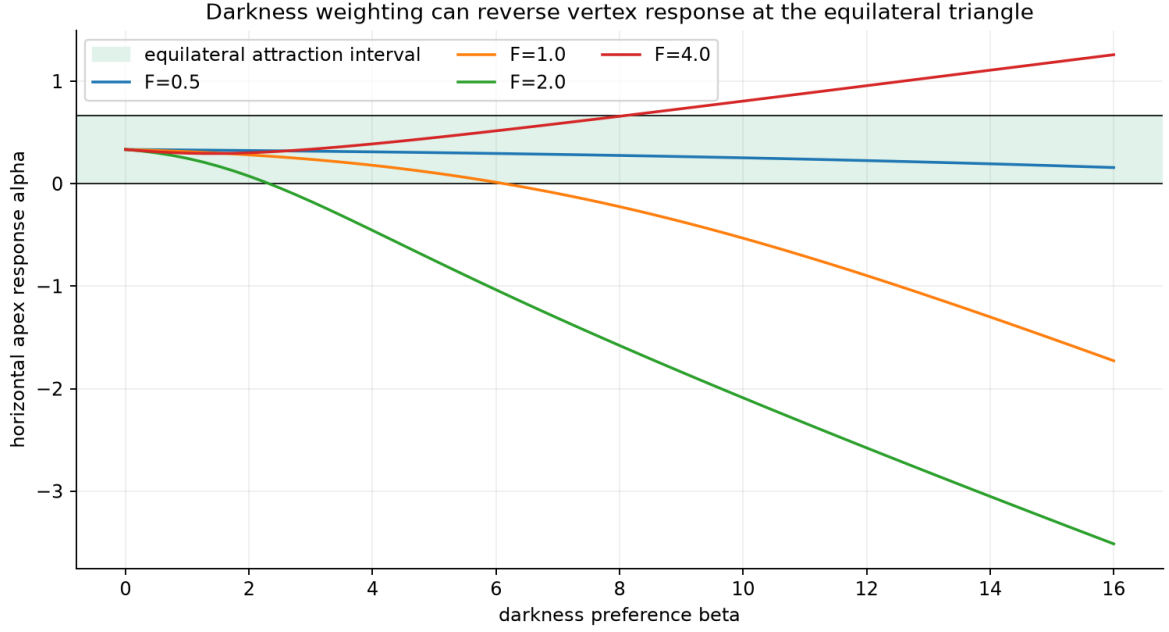


Figure 2. Archived equilateral darkness-response calculations as detector strength changes. A positive smooth measurement density guarantees an interior center, but does not guarantee that this center follows every moving vertex.

7.5 Nonlinear fluorescence and Gaussian image centroids

7.5.1 Linear imaging preserves the mass centroid

Blur the uniformly luminous triangle with a centered isotropic Gaussian of width $s = \sigma L$:

$$I_s(p) = \int_T \exp\left(-\frac{|p-q|^2}{2s^2}\right) dq.$$

The image extends over the whole plane. Its linear-detection centroid is G : each source point q contributes a centered Gaussian whose integral is constant and whose first moment is that constant times q . Integrating over q reproduces the uniform area first moment. This is the optical analogue of a uniform lamina's mass centroid.

If the detector response or fluorescence rate is proportional to I_s squared, the mean generally changes. Expanding the square introduces a pair of source points q, r . Complete the square:

$$|p-q|^2 + |p-r|^2 = 2\left|p - \frac{q+r}{2}\right|^2 + \frac{|q-r|^2}{2}.$$

The p -dependent Gaussian is centered at $(q+r)/2$. Integrating it over the entire plane gives the exact pair formula

$$\text{Opt}(\sigma) = \frac{\int_T \int_T \frac{q+r}{2} e^{-|q-r|^2/(4s^2)} dq dr}{\int_T \int_T e^{-|q-r|^2/(4s^2)} dq dr}.$$

This removes the need to truncate an infinite camera plane numerically. Interchanging q and r also rewrites it as the aperture mean weighted by the positive density $q \rightarrow \int \exp(-|q-r|$

squared/(4s squared))dr. The atlas contours display that equivalent aperture density; they are not pictures of the full-plane squared image. Its positive first moment proves that Opt lies strictly inside T even though the actual image extends outside it.

7.5.2 Narrow and broad blur, and the inertia tensor

For very broad blur the pair weight tends uniformly to one, so Opt tends to G. For very narrow blur, source pairs must be close. Away from a boundary layer the integrated pair weight is nearly constant; after normalization the limiting aperture density is again uniform. Thus both sigma endpoints give X2, while the intermediate family can depart from it. The sampled widths are 0.1,0.3,1.

The previous mass chapter has a useful exact connection. Normalize the Gaussian as a probability density. Linear imaging is the distribution of the sum of an independent uniform aperture point and a centered Gaussian displacement. Therefore its mean is G and its covariance is the lamina covariance plus s squared times the identity. The principal axes are preserved by isotropic blur. Their relation to the Steiner inellipse carries over to this second-moment measurement, although the covariance contour's scale is not the Steiner inellipse's boundary scale. Quadratic detection changes the weighting and has no such automatic covariance-addition rule.

The parameter sigma must be held fixed relative to L. A blur width fixed in laboratory millimeters while triangles are magnified is a different protocol. Similarly a finite camera crop can shift the full-plane centroid; the pair reduction corresponds to the uncropped measurement.

7.6 Solid-angle illumination from above the plane

7.6.1 Why the kernel contains a cosine factor

Place an isotropic point source at height h above transverse position p. A small planar receiver patch dq at q subtends solid angle cos(theta)dq/R squared, where R squared=|q-p| squared+h squared and cos(theta)=h/R. The total fraction of isotropic radiation intercepted by the triangle is its solid angle divided by 4pi. Maximizing intercepted fraction therefore means maximizing

$$\Omega(p, h) = \int_T \frac{h}{(|q - p|^2 + h^2)^{3/2}} dq.$$

Fix h=eta L. The atlas uses eta=0.1,1,10, denoted Solid0.1, Solid1.0, Solid10.0. An exact stable atan2 formula evaluates a triangle's solid angle from the three source-to-vertex vectors. It avoids discretizing a sharply peaked area kernel near the plane. Numerical constrained optimization then selects a maximizing point.

Projection onto a convex triangle decreases distance to every point q in it, so an exterior global maximum is unnecessary. At a boundary point, the derivative in an inward normal direction is strictly positive: the gradient is a positive weighted average of q-p, and the triangle has positive area inward of the supporting line. Hence every global maximum is interior. Existence and interiority do not establish a unique maximum for every height and triangle. The numerical point selector is retained with that qualification; a general uniqueness theorem is not supplied here.

7.6.2 Two limits connect optics to earlier chapters

For large height expand the kernel as h to the power minus two times $[1-3|q-p|^2/(2h^2)+\dots]$. The first p -dependent term asks us to minimize the mean squared distance to a uniform triangular lamina. Chapter 2 shows that its unique minimizer is G . Thus the large-height maximizer tends to X_2 .

For a point well inside T at small h , the full plane subtends 2π . The missing exterior solid angle has first-order coefficient equal to the dimensionless normal field $B(p)$ of the unit-current triangular loop from Chapter 3:

$$\Omega(p, h) = 2\pi - hB(p) + O(h^3).$$

One can derive the coefficient by integrating the exterior radial tail in polar coordinates: it is integral $d\theta/R_p(\theta)$. Resolving the angular sectors side by side gives exactly the finite-edge magnetic expression. Minimizing that coefficient selects M_1 . Boundary-distance estimates prevent a maximizing sequence from escaping to the boundary in this limit. Thus the small-height limit is M_1 , rather than the incenter suggested by a nearest-edge-only argument. The whole surrounding angular geometry contributes.

These limits do not say a finite-height center equals either endpoint. They also do not transfer the global nonattractivity of M_1 to every positive ϵ : the parameter-dependent shape response must be analyzed in its own right.

7.7 Renormalized inverse-square illumination

7.7.1 A different scalar source model

Suppose a uniform planar source contributes scalar fluence proportional to inverse squared distance, with isotropic local receiving response. This is not the same as irradiance on a coplanar planar detector, whose projection factor would change the physical model. At an interior observation point, integral $|q-p|^{-2}$ to the power minus two diverges logarithmically. Exclude a circular disk of radius ϵ around p and integrate radially:

$$\int_{T \setminus B_\epsilon(p)} \frac{dq}{|q-p|^2} = 2\pi \log(1/\epsilon) + \int_0^{2\pi} \log R_p(\theta) d\theta.$$

Subtracting the universal first term leaves the finite part $\Phi(p)$. Choosing another common length inside the logarithm adds a constant and cannot change its maximizer. The renormalized illuminating center I_{illum} maximizes this finite part. A common circular exclusion is part of the physical regularization convention.

7.7.2 A uniqueness argument for the finite-part center

There is an elementary concavity proof on a convex polygon. Let $d_i(p)$ be the inward distance from p to side i and n_i its outward unit normal. Along direction $e(\theta)$, the first side hit gives

$$\log R_p(\theta) = \min_{i: n_i \cdot e(\theta) > 0} [\log d_i(p) - \log(n_i \cdot e(\theta))].$$

Each $\log d_i$ is concave because d_i is positive affine. The minimum of concave functions is concave, as is their angular integral. For strictness take two distinct interior points. At their

midpoint every side has a positive angular interval on which it is hit first. At least one such side has a normal not perpendicular to the displacement of the points. On that interval $\log d_i$ has a strict concavity inequality. Comparing the minimum with that active term preserves a strict inequality after angular integration. Hence Φ is strictly concave.

As p approaches the boundary, a positive angular sector has ray lengths tending to zero, so Φ tends to minus infinity. Continuity on interior compact sets and strict concavity therefore give a unique interior global maximum. This upgrades the older atlas's conditional stationary-root description. It is a direct derivation for the stated finite-part model, not a uniqueness claim about all illumination kernels.

7.7.3 Stationarity and a new numerical response survey

Translate the excluded disk with p and differentiate the finite part. The circular singular contribution cancels by symmetry; the outer boundary term remains. If θ_i is the angle subtended by side i , length a_i , and distance d_i , the line integral of inverse squared distance over that side is θ_i/d_i . Thus

$$\nabla\Phi(p) = - \sum_i \frac{\theta_i}{d_i} n_i.$$

Because the side normals satisfy $\sum a_i n_i = 0$ and span a two-dimensional space, stationarity is equivalent to the three values $\theta_i/(a_i d_i)$ being equal. Solve two independent differences in positive barycentric coordinates. Strict concavity proves that an exact stationary point is the unique maximum; a small floating-point residual is still not a coordinate enclosure.

The new 100-shape response screen finds a negative vertex response for Illum. Direct finite motions with successively halved step preserve that negative sign. The three sampled solid-angle families produce no negative response in this screen. The individual entries report the actual values and the limits of this evidence. No interval certificate is claimed for these optical calculations.

The Illum witness was independently recomputed by solving the exact angular stationary equations at 45 and 75 decimal digits. Both precisions give finite-motion ratios -0.103971291593, -0.101927230216 and -0.100920345465 for the three decreasing steps. The strict-concavity proof identifies the stationary solution as the desired maximum. High precision still does not provide outward-rounded enclosures of the roots or signs; the evidence is explicitly classified as numerical.

7.8 New reference calculations and inherited shape surveys

Selector	x	y	Last change / D
Dark0.5b1.0	1.3402234	1.0044145	7.59e-16
Dark0.5b8.0	1.4333150	1.0318508	9.44e-16
Dark1.0b1.0	1.3664001	1.0133786	9.16e-16
Dark1.0b8.0	2.2701637	0.8607302	6.6e-16
Dark2.0b1.0	1.4710809	1.0049869	5.42e-16
Dark2.0b8.0	3.0310394	0.4690223	1.15e-14
Dark4.0b1.0	1.4829305	1.0067498	6.85e-12
Dark4.0b8.0	1.8057848	1.2627775	1.37e-07
Persist1.0	1.3757932	1.0118755	5.35e-16
Persist8.0	2.3589100	0.8203681	1.05e-15
Opt0.1	1.2831431	0.9998646	8.58e-16
Opt0.3	1.2976122	0.9988912	8.99e-16
Opt1.0	1.3284476	0.9997030	5.96e-16
Illum	1.1320074	0.9959148	not enclosed
Solid0.1	1.1146650	0.9983532	not enclosed
Solid1.0	1.2920118	0.9979634	not enclosed
Solid10.0	1.3328232	0.9999671	not enclosed

All values use $A=(0,0)$, $B=(4,0)$, $C=(0,3)$. A last-change value compares two discretizations; it is neither an error bound nor the number of reliable printed digits. Exact and numerical location statements are identified separately in each entry.

The new reference values all use the same 3-4-5 triangle as earlier chapters. ETC rankings still use the inherited 100 shapes in the ordered angle region and their 40 holdouts. The contour atlas combines the new reference field with previously archived acute and obtuse fields. Every panel identifies its triangle; figures use illustrative discretizations, while the reference table and evidence records retain the refinement results.

All comparisons use a frozen ETC snapshot rather than a presumed complete classification of physically defined centers. The main score is RMS Euclidean location error divided by the triangle diameter, over 100 shapes distributed by equal-area sampling in the ordered angle region $\theta_1 \leq \theta_2 \leq \theta_3$. Forty separate shapes are held out for validation. Uniformity refers to area in angle space, equivalently distances to the sides of an equilateral angle diagram; it is not uniformity in side lengths or Cartesian apex coordinates.

The top candidates are then compared by derivatives along isosceles and right-triangle families and by their vertex-response margins. A fingerprint derivative RMS is dimensionless in the stated normalized families; a location RMS is normalized by diameter. They are listed separately, not combined into an unexplained weighted score. A leading location neighbor need not be the leading response neighbor, and proximity to an ETC center is not an identity theorem.

For a center C and a vertex V_i , write J_i for the two-by-two derivative of C with respect to that vertex. An infinitesimal displacement d has nonnegative dot product with the induced center displacement precisely when $d^T J_i d$ is nonnegative. Only the symmetric part contributes. Thus local vertex attractivity requires the smaller eigenvalue of $(J_i + J_i^T)/2$ to be nonnegative for all three vertices. A finite witness instead reports the ratio $d \cdot [C(T+d) - C(T)] / |d|^2$. A strictly negative rigorously enclosed ratio disproves the global property; an unvalidated negative floating-point ratio is numerical evidence requiring resolution checks.

Maximum-location errors, spectral crossings, quadrature errors and cancellation can all corrupt finite differences. Passing sampled triangles establishes no universal theorem. Missing symmetry-locus expressions remain missing; the evaluator does not invent values for a potentially removable

singularity. The tables retain evaluated counts, and the figures compare two leading location candidates with X1, X2 and X10. No claim of transcendence follows merely because an elementary center function is unavailable.

7.9 Thin limits, local derivatives and symbolic approximations

Collapse the third vertex to (x, ϵ) over base $[-1, 1]$. Normalized area becomes a longitudinal tent density $h_x(t)$, equal to $(t+1)/(x+1)$ on the left of x and $(1-t)/(1-x)$ on the right. Its integral over $[-1, 1]$ is one. Squared spatial separations tend to longitudinal squared separations, and L squared tends to $2+2x$ squared/3. The Fresnel amplitudes, persistence averages and Gaussian pair integrals therefore reduce to explicit one-dimensional weighted integrals at fixed finite parameters.

For the vertical fingerprint divided by epsilon, conditional area at longitudinal t has mean height $h_x(t)/2$. This supplies the rescaled vertical moment once the limiting optical weight has been calculated. Bounded-integrand dominated convergence justifies these finite-parameter moment limits. An endpoint profile at $x=1$ is one-sided; formulae containing division by $1-x$ must be interpreted by that limit, not evaluated literally.

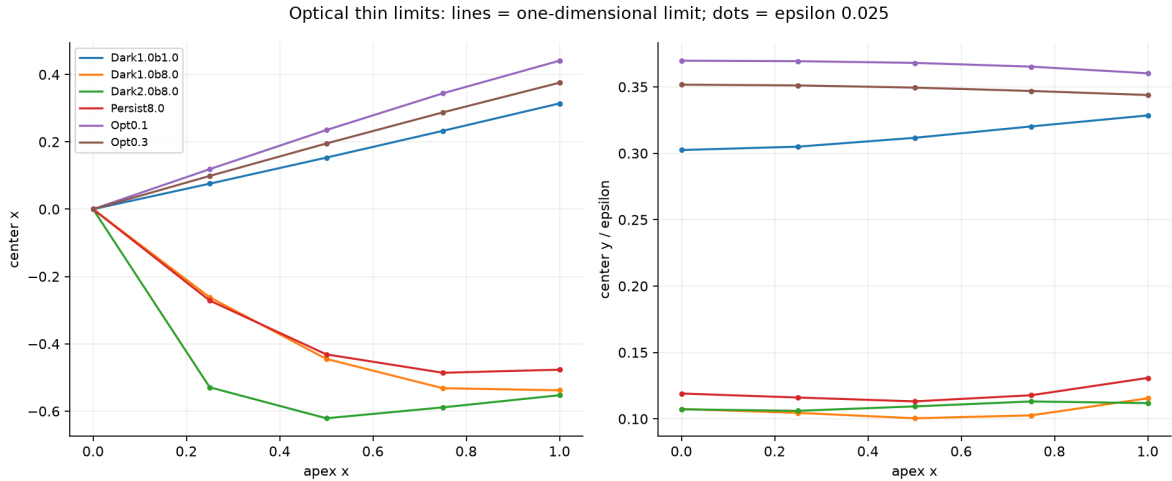


Figure 3. Archived two-dimensional thin experiments and reduced one-dimensional predictions. These are moment limits at fixed optical parameters, not assertions about an arbitrarily selected intensity minimum.

The nearly equilateral expansions have the reflection parities and trace constraint described in earlier chapters. Positive barycentric surrogates enforce location and permutation symmetry by construction. The inherited six-coefficient fits report held-out errors in each entry. A simple symbolic approximation is valuable for quick plotting, but the physical integral remains the definition; fitting a location does not prove a match with its closest ETC formula or settle shape-response signs.

For these optical fits, put $ell_i = a_i / (a+b+c)$, $u_i = ell_i - 1/3$, $v_i = ell_i$ squared minus their mean squared value, and $z = 3 \sqrt{3} K/s$ squared. Form P_i as the sum, for $k=0,1,2$, of z to the power k times $(A_k u_i + B_k v_i)$. The fitted barycentric weights are $\exp(P_i)$ divided by their sum. This explicit six-coefficient convention uses the uniform baseline; the archived coefficients and their training/holdout errors complete the formula. A parameter family not fitted in this way, such as Illum, is not assigned those coefficients by analogy.

The illumination extensions previously lacked a full vertex-response comparison. This chapter

supplies a common 100-shape screen and separate family-derivative and ETC-response discrepancies. Numerical optimizer sensitivity at large height is retained; a nearly flat solid-angle field can make small finite differences unreliable beyond the digits justified by the step comparison.

7.10 Individual atlas entries

7.10.1 Dark0.5b1.0: Darkness mean, F=0.5, beta=1

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_{0.5}(p)} dp}{\int_T e^{-1J_{0.5}(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (1.3402234, 1.0044145); normalized barycentrics (0.3301393, 0.3350558, 0.3348048). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 7.59e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.22408e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

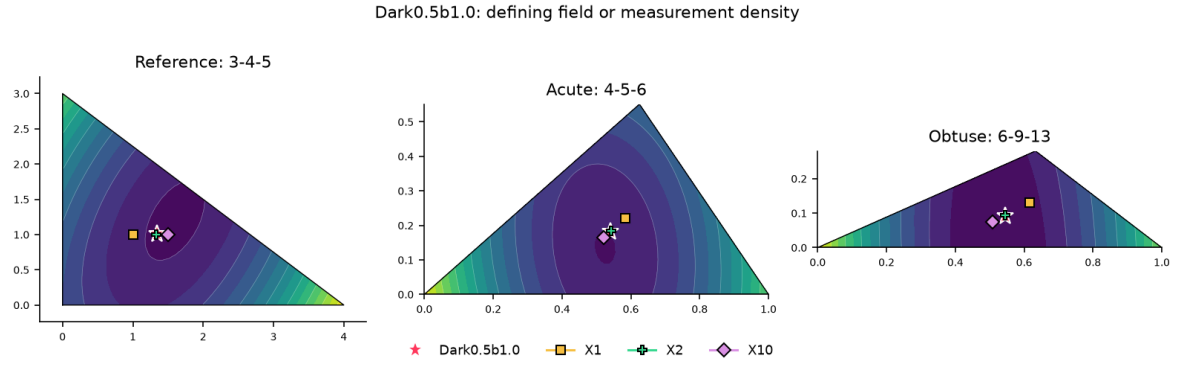
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X16900	0.000398843	0.000410108	0.000674013
X50409	0.000399888	0.000412107	0.00067411
X17514	0.000408841	0.000421558	0.000674114
X19334	0.000428256	0.000433851	0.000673623
X16457	0.000431785	0.000445196	0.000674118

For location winner X16900, the separately computed derivative-fingerprint RMS discrepancy is 0.000711612, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.000625179, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.323926. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

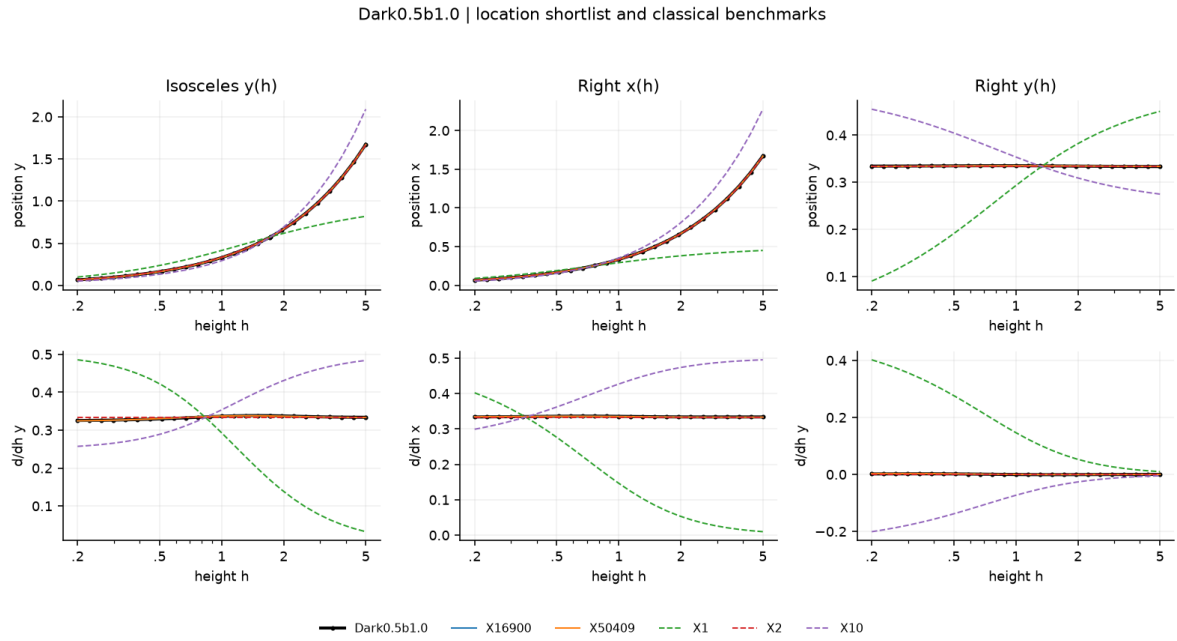
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000151723 diameter and maximum error 0.000398286. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=0.3285671$, $\gamma=0.3380995$, $\beta=0.0006350$, $\delta=0.0038102$, $\epsilon=-0.0010583$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 1a. Dark0.5b1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 1b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.2 Dark0.5b8.0: Darkness mean, F=0.5, beta=8

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_{0.5}(p)} dp}{\int_T e^{-8J_{0.5}(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (1.4333150, 1.0318508); normalized barycentrics (0.2977210, 0.3583288, 0.3439503). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 9.44e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.66537e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

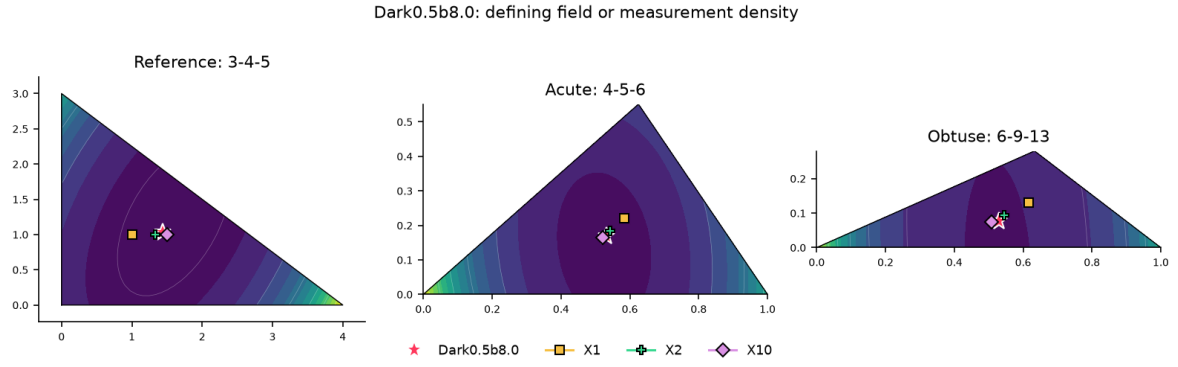
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X52788	0.00205648	0.00194278	0.00287359
X7914	0.00209989	0.00205162	0.00359773
X59214	0.00378229	0.00390792	0.00673745
X71048	0.0039051	0.00426971	0.00892666
X51780	0.00392372	0.00380993	0.00592815

For location winner X52788, the separately computed derivative-fingerprint RMS discrepancy is 0.00588016, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00548744, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.201035. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

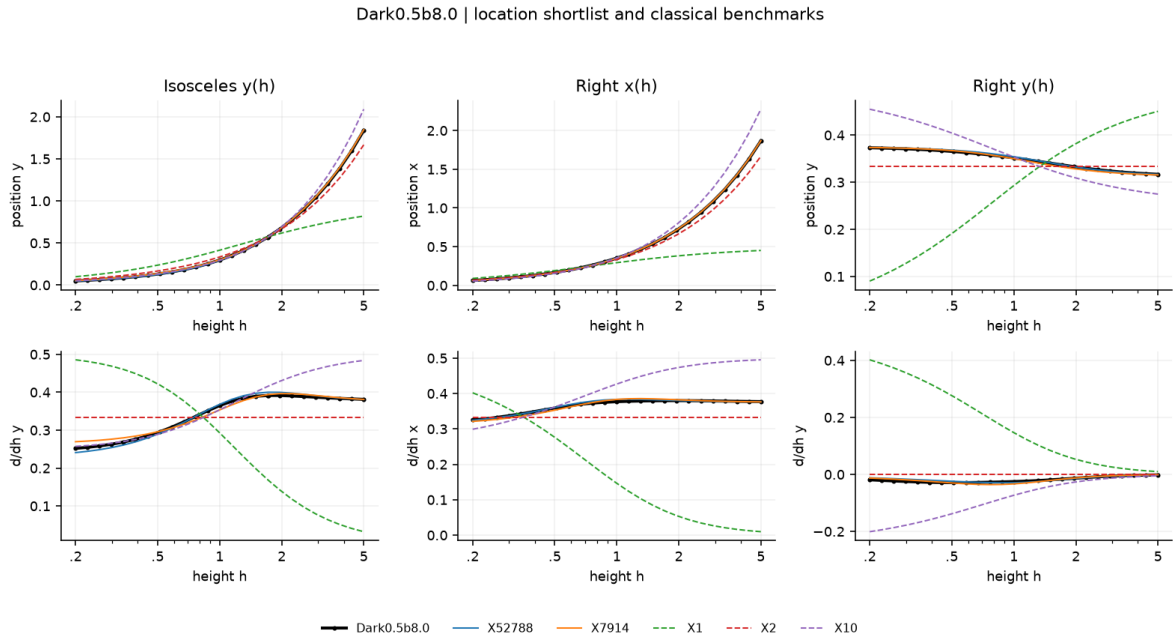
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000854214 diameter and maximum error 0.00229416. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=0.2752008$, $\gamma=0.3914649$, $\beta=0.0365326$, $\delta=0.0320779$, $\eta=0.0014859$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 2a. Dark0.5b8.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 2b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.3 Dark1.0b1.0: Darkness mean, F=1, beta=1

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_1(p)} dp}{\int_T e^{-1J_1(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (1.3664001, 1.0133786); normalized barycentrics (0.3206071, 0.3416000, 0.3377929). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 9.16e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.3323e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

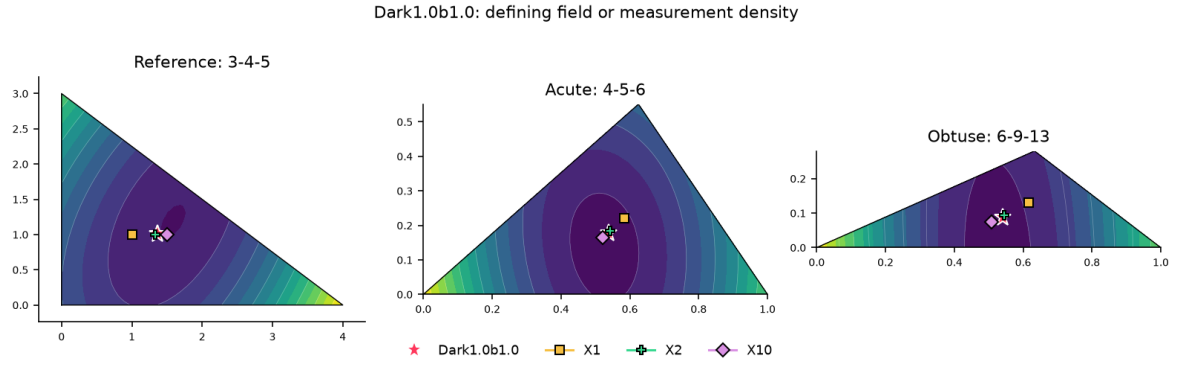
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X17210	0.00248291	0.00279546	0.00887571
X17248	0.00255032	0.002856	0.00887575
X17306	0.00274505	0.00310321	0.00906258
X19874	0.002855	0.00318261	0.00851693
X19853	0.00288935	0.00316347	0.00851111

For location winner X17210, the separately computed derivative-fingerprint RMS discrepancy is 0.00675218, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00947121, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.285039. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

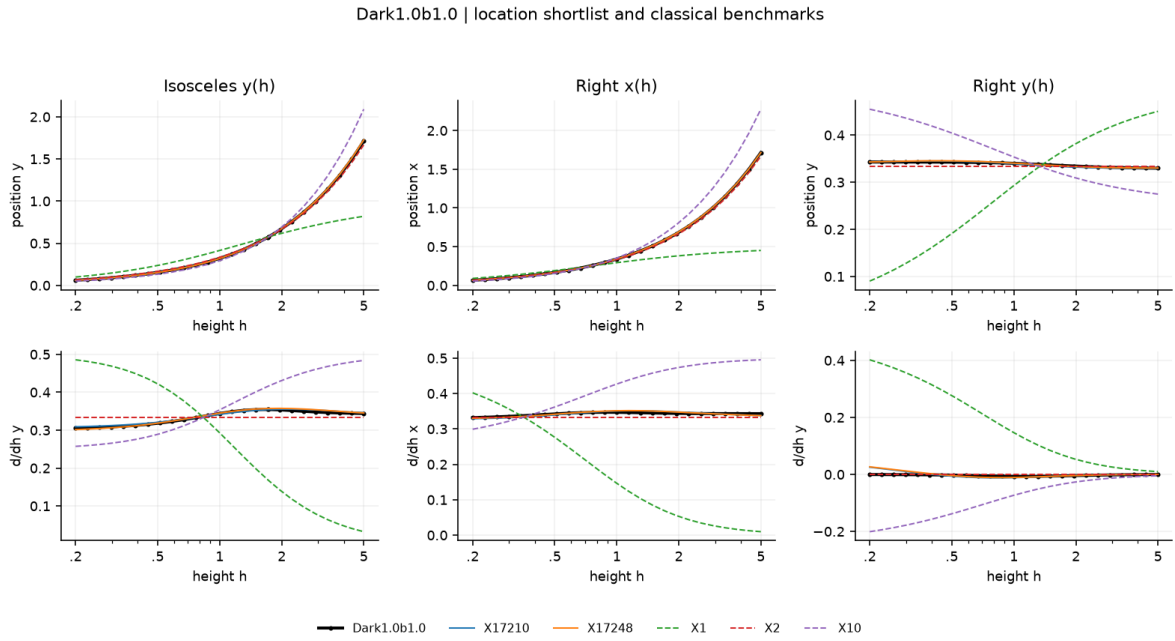
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.0002598 diameter and maximum error 0.000718563. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=0.3122741$, $\gamma=0.3543923$, $\beta=0.0097381$, $\delta=0.0133689$, $\eta=-0.0012097$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 3a. Dark1.0b1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 3b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.4 Dark1.0b8.0: Darkness mean, F=1, beta=8

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_1(p)} dp}{\int_T e^{-8J_1(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (2.2701637, 0.8607302); normalized barycentrics (0.1455490, 0.5675409, 0.2869101). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 6.6e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. Stable negative finite-response evidence; for F=1 and F=2 there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 4.49698e-15 diameter. Coarse screen: 86 sampled triangles with a negative response; those counts are not established continuum counterexamples.

Spatial quadrature order	Finite-motion ratio
22	-1.077928084
30	-1.077928084
38	-1.077928084
46	-1.077928084

The archived finite displacement is evaluated with increasing spatial quadrature. These stable signs have not been enclosed by interval integration.

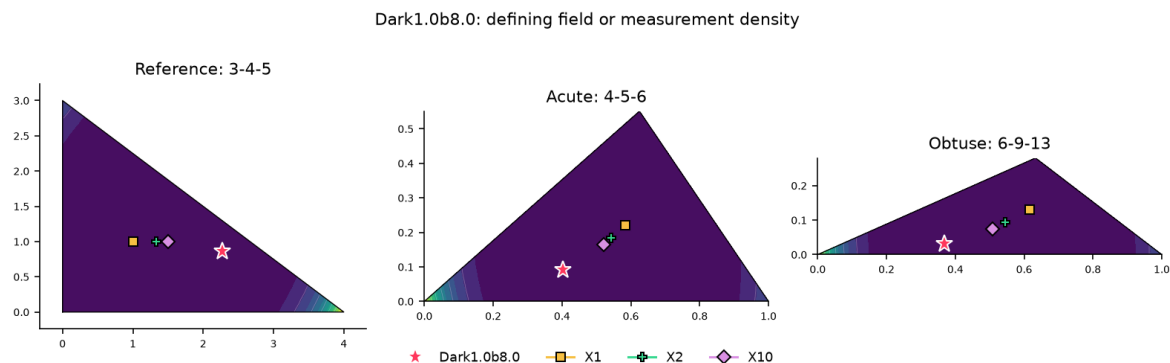
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X7908	0.0170815	0.0170995	0.0253693
X29577	0.0197341	0.0196752	0.0316561
X21712	0.0252998	0.0241877	0.0503979
X7895	0.0255156	0.0258081	0.0380174
X70167	0.0267667	0.025681	0.040528

For location winner X7908, the separately computed derivative-fingerprint RMS discrepancy is 0.0533596, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0409545, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.946305. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

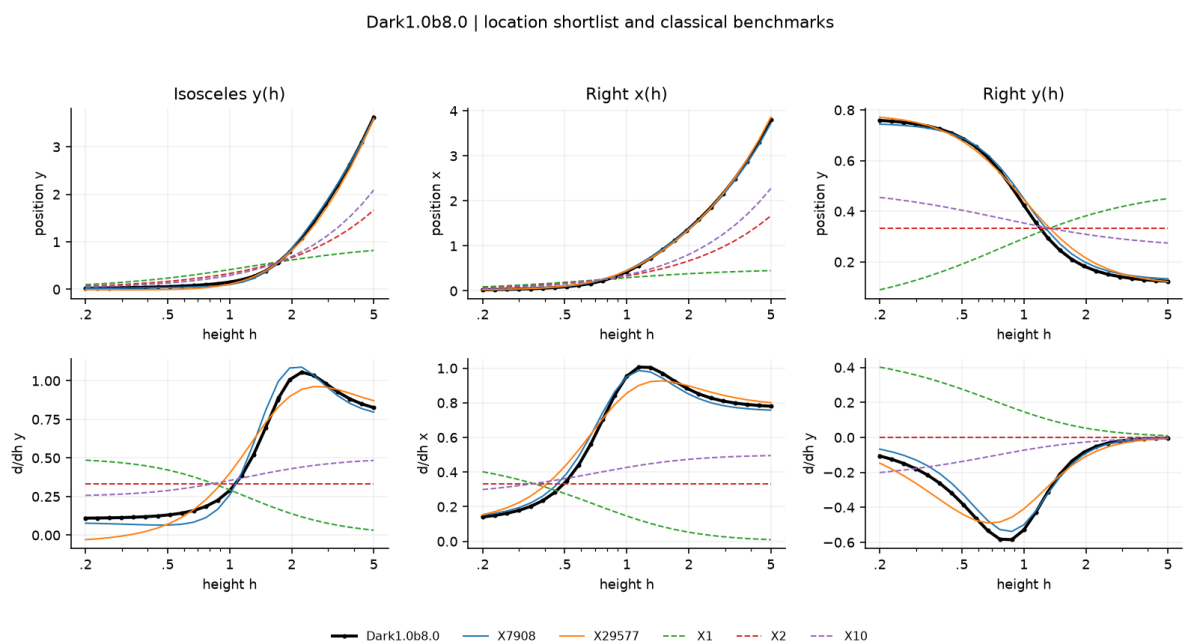
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.00715627 diameter and maximum error 0.0191956. Baseline: G; powers start at $z^{\wedge}0$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=-0.2245172$, $\gamma=0.8911746$, $\beta=1.0067080$, $\delta=-0.0202747$, $\eta=0.3423575$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 4a. Dark1.0b8.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 4b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.5 Dark2.0b1.0: Darkness mean, F=2, beta=1

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_2(p)} dp}{\int_T e^{-1J_2(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (1.4710809, 1.0049869); normalized barycentrics (0.2972341, 0.3677702, 0.3349956). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 5.42e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 2.04546e-12 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

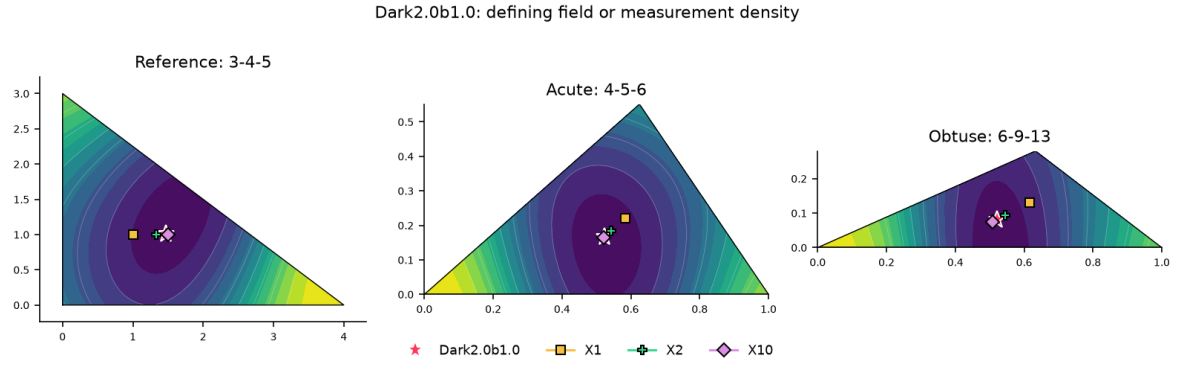
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X55742	0.00397104	0.00412169	0.00675869
X16988	0.00434193	0.00418931	0.00855425
X7944	0.00439794	0.00428737	0.00668291
X60279	0.00456159	0.00468694	0.00752855
X32781	0.0054231	0.00576068	0.0123999

For location winner X55742, the separately computed derivative-fingerprint RMS discrepancy is 0.0162517, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0140685, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.159649. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

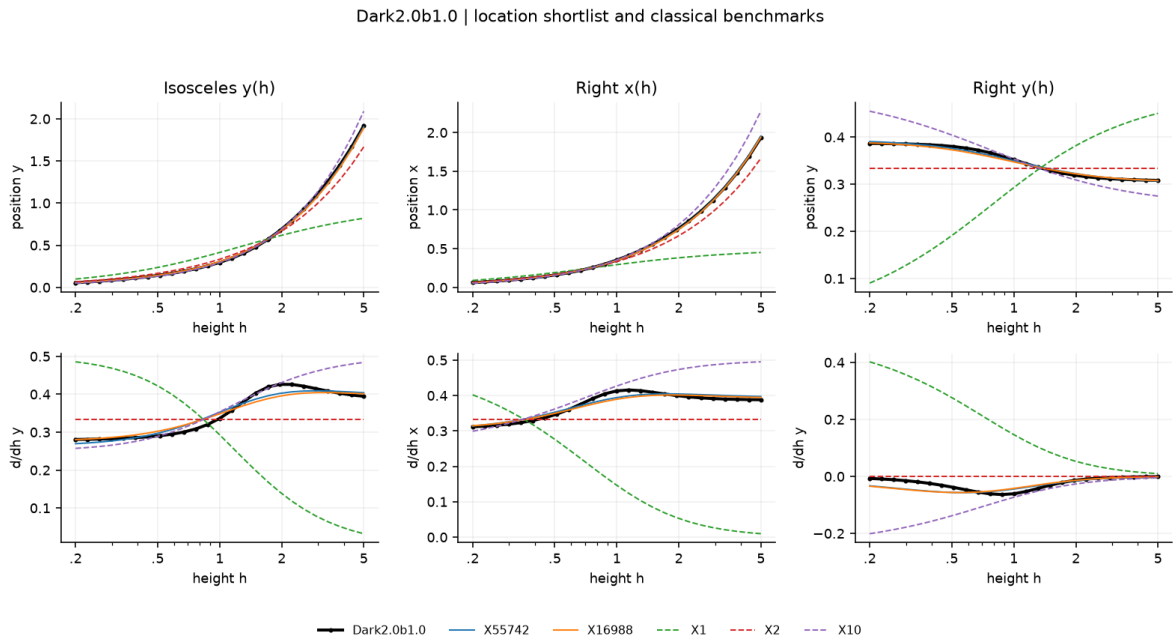
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.00167577 diameter and maximum error 0.00319536. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=0.2469353$, $\gamma=0.4197300$, $\beta=0.0962961$, $\delta=0.0266736$, $\eta=0.0232114$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 5a. Dark2.0b1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 5b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.6 Dark2.0b8.0: Darkness mean, F=2, beta=8

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_2(p)} dp}{\int_T e^{-8J_2(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (3.0310394, 0.4690223); normalized barycentrics (0.0858994, 0.7577598, 0.1563408). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 1.15e-14 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. Stable negative finite-response evidence; for F=1 and F=2 there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 4.19358e-07 diameter. Coarse screen: 47 sampled triangles with a negative response; those counts are not established continuum counterexamples.

Spatial quadrature order	Finite-motion ratio
22	-2.878897248
30	-2.878897254
38	-2.878897254
46	-2.878897254

The archived finite displacement is evaluated with increasing spatial quadrature. These stable signs have not been enclosed by interval integration.

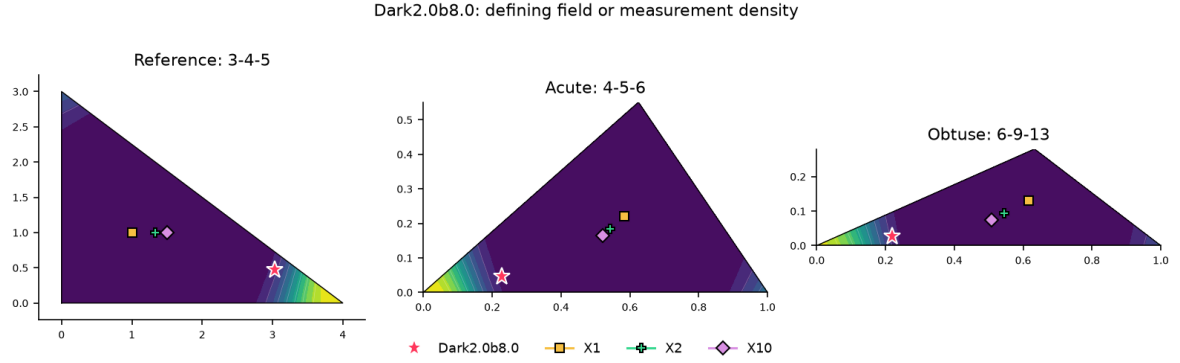
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X69149	0.0756778	0.0731979	0.147084
X30153	0.0781078	0.0784059	0.121257
X31000	0.0798451	0.080473	0.169421
X39107	0.0832551	0.0823693	0.142939
X69258	0.0874826	0.0855907	0.138821

For location winner X69149, the separately computed derivative-fingerprint RMS discrepancy is 0.392071, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.405709, using 300/300 finite vertex margins. The candidate's minimum screened margin is -1.53447. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

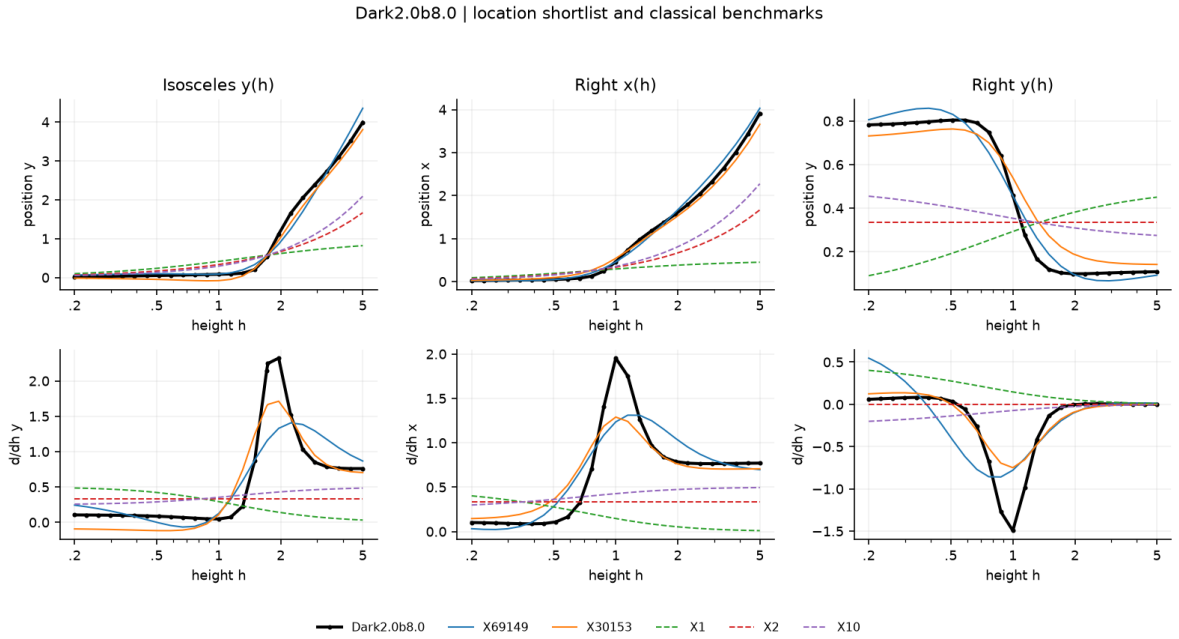
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.0525595 diameter and maximum error 0.132877. Baseline: G; powers start at $z^{\wedge}0$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=-1.5787566$, $\gamma=2.2453917$, $\beta=5.1594711$, $\delta=-0.9246733$, $\epsilon=2.0293237$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 6a. Dark2.0b8.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 6b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.7 Dark4.0b1.0: Darkness mean, F=4, beta=1

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_4(p)} dp}{\int_T e^{-1J_4(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (1.4829305, 1.0067498); normalized barycentrics (0.2936841, 0.3707326, 0.3355833). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 6.85e-12 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.06147e-07 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

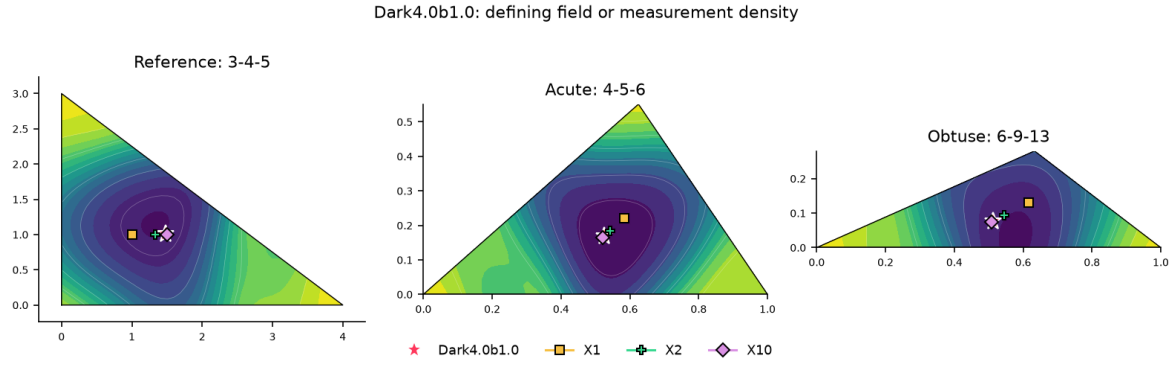
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X20582	0.00452696	0.00421982	0.00756461
X60728	0.00464532	0.00461379	0.0077614
X60643	0.00481028	0.00455362	0.00927948
X29677	0.00488461	0.00468308	0.0107338
X31242	0.00528237	0.00487004	0.0105229

For location winner X20582, the separately computed derivative-fingerprint RMS discrepancy is 0.024647, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0198938, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.0944977. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

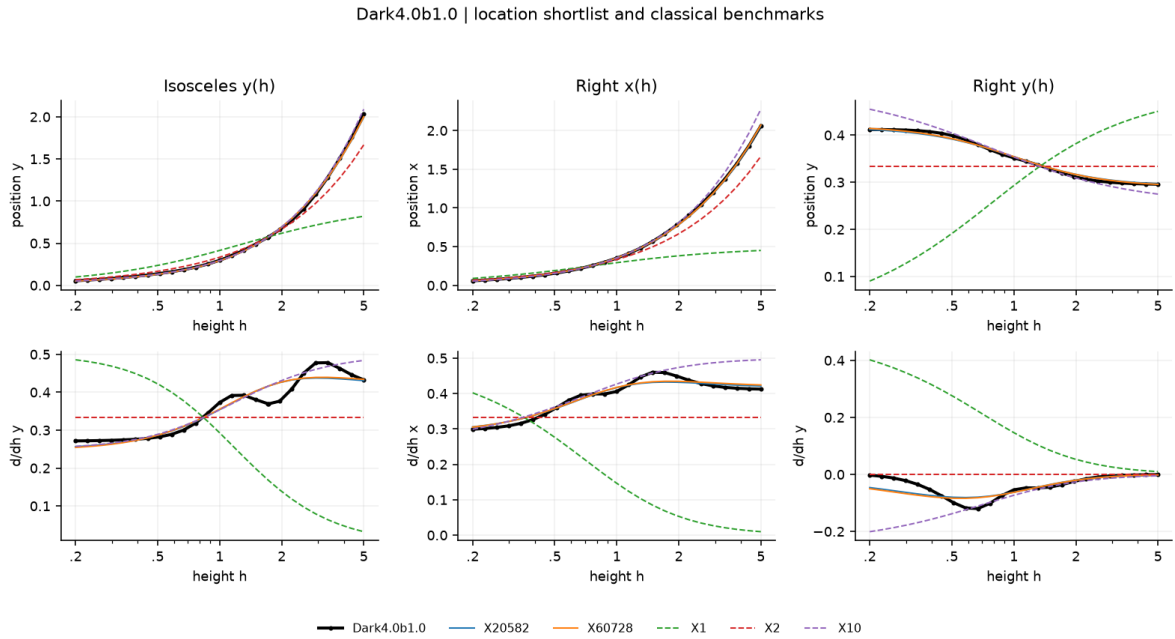
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.00289262 diameter and maximum error 0.00575188. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=0.2971694$, $\gamma=0.3694967$, $\beta=0.0077852$, $\delta=0.0274358$, $\eta=-0.0065607$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 7a. Dark4.0b1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 7b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.8 Dark4.0b8.0: Darkness mean, F=4, beta=8

Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_4(p)} dp}{\int_T e^{-8J_4(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize.

Reference calculation. In the 3-4-5 triangle: (1.8057848, 1.2627775); normalized barycentrics (0.1276280, 0.4514462, 0.4209258). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 1.37e-07 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. Stable negative finite-response evidence; for F=1 and F=2 there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 6.07062e-05 diameter. Coarse screen: 58 sampled triangles with a negative response; those counts are not established continuum counterexamples.

Spatial quadrature order	Finite-motion ratio
22	-1.607577455
30	-1.607650003
38	-1.607640705
46	-1.607641368

The archived finite displacement is evaluated with increasing spatial quadrature. These stable signs have not been enclosed by interval integration.

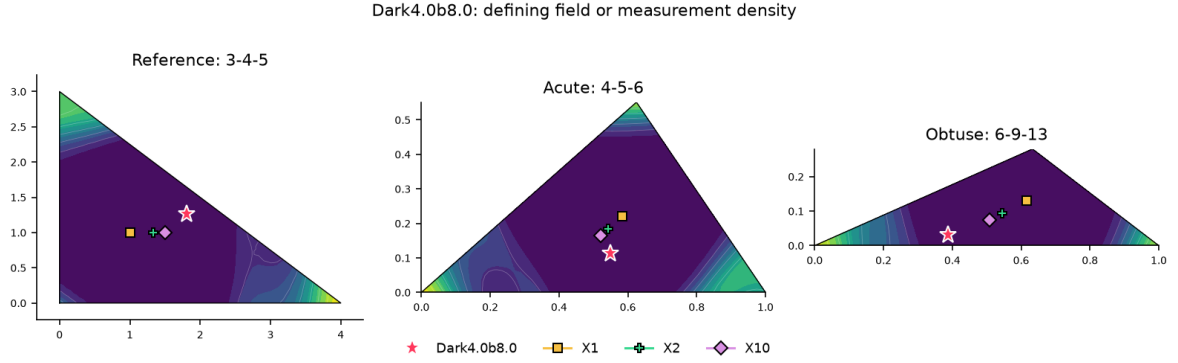
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X21716	0.0529475	0.0452806	0.130253
X29577	0.0547327	0.0454713	0.127296
X60637	0.0553884	0.0500769	0.105513
X50081	0.0566498	0.0489644	0.109089
X44772	0.0574232	0.0489812	0.113943

For location winner X21716, the separately computed derivative-fingerprint RMS discrepancy is 0.405211, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.292024, using 300/300 finite vertex margins. The candidate's minimum screened margin is -1.43232. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

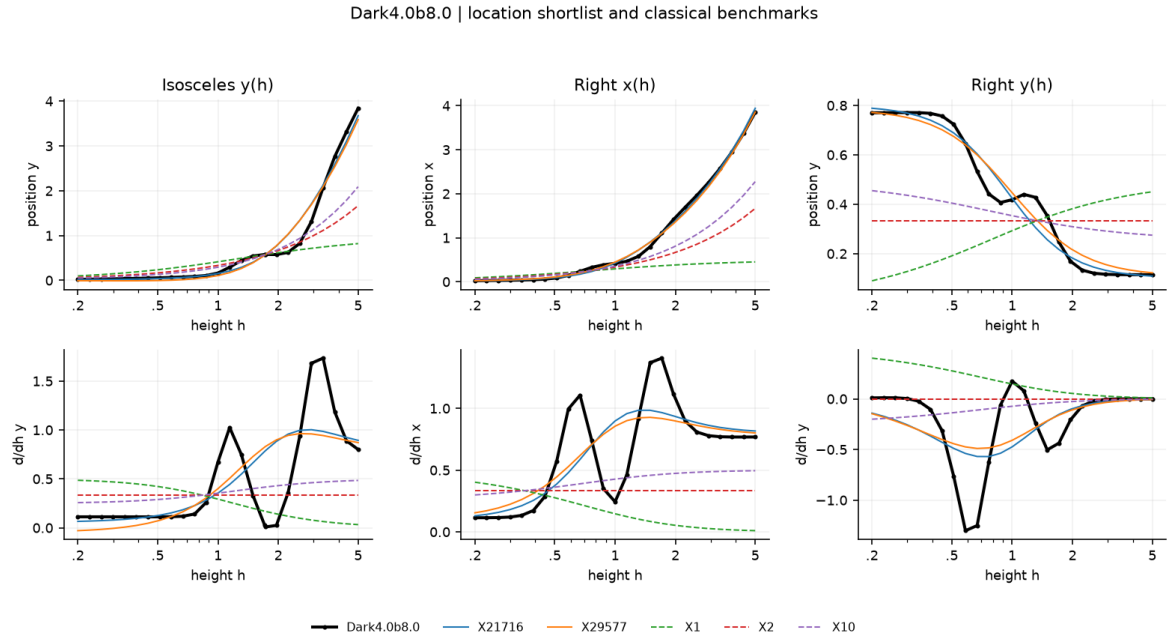
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.0350119 diameter and maximum error 0.0716454. Baseline: G; powers start at $z^{\wedge}0$. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by the nonvanishing proof in Section 7.3. Wavelength independence requires adjusting screen distance to hold F fixed.

The archived numerical almost-equilateral coefficients are $\alpha=0.6575741$, $\gamma=0.0090980$, $\beta=-0.8579602$, $\delta=0.1482640$, $\epsilon=-0.3356103$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 8a. Dark4.0b8.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 8b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.9 Persist1.0: Log-F averaged darkness mean, beta=1

Average normalized intensity uniformly in log Fresnel number from one half to two, then apply the darkness response and take its mean.

$$\bar{J} = \frac{1}{2 \log 2} \int_{1/2}^2 J_F \frac{dF}{F}, \quad C = \int p e^{-1\bar{J}} / \int e^{-1\bar{J}}$$

Add Gauss quadrature in log F to the nested spatial integrals. Normalize intensity separately at every F before averaging.

Reference calculation. In the 3-4-5 triangle: (1.3757932, 1.0118755); normalized barycentrics (0.3187599, 0.3439483, 0.3372918). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 5.35e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive weight gives a unique strictly interior moment.

Attractivity. No violation in the retained screen; global attractivity remains open.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.33342e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

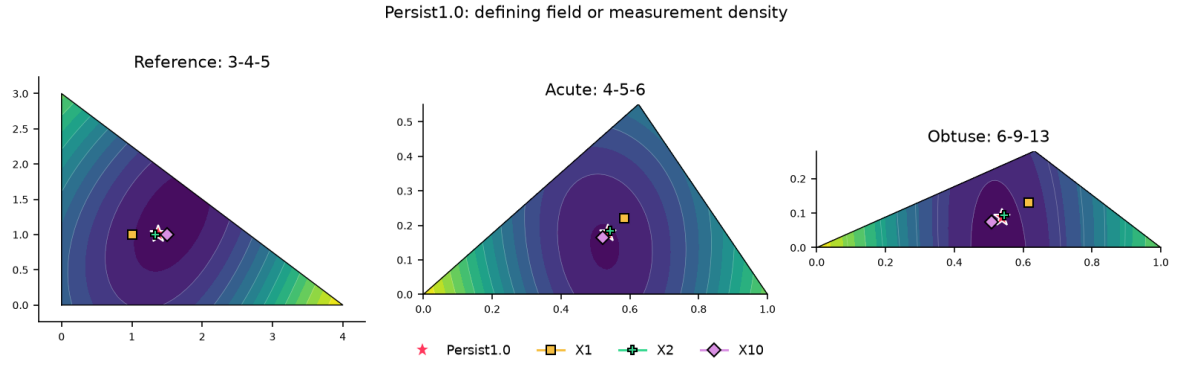
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X61001	0.00181575	0.00171806	0.00272291
X19870	0.00326469	0.00383382	0.0126299
X19872	0.00351643	0.00362872	0.00563702
X27026	0.0036167	0.00402515	0.0120881
X27270	0.00362592	0.00411315	0.0131659

For location winner X61001, the separately computed derivative-fingerprint RMS discrepancy is 0.00626391, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00802767, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.3. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

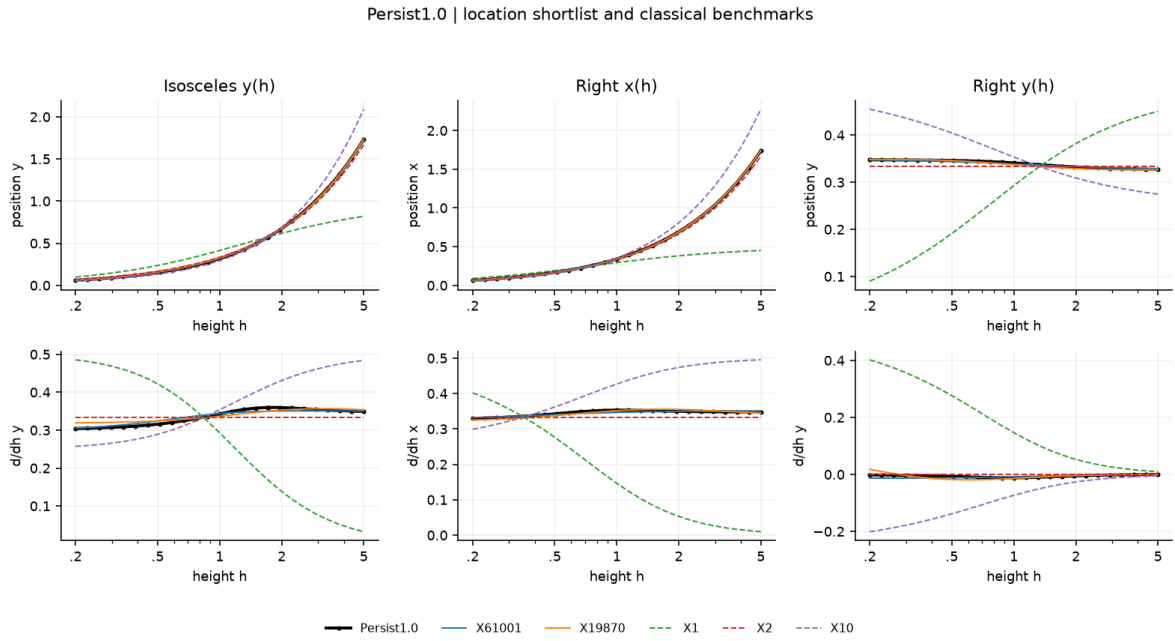
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000190571 diameter and maximum error 0.000377302. Baseline: G; powers start at z^0 . Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Averaging intensity first differs from averaging centers or exponential weights. Both the interval and response law are declared modeling choices.

The archived numerical almost-equilateral coefficients are $\alpha=0.3068132$, $\gamma=0.3598530$, $\beta=0.0174937$, $\delta=0.0142203$, $\eta=0.0010919$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 9a. Persist1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 9b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.10 Persist8.0: Log-F averaged darkness mean, beta=8

Average normalized intensity uniformly in log Fresnel number from one half to two, then apply the darkness response and take its mean.

$$\bar{J} = \frac{1}{2 \log 2} \int_{1/2}^2 J_F \frac{dF}{F}, \quad C = \int p e^{-8\bar{J}} / \int e^{-8\bar{J}}$$

Add Gauss quadrature in log F to the nested spatial integrals. Normalize intensity separately at every F before averaging.

Reference calculation. In the 3-4-5 triangle: (2.3589100, 0.8203681); normalized barycentrics (0.1368165, 0.5897275, 0.2734560). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 1.05e-15 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The positive weight gives a unique strictly interior moment.

Attractivity. Stable negative finite-displacement numerical witnesses; a validated continuum sign certificate remains open.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.67301e-12 diameter. Coarse screen: 75 sampled triangles with a negative response; those counts are not established continuum counterexamples.

Spatial quadrature order	Finite-motion ratio
22	-1.174003487
30	-1.174003487
38	-1.174003487
46	-1.174003487

The archived finite displacement is evaluated with increasing spatial quadrature. These stable signs have not been enclosed by interval integration.

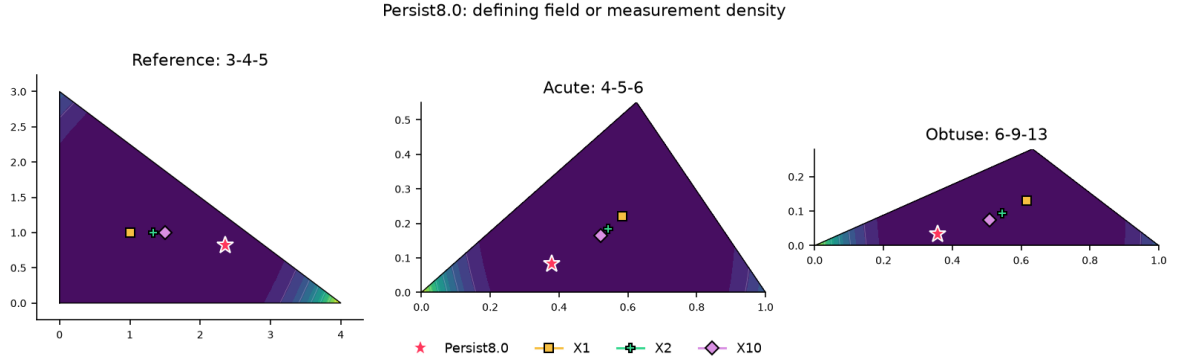
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X7908	0.0158594	0.0155423	0.0252555
X7947	0.0260439	0.0255323	0.0350615
X7895	0.0276325	0.0265539	0.0454957
X17267	0.0300792	0.0332081	0.0693114
X31137	0.030545	0.0302096	0.0664469

For location winner X7908, the separately computed derivative-fingerprint RMS discrepancy is 0.0642595, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0738032, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.946305. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

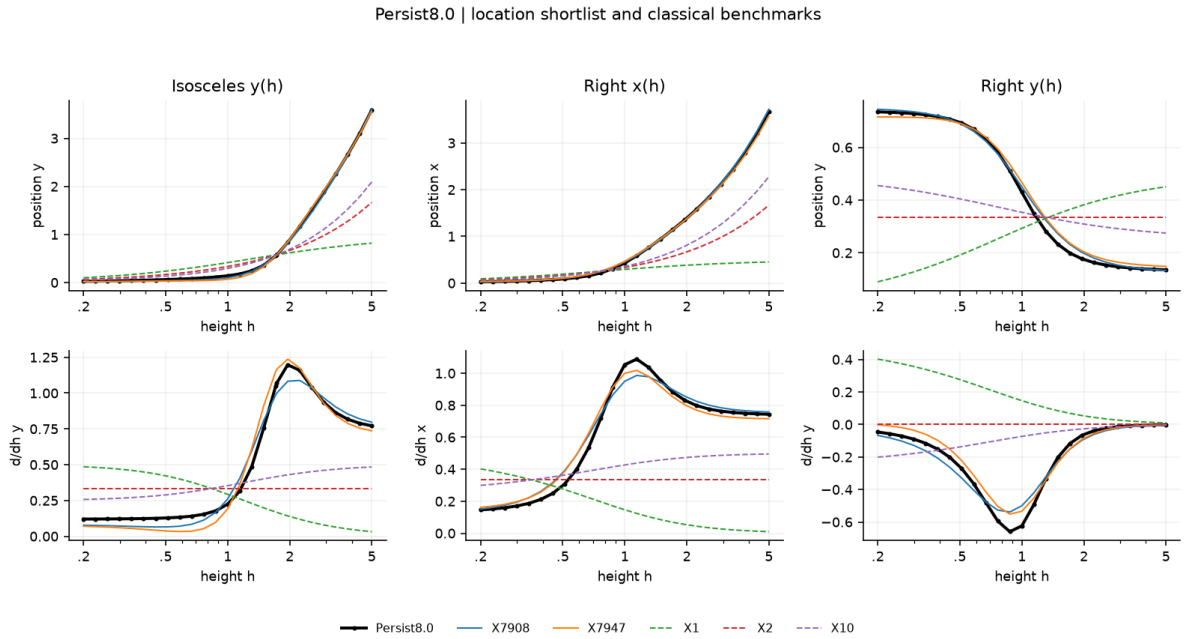
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.013236 diameter and maximum error 0.0325144. Baseline: G; powers start at z^0 . Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Averaging intensity first differs from averaging centers or exponential weights. Both the interval and response law are declared modeling choices.

The archived numerical almost-equilateral coefficients are $\alpha=-0.4040532$, $\gamma=1.0707075$, $\beta=1.4201725$, $\delta=-0.0715712$, $\eta=0.4973449$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 10a. Persist8.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 10b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.11 Opt0.1: Nonlinear fluorescence mean, sigma=0.1

Blur a uniform luminous triangle by a Gaussian of width sigma times L and measure the full-plane centroid after quadratic intensity detection.

$$C = \frac{\iint (q+r) e^{-|q-r|^2/(4(0.1L)^2)} dqdr}{2 \iint e^{-|q-r|^2/(4(0.1L)^2)} dqdr}$$

Use the exact Gaussian pair reduction and smooth double triangle quadrature. This avoids truncating a full-plane image integral.

Reference calculation. In the 3-4-5 triangle: (1.2831431, 0.9998646); normalized barycentrics (0.3459260, 0.3207858, 0.3332882). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 8.58e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The pair formula is a positive mean of points inside the triangle and gives strict interiority.

Attractivity. No violation in the retained 100-shape screen; global attractivity is open.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.22174e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

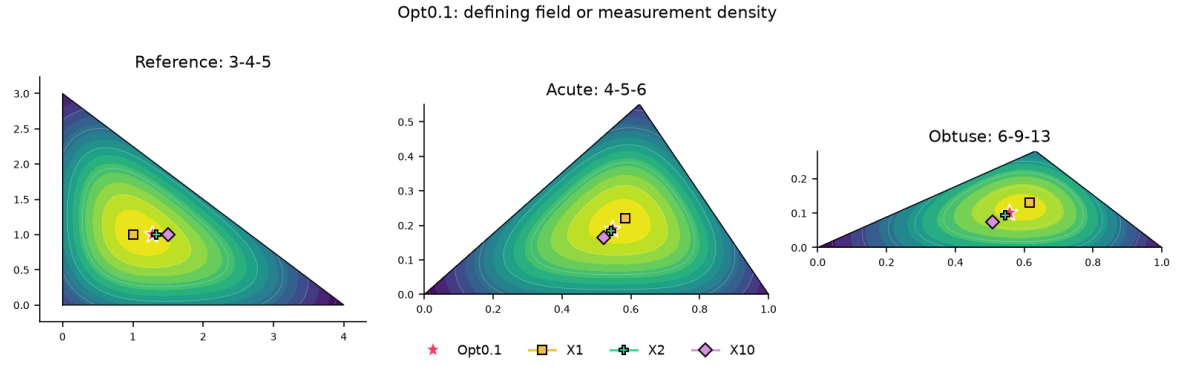
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X5550	0.00184896	0.00195762	0.00585596
X19812	0.00197125	0.00210614	0.00593987
X30598	0.00238199	0.00247934	0.00594165
X28617	0.00251443	0.00250064	0.00534206
X29603	0.0026391	0.00272737	0.00602627

For location winner X5550, the separately computed derivative-fingerprint RMS discrepancy is 0.00609866, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.0110103, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.272775. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

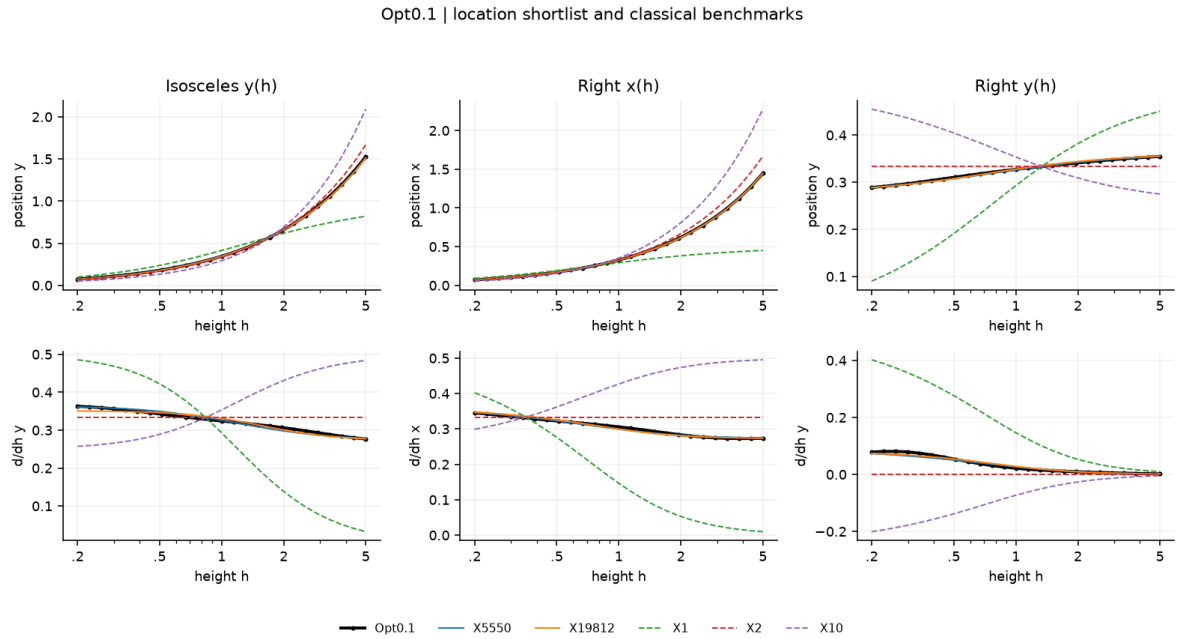
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000576869 diameter and maximum error 0.00139388. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Both very small and very large sigma tend to X(2). The contours show the equivalent positive aperture density, whose mean equals the full-plane quadratic fluorescence centroid.

The archived numerical almost-equilateral coefficients are $\alpha=0.3561798$, $\gamma=0.3104872$, $\beta=-0.0295467$, $\delta=-0.0050121$, $\eta=-0.0081780$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 11a. Opt0.1 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 11b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.12 Opt0.3: Nonlinear fluorescence mean, sigma=0.3

Blur a uniform luminous triangle by a Gaussian of width sigma times L and measure the full-plane centroid after quadratic intensity detection.

$$C = \frac{\iint (q+r) e^{-|q-r|^2/(4(0.3L)^2)} dqdr}{2 \iint e^{-|q-r|^2/(4(0.3L)^2)} dqdr}$$

Use the exact Gaussian pair reduction and smooth double triangle quadrature. This avoids truncating a full-plane image integral.

Reference calculation. In the 3-4-5 triangle: (1.2976122, 0.9988912); normalized barycentrics (0.3426332, 0.3244031, 0.3329637). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 8.99e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The pair formula is a positive mean of points inside the triangle and gives strict interiority.

Attractivity. No violation in the retained 100-shape screen; global attractivity is open.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.44995e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

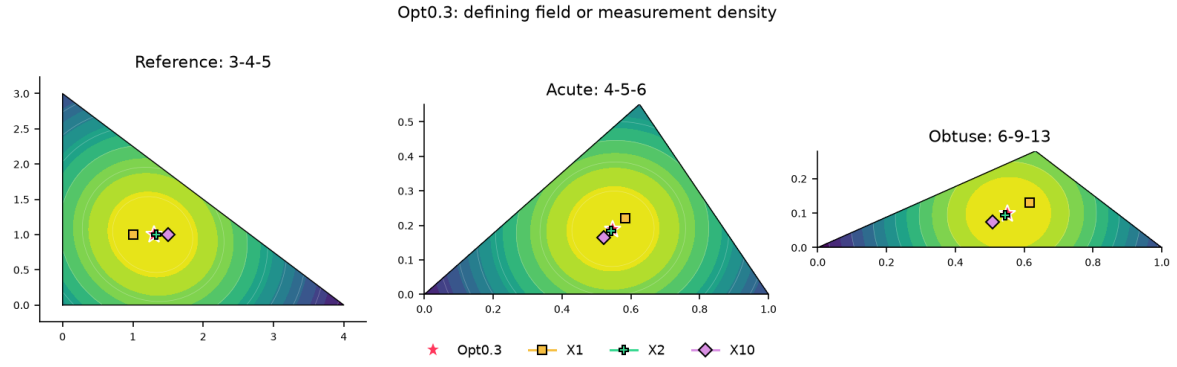
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X55752	0.000225097	0.00023181	0.000333967
X51127	0.000375559	0.000383664	0.000643488
X62940	0.00120084	0.00126771	0.00204974
X55751	0.00123596	0.00129665	0.00204995
X55753	0.00142757	0.00146028	0.00188731

For location winner X55752, the separately computed derivative-fingerprint RMS discrepancy is 0.000826633, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.000464989, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.303016. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

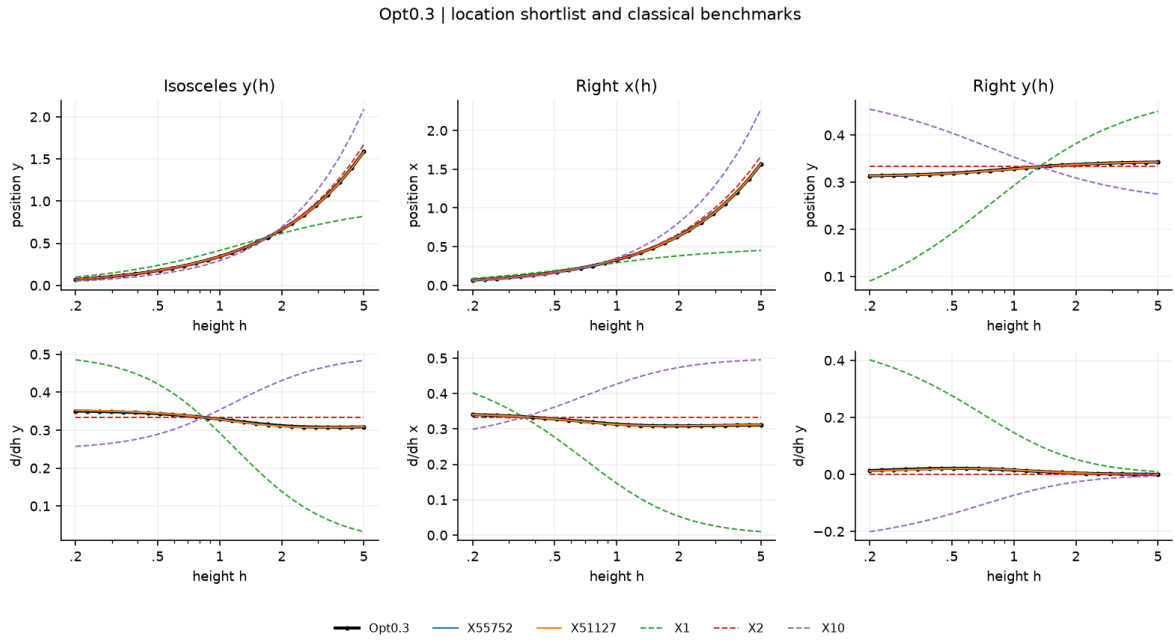
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 0.000312618 diameter and maximum error 0.000735572. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Both very small and very large sigma tend to X(2). The contours show the equivalent positive aperture density, whose mean equals the full-plane quadratic fluorescence centroid.

The archived numerical almost-equilateral coefficients are $\alpha=0.3521509$, $\gamma=0.3145161$, $\beta=-0.0231925$, $\delta=-0.0046999$, $\eta=-0.0061644$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 12a. Opt0.3 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 12b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.13 Opt1.0: Nonlinear fluorescence mean, sigma=1

Blur a uniform luminous triangle by a Gaussian of width sigma times L and measure the full-plane centroid after quadratic intensity detection.

$$C = \frac{\iint (q+r) e^{-|q-r|^2/(4(1L)^2)} dqdr}{2 \iint e^{-|q-r|^2/(4(1L)^2)} dqdr}$$

Use the exact Gaussian pair reduction and smooth double triangle quadrature. This avoids truncating a full-plane image integral.

Reference calculation. In the 3-4-5 triangle: (1.3284476, 0.9997030); normalized barycentrics (0.3346538, 0.3321119, 0.3332343). Method: spatial quadrature 34 to 46; log-frequency order 16 to 24. Last position change is 5.96e-16 diameter. This is a successive-change diagnostic, not a certified error bound.

Location and well-posedness. The pair formula is a positive mean of points inside the triangle and gives strict interiority.

Attractivity. No violation in the retained 100-shape screen; global attractivity is open.

Inherited resolution record: 100/100 training shapes and 40/40 holdouts meet the declared successive-change tolerance. Largest training change: 1.78415e-15 diameter. Coarse screen: 0 sampled triangles with a negative response; those counts are not established continuum counterexamples.

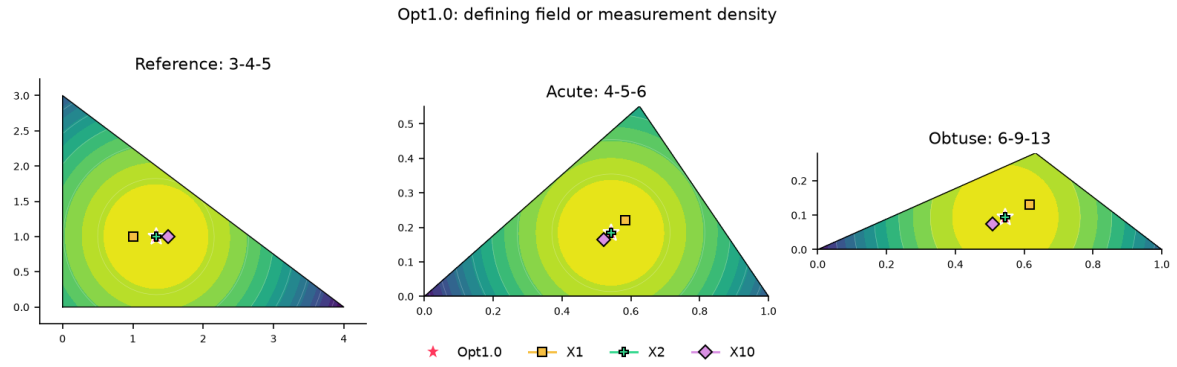
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X71328	0.00112595	0.00121855	0.00252875
X39331	0.00129455	0.00137101	0.00264849
X53004	0.00146708	0.00155034	0.00266789
X19865	0.00150932	0.00145142	0.00236999
X25982	0.00153786	0.00156757	0.00266485

For location winner X71328, the separately computed derivative-fingerprint RMS discrepancy is 0.00227818, using 51/51 finite family values. The vertex-response margin RMS discrepancy is 0.00280362, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.32428. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

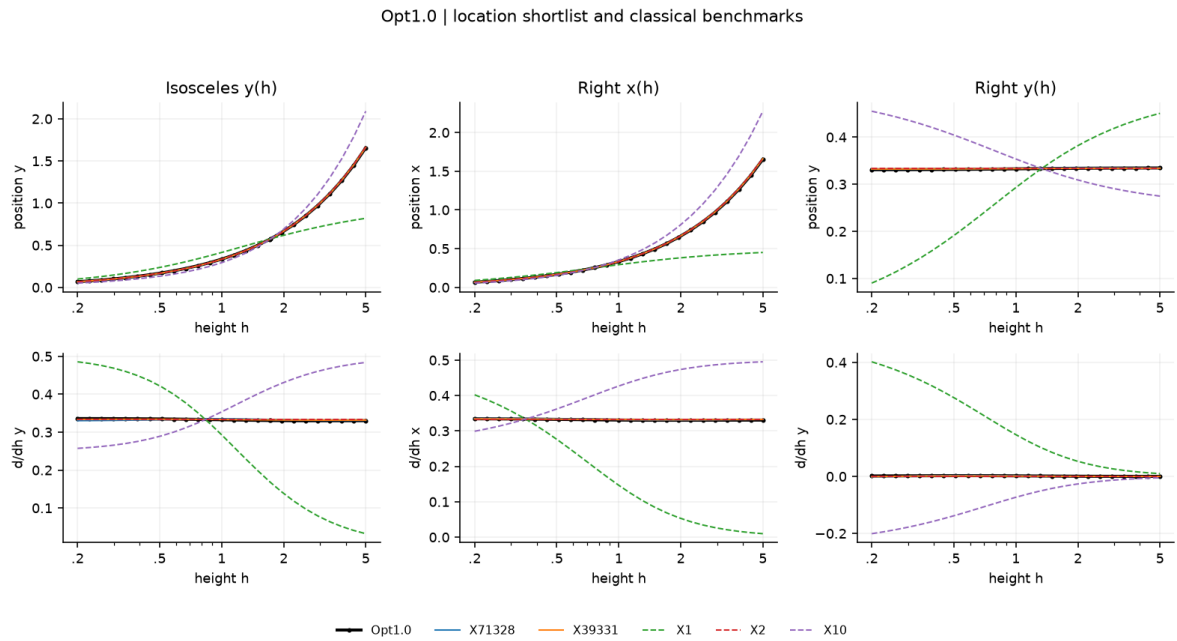
Symbolic approximation. The six-coefficient positive-barycentric model has held-out RMS error 3.35444e-05 diameter and maximum error 7.7101e-05. Baseline: G; powers start at $z \hat{=}$ 0. Coefficients and the precise feature rule are in `mechanical_optical/symbolic_models.json` and `mech_opt_fit.py`; positivity is built in, attractivity is not.

Further fingerprint information. Both very small and very large sigma tend to X(2). The contours show the equivalent positive aperture density, whose mean equals the full-plane quadratic fluorescence centroid.

The archived numerical almost-equilateral coefficients are $\alpha=0.3359825$, $\gamma=0.3306842$, $\beta=-0.0030853$, $\delta=-0.0007516$, $\eta=-0.0007780$. They refer to the local expansion defined above; discretization and finite-step uncertainty remain, so these are not exact Taylor constants.



Atlas figure 13a. Opt1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 13b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.14 Illum: Renormalized inverse-square illumination center

Measure the renormalized inverse-square illumination received from a uniform triangular source, subtracting the universal local logarithmic divergence.

$$C = \arg \max_{p \in T} \int_{S^1} \log R_p(\theta) d\theta$$

Solve the equal edge-angle/area stationarity equations in positive barycentric coordinates. The illustrative contour uses angular ray quadrature, independently of the root equations.

Reference calculation. In the 3-4-5 triangle: (1.1320074, 0.9959148); normalized barycentrics (0.3850266, 0.2830018, 0.3319716). Method: stationary root / solid-angle optimizer. No independent coordinate error enclosure is asserted.

Location and well-posedness. Unique interior global maximum of the strictly concave logarithmic ray-length finite part; see Section 7.7. It diverges to minus infinity on approach to the boundary.

Attractivity. Numerical evidence of nonattractivity: the new 100-shape screen has a negative symmetric-Jacobian margin and the retained finite motion stays negative as its step is halved twice. No outward-rounded coordinate or continuum sign enclosure is supplied.

The minimum sampled local margin was -0.099924273, at shape 36. The same selected direction gives:

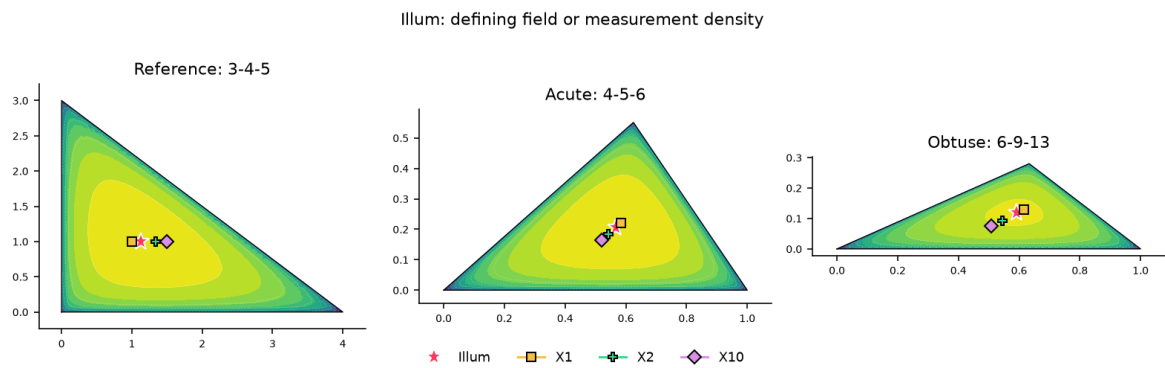
Motion / diameter	Finite-motion ratio
0.002	-0.103971292
0.001	-0.101927230
0.0005	-0.100920345

All witness coordinates and root/optimizer settings are stored. Positive values only screen these sampled motions; for Illum the strict-concavity proof identifies the intended root but does not enclose the numerical sign.

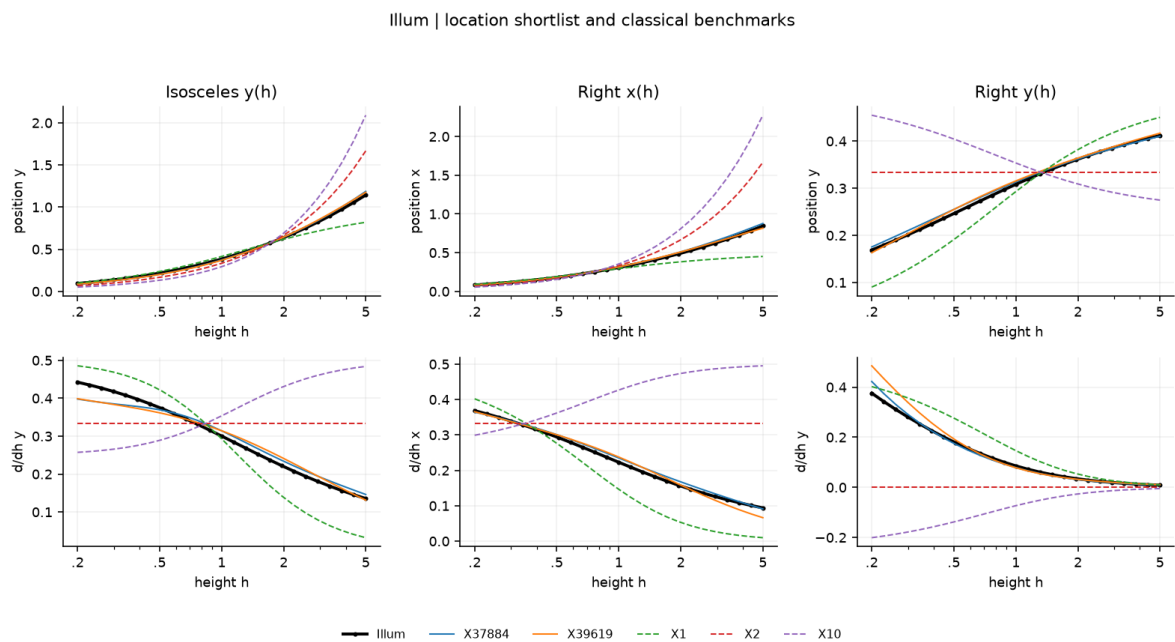
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X37884	0.00832964	0.00787365	0.0211768
X39619	0.0109522	0.0103589	0.0266296
X29622	0.0110047	0.0144097	0.0557655
X63628	0.0111363	0.00916273	0.0254759
X26102	0.0111604	0.00963877	0.0256217

For location winner X37884, the separately computed derivative-fingerprint RMS discrepancy is 0.0179218, using 50/50 finite family values. The vertex-response margin RMS discrepancy is 0.0389546, using 300/300 finite vertex margins. The candidate's minimum screened margin is -0.243833. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. This is distinct from solid-angle illumination and M1. The common-angle location rankings and the newly added derivative/vertex-response comparisons supersede the earlier pilot metric. No full thin or almost-equilateral analytic expansion is asserted for this selector.



Atlas figure 14a. Illum in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 14b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.15 Solid0.1: Solid-angle illumination maximum, height/L=0.1

Place an isotropic point source above the plane at a fixed dimensionless height and maximize the fraction of rays intercepted by the triangular receiver.

$$C = \arg \max_p \int_T \frac{0.1L}{(|q-p|^2 + (0.1L)^2)^{3/2}} dq$$

Evaluate the exact triangle solid-angle atan2 formula and optimize in barycentric coordinates. Hold height/L fixed when comparing different triangle shapes.

Reference calculation. In the 3-4-5 triangle: (1.1146650, 0.9983532); normalized barycentrics (0.3885493, 0.2786663, 0.3327844). Method: stationary root / solid-angle optimizer. No independent coordinate error enclosure is asserted.

Location and well-posedness. Every global maximum is interior by projection and the strictly positive inward boundary derivative. Global uniqueness for all shapes and heights is not established here; the numerical point selector remains qualified.

Attractivity. No failure found in the retained 100-shape screen at this height/L; global attractivity and universal maximum uniqueness remain open. The family has a separate analytical nonattractivity result for all sufficiently small heights, inherited from the certified M1 limit. This does not classify the present parameter. See Appendix D.

The minimum sampled local margin was 0.079223321, at shape 2. The same selected direction gives:

Motion / diameter	Finite-motion ratio
0.002	0.079225088
0.001	0.079221361
0.0005	0.079218981

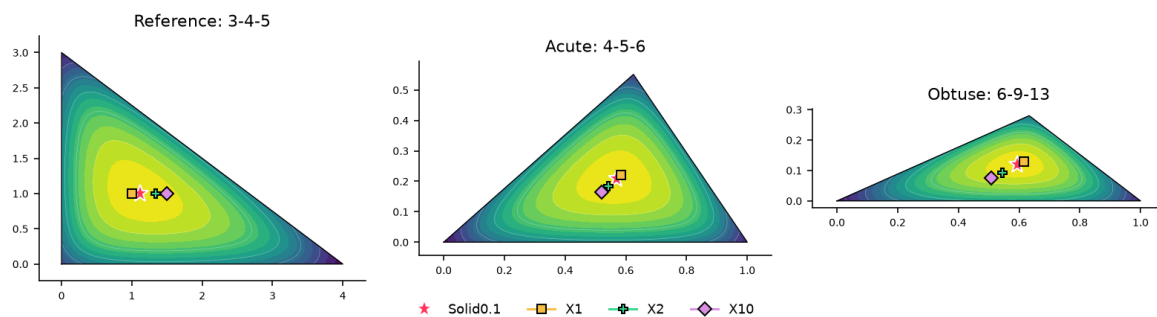
All witness coordinates and root/optimizer settings are stored. Positive values only screen these sampled motions; for Illum the strict-concavity proof identifies the intended root but does not enclose the numerical sign.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X41815	0.00446418	0.00490645	0.015729
X17394	0.00450557	0.00491539	0.015666
X64667	0.00460432	0.00490781	0.0157242
X38314	0.00471235	0.00501187	0.0155363
X40439	0.0049891	0.00547369	0.0159205

For location winner X41815, the separately computed derivative-fingerprint RMS discrepancy is 0.00974643, using 50/50 finite family values. The vertex-response margin RMS discrepancy is 0.0321727, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.108538. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

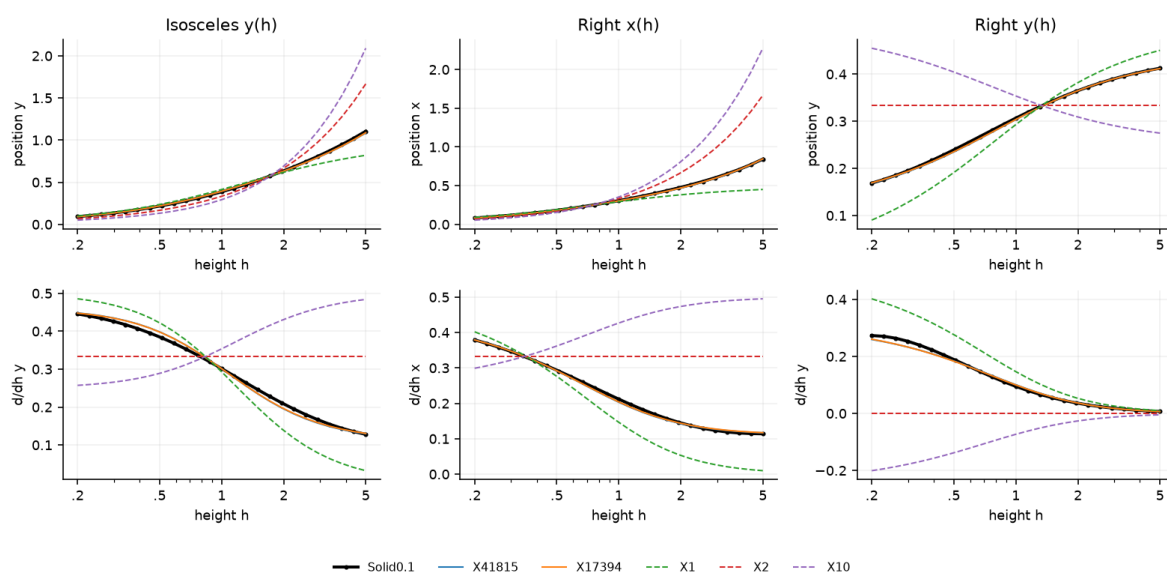
Further fingerprint information. As height/L tends to zero the center approaches M1; as it tends to infinity it approaches X(2). These endpoint statements do not identify an arbitrary finite-height center with either endpoint.

Solid0.1: defining field or measurement density



Atlas figure 15a. Solid0.1 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.

Solid0.1 | location shortlist and classical benchmarks



Atlas figure 15b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.16 Solid1.0: Solid-angle illumination maximum, height/L=1

Place an isotropic point source above the plane at a fixed dimensionless height and maximize the fraction of rays intercepted by the triangular receiver.

$$C = \arg \max_p \int_T \frac{1L}{(|q-p|^2 + (1L)^2)^{3/2}} dq$$

Evaluate the exact triangle solid-angle atan2 formula and optimize in barycentric coordinates. Hold height/L fixed when comparing different triangle shapes.

Reference calculation. In the 3-4-5 triangle: (1.2920118, 0.9979634); normalized barycentrics (0.3443426, 0.3230029, 0.3326545). Method: stationary root / solid-angle optimizer. No independent coordinate error enclosure is asserted.

Location and well-posedness. Every global maximum is interior by projection and the strictly positive inward boundary derivative. Global uniqueness for all shapes and heights is not established here; the numerical point selector remains qualified.

Attractivity. No failure found in the retained 100-shape screen at this height/L; global attractivity and universal maximum uniqueness remain open. The family has a separate analytical nonattractivity result for all sufficiently small heights, inherited from the certified M1 limit. This does not classify the present parameter. See Appendix D.

The minimum sampled local margin was 0.299696502, at shape 4. The same selected direction gives:

Motion / diameter	Finite-motion ratio
0.002	0.299694339
0.001	0.299689516
0.0005	0.299690076

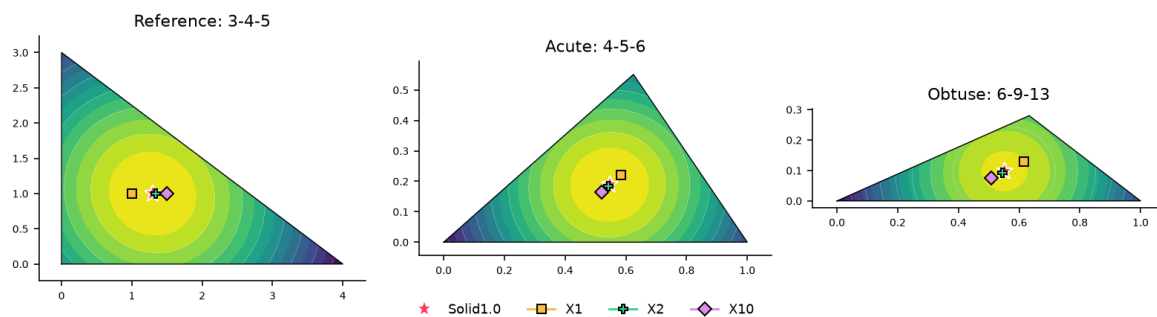
All witness coordinates and root/optimizer settings are stored. Positive values only screen these sampled motions; for Illum the strict-concavity proof identifies the intended root but does not enclose the numerical sign.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X55753	0.000107572	0.000111482	0.000148009
X60182	0.000344995	0.000356144	0.000648984
X51127	0.00147355	0.00153252	0.00230105
X55752	0.00153922	0.00159415	0.0023012
X55754	0.00165979	0.00171924	0.00250206

For location winner X55753, the separately computed derivative-fingerprint RMS discrepancy is 0.000184487, using 50/50 finite family values. The vertex-response margin RMS discrepancy is 0.000164885, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.299884. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

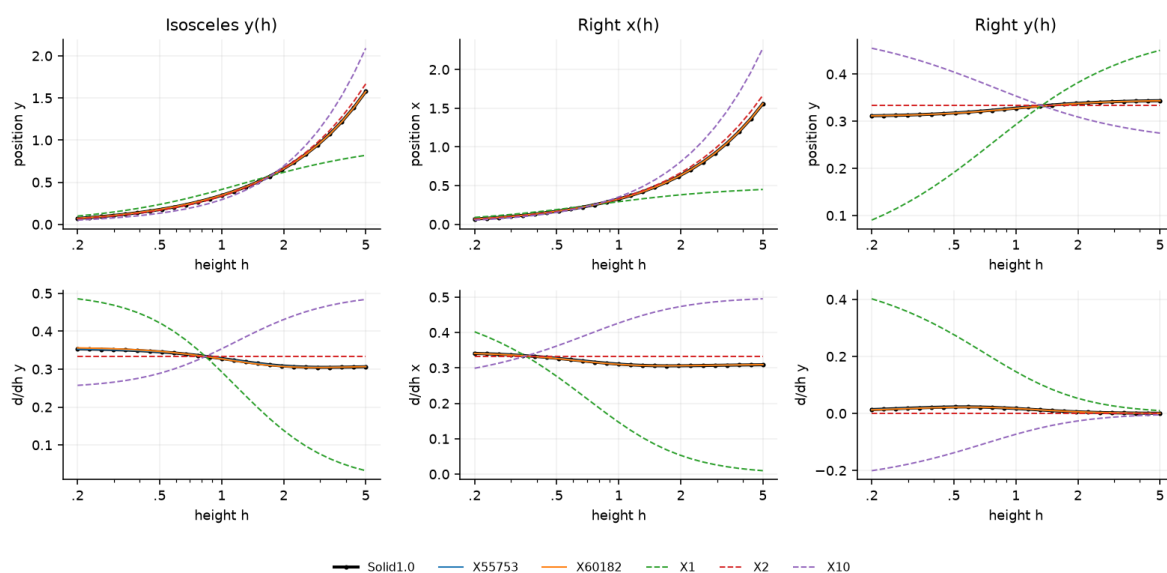
Further fingerprint information. As height/L tends to zero the center approaches M1; as it tends to infinity it approaches X(2). These endpoint statements do not identify an arbitrary finite-height center with either endpoint.

Solid1.0: defining field or measurement density



Atlas figure 16a. Solid1.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.

Solid1.0 | location shortlist and classical benchmarks



Atlas figure 16b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.10.17 Solid10.0: Solid-angle illumination maximum, height/L=10

Place an isotropic point source above the plane at a fixed dimensionless height and maximize the fraction of rays intercepted by the triangular receiver.

$$C = \arg \max_p \int_T \frac{10L}{(|q-p|^2 + (10L)^2)^{3/2}} dq$$

Evaluate the exact triangle solid-angle atan2 formula and optimize in barycentric coordinates. Hold height/L fixed when comparing different triangle shapes.

Reference calculation. In the 3-4-5 triangle: (1.3328232, 0.9999671); normalized barycentrics (0.3334718, 0.3332058, 0.3333224). Method: stationary root / solid-angle optimizer. No independent coordinate error enclosure is asserted.

Location and well-posedness. Every global maximum is interior by projection and the strictly positive inward boundary derivative. Global uniqueness for all shapes and heights is not established here; the numerical point selector remains qualified.

Attractivity. No failure found in the retained 100-shape screen at this height/L; global attractivity and universal maximum uniqueness remain open. The family has a separate analytical nonattractivity result for all sufficiently small heights, inherited from the certified M1 limit. This does not classify the present parameter. See Appendix D.

The minimum sampled local margin was 0.332625702, at shape 17. The same selected direction gives:

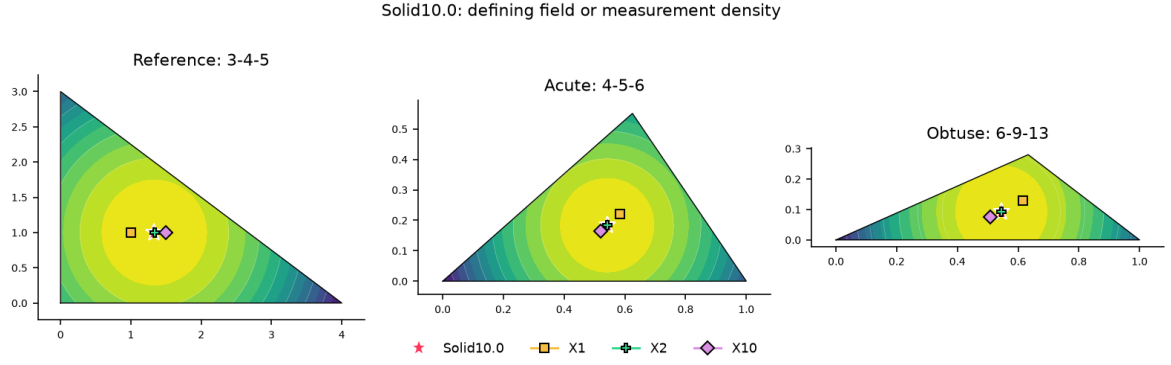
Motion / diameter	Finite-motion ratio
0.002	0.332948316
0.001	0.332996889
0.0005	0.332818693

All witness coordinates and root/optimizer settings are stored. Positive values only screen these sampled motions; for Illum the strict-concavity proof identifies the intended root but does not enclose the numerical sign.

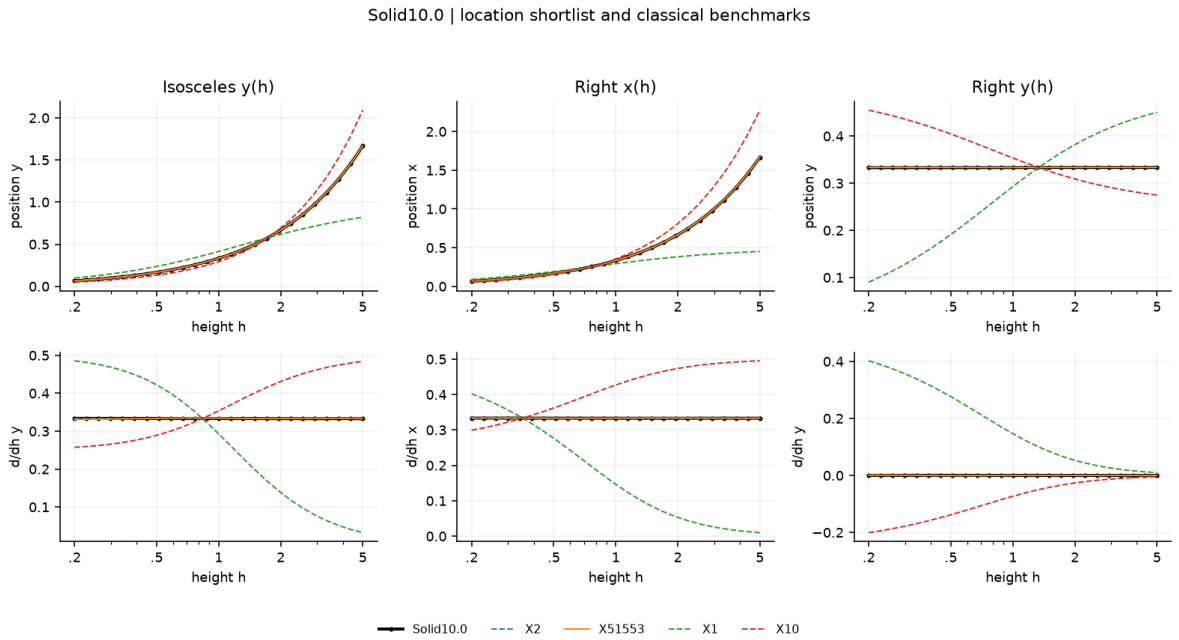
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X2	0.000186889	0.000193339	0.000276859
X51553	0.000422911	0.000412297	0.000578434
X53004	0.000428889	0.000413424	0.000850339
X39331	0.000444343	0.000462781	0.000889529
X16901	0.000494605	0.000506876	0.000881667

For location winner X2, the separately computed derivative-fingerprint RMS discrepancy is 0.000288897, using 50/50 finite family values. The vertex-response margin RMS discrepancy is 0.000336282, using 300/300 finite vertex margins. The candidate's minimum screened margin is 0.333333. These numbers compare numerical behavior; none establishes an exact identification or a global response theorem.

Further fingerprint information. As height/L tends to zero the center approaches M1; as it tends to infinity it approaches X(2). These endpoint statements do not identify an arbitrary finite-height center with either endpoint.



Atlas figure 17a. Solid10.0 in the reference 3-4-5 triangle, an acute 4-5-6 triangle and an obtuse 6-9-13 triangle. Red star: selected point; comparison markers: X1, X2, X10. Each field panel is normalized separately. The plotted fields are illustrative, not interval enclosures.



Atlas figure 17b. Archived isosceles and right-triangle position curves and height derivatives. Comparisons include two leading ETC location neighbors and X1, X2, X10. Gaps represent undefined or excluded values; finite differences near degeneracies and unresolved maximum locations require the qualifications in the text.

7.11 Remaining optical proposals and the limits of the inventory

An elevated source can instead maximize the weakest illumination of the three vertices. At fixed height, a monotone radial law reduces that maximin problem to minimizing the greatest distance to the vertices. The answer is the minimum-enclosing-circle center: the circumcenter for an acute triangle, the midpoint of the hypotenuse for a right triangle, and the midpoint of the longest side for an obtuse triangle. It is a natural piecewise classical construction, not the circumcenter continued outside every obtuse triangle. If height is also optimized, the height objective must be stated; different radiometric laws need not select the same height.

A triangular total-internal-reflection guide adds a refractive-index ratio and a launch-direction distribution. Passing a first-wall incidence test does not prove indefinite guidance after later

reflections. The existing preliminary proposal is therefore an experiment requiring a complete dynamical definition, not a fully investigated parameter-free center. Its omission from numerical rankings is deliberate and is recorded in the coverage ledger.

A Fraunhofer central bright direction is perpendicular to the uniformly phased aperture. It gives no preferred point within the aperture plane without an additional convention. Likewise a complex zero outside the triangle may have multiple symmetry-related branches and cannot be turned into a canonical center by selecting whichever root an optimizer finds first.

Together with the earlier chapters, these distinctions complete the physical inventory used by the atlas: classical identities, well-defined new means and maxima, parameter families, and proposals that do not yet define a unique intrinsic point. They are different kinds of outcome. The book preserves unsuccessful proposals when their failure teaches us how to formulate a better experiment.

7.12 Demonstrations and reproducibility

Project the aperture intensity onto a calibrated camera, display the raw image next to its exponential darkness image, and overlay their centroids. Vary beta while leaving F fixed. This makes the response law visible and shows why a strong darkness center can move sharply even though every pixel weight remains positive. Move the screen as wavelength changes to test the fixed-F protocol; keep it fixed in a second run to show that this is a different experiment.

For Gaussian imaging, compare a linear detector and a squared-intensity response. The linear mean stays at G; the nonlinear mean follows $\text{Opt}(\sigma)$. A sufficiently large camera window or the exact pair reduction is needed for the full-plane definition. An elevated point source and triangular receiver demonstrate the solid-angle family, approaching the magnetic minimum at low height and the lamina centroid at high height under the stated model.

All coefficients, quadrature orders, shape samples, figures and witness inputs are included in the handoff. The full ETC database remains an external licensed input; the package includes selected formulas and the existing finite-candidate ranking results. The chapter's nonvanishing proof, positive-moment certificates and strict-concavity argument do not rely on a numerical search.

Appendix A

Steiner geometry, inertia and related ETC structures

A.1 Three proofs about the Steiner ellipse

A.1.1 Maximum area without advanced convex geometry

Normalize an equilateral triangle so its incircle is the unit circle centered at zero. Write its sides as $n_i \cdot x \leq 1$, where the three outward unit normals satisfy $\sum_i n_i = 0$ and $\sum_i n_i n_i^T = (3/2)\mathbf{1}$. Any contained ellipse has center c and a symmetric positive definite square-root shape matrix A : it consists of $c + Au$ with $|u| \leq 1$.

Containment requires $n_i \cdot c + |An_i| \leq 1$. Summing eliminates c , giving $\sum_i |An_i| \leq 3$. Since $n_i^T An_i \leq |An_i|$, it follows that $(3/2) \operatorname{tr} A \leq 3$. The arithmetic-geometric mean inequality gives $\det A \leq (\operatorname{tr} A/2)^2 \leq 1$. Its area is $\pi \det A$, no larger than that of the incircle. Equality forces $A = \mathbf{1}$, and the three side inequalities then force $c = 0$.

Affine maps multiply every candidate ellipse area by the same determinant. Therefore the affine image of that circle, the Steiner inellipse, is uniquely maximal for every triangle. This proof also explains why the optimum has midpoint contacts and why it is a statement about the triangle as a convex set, independent of what mass might later be put on its boundary.

A.1.2 Minimum enclosing ellipse

In an equilateral triangle with vertices v_i at radius R , $\sum_i v_i = 0$ and $\sum_i v_i v_i^T = (3R^2/2)\mathbf{1}$. If an ellipse centered at c has inverse shape matrix B and contains all vertices, $\sum (v_i - c)^T B (v_i - c) \leq 1$ over the three vertices. This gives $(R^2/2) \operatorname{tr} B + c^T B c \leq 1$. Hence $\det B \leq (\operatorname{tr} B/2)^2 \leq R^{-4}$, so its area $\pi/\sqrt{\det B}$ is at least πR^2 . Equality requires the circumcircle centered at zero.

An affine map again preserves the extremizer relative to all candidates, proving the Steiner circumellipse result. The equilateral circumradius is twice its inradius, so the shape matrix of the circumellipse is four times the inellipse's, and its area is four times as large. Its center remains G .

A.1.3 Foci, derivatives of a cubic, and oblique prism sections

Represent a centered vertex by $z_i = x_i + iy_i$, so $\sum z_i = 0$. The vertex polynomial is $P(z) = z^3 + s_2 z - s_3$, with $s_2 = -\frac{1}{2} \sum z_i^2$. Thus $P'(z) = 0$ means

$$z^2 = \frac{1}{6} \sum_i z_i^2.$$

For any ellipse shape matrix Q , its two centered complex foci satisfy $z^2 = Q_{xx} - Q_{yy} + 2iQ_{xy}$. This follows by rotating to principal axes: there the right side is the difference of squared semiaxes, and rotating back multiplies it by the square of the unit complex rotation. Substituting $Q = \sum v_i v_i^T / 6$ proves that the derivative roots are the Steiner foci.

There is a spatial visualization from the referenced discussion. Intersect a right prism over an equilateral triangle by an oblique plane. Projection along the prism maps the equilateral base affinely onto the section triangle. The inscribed and circumscribed circular cylinders intersect the cutting plane in its Steiner inellipse and circumellipse. Any triangle similarity class can arise: after an overall scale is removed, a positive two-dimensional metric has eigenvalues 1 and $1 + s^2$, exactly the metric $\mathbf{1} + uu^T$ of a tilted graph plane with $|u| = s$.

This construction explains the affine geometry. It does not preserve uniform wire length: an oblique section stretches the three side directions differently. That is precisely the obstruction encountered in Section 4. The discussion is restricted to two-dimensional triangular sections; higher-dimensional simplex analogies require additional conditions and are not used here.

A.2 Steiner geometry, inertia and higher ETC relations

A.2.1 Constructing the inellipse from the lamina covariance

Recall the normalized covariance of a uniform triangular lamina about G :

$$\Sigma = \frac{1}{S} \int_T (p - G)(p - G)^T dA = \frac{1}{12} \sum_{V=A,B,C} (V - G)(V - G)^T.$$

This is the second-moment tensor used in the preceding mass chapter. The planar mechanical inertia tensor is $\text{tr}(\Sigma)I - \Sigma$ after normalization by mass. It has the same principal directions, with the corresponding eigenvalues interchanged.

Start with an equilateral triangle. Its incircle is tangent to the three side midpoints. An invertible affine map taking that triangle to T takes the circle to an ellipse and takes side midpoints to side midpoints. This gives the Steiner inellipse. Its center is G because affine maps preserve centroids. The construction also establishes uniqueness by pulling any competing midpoint-tangent ellipse back to the equilateral configuration.

Let $Q = 2\Sigma$. The inellipse has the equation

$$(p - G)^T Q^{-1} (p - G) = 1, \quad Q = \frac{1}{6} \sum_{V=A,B,C} (V - G)(V - G)^T.$$

The factor can be checked in the equilateral case and then transported by the affine covariance law $Q \mapsto LQL^T$. If the eigenvalues of Q are $q_1 \geq q_2 > 0$, the semiaxes are their square roots. The major principal direction is a unit eigenvector e_1 for q_1 . The foci are

$$F_{\pm} = G \pm \sqrt{q_1 - q_2} e_1.$$

Notice what is and is not affine-invariant. Tangency to side midpoints and the center G are preserved by affine maps. Euclidean lengths and foci are not obtained by applying an arbitrary

affine map to the old foci. Instead one must recompute the eigenvalues and directions of Q after transforming the triangle.

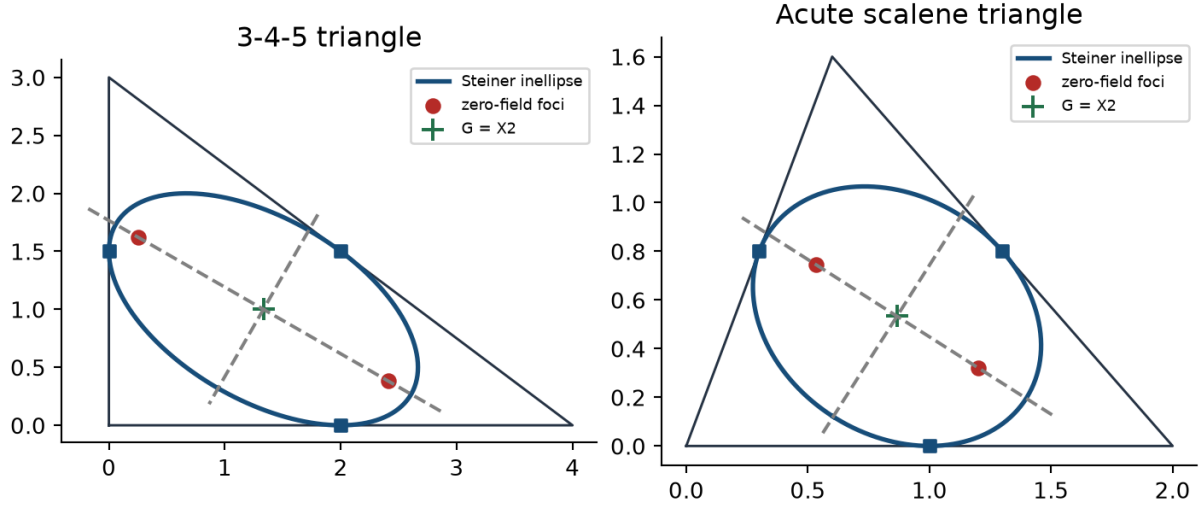


Figure 5. Side-midpoint contacts, centroid, principal axes and the two foci in two triangles. The foci are precisely the zero-field points for three equal axial vertex currents. The same principal directions arise from the uniform lamina's inertia tensor, although the physical source in the magnetic experiment is concentrated at the vertices.

A.2.2 A short covariance proof of the cubic-root identification

Translate the complex vertices so that their mean is zero, and write them as $w_i = x_i + iy_i$. The quadratic discriminant in the formula for the zeros of Q prime is proportional to the complex second moment $\sum w_i^2$. If a focus displacement from G is represented by the complex number f , the cubic derivative gives

$$f^2 = \frac{1}{6} \sum_{i=1}^3 w_i^2.$$

On the other hand, the real matrix Q above satisfies

$$Q_{xx} - Q_{yy} + 2iQ_{xy} = \frac{1}{6} \sum_{i=1}^3 w_i^2.$$

Its right-hand side has modulus $q_1 - q_2$ and argument twice the major-axis angle. Taking its two square roots therefore gives exactly the two displacement vectors of the ellipse foci. This proves the magnetic identification directly from the covariance formula and elementary complex algebra. Marden's theorem gives the same result in its classical polynomial language [12, 13].

For the 3-4-5 triangle,

$$\Sigma_{xx} = 8/9, \quad \Sigma_{xy} = \Sigma_{yx} = -1/3, \quad \Sigma_{yy} = 1/2, \quad Q = 2\Sigma.$$

The eigenvalues of Q are $(25 + \sqrt{193})/18$ and $(25 - \sqrt{193})/18$. Thus the squared focal distance is $\sqrt{193}/9$. Its direction is the major eigenvector, which reproduces the two numerical minima in Section 5. This offers an independent check that does not evaluate a magnetic integral.

The covariance of three equal point masses at the vertices is four times the uniform-lamina covariance. It consequently gives the same principal directions. The covariance of a uniform perimeter wire is a different tensor and does not in general yield these same foci. Uniformity with respect to different source measures must not be conflated.

A.2.3 Isogonal conjugates and why a pair is natural

Two rays from a vertex are isogonal when reflection in the internal angle bisector takes one to the other. The two foci of an ellipse tangent to the sides of a triangle form an isogonal pair. One way to see this is to reflect a focus across a tangent line: the reflected point lies on the circle centered at the other focus whose radius is the sum of focal distances on the ellipse. The two tangents from a triangle vertex therefore impose the equal-angle reflection relation.

For the Steiner inellipse, the additional fact that the midpoint of the foci is G singles out the corresponding pair. In normalized or homogeneous barycentrics, isogonal conjugation takes

$$(u : v : w) \longmapsto (a^2/u : b^2/v : c^2/w).$$

This is a useful independent numerical test for the magnetic zeros. It does not mean either focus is generally a single ETC X center. The bicentric entries $P(118)$, $U(118)$ record the pair; their midpoint is $X2$ [1]. Degenerate circular symmetry makes the distinction between the two members disappear.

A.2.4 The circumellipse, complements and the axis centers

The Steiner circumellipse is the ellipse centered at G and passing through A , B and C . It is obtained from the inellipse by the homothety $p \mapsto G - 2(p - G)$. Its axes are twice as long and its real foci are the images of the inellipse's real foci. Those circumellipse foci are the Bickart pair $P(116)$, $U(116)$. The map is the anticomplement map; its inverse is the complement map. The conventional names of the two members swap under this negative dilation [1].

The common major axis of the Steiner ellipses is the central line through $X2$ and $X1341$, and the minor axis passes through $X2$ and $X1340$, in generic configurations where these line descriptions are nondegenerate. The ETC describes $X1341$ and $X1340$ as the external and internal centers of similitude of the circumcircle and the Brocard circle. Equivalently, they are the intersections of the Brocard axis with the major and minor Steiner axes [1].

For orientation, the Brocard axis is the line through the circumcenter $X3$ and the symmedian point $X6$. The Brocard circle has diameter joining those two points. A center of similitude of two circles is a center of a homothety taking one circle to the other. The external and internal choices use the difference and sum of their radii. These constructions explain why points with high ETC indices can encode simple axis information already contained in an inertia tensor.

At an equilateral triangle both Steiner axes are undetermined because Q is a scalar matrix. The tensor and ellipse remain well-defined, but an axis direction or a line through two coincident auxiliary centers should not be assigned an artificial numerical value. Appendix relations involving generic axes must be read with this symmetry caveat.

A.2.5 $X360$ and the limits of algebraic expectations

The angle-weighted point $M3$ has barycentrics $\alpha:\beta:\gamma$, so its first center function may be written as the arccosine from the cosine rule. The existence of this entry, $X360$, already shows that it is unsafe to equate ETC membership with a short rational or radical formula.

Conversely, a numerical neighbor with a radical expression need not be identical to an integral-defined magnetic mean.

For the normalized isosceles family, M3 has an exact fingerprint:

$$M3_y(h) = \frac{2h}{\pi} \arctan(1/h), \quad M3'_y(h) = \frac{2}{\pi} \left[\arctan(1/h) - \frac{h}{1+h^2} \right].$$

For the right family used in the atlas,

$$M3(h) = \left(\frac{h \arctan(1/h)}{\pi}, \frac{\arctan h}{\pi} \right).$$

At the equilateral triangle, the apex response coefficients are $1/3 + \sqrt{3}/(2\pi)$ horizontally and $1/3 - \sqrt{3}/(2\pi)$ vertically. For the thin family with fixed x strictly below 1, M3 approaches the apex and its vertical coordinate divided by epsilon tends to 1. At the endpoint $x=1$ that ratio tends to $1/2$. This is a simple exact example of the nonuniform thin limits that numerical fingerprints must distinguish.

A.3 Steiner inellipse and affine changes of the experiment

For uniform area mass, let Sigma be the centered second-moment matrix from the first chapter. If v_i are vertex vectors relative to G , then $\text{Sigma} = (1/12) \sum v_i v_i^T$. The Steiner inellipse can be written as $(x-G)^T \text{Sigma}^{-1} (x-G) \leq 2$. Its axes are the inertia axes, its center is G , and its boundary touches the three side midpoints. These facts describe geometry and uniform mass moments.

Set $x=A+By$ under a general invertible affine map. The gradient becomes B^{-T} times the y gradient. The tension energy transforms into an integral with tensor $B^{-T} B^{-1}$. Thus isotropic membrane energy in one coordinate system becomes anisotropic in the other unless B is a rotation times a scalar. The triangular isotropic PDE cannot be reduced to a universal affine image of the equilateral polynomial while leaving the physical tension isotropic.

For an elliptic plate, a similarly useful exact clamped comparison is $w=C(1-x^2/a^2-y^2/b^2)^2$. Its value and normal derivative vanish at the boundary. Its biharmonic is a constant, so C can be chosen to match any uniform load. All maximum and symmetric first-moment selectors lie at the ellipse center. Again, that special result comes from the elliptical boundary and central symmetry, not from the covariance of a triangle.

The skeleton inertia ellipse from the first chapter generally lacks a homothetic inscribed member tangent to all three sides. A triangular wire frame can support a sheet, but its mass distribution is not the sheet's displacement or bending-energy distribution. None of the frame's inertia axes supplies an exact general membrane-center formula.

Appendix B

Inellipses, circumellipses and the principal axes of a skeleton

The Steiner inellipse encodes area covariance with exceptional simplicity. A skeleton also has a covariance ellipse, but rescaling it does not reproduce the geometric inellipse or circumellipse at the skeleton centroid in a scalene triangle. The derivation below makes this distinction precise.

B.1 The unique inellipse with a prescribed center

Let $D = \alpha A + \beta B + \gamma C$ with positive coordinates summing to one, and $H = [B - A \ C - A]$. In affine reference coordinates an ellipse centered at (β, γ) has shape matrix

$$S = \begin{pmatrix} \beta^2 & (\alpha^2 - \beta^2 - \gamma^2)/2 \\ (\alpha^2 - \beta^2 - \gamma^2)/2 & \gamma^2 \end{pmatrix}.$$

Tangency to $x = 0$, $y = 0$, $x + y = 1$ forces these entries, since the extent in direction n is $\sqrt{n^T S n}$. Also

$$\det S = \frac{(1 - 2\alpha)(1 - 2\beta)(1 - 2\gamma)}{4}, \quad Q_i = H S H^T.$$

There is exactly one positive-definite solution inside the medial triangle, a degenerate segment on its boundary, and no inellipse outside. This independently derives the geometric theorem in [30].

At a fixed center its axes are fixed. Across all centers any orientation and semiaxis ratio greater than one can be realized: transform the triangle by the inverse of the desired stretch, construct its incircle and transform back. At the incenter the resulting circle has no distinguished major axis.

B.2 The circumellipse with the same center

Put $u = \alpha(1 - 2\alpha)$, $v = \beta(1 - 2\beta)$, $w = \gamma(1 - 2\gamma)$ and

$$T_D = \begin{pmatrix} w & (v + w - u)/2 \\ (v + w - u)/2 & v \end{pmatrix}.$$

The centered circumconic is $u y z + v z x + w x y = 0$ in homogeneous barycentric observation coordinates. Translation to D gives $\alpha\beta\gamma - \xi^T T_D \xi = 0$. Within the medial triangle,

$$Q_o = \alpha\beta\gamma H T_D^{-1} H^T, \quad \det T_D = \det S.$$

The ellipse area ratio is therefore

$$\rho(D) = \frac{\text{area}(E_o)}{\text{area}(E_i)} = \frac{4\alpha\beta\gamma}{(1-2\alpha)(1-2\beta)(1-2\gamma)}.$$

At G , $Q_o = 4Q_i$, recovering the Steiner pair.

B.3 A cubic records axis misalignment

For counterclockwise vertices, define

$$F(D) = a^2\beta\gamma(\beta - \gamma) + b^2\gamma\alpha(\gamma - \alpha) + c^2\alpha\beta(\alpha - \beta).$$

Matrix multiplication gives

$$[Q_i, Q_o]_{12} = -\mathcal{A}\rho(D)F(D).$$

Symmetric matrices have common principal directions exactly when they commute. Away from circles, the unordered axis pairs therefore agree on $F = 0$ within the medial triangle. The polynomial is the Thomson cubic K002 [31]. This is a common-center locus, not a common-perspector statement.

Let g_i, g_o be eigenvalue gaps and $\delta = \phi_o - \phi_i$ the major-axis angle difference. Then

$$\sin(2\delta) = -\frac{2\mathcal{A}\rho(D)F(D)}{g_i g_o}.$$

With

$$B_D = \text{tr}(Q_i Q_o) - \frac{1}{2} \text{tr}(Q_i) \text{tr}(Q_o), \quad \delta = \frac{1}{2} \text{atan2}(-\mathcal{A}\rho(D)F(D), B_D),$$

the sign and cosine distinguish agreeing major axes from interchanged major/minor axes. A circle has no major direction, so the divided formula is then inapplicable.

The cubic tangent is not generally an ellipse axis. At a regular point D_0 where noncircular major axes agree, a Cartesian displacement h gives, in radians,

$$\delta(D_0 + h) = -\frac{\mathcal{A}\rho(D_0)}{g_i(D_0)g_o(D_0)} \nabla F(D_0) \cdot h + O(|h|^2).$$

This is a relative-rotation sensitivity. An absolute axis angle comes from $\phi(Q) = \frac{1}{2} \text{atan2}(2Q_{12}, Q_{11} - Q_{22})$.

B.4 The exact skeleton identity at X10

Uniform arclength gives $W = X_{10}$ with

$$(\alpha, \beta, \gamma) = \frac{(b+c, c+a, a+b)}{2P}.$$

An edge with vector $d = V - U$ and midpoint m contributes $|d| [dd^T/12 + (m - W)(m - W)^T]/P$ to covariance. Its affine-reference sum is

$$C_{\text{ref}} = \begin{pmatrix} 2\beta/3 - \beta^2 & (1-2\alpha)/6 - \beta\gamma \\ (1-2\alpha)/6 - \beta\gamma & 2\gamma/3 - \gamma^2 \end{pmatrix} = \frac{1}{3}S + \frac{2}{3} \text{adj } T_D.$$

Using $\det T_D = abc/(4P^3)$ at W proves

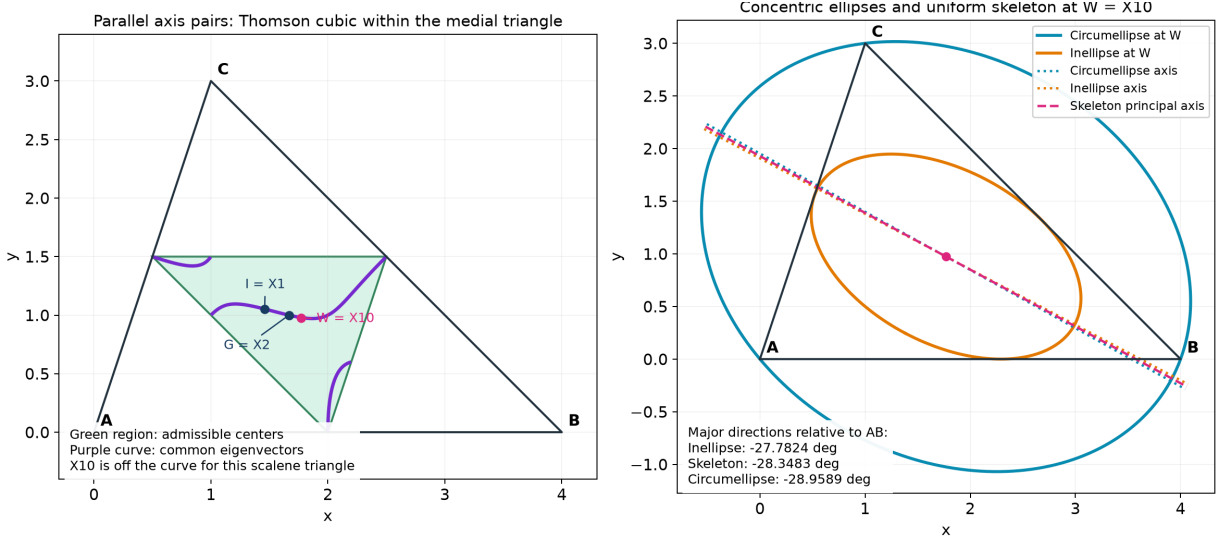
$$\Sigma_{\text{wire}} = \frac{1}{3}Q_i + \frac{4abc}{3(a+b)(b+c)(c+a)}Q_o.$$

Both coefficients are positive. The traceless vector $(Q_{11} - Q_{22}, 2Q_{12})$ points along twice the major-axis angle. Positive addition puts the wire's major covariance direction between the two ellipse directions along their shorter angular interval.

Moreover,

$$F(W) = \frac{(a-b)(b-c)(c-a)}{8P}.$$

For scalene triangles it is nonzero, so all three axis pairs differ. The in-plane inertia tensor is total mass times $\text{tr}(\Sigma_{\text{wire}})\text{Id} - \Sigma_{\text{wire}}$: the same axes with reversed eigenvalue order.



Conic-axis illustration. For the acute triangle $(0,0), (4,0), (1,3)$, the inellipse, skeleton and circumellipse major directions at X_{10} are -27.7824° , -28.3483° and -28.9585° relative to AB . The covariance identity's numerical residual is below $3 \cdot 10^{-16}$; its proof is the algebra above.

B.5 A physical consequence

For area mass, a scalar multiple of covariance gives the midpoint-tangent Steiner inellipse. For scalene wire mass, no scalar multiple gives either geometric ellipse centered at W : their orientations differ already. A wire covariance ellipse and a unique geometric inellipse at W both exist. They answer different selection questions.

Appendix C

Capacity and affine moment identities

C.1 Small-electrode capacity in higher dimensions

Place a spherical electrode of radius ϵ about p inside a grounded cavity. Near p , the unnormalized Green function equals $r^{2-n} - R_\Omega(p) + O(r)$ for $n \geq 3$. To make the small electrode approximately equipotential, its monopole coefficient is inversely proportional to $\epsilon^{2-n} - R_\Omega(p)$. Up to positive physical constants, its capacitance has expansion

$$C_\epsilon(p) = \epsilon^{n-2} \left[1 + R_\Omega(p)\epsilon^{n-2} + o(\epsilon^{n-2}) \right].$$

The leading term is the free-space capacitance of the small sphere and does not depend on its position. The first boundary-dependent term is minimized at the Robin minimum. In two dimensions the corresponding expression is $1/\log[r_T(p)/\epsilon]$. These asymptotics give a direct physical interpretation of the least-capacity center without requiring an experiment with an inverted conductor.

The probe must stay a positive distance from the cavity boundary during the expansion. Optimizing a large probe, holding its shape fixed rather than its radius, or allowing contact with the boundary creates different constrained problems. No claim is made that their finite-size centers coincide exactly with the small-probe limit.

C.2 Affine first-moment identities

For a smooth clamped w , the boundary traces w_x and w_y vanish. First, the integral of Δw times w_x is zero: integrate once to obtain minus one half the integral of the x derivative of $|\text{grad } w|^2$, whose boundary term is zero. The same holds with w_y . Approximation by smooth clamped functions extends the identity to the natural energy setting when the products are defined.

Write $\det(D^2 w)$ as the divergence of $(w_x w_{yy}, -w_x w_{xy})$. Multiplication by an affine ℓ and integration leaves $-\ell_x$ times the integral of $w_x w_{yy}$ plus ℓ_y times the integral of $w_x w_{xy}$. The second integral is one half the integral of the y derivative of w_x^2 and vanishes. Integrate the first in y to get minus the integral of $w_{xy} w_y$; that is minus one half the integral of the x derivative of w_y^2 and also vanishes. The same zero gradient traces remove every boundary term.

For constant load, testing Delta squared $w=q$ against $ell\ w$ then gives the energy/volume identity of Section 10. Taking $ell=1$ supplies the total mass of each weight; taking $ell=x$ or y supplies its first moment. This is why the result identifies a centroid even though the energy contours depend on Poisson ratio.

Appendix D

Attractivity: proofs, certificates and current limits

This appendix supersedes earlier positive-candidate wording for E0. A failed finite screen and a rigorous finite counterexample are different evidence. The four exact classical points below exhaust the general positive proofs claimed for the studied selectors. Their physical aliases add realizations, not new independent theorems.

D.1 Four exact attractivity results

All derivatives below keep the other two vertices fixed. The triangle inequalities are strict. Integrating the positive quadratic form along any nondegenerate straight vertex path gives the finite-motion result.

D.1.1 The centroid

$D_A G = \mathbf{1}/3$, so $h \cdot [G(A+h) - G(A)] = |h|^2/3 > 0$ for $h \neq 0$. Its physical aliases include uniform lamina mean, equal vertex-mass mean, prescribed uniform area-charge mean, and the reflecting Brownian stationary mean.

D.1.2 The incenter

Set $A = 0$, $p = (B - A)/c$, $q = (C - A)/b$. Use an orthonormal frame along the internal angle bisector, so $p = (\cos t, \sin t)$ and $q = (\cos t, -\sin t)$, where $t = \theta_A/2$. The incenter is $(2bc \cos t/P, 0)$. Direct differentiation of $I = (aA + bB + cC)/P$ gives

$$D_A I = \frac{a}{P} \mathbf{1} - \frac{(B - I)q^T + (C - I)p^T}{P}.$$

Put $m = b + c$ and $d = b - c$. The cosine law gives $\sin^2 t = (a^2 - d^2)/(m^2 - d^2)$. The diagonal entries of the symmetric part are

$$(S_A)_{11} = \frac{m(a^2 - d^2)}{(a + m)(m^2 - d^2)}, \quad (S_A)_{22} = \frac{a + m \sin^2 t}{P}.$$

Its off-diagonal entries vanish; the asymmetry of b, c only produces an antisymmetric part of the full derivative. Since $|d| < a < m$, both displayed eigenvalues are positive. The incenter is strictly attractive. The same theorem applies to the restricted three-face deepest-apex construction.

D.1.3 The uniform-wire mean

Set $u = p \cdot q$. Differentiating the length-weighted midpoint mean gives

$$S_A(W) = \frac{b+c}{2P} \mathbf{1} - \frac{bc}{2P^2} (pp^T + qq^T) + \frac{c(a+c) + b(a+b)}{4P^2} (pq^T + qp^T).$$

Directions $p+q$ and $p-q$ are orthogonal eigenvectors. Their eigenvalues are

$$\lambda_+ = \frac{2P(b+c) + (1+u)[(b-c)^2 + a(b+c)]}{4P^2},$$

$$\lambda_- = \frac{(b+c)(1+u)}{4P}.$$

Both are positive because $-1 < u < 1$. Thus X_{10} is strictly attractive, including its specified thin-wire charge and loop-field cutoff aliases.

D.1.4 The angle-weighted vertex mean

For counterclockwise vertices put $u = (A-B)/|A-B|$, $v = (A-C)/|A-C|$ and let R rotate 90° . Differentiating angle functions and their weighted mean gives

$$D_A X_{360} = \frac{1}{\pi} [\theta_A \mathbf{1} - u(Ru)^T + v(Rv)^T].$$

The symmetric part has eigenvalues $[\theta_A \pm \sin \theta_A]/\pi$. For $0 < \theta_A < \pi$, both are strictly positive. This proves attractivity of M3 for its equal-current equal-core interior-energy convention. Very thin angles make the smaller eigenvalue tiny; numerical differentiation would be a weaker argument.

D.2 Why the loop minimum is unique

Let $R_p(\theta)$ be the distance from an interior point to the boundary along direction e_θ . Up to a positive constant, the loop field is

$$b(p) = \int_0^{2\pi} \frac{1}{R_p(\theta)} d\theta.$$

One can obtain this identity from the boundary Biot-Savart integral, or as the small-height solid-angle coefficient. If $d_i(p)$ is side distance,

$$\frac{1}{R_p(\theta)} = \max_i \frac{\max(n_i \cdot e_\theta, 0)}{d_i(p)}.$$

Each positive reciprocal affine-distance function is convex in p ; the maximum and integral preserve convexity. For any nonzero direction of motion, at least one side distance changes. That side is first hit on an open angular interval, where its reciprocal has strictly positive second derivative. Thus b is strictly convex. It diverges at the boundary and has exactly one interior minimum. This proves that M1 is a point rule; it does not prove attractive response to moving vertices.

D.3 A finite-motion certificate for E0

The explicit witness uses

$$A = (0, 0), \quad B = (1, 0), \quad C = (0.9963640131614389, 0.0015991738862318396),$$

and $h = 10^{-5}(-0.9997966839583786, -0.02016409546273338)$. Both triangles are nondegenerate. The project certificate encloses

$$\frac{h \cdot [\text{E0}(A + h, B, C) - \text{E0}(A, B, C)]}{|h|^2} \in [-3.13100322614, -3.13100322613] 10^{-7}.$$

Outward-rounded incomplete-beta series evaluations, a degree-250 local series, a degree-160 continuation and explicit remainder bounds produce the enclosure. The finite-motion margin is small, but its entire interval is negative. This single witness disproves global attractivity. It does not say that E0 responds negatively on most triangles. Rounded printed center coordinates alone would not reproduce the certificate's sign; use the bundled interval record.

D.4 Why large removal eventually fails

Here is the mechanism behind the analytical family result. Put $\varepsilon = \mu^{-1/2}$ and consider $A = (0, 0), B = (1, 0), C = (x, \varepsilon\tau)$ with $1/2 < x < 1$. Rescale $y = \varepsilon t$ and write $u = \varepsilon^2 w$. Apart from an overall factor, the energy becomes

$$\mathcal{J}_\varepsilon[w] = \int_{D_\tau} \left[\frac{\varepsilon^2}{2} |w_s|^2 + \frac{1}{2} |w_t|^2 + \frac{\kappa_\varepsilon^2}{2} w^2 - w \right] ds dt, \quad \kappa_\varepsilon^2 = L_\varepsilon^{-2} \rightarrow k^2.$$

The horizontal derivative vanishes in the limit. On a vertical interval of height H , solve $-w_{tt} + k^2 w = 1$, $w(0) = w(H) = 0$:

$$w(t) = k^{-2} \left[1 - \frac{\cosh(k(t - H/2))}{\cosh(kH/2)} \right].$$

Its column mass is $F(H) = H/k^2 - 2 \tanh(kH/2)/k^3$. The triangular height is $\tau q(s)$, where $q(s) = s/x$ to the left of the apex and $q(s) = (1 - s)/(1 - x)$ to its right. Changing variables on each side shows

$$c_x(\tau) = 1 - x + (2x - 1)R_k(\tau), \quad R_k(\tau) = \frac{\int_0^1 qF(\tau q) dq}{\int_0^1 F(\tau q) dq}.$$

Since $F(H) \sim H^3/12$ for small H and $F(H) \sim H/k^2$ for large H , $R_k(\tau) \rightarrow 4/5$ as $\tau \downarrow 0$ and $R_k(\tau) \rightarrow 2/3$ as $\tau \rightarrow \infty$. For $x > 1/2$ choose fixed τ_1, τ_2 giving $c_x(\tau_2) - c_x(\tau_1) = -\delta < 0$.

Move the apex from $(x, \varepsilon\tau_1)$ to $(x + \varepsilon, \varepsilon\tau_2)$. The horizontal motion is positive, but the horizontal center change tends to $-\delta$. The vertical contribution is $O(\varepsilon^2)$, so the dot product is $-\varepsilon\delta + o(\varepsilon) < 0$. Energy comparison controls convergence; for the vertical limiting solution with κ_ε ,

$$\|(w_\varepsilon - w_0)_t\|_2^2 + \kappa_\varepsilon^2 \|w_\varepsilon - w_0\|_2^2 \leq \varepsilon^2 \|(w_0)_s\|_2^2.$$

This is a family theorem for every sufficiently large μ , with triangles depending on μ . It supplies no explicit threshold. The numerical $\mu = 10^4$ witness is supporting evidence, not a certified threshold value.

D.5 The M6 candidate and the moderate-removal family

M6 has no negative witness retained after refinement. The earlier 100-, 284- and 2,048-shape investigations were followed by 512 extreme shapes and 100 repeated shapes using complex-step sensitivities. Ten difficult shapes kept positive margins through quadrature order 256. Initial subtraction-based negative values disappeared under refinement. This is a strong positive conjecture, not a uniform lower bound over all triangles.

For Abs(0), Abs(1), Abs(10) and Abs(100), 612-shape direct-Ritz sensitivity screens and selected degree 14, 20 and 28 refinements remain positive. At removal 10^4 , independently refined Ritz and quadratic FEM give negative finite motions; no interval sign at that exact parameter is claimed. The preceding analytical argument proves failure for all sufficiently large parameters without bounding a threshold. A fixed-shape limit to X2 does not imply uniformly attractive behavior over thin triangles.

D.6 Small-height solid-angle centers inherit an M1 failure

Suppose $\mathcal{C}_t(T)$ tends to $\mathcal{C}_0(T)$ on each of two fixed triangles connected by an admissible vertex motion h . If $h \cdot [\mathcal{C}_0(T+h) - \mathcal{C}_0(T)] < 0$, the same dot product is negative for all sufficiently small t . Only convergence on those two triangles is needed; convergence of shape derivatives is unnecessary.

For a point p inside the aperture, let $R_p(\varphi)$ be its ray distance to the boundary. Polar integration gives

$$\Omega_h(p) = 2\pi - h \int_0^{2\pi} \frac{d\varphi}{\sqrt{R_p(\varphi)^2 + h^2}}.$$

On compact interior sets the coefficient tends uniformly to the loop-field objective defining M1. A sequence approaching the boundary has an unbounded coefficient as $h \downarrow 0$: a fixed angular sector contributes at least a positive constant divided by $\sqrt{d^2 + h^2}$. Comparison with a fixed interior point keeps maximizers away from the boundary. The unique M1 minimum therefore attracts every global solid-angle maximizing sequence.

Take the two triangles of the certified M1 negative finite motion and put $h_{\text{physical}} = \eta L(T)$ in each experiment. Both physical heights tend to zero as $\eta \downarrow 0$. The closure argument proves nonattractivity for every sufficiently small positive η , even if an arbitrary finite height admits several global maximizers. It gives no certified finite threshold. The sampled heights 0.1, 1, 10 are not reclassified by this theorem.

D.7 A practical classification ledger

The center index gives the individual records. Positive first-moment densities prove interiority, not attraction. The continuum conclusions for F1, Q0m and several plate maxima remain sensitive to maximum-location refinement or unproved global uniqueness. Stable negative numerical responses are retained as evidence rather than promoted to interval proofs.

Appendix E

Algorithms, reproduction and open problems

E.1 Reproducibility, numerical safeguards, and open work

The companion contains editable Markdown and LaTeX, the generated figures in PNG and SVG, the reference JSON, full reference charge arrays, source code, selected ETC expressions, and the inherited atlas data with their original diagnostics. All quantities needed to reproduce the displayed reference calculations are supplied; the full external ETC database is not duplicated. A hash manifest detects accidental changes in the package.

The reference calculations can be run using Python, NumPy, SciPy, mpmath and Matplotlib. The PDF renderer additionally uses ReportLab and Pillow. `mass_charge_compute.py` generates conductor and prism data. `mass_charge_figures.py` draws the illustrations and comparisons. `mass_charge_build.py` assembles the chapter and evidence tables. `mass_charge_verify.py` checks exact moments, tangencies, covariance transformations, special cases, numerical symmetries and independent potential residuals. The README gives the commands and explains the separate optional rendering dependencies.

For the planar conductor, the linear system includes an explicit charge constraint. For the three-dimensional plate, exact self-panel integration is essential. For the prism, both panel refinement and frequency quadrature must be checked; a small optimizer tolerance does not imply an equally accurate physical center. Conformal calculations are checked against the isosceles-right special value and against vertex relabeling. All data tables distinguish the last mesh change from a rigorous error enclosure.

The delivered checks recover the isosceles-right conformal center coordinate 0.301121610841322 to double precision, and a separate logarithmic boundary solver locates the reference E1 within 0.0000588 of the conformal value. Off-collocation potential residuals are about 0.000248 maximum for E2 and 0.0111 maximum for E3, in their respective unit-charge kernel normalizations; their RMS values are 0.0000178 and 0.00234. These residuals expose the remaining panel error even though the solved collocation equations have much smaller residuals. Increasing the prism frequency range and node count leaves its objective unchanged to about fourteen digits at fixed panels; panel refinement remains the significant error source in that test.

The existing atlas supports location comparisons, not a theorem identifying E1, E2 or E3 with an ETC entry. It also contains exploratory fits to simple center functions. This chapter favors the defining integral or map over a fitted symbolic expression unless a precise domain and error criterion are supplied. The first-order equilateral coefficients already offer useful local symbolic approximations, but a fit near equilateral triangles cannot certify behavior near degeneracy.

Useful remaining research includes certified coordinate and derivative bounds for E3, interval validation of the L3/L4 witnesses, and explicit finite-thickness corner error bounds. The global E1 and E2 attractivity question is settled negatively by the new certificates. The finite-sample fingerprints and ETC comparisons for L3/L4 are now present; classification of all higher dimensions and the bounded-cavity thickness family remains open.

E.2 Reproducibility, evidence and exercises

E.2.1 What the companion contains

The chapter companion contains the editable chapter text, figure sources and rendered figures, raw reference resolutions, 100-shape and 40-holdout data, fingerprints and their derivatives, thin-family data, near-equilateral derivative matrices, ETC shortlists and finite-motion records. Python scripts reproduce the analytical-field evaluations, quadratures, searches and figures. The earlier M1 interval certificate and its verifier are included with their original provenance.

No complete new ETC database is embedded in the small chapter companion. For a full fresh ETC traversal, use the database supplied with the project's expanded handoff or set the documented external path. Existing figures, shortlists and this PDF can be inspected without it. The source WXF snapshot hash is recorded in the companion README; unsupported entries and cache row mappings remain explicit.

The fresh numerical work uses exact segment formulas and corner-adapted quadrature. Its error estimates remain empirical refinement changes. M1 retains its interval-certified negative witness; M3 has an analytical attractivity theorem. M2, M4, M5 and M7 have refined numerical negative witnesses, while M6 remains open. Full-snapshot location rankings are supplemented by explicit derivative-fingerprint and vertex-response comparisons for the top ten candidates in the atlas update data.

E.2.2 Suggested laboratory demonstrations

For the planar loop, drive a triangular wire with a low steady current and use a three-axis Hall probe on a fixed-height scanning stage. Record the normal component with its sign before computing squared-field means. Repeat at several heights and compare the normalized mean with X10. Finite wire thickness, probe averaging and lead fields set the effective regularization, so their dimensions should be recorded rather than inferred from a fit.

For the axial vertex experiment, three parallel wires through a rigid triangular template approximate the local prism model. Equal currents are essential for the Steiner identification. Map the transverse field near the predicted foci and compare the two zeros with an independently drawn Steiner inellipse. Keep the return conductors far enough away for the local approximation, and measure their residual field if quantitative agreement is required.

The surface model can be approximated by many parallel wires distributed uniformly by perimeter arclength. The volume model uses wires distributed uniformly by cross-sectional area. In both approximations the current per wire must represent its source cell's measure. Equal counts per side generally do not approximate a uniform sheet. A simulation can first compare these discrete constructions with the exact edge-integral field, making the continuum limit visible before a physical build.

An energy centroid is a mean of measured B squared, not a point a test charge will automatically seek. A clear demonstration is therefore to construct a digital weighted sheet from the field map and show its computed balance point alongside X1, X2 and X10. The distinction between

minimum and mean then becomes visible rather than purely verbal.

E.2.3 Exercises with checkpoints

1. Derive the infinite-wire integral in Section 4 by substituting $z' = r \tan \theta$. Check that the resulting field is tangent to circles and that its line integral around a circle is $\mu_0 I$.
2. Derive the loop's finite-edge formula without assuming that the perpendicular foot lies on the segment. Explain why the sign remains positive at any point inside a convex triangle.
3. Replace the equal vertex currents by positive fractions w_A, w_B, w_C summing to one. Show that the interior energy cutoff mean has barycentrics $w_A^2 \alpha : w_B^2 \beta : w_C^2 \gamma$. Explain why the equal-current Steiner theorem no longer applies unchanged.
4. Check the M3 derivative matrix in a coordinate system where one unit vector is horizontal. Its traceless part has eigenvalues plus and minus sin alpha. Integrate the positive quadratic form to obtain the finite-motion attractivity inequality.
5. Derive the first far-field correction for a general compact transverse current measure. Verify both disk-limit means and explain why the magnitude mean retains a dependence on the disk center.
6. Calculate the covariance matrix of the reference triangle by direct double integration. Diagonalize twice that matrix and recover the two Steiner foci. Check their midpoint and isogonal relation numerically.
7. On an equilateral triangle, compare the three axial current models. Symmetry places all single-valued means at G, but it does not make their fields identical. Explain the double zero for the source-free vertex and surface models and the linear leading field for the uniform-volume model.
8. Repeat a fingerprint with twice the quadrature order and half the finite-difference step. Identify which digits survive both changes. Explain why a positive response screen on a finite sample does not prove attractivity on all triangles.

E.3 Well-posedness ledger and reproduction

Construction	What is established	What remains open
Tetrahedral sheet / X1	Exact global restricted optimizer; strictly attractive	Not the unrestricted fixed-area sheet
F1 / Mem(2)	Unique interior maximum; numerical negative candidates	Refinement-sensitive half-step; continuum attractivity conclusion
Mem(p), $p > 1$	Unique field and unique interior maximum	Reliable shape derivatives and global attractivity
Positive membrane means	Unique positive moment; interiority	Global attractivity
Clamped plate maxima	Unique field; interior maximizing set	Unique maximizing point and global attractivity
Buckling mode selectors	Defined on a simple mode, or with a full projector convention	Global spectral/maximum uniqueness; attractivity
Plate means	Positive-density interior moments	Global attractivity; slow diagonal convergence
Membrane probe limit / E1	Conformal-radius identity under a common small footprint	Finite-footprint shape family requires its own survey
Only total area, unrestricted sheet	Energy unbounded below	No center exists for the stated model

The companion contains the authored Markdown, an editable LaTeX chapter, the reviewed PDF, PNG and SVG figures, new 3-4-5 reference records, targeted finite-motion refinements, the inherited ETC and fingerprint data, and verification scripts. Run the chapter's computation, figure and build scripts from the project root. Figure generation uses stored fields; it does not repeat the entire ETC search. Raw data keep historical resolution flags rather than deleting difficult shapes from the analysis.

The supplied chain-and-sheet LaTeX is retained unchanged as a source. This chapter uses its variational restrictions explicitly. The master atlas records the tetrahedral and shallow-sheet constructions as aliases of X1 and F1, and the membrane point-probe limit as E1. It upgrades p -torsion maximum uniqueness using [17], while retaining the unresolved attractivity and plate-maximizer questions. No new ETC identity is claimed from numerical proximity.

E.4 A unified reproduction map

The original chapter directories contain the complete saved fields, fingerprints and comparison inputs. The monograph bundle includes the typeset source and every displayed image, the atlas catalogue and selected ETC expressions, four interval records and the dated attractivity addendum. The full vendor WXF snapshot is an external input and is not redistributed.

Run numerical research scripts from the original project root with their documented scientific dependencies. The six-chapter companion's requirements and verification scripts describe that environment. The monograph's source package is self-sufficient for typesetting but does not claim that every original research computation can be regenerated from the figure-only assets. Keep those two purposes separate.

The browser explorer is a free static companion. It displays 66 selectors, classical objects, seven fingerprints, local Taylor records and independently selectable ETC neighbors for each PITC. Most displayed PITC positions are atlas interpolations; M1 and Illum have direct floating-point solvers. Interpolation is not a new continuum solve, an error enclosure or an attractivity proof. Its ZIP contains GitHub Pages deployment instructions.

E.5 Questions for further study

Can a global sign theorem be proved for M6? Can a continuum derivative enclosure settle F1 and Q0m? Which spectral clusters give smooth quantum centers through all relevant crossings? What finite-thickness convergence rates hold at polygonal conductor corners? Can ETC matching use full symmetric vertex Jacobians with justified uncertainty weights? These are mathematical projects; adding more finite samples alone cannot answer every one of them.

Appendix F

Center index and evidence ledger

Each entry links to its primary treatment. Exact aliases, parameter choices and evidence levels are retained; neighboring ETC points are location comparisons, not identities.

Q0m — Quantum ground-state maximum

Treatment: Chapter 5, page 137.

A candidate negative finite response appears only after substantial refinement; its small magnitude and maximum-location sensitivity leave the rigorous conclusion open.

Q0 — Quantum ground-state expectation

Treatment: Chapter 5, page 139.

Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Q1 — Quantum first excited-state expectation

Treatment: Chapter 5, page 141.

Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Q2 — Quantum second excited-state expectation

Treatment: Chapter 5, page 143.

Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Q3 — Quantum third excited-state expectation

Treatment: Chapter 5, page 145.

Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Q4 — Quantum fourth excited-state expectation

Treatment: Chapter 5, page 147.

Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

E0 — Equal-exit conformal center

Treatment: Chapter 5, page 149.

Not globally vertex-attractive: a new outward-rounded interval certificate encloses a finite-motion ratio near $-3.13100322614e-7$. This supersedes the previous 100-shape no-failure screen. See `output/attractivity_candidates/E0_interval_certificate.json` and its proof note.

E1 — Inner conformal / least-capacity center

Treatment: Chapter 2, page 35.

Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.02086913).

E2 — Outer conformal / planar charge centroid

Treatment: Chapter 2, page 36.

Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.04665096), independently evaluated with the exterior Schwarz-Christoffel map.

E3 — Three-dimensional plate charge centroid

Treatment: Chapter 2, page 37.

Refined numerical evidence of nonattractivity: negative finite motions persist on positive-charge anisotropic meshes up to 3,180 panels. The finest halved-step ratio is -0.55181093 . A continuum interval certificate remains open.

E4 — Four-dimensional plate charge centroid

Treatment: Chapter 2, page 10. Exact alias under the stated convention: X2.

Strictly globally vertex-attractive through the exact uniform-thickness limiting identity with X2.

S3 — 3-dimensional skeleton charge centroid

Treatment: Chapter 2, page 10. Exact alias under the stated convention: X10.

Strictly globally vertex-attractive through the exact uniform-thickness limiting identity with X10.

S4 — 4-dimensional skeleton charge centroid

Treatment: Chapter 2, page 10. Exact alias under the stated convention: X10.

Strictly globally vertex-attractive through the exact uniform-thickness limiting identity with X10.

M1 — Loop-field minimum

Treatment: Chapter 3, page 60.

Not globally attractive: the earlier interval-certified negative finite vertex motion remains the rigorous witness.

F1 — Fluid torsion / linear membrane maximum

Treatment: Chapter 6, page 91.

Negative numerical evidence, with unresolved refinement sensitivity. The inherited finite-element witness is negative at subdivisions 128, 192 and 256. Independent Ritz degrees 20 and 32 retain both negative steps, but degree 44 changes the half-step sign. The data do not supply a converged, validated continuum counterexample; global nonattractivity is not certified.

H0.01 — Heat maximum at dimensionless time 0.01

Treatment: Chapter 6, page 164.

The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

H0.03 — Heat maximum at dimensionless time 0.03

Treatment: Chapter 6, page 166.

The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

H0.1 — Heat maximum at dimensionless time 0.1

Treatment: Chapter 6, page 168.

The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

H0.3 — Heat maximum at dimensionless time 0.3

Treatment: Chapter 6, page 170.

The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

H1.0 — Heat maximum at dimensionless time 1.0

Treatment: Chapter 6, page 172.

The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

Pgrav — Gravity-loaded clamped plate maximum

Treatment: Chapter 4, page 107.

Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Bmode — First buckling-mode maximum

Treatment: Chapter 4, page 109.

Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Bmean — First buckling-mode expectation

Treatment: Chapter 4, page 111.

Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Compliance — Maximum point compliance / thermal variance

Treatment: Chapter 4, page 113.

Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

FluctMean — Thermal fluctuation-density centroid

Treatment: Chapter 4, page 115.

Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Energy0.3 — Bending-energy centroid

Treatment: Chapter 4, page 117.

Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Pheat0.5 — Compressed, gravity-biased plate at fraction 0.5

Treatment: Chapter 4, page 119.

Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Pheat0.9 — Compressed, gravity-biased plate at fraction 0.9

Treatment: Chapter 4, page 121.

Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Mem1.5 — Nonlinear membrane maximum, exponent 1.5

Treatment: Chapter 4, page 95.

Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

MemMean1.5 — Nonlinear membrane density mean, exponent 1.5

Treatment: Chapter 4, page 101.

Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Mem3.0 — Nonlinear membrane maximum, exponent 3.0

Treatment: Chapter 4, page 97.

Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

MemMean3.0 — Nonlinear membrane density mean, exponent 3.0

Treatment: Chapter 4, page 103.

Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Mem4.0 — Nonlinear membrane maximum, exponent 4.0

Treatment: Chapter 4, page 99.

Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

MemMean4.0 — Nonlinear membrane density mean, exponent 4.0

Treatment: Chapter 4, page 105.

Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Abs0.0 — Production–removal concentration mean, $\mu=0$

Treatment: Chapter 6, page 93.

Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry.

Abs1.0 — Production–removal concentration mean, $\mu=1$

Treatment: Chapter 6, page 176.

Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry.

Abs10.0 — Production–removal concentration mean, $\mu=10$

Treatment: Chapter 6, page 178.

Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full

Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry.

Abs100.0 — Production–removal concentration mean, $\mu=100$

Treatment: Chapter 6, page 180.

Global attractivity remains open for this parameter. No negative response was found in the new 612-shape direct-Ritz-sensitivity screen; selected cases were refined at degrees 14,20,28. The full Abs family is not globally attractive for all parameters: a thin-domain argument proves failure for every sufficiently large μ , and $\mu=10000$ has independent negative Ritz/FEM evidence. That does not refute this entry.

Bio1.1 — Logistic population mean, $\kappa=1.1$

Treatment: Chapter 6, page 182.

A strong finite numerical counterexample is independently reproduced by quadratic FEM; no interval certificate is supplied.

Bio2.0 — Logistic population mean, $\kappa=2$

Treatment: Chapter 6, page 184.

Global attractivity is unresolved; numerical screening is not a theorem.

Bio10.0 — Logistic population mean, $\kappa=10$

Treatment: Chapter 6, page 186.

Global attractivity is unresolved; numerical screening is not a theorem.

Dark0.5b1.0 — Darkness mean, $F=0.5$, $\beta=1$

Treatment: Chapter 7, page 200.

No negative response was found in the retained screen; a global proof is not supplied.

Dark0.5b8.0 — Darkness mean, $F=0.5$, $\beta=8$

Treatment: Chapter 7, page 202.

No negative response was found in the retained screen; a global proof is not supplied.

Dark1.0b1.0 — Darkness mean, $F=1$, $\beta=1$

Treatment: Chapter 7, page 204.

No negative response was found in the retained screen; a global proof is not supplied.

Dark1.0b8.0 — Darkness mean, $F=1$, $\beta=8$

Treatment: Chapter 7, page 206.

Stable negative finite-response evidence; for $F=1$ and $F=2$ there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Dark2.0b1.0 — Darkness mean, $F=2$, $\beta=1$

Treatment: Chapter 7, page 208.

No negative response was found in the retained screen; a global proof is not supplied.

Dark2.0b8.0 — Darkness mean, $F=2$, $\beta=8$

Treatment: Chapter 7, page 210.

Stable negative finite-response evidence; for $F=1$ and $F=2$ there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Dark4.0b1.0 — Darkness mean, $F=4$, $\beta=1$

Treatment: Chapter 7, page 212.

No negative response was found in the retained screen; a global proof is not supplied.

Dark4.0b8.0 — Darkness mean, $F=4$, $\beta=8$

Treatment: Chapter 7, page 214.

Stable negative finite-response evidence; for $F=1$ and $F=2$ there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Persist1.0 — Log- F averaged darkness mean, $\beta=1$

Treatment: Chapter 7, page 216.

No violation in the retained screen; global attractivity remains open.

Persist8.0 — Log- F averaged darkness mean, $\beta=8$

Treatment: Chapter 7, page 218.

Stable negative finite-displacement numerical witnesses; a validated continuum sign certificate remains open.

Opt0.1 — Nonlinear fluorescence mean, $\sigma=0.1$

Treatment: Chapter 7, page 220.

No violation in the retained 100-shape screen; global attractivity is open.

Opt0.3 — Nonlinear fluorescence mean, $\sigma=0.3$

Treatment: Chapter 7, page 222.

No violation in the retained 100-shape screen; global attractivity is open.

Opt1.0 — Nonlinear fluorescence mean, $\sigma=1$

Treatment: Chapter 7, page 224.

No violation in the retained 100-shape screen; global attractivity is open.

Illum — Renormalized inverse-square illumination center

Treatment: Chapter 7, page 226.

Numerical evidence of nonattractivity: the new 100-shape screen has a negative symmetric-Jacobian margin and the retained finite motion stays negative as its step is halved twice. No outward-rounded coordinate or continuum sign enclosure is supplied.

Solid0.1 — Solid-angle illumination maximum, $\text{height}/L=0.1$

Treatment: Chapter 7, page 228.

No failure found in the retained 100-shape screen at this height/L ; global attractivity and universal maximum uniqueness remain open. The family has a separate analytical nonattractivity result for all sufficiently small heights, inherited from the certified M1 limit. This does not classify the present parameter.

Solid1.0 — Solid-angle illumination maximum, $\text{height}/L=1$

Treatment: Chapter 7, page 230.

No failure found in the retained 100-shape screen at this height/L ; global attractivity and universal maximum uniqueness remain open. The family has a separate analytical nonattractivity result for all sufficiently small heights, inherited from the certified M1 limit. This does not classify the present parameter.

Solid10.0 — Solid-angle illumination maximum, $\text{height}/L=10$

Treatment: Chapter 7, page 232.

No failure found in the retained 100-shape screen at this height/L ; global attractivity and universal maximum uniqueness remain open. The family has a separate analytical nonattractivity

result for all sufficiently small heights, inherited from the certified M1 limit. This does not classify the present parameter.

M2 — Vertex-current magnitude mean

Treatment: Chapter 3, page 62.

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -1.61118595 and -1.61648215. No interval certificate is asserted.

M3 — Vertex-current energy limit

Treatment: Chapter 3, page 64. Exact alias under the stated convention: X360.

Strictly globally vertex-attractive: $M3 = X360$ exactly, with symmetric vertex-derivative eigenvalues $(\alpha \pm \sin \alpha)/\pi$.

M4 — Surface-current magnitude mean

Treatment: Chapter 3, page 66.

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.11778058 and -0.11804977. No interval certificate is asserted.

M5 — Surface-current energy mean

Treatment: Chapter 3, page 68.

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.72777353 and -0.72897714. No interval certificate is asserted.

M6 — Volume-current magnitude mean

Treatment: Chapter 3, page 70.

Global attractivity remains open. In addition to the earlier 100/284/2048-shape investigations, 512 new extreme shapes and 100 repeated shapes passed a complex-step response screen; ten difficult shapes retained positive signs through quadrature order 256. Initial subtraction-based negative values failed refinement and are not accepted as witnesses. No uniform positive lower bound or global theorem is claimed.

M7 — Volume-current energy mean

Treatment: Chapter 3, page 72.

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.10211640 and -0.10259846. No interval certificate is asserted.

L3 — Three-dimensional prism Robin center

Treatment: Chapter 2, page 38.

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01582971 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

L4 — Four-dimensional prism Robin center

Treatment: Chapter 2, page 39.

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01102191 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Aliases and extensions outside the 66-entry list

S2 is E2; Mem(2) is F1; MemMean(2) is Abs(0). The tetrahedral sheet gives X1. Reflecting equilibrium, complete high-energy spectral averaging and several controlled limits give X2. The two equal-current field zeros are an unordered Steiner-focal pair. Complete quantum projectors, canonical thermal means, conditioned Brownian means and other parameter extensions are defined in their chapters; they are not claimed to have a fresh complete ETC survey.

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